Positive Lower Density of Simultaneous Primitive Roots

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Abstract

Let A be a nonempty finite set of integers at least 2. We prove that at least $c_A x/\log x$ primes in every sufficiently large interval $(x, 2x]$ make all elements of A primitive roots, provided that no odd-cardinality subset of A has square product. We also prove the converse obstruction and deduce the corresponding result for any finite set of prime bases.

1 The main theorem

For p prime and $p \nmid a$, the multiplicative order $\mathrm{ord}_p(a)$ is the smallest positive integer k for which $a^k \equiv 1 \pmod p$. Thus a is a primitive root modulo p exactly when $\mathrm{ord}_p(a) = p-1$. For a finite set $A \subset \mathbb{N}_{\geq 2}$, write

$$N_A(x) = \#\left\{p \text{ prime}: \begin{array}{l} x < p \leq 2x, \quad p \nmid a \text{ for every } a \in A, \\ \mathrm{ord}_p(a) = p-1 \text{ for every } a \in A \end{array}\right\}.$$

Theorem 1.1. *Let $A \subset \mathbb{N}_{\geq 2}$ be nonempty and finite. Suppose that*

$$\prod_{a \in B} a \text{ is not a square for every } B \subseteq A \text{ with } |B| \text{ odd.} \tag{1.1}$$

There are constants $c_A > 0$ and $x_A \geq 2$ such that

$$N_A(x) \geq c_A \frac{x}{\log x} \qquad (x \geq x_A).$$

If (1.1) *fails, no odd prime makes every element of A a primitive root.*

We first construct many primes p for which every base is a quadratic nonresidue and $p-1$ has one very large prime factor. We then bound the candidates at which some base has smaller order. The candidate count decreases more slowly than the bound for these exceptions; choosing a fixed cutoff leaves a positive lower bound. All logarithms are natural. Letters used as indices of prime sums range over primes, and constants may depend on parameters already fixed.

2 Analytic theorems used in the proof

We state two analytic theorems in the precise forms needed here. The counting estimates needed for the main theorem are derived from them in the following sections. Their proofs are developed in the two later parts entitled Proof of Theorem 2.1 and Proof of Theorem 2.2.

Fix $1/2 < \alpha < \beta < 1$, and put

$$f_x(n) = \sum_{x^\alpha < Q \leq x^\beta} \mathbf{1}_{Q \mid n}, \qquad C_x = \sum_{x^\alpha < Q \leq x^\beta} \frac{1}{Q}.$$

Theorem 2.1 (Prime factor statistic in a fixed residue class)**.** *For every fixed integer $H \geq 1$, every reduced residue $b \pmod H$, and every nonnegative smooth function Φ with compact support in $(0, \infty)$,*

$$\sum_{\substack{p \text{ prime} \\ p \equiv b \ (H)}} \Phi\Big(\frac{p-1}{x}\Big) \left(f_x(p-1) - C_x\right) = o(x/\log x). \tag{2.1}$$

The modulus, residue, exponents, and function are fixed before x tends to infinity through real values.

Theorem 2.2 (Uniform Hecke zero-free region)**.** *Put $\delta = 10^{-6}$. Let F be a cyclotomic number field containing μ_{12}, and let χ be a finite-order Hecke character of F. The meromorphic continuation of $L_F(s, \chi)$ has no zero in*

$$\Re s > 1 - \delta.$$

The principal character is included, with its pole at $s = 1$ permitted. The same width δ applies to all fields and characters in the statement, without a height or conductor cutoff.

We also use the prime number theorem in each fixed reduced residue class and Mertens' formula

$$\sum_{p \leq y} \frac{1}{p} = \log\log y + M + o(1).$$

In particular,

$$C_x \longrightarrow \theta := \log(\beta/\alpha) > 0. \tag{2.2}$$

For the remainder of the proof fix

$$\eta = \frac{\delta}{2}, \qquad \alpha = 1 - \eta, \qquad \beta = 1 - \frac{\eta}{2}. \tag{2.3}$$

These choices satisfy $0 < \eta < \delta$ and $2\eta < 1$.

3 A supply of candidate primes

Choose once and for all a smooth function Φ such that

$$0 \leq \Phi \leq 1, \qquad \operatorname{supp}\Phi \subset (5/4, 7/4), \qquad m := \int_0^\infty \Phi(t)\,\mathrm{d}t > 0.$$

Lemma 3.1. *For every fixed reduced residue $b \pmod H$,*

$$\sum_{\substack{p \equiv b \ (H) \\ f_x(p-1)=1}} \Phi((p-1)/x) = \left(\frac{m\theta}{\varphi(H)} + o(1)\right) \frac{x}{\log x}. \tag{3.1}$$

Every prime counted with nonzero weight lies in $(x, 2x]$ and has

$$p - 1 = Qr, \qquad x^{1-\eta} < Q \leq x^{1-\eta/2}, \qquad r < 2x^{\eta}, \tag{3.2}$$

for all sufficiently large x.

Proof. The prime number theorem in the fixed class, followed by partial summation, gives

$$\sum_{p \equiv b \ (H)} \Phi((p-1)/x) = \left(\frac{m}{\varphi(H)} + o(1)\right) \frac{x}{\log x}.$$

Indeed p on the support is comparable to x, so its logarithm is $(1+o(1))\log x$, and changing p to $p-1$ does not affect the leading term. Equations (2.1) and (2.2) therefore imply

$$\sum_{p\equiv b\ (H)} \Phi((p-1)/x) f_x(p-1) = \left(\frac{m\theta}{\varphi(H)} + o(1)\right)\frac{x}{\log x}.$$

Since $2\alpha > 1$, eventually $x^{2\alpha} > 2x$. Two distinct prime factors greater than x^α cannot both divide a positive integer smaller than $2x$. Thus $f_x(p-1)$ is either 0 or 1 on the support of Φ, proving (3.1).

If the weight is nonzero, then $5x/4 < p-1 < 7x/4$. For $x \geq 4$ this implies $x < p \leq 2x$. When $f_x(p-1) = 1$, let Q be the unique prime factor in its defining interval and write $p-1 = Qr$. The bounds on Q, together with $p-1 < 2x$, give $r < 2x^{1-\alpha} = 2x^\eta$. This is (3.2). □

Because $0 \leq \Phi \leq 1$, a lower bound for this weighted count is also a lower bound for the number of candidates. Formula (3.1) can be added over any fixed finite set of reduced classes; the number of classes is fixed before taking the limit.

4 A uniform bound for power residue exceptions

For $a \geq 2$ and a prime $q > a$, put

$$K_{a,q} = \mathbb{Q}(\mu_q, a^{1/q}),$$

and define

$$B_{a,q}(x) = \#\left\{p \text{ prime}: \ x < p \leq 2x, \quad p \nmid aq, \ q \mid p-1, \quad a^{(p-1)/q} \equiv 1 \pmod p\right\}.$$

We shall prove

$$B_{a,q}(x) \ll_a \frac{x}{q(q-1)\log x} + \frac{x^{1-\delta}\log(2q)}{\log x}. \tag{4.1}$$

The constant is independent of q.

Lemma 4.1. *For every $a \geq 2$ and prime $q > a$, the degree $n_{a,q}$ and absolute discriminant $D_{a,q}$ of $K_{a,q}$ satisfy*

$$n_{a,q} = q(q-1), \qquad \log D_{a,q} + n_{a,q} \ll_a q(q-1)\log(2q).$$

Moreover, $\zeta_{K_{a,q}}$ has no zero in $\Re s > 1-\delta$.

Proof. Let $E = \mathbb{Q}(\mu_q)$ and choose a prime $\ell \mid a$. Since $q > a$, we have $q \neq \ell$ and $0 < v_\ell(a) < q$. The prime ℓ is unramified in E, so the valuation of a at each prime above ℓ is not divisible by q. Consequently a is not a qth power in E. Kummer theory gives $[K_{a,q} : E] = q$, and hence $[K_{a,q} : \mathbb{Q}] = q(q-1)$.

The cyclotomic discriminant is q^{q-2}. The relative discriminant divides the discriminant of $X^q - a$, so the discriminant tower formula gives

$$D_{a,q} \leq q^{q(q-2)}(q^q a^{q-1})^{q-1}.$$

Taking logarithms proves the required bound. This calculation also shows that rational primes not dividing aq are unramified.

Put $F = \mathbb{Q}(\mu_{12q})$ and $T = K_{a,q}F$. Both T/F and $T/K_{a,q}$ are abelian. Artin factorization over F gives

$$\zeta_T(s) = \prod_\chi L_F(s,\chi),$$

where the factors are the Hecke functions corresponding to the characters of $\mathrm{Gal}(T/F)$. Every factor is covered by Theorem 2.2, so ζ_T has no zero in the stated half-plane. Factorization over $K_{a,q}$ gives

$$\zeta_T(s) = \zeta_{K_{a,q}}(s) \prod_{\psi\neq 1} L_{K_{a,q}}(s,\psi).$$

The nonprincipal factors on the right are entire. A zero of $\zeta_{K_{a,q}}$ away from 1 would therefore give a zero of ζ_T, a contradiction. At 1, both zeta functions have a simple pole, rather than a zero. □

Lemma 4.2 (Smoothed prime ideal bound)**.** *Let K be a number field of degree n and absolute discriminant D. Suppose that ζ_K has no zero in $\Re s > 1-\delta$. For a fixed nonnegative $g \in C_c^\infty((0,\infty))$ and $x \geq 2$,*

$$\Theta_{K,g}(x) := \sum_{\mathfrak{p}} \sum_{j\geq 1} \log N\mathfrak{p}\, g\left(\frac{(N\mathfrak{p})^j}{x}\right) \ll_g x + x^{1-\delta}(\log D + n). \tag{4.2}$$

The constant is independent of K and x.

Proof. Write

$$G(s) = \int_0^\infty g(t) t^{s-1}\, \mathrm{d}t.$$

Integration by parts gives $G(s) \ll_{g,A} (1+|\Im s|)^{-A}$ on every fixed vertical strip, for every $A > 0$. Mellin inversion gives

$$\Theta_{K,g}(x) = \frac{1}{2\pi i} \int_{\Re s = 2} -\frac{\zeta_K'}{\zeta_K}(s) x^s G(s)\, \mathrm{d}s.$$

We move the contour to $\Re s = -1/2$. If (r_1, r_2) is the signature of K, the residues yield

$$\Theta_{K,g}(x) = xG(1) - \sum_\rho x^\rho G(\rho) - (r_1 + r_2 - 1)G(0) + O_g(x^{-1/2}(\log D + n)), \tag{4.3}$$

where the sum runs over nontrivial zeros with multiplicity. We recall the estimates that both justify the shift and keep the constant uniform in K.

On $\Re s = -1/2$, the functional equation and Stirling's formula give

$$\left|\frac{\zeta_K'}{\zeta_K}(s)\right| \ll \log D + n\log(2+|\Im s|).$$

The Euler series on the reflected line $\Re(1-s) = 3/2$ is $O(n)$; the remaining terms are the discriminant and the n gamma factors. The zero-counting estimate on unit intervals is

$$\#\{\rho : k \leq \Im\rho < k+1\} \ll \log D + n\log(2+|k|), \qquad k \in \mathbb{Z}.$$

It follows from the completed zeta function's Hadamard product and its functional equation. The implied constant is absolute. For each fixed field, horizontal contour edges can therefore be chosen tending to infinity and avoiding zeros by an inverse polynomial distance. The logarithmic derivative is polynomially bounded on those edges, and the rapid decay of G makes their integrals tend to zero. This proves (4.3).

Every nontrivial zero satisfies $\Re\rho \leq 1-\delta$. Applying the unit-interval count and the decay of G gives

$$\begin{aligned}\sum_\rho |x^\rho G(\rho)| &\ll_g x^{1-\delta} \sum_{k\in\mathbb{Z}} (1+|k|)^{-3}(\log D + n\log(2+|k|)) \\ &\ll_g x^{1-\delta}(\log D + n).\end{aligned}$$

The residue at 0 is $O_g(n)$. These estimates in (4.3) prove (4.2). □

Proof of (4.1). Choose a fixed $g \geq 0$ with $g \geq 1$ on $[1,2]$. If $p \nmid aq$, the conditions defining $B_{a,q}(x)$ are equivalent to $X^q - a$ splitting completely over $\mathbb{F}_p$: the congruence $q \mid p-1$ supplies all qth roots of unity, and $a^{(p-1)/q} = 1$ says that a is a qth power in $\mathbb{F}_p^\times$. Thus p splits completely in $K_{a,q}$. Each

such $p \in (x, 2x]$ gives $n_{a,q}$ prime ideals of norm p, and contributes at least $n_{a,q} \log x$ to $\Theta_{K_{a,q},g}(x)$. By positivity and Lemmas 4.1–4.2,

$$\begin{aligned} B_{a,q}(x) &\le \frac{\Theta_{K_{a,q},g}(x)}{n_{a,q} \log x} \\ &\ll_g \frac{x}{n_{a,q} \log x} + \frac{x^{1-\delta}(\log D_{a,q} + n_{a,q})}{n_{a,q} \log x} \\ &\ll_a \frac{x}{q(q-1) \log x} + \frac{x^{1-\delta} \log(2q)}{\log x}. \end{aligned}$$

This proves (4.1). □

Lemma 4.3 (Summed exceptions)**.** *For a fixed nonempty finite $A \subset \mathbb{N}_{\ge 2}$, there is $C_A > 0$ such that, for every fixed integer $Y \ge \max(3, \max A)$,*

$$\sum_{a \in A} \sum_{Y < q \le 2x^\eta} B_{a,q}(x) \le \left(\frac{C_A}{Y} + o_{A,Y}(1) \right) \frac{x}{\log x}. \tag{4.4}$$

The constant C_A is independent of Y.

Proof. For the main terms in (4.1),

$$\sum_{q > Y} \frac{1}{q(q-1)} \le \sum_{n=Y+1}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = \frac{1}{Y}.$$

For the error terms, there are at most $2x^\eta$ summands, and $\log(2q) \ll_\eta \log x$ in the stated range. Hence their total is $O_A(x^{1-\delta+\eta}) = o(x/\log x)$, because $\eta < \delta$. Summing over the fixed finite set A proves the claim. □

5 Choosing the quadratic signs

Lemma 5.1. *If A satisfies* (1.1)*, there are an integer $H_0 \ge 1$ and a reduced residue $b_0 \pmod{H_0}$ such that every prime in this class makes each $a \in A$ a quadratic nonresidue. Moreover, no odd prime dividing H_0 divides $p - 1$.*

Proof. Let S contain $2, 3$ and all prime divisors of $\prod_{a \in A} a$. For $a \in A$, let $v_a \in \mathbb{F}_2^S$ be the vector of its prime exponents modulo 2. If $\sum_{a \in B} v_a = 0$, then $\prod_{a \in B} a$ is a square. By (1.1), $|B|$ must be even. Consequently the rule $v_a \mapsto 1$ extends linearly to a well-defined functional on the span of the v_a. Extend it to all of $\mathbb{F}_2^S$ and write it as $v \mapsto v \cdot z$. Put $s_\ell = (-1)^{z_\ell}$.

Choose a residue modulo 24 according to

(s_2, s_3)	$(-1,+1)$	$(+1,+1)$	$(-1,-1)$	$(+1,-1)$
$p \bmod 24$	11	23	5	17.

This makes $(2/p) = s_2$, $(3/p) = s_3$, and $p \equiv 2 \pmod 3$. It also fixes $\tau = (-1/p)$. For each prime $\ell \in S$ with $\ell \ge 5$, choose $r_\ell \not\equiv 0, 1 \pmod \ell$ such that

$$\left(\frac{r_\ell}{\ell} \right) = s_\ell \tau^{(\ell-1)/2}.$$

This is possible because there is both a nonresidue and a nonzero square other than 1 modulo every prime $\ell \ge 5$. The Chinese remainder theorem gives a reduced class b_0 modulo

$$H_0 = 24 \prod_{\substack{\ell \in S \\ \ell \ge 5}} \ell.$$

Quadratic reciprocity now gives $(\ell/p) = s_\ell$ for all $\ell \in S$. Therefore

$$\left(\frac{a}{p}\right) = (-1)^{v_a \cdot z} = -1 \qquad (a \in A).$$

The choices also give $p \not\equiv 1 \pmod{\ell}$ for every odd $\ell \mid H_0$. □

Fix an integer $Y \geq \max(3, \max A)$. Retain the class from Lemma 5.1. For every odd prime $\ell \leq Y$ not dividing H_0, permit all residues except 0 and 1 modulo ℓ. The Chinese remainder theorem gives a finite union of reduced classes, whose relative density among primes is

$$\lambda_{A,Y} = \frac{1}{\varphi(H_0)} \prod_{\substack{3 \leq \ell \leq Y \\ \ell \nmid H_0}} \frac{\ell - 2}{\ell - 1}. \tag{5.1}$$

Every prime in this union makes all bases nonresidues and satisfies

$$\ell \nmid p - 1 \quad \text{for every odd prime } \ell \leq Y. \tag{5.2}$$

Lemma 5.2. *If $T \geq 1$ is an integer and $Y = 2T^2 + 1$, then*

$$\lambda_{A,Y} \geq \frac{1}{2\varphi(H_0)T}.$$

Proof. Adding the omitted factors indexed by odd integers decreases the product in (5.1). Thus

$$\varphi(H_0)\lambda_{A,Y} \geq \prod_{j=1}^{T^2} \frac{2j-1}{2j} = \frac{\binom{2T^2}{T^2}}{4^{T^2}}.$$

For $u_n = \binom{2n}{n}/4^n$ we have $u_1 = 1/2$ and

$$\frac{u_{n+1}}{u_n} = \frac{2n+1}{2n+2} \geq \sqrt{\frac{n}{n+1}}.$$

Induction gives $u_n \geq 1/(2\sqrt{n})$. Take $n = T^2$. □

6 Removing the order obstructions

For $p \nmid a$ put $h_a(p) = (p-1)/\operatorname{ord}_p(a)$. This is a positive integer, and a is a primitive root exactly when $h_a(p) = 1$.

Lemma 6.1. *If q is prime and $q \mid p - 1$, then*

$$q \mid h_a(p) \quad \Longleftrightarrow \quad a^{(p-1)/q} \equiv 1 \pmod{p}.$$

If a is a quadratic nonresidue modulo the odd prime p, then $h_a(p)$ is odd.

Proof. Writing $d = \operatorname{ord}_p(a)$, the congruence is equivalent to $d \mid (p-1)/q$, which is equivalent to $q \mid (p-1)/d$. For $q = 2$, the congruence is excluded by Euler's criterion for a quadratic nonresidue. □

Lemma 6.2. *For fixed $a \geq 2$, $x > 1$ and $U \geq 1$,*

$$\#\{p > x : p \text{ prime},\ p \nmid a,\ \operatorname{ord}_p(a) \leq U\} \ll_a \frac{U^2}{\log x}.$$

Proof. Every such prime divides the positive integer

$$M_U = \prod_{1 \le j \le \lfloor U \rfloor} (a^j - 1).$$

Moreover,

$$\log M_U \le (\log a) \sum_{j \le U} j \ll_a U^2.$$

If there are r distinct primes in question, their product divides M_U and exceeds x^r. Hence $r \log x \le \log M_U$, proving the claim. The same conclusion holds when $M_U = 1$. □

Lemma 6.3. *Consider candidate primes satisfying* (3.2), (5.2), *and* $(a/p) = -1$ *for every* $a \in A$. *The number at which some base fails to be a primitive root is*

$$\le \left(\frac{C_A}{Y} + o_{A,Y}(1) \right) \frac{x}{\log x},$$

where $C_A > 0$ *is independent of the fixed* Y.

Proof. For sufficiently large x, none of the primes in question divides a base. If $h_a(p) > 1$, choose a prime $q \mid h_a(p)$. By Lemma 6.1, q is odd. By (5.2), $q > Y$. There are two cases.

If $q \neq Q$, then $q \mid r$, because $p - 1 = Qr$. Thus $q < 2x^\eta$, and the defining congruence in Lemma 6.1 puts p among the primes counted by $B_{a,q}(x)$. Lemma 4.3 bounds all these exceptions by $(C_A/Y + o_{A,Y}(1))x/\log x$.

If $q = Q$, then

$$\operatorname{ord}_p(a) \le \frac{p-1}{Q} = r < 2x^\eta.$$

Lemma 6.2 bounds this case by $O_a(x^{2\eta}/\log x) = o(x/\log x)$, since $2\eta < 1$. Sum over the fixed finite set A. This changes only the constant and completes the proof. □

7 Proof of the main theorem

Proof of the sufficient direction of Theorem 1.1. Construct H_0 as in Lemma 5.1, and put $M = m\theta > 0$. Choose an integer $T \ge 1$ sufficiently large that

$$T \ge \frac{4C_A\varphi(H_0)}{M}, \qquad Y = 2T^2 + 1 \ge \max(3, \max A). \tag{7.1}$$

Keep T and Y fixed.

Apply Lemma 3.1 to each of the finitely many classes in (5.1). Their total candidate weight is

$$(M\lambda_{A,Y} + o_{A,Y}(1)) \frac{x}{\log x}.$$

Every candidate satisfies all the conditions of Lemma 6.3. Since $0 \le \Phi \le 1$, removing those exceptions loses no more weight than their number. Therefore

$$N_A(x) \ge \left(M\lambda_{A,Y} - \frac{C_A}{Y} + o_{A,Y}(1) \right) \frac{x}{\log x}. \tag{7.2}$$

By Lemma 5.2 and (7.1),

$$M\lambda_{A,Y} \ge \frac{M}{2\varphi(H_0)T}, \qquad \frac{C_A}{Y} < \frac{C_A}{2T^2} \le \frac{M}{8\varphi(H_0)T}.$$

For all sufficiently large x, the absolute value of the little-oh term in (7.2) is at most $M/(8\varphi(H_0)T)$. Thus

$$N_A(x) \ge \frac{M}{4\varphi(H_0)T} \frac{x}{\log x}.$$

Set $c_A = M/(4\varphi(H_0)T) > 0$ and increase x_A if necessary. All choices depending on A were fixed before taking x large. □

Proof of the necessary direction of Theorem 1.1. If (1.1) fails, choose an odd-cardinality $B \subseteq A$ whose product is a square. Suppose that every $a \in A$ were a primitive root modulo some odd prime p. None is divisible by p, and every one has Legendre symbol -1. Consequently

$$1 = \left(\frac{\prod_{a\in B} a}{p}\right) = \prod_{a\in B}\left(\frac{a}{p}\right) = (-1)^{|B|} = -1,$$

a contradiction. □

8 Consequences

Corollary 8.1. *For every A satisfying Theorem 1.1,*

$$\liminf_{X\to\infty} \frac{\#\{p \le X : p \text{ prime},\ p \nmid a, \operatorname{ord}_p(a) = p-1 \text{ for all } a \in A\}}{\pi(X)} > 0.$$

Proof. The numerator is at least $N_A(X/2)$. The theorem and $\pi(X) \sim X/\log X$ give a positive lower bound for the ratio. □

Corollary 8.2 (Prime bases)**.** *For every nonempty finite set A of positive primes, there are $d_A > 0$ and $y_A > 1$ such that*

$$\#\{p \text{ prime} : x < p < 2x,\ p \nmid a, \operatorname{ord}_p(a) = p-1 \text{ for all } a \in A\} \ge d_A \frac{x}{(\log x)^2} \qquad (x \ge y_A).$$

Proof. A nonempty product of distinct primes is squarefree and greater than 1, hence not a square. The set A therefore satisfies (1.1). For large real x, put $n = \lfloor x \rfloor$. Then $n \ge x/2$, $n \ge x_A$, and $\log x \ge 1$. If $n < p \le 2n$ is prime, then $p \ge n+1 > x$ and $p < 2n \le 2x$, because $2n$ is composite. Hence

$$\begin{aligned}\#\{p \in (x, 2x) : p \text{ is good for all bases in } A\} &\ge c_A \frac{n}{\log n} \\ &\ge \frac{c_A}{2}\frac{x}{\log x} \ge \frac{c_A}{2}\frac{x}{(\log x)^2}.\end{aligned}$$

Take $d_A = c_A/2$ and increase the starting point. □

For $A = \{2, 3\}$, the only odd-cardinality subsets are the two singletons, so the square condition holds. The set $\{3, 6\}$ also satisfies it. In contrast, $\{2, 3, 6\}$ fails the condition, since $2 \cdot 3 \cdot 6 = 36$. Thus no odd prime works simultaneously for all three of these bases.

Proof of Theorem 2.1

The proof below develops the marked correlation estimates, the sieve extraction and the extension to a fixed residue class. We use the prime number theorem, Siegel–Walfisz and Mertens as known theorems. The nonstandard correlation and extraction steps are proved in the following sections. The notation is local to each construction; in particular the growing determinant padding and the fixed graph padding are different parameters.

9 Notation and preliminary estimates

All factor variables are positive integers unless another domain is specified. We write $P^+(n)$ and $P^-(n)$ for the largest and least prime factors of $n > 1$, with $P^+(1) = 1$ and $P^-(1) = \infty$. An integer is y-smooth if $P^+(n) \le y$, and is rough above y if $P^-(n) > y$. We use φ for Euler's function, τ for the divisor function, and

$$e(t) = \exp(2\pi i t), \qquad (z)_j = z(z-1)\cdots(z-j+1), \qquad (z)_0 = 1.$$

We write $\mu(n) = 0$ if n is divisible by a prime square, and $\mu(n) = (-1)^r$ if n is a product of r distinct primes. The symbol ω will count distinct prime divisors in the specified prime groups; an unrestricted distinct-prime count will be identified when it occurs. In particular, multiplicities in an integer do not increase its group count.

Throughout the analytic argument,

$$L = \log x, \qquad T = \log L, \qquad W = \exp(\sqrt{L}), \tag{9.1}$$

and x tends to infinity. A dyad is an interval $[Y, 2Y)$, or a fixed constant enlargement of one; subinterval restrictions will be allowed explicitly. We write $n \asymp Y$ when n is bounded above and below by fixed positive multiples of Y.

Convention 9.1 (Parameters and uniformity)**.** Every constant is fixed before $x \to \infty$. A bound $L^{O(1)}$ has an exponent independent of x. Its dependence on earlier fixed parameters is permitted unless explicitly excluded. For a two-sided size statement $Y = xL^{O(1)}$, we mean $|\log(Y/x)| = O(T)$. Arbitrary logarithmic accuracy means a bound $O_A(L^{-A})$ for every desired fixed A, with the auxiliary choices made in the specified order.

The geometric parameters used in the final sieve application are chosen first. The required Type II accuracy is chosen next. Inside the analytic argument the physical and ideal norm targets precede the comparison kernels; the mark length J is chosen after those kernels. The strength of the character and Mellin discrepancy is chosen after the graph and Fourier parameters, and the precision of the cofactor proxies is chosen last. Each result below specifies the uniformity needed to respect this order.

All Hilbert spaces are complex. An operator defined by transitions acts on a function at the target and sums or integrates its weighted values at the input. Symmetrization of a list means the orthogonal projection that averages all permutations of its coordinates.

9.1 Classical prime estimates

We use the prime number theorem with an error smaller than any fixed negative power of the logarithm, its Siegel–Walfisz extension to small moduli, and Mertens' estimates

$$\sum_{p \le y} \frac{1}{p} = \log\log y + O(1), \qquad \prod_{p \le y} \left(1 - \frac{1}{p}\right) \sim \frac{e^{-\gamma_E}}{\log y}. \tag{9.2}$$

Here and below sums or products indexed by p are over primes. The following forms of Siegel–Walfisz and the multiplicative large sieve will be used. The former follows from its von Mangoldt

formulation by partial summation; see the Siegel–Walfisz theorem. For the latter see the classical large-sieve and derivative estimates.

Theorem 9.2 (Siegel–Walfisz)**.** *For fixed $A, C > 0$, uniformly for $(a, r) = 1$ and $1 \le r \le (\log y)^C$,*

$$\#\{p \le y : p \equiv a \pmod r\} = \frac{\mathrm{Li}(y)}{\varphi(r)} + O_{A,C}\Big(\frac{y}{(\log y)^A}\Big).$$

The constants, which need not be effective, are independent of y, a, r.

Theorem 9.3 (Multiplicative large sieve)**.** *Let I be a real interval of length H, let a_n be complex numbers supported on $I \cap \mathbb{Z}$, and let $R \ge 1$. Then*

$$\sum_{r \le R} \frac{r}{\varphi(r)} \sideset{}{^*}\sum_{\chi \bmod r} \Big| \sum_{n \in I \cap \mathbb{Z}} a_n \chi(n) \Big|^2 \ll (H + 1 + R^2) \sum_n |a_n|^2,$$

where the asterisk restricts the sum to primitive Dirichlet characters.

We will use absolute character estimates on intervals that may be very short. The next consequence records precisely what is required; it does not require a relative prime asymptotic on every such interval.

Corollary 9.4 (Harmonic character estimates)**.** *Fix $c, C, B, A > 0$. If χ is nonprincipal modulo $r \le L^B$, then, uniformly for real $|t| \le L^B$ and intervals $I \subset [\exp(L^c), x^C]$,*

$$\sum_{p \in I} \frac{\chi(p) p^{it}}{p} \ll_{c,C,B,A} L^{-A}.$$

The same conclusion holds for a nonprincipal character induced from a smaller modulus.

Proof. Writing $A_\chi(y) = \sum_{p \le y} \chi(p)$, sum Theorem 9.2 against χ over reduced residue classes. The main terms cancel, and the loss from the number of classes is at most L^B. Because $\log y \ge L^c$, the permitted moduli are bounded by a fixed power of $\log y$. Thus $A_\chi(y) \ll_K L^B y (\log y)^{-K}$ for every fixed K. Partial summation on $I = (a, b]$ against y^{-1+it} gives boundary terms of size $O_K(L^B (\log a)^{-K})$ and an integral bounded by

$$C_K L^B (1 + |t|) \int_a^b \frac{dy}{y (\log y)^K} \ll_K L^{2B} (\log a)^{1-K}.$$

Choose K sufficiently large. The argument is uniform in both endpoints and also covers an empty interval. An induced nonprincipal character is itself nonprincipal on the modulus on which it is used, so the same argument applies. □

9.2 Divisors, Dirichlet polynomials, and smooth separation

Lemma 9.5 (Fixed divisor moments)**.** *For each fixed nonnegative integer k,*

$$\sum_{n \le Y} \frac{\tau(n)^k}{n} \ll_k (\log(2Y))^{C_k}, \qquad \sum_{n \le Y} \tau(n)^k \ll_k Y (\log(2Y))^{C_k}$$

for a constant C_k. Consequently, a convolution of a fixed number of sequences bounded by fixed logarithmic powers has coefficient-square sum $U L^{O(1)}$ on a dyad of size $U \le x^{O(1)}$. If its coefficients include the reciprocal of the product index, this bound is $L^{O(1)}/U$ instead.

Proof. Positivity and unique factorization bound the harmonic sum by

$$\prod_{p\le Y}\sum_{a\ge 0}\frac{(a+1)^k}{p^a}=\prod_{p\le Y}\left(1+\frac{2^k}{p}+O_k(p^{-2})\right)\ll_k (\log(2Y))^{C_k},$$

using Equation (9.2). The counting bound follows by multiplying by Y. If there are b convolution factors, their coefficient at n has absolute value at most a fixed logarithmic power times the number of ordered b-factorizations of n. The latter is at most $\tau(n)^{b-1}$: choose the first $b-1$ divisors, which determine the last factor. Apply the counting moment bound with $k=2(b-1)$. On a dyad the reciprocal index is comparable to $1/U$. □

The next coefficient-independent estimate is a weaker elementary form of the Dirichlet-polynomial mean-value theorem; compare the Dirichlet polynomial mean-value theorem.

Lemma 9.6 (Mean squares)**.** *If $P(t)=\sum_{n\le Y}c_n n^{it}$ and I is an interval of length H, then*

$$\int_I |P(t)|^2\,dt\ll (H+Y\log(2Y))\sum_n |c_n|^2.$$

For a set $\mathcal{T}$ of real points at mutual distance at least one, contained in an interval of length H, one also has

$$\sum_{t\in\mathcal{T}}|P(t)|^2\ll (H+1+Y\log(2Y))(\log(2Y))^2\sum_n |c_n|^2. \tag{9.3}$$

The constants are absolute, independently of any factorization used to form the coefficients c_n.

Proof. Integration gives the diagonal $H\sum|c_n|^2$. For $n\ne m$, the absolute value of the exponential integral is at most $2/|\log(n/m)|\ll Y/|n-m|$. Apply $2|c_nc_m|\le |c_n|^2+|c_m|^2$ and sum $1/|n-m|$ to obtain the first bound. On the unit interval centered at t, the elementary one-dimensional Sobolev inequality bounds $|P(t)|^2$ by a constant times the integral of $|P|^2+|P'|^2$. These unit intervals have bounded overlap. The derivative has coefficients $i(\log n)c_n$, so the first bound applied twice gives Equation (9.3). □

We also use two elementary exponential-sum estimates in the following forms; see the classical large-sieve and derivative estimates.

Lemma 9.7 (Derivative tests)**.** *Let f be real-valued on an interval containing H consecutive integers.*

1. *If f' is monotone and stays in $[j+\lambda, j+1-\lambda]$ for some integer j and $0<\lambda\le 1/2$, then $\sum_n e(f(n))\ll\lambda^{-1}$.*
2. *If f is twice continuously differentiable and $\lambda\le|f''|\le C\lambda$ throughout the interval, then $\sum_n e(f(n))\ll_C H\lambda^{1/2}+\lambda^{-1/2}$.*

Both estimates hold on every subinterval on which the stated hypotheses hold.

Lemma 9.8 (Fourier separation)**.** *Let F be smooth, supported in a fixed bounded box in $\mathbb{R}^b$, where b is fixed. Suppose that for every nonnegative integer j its derivatives of order j are bounded by $C_jL^{C(j+1)}$. Then Fourier inversion separates the variables with integrated absolute coefficient mass $L^{O(1)}$. There is a fixed $C'>0$, depending on b, C, such that restricting each Fourier frequency to absolute value at most $L^{C'}$ leaves an error $O_A(L^{-A})$ for every fixed A. The exponent C' may be fixed before the desired A.*

Proof. Repeated integration by parts gives a bound for $\widehat{F}(\xi)$ by a derivative norm times an arbitrary fixed negative power of $1+|\xi|$. Taking more than b derivatives first makes this bound integrable, with a fixed power of L. For the tail, take $j>b$ derivatives and integrate outside $|\xi|\ge L^{C'}$. The resulting bound is at most a constant depending on j times

$$L^{C(j+1)-C'(j-b)}.$$

Fix $C' > C$ sufficiently large for the chosen Fourier convention and box. Increasing j then supplies any prescribed negative power of L. Fourier inversion expresses the integrand as a product of one-variable phases. Applying this in normalized logarithmic coordinates gives the corresponding Mellin separation on fixed dyads. Fixed-dimensional smooth cutoffs localized at logarithmic precision are included in the derivative hypothesis. □

9.3 An elementary weighted sieve

We record a form with explicit coefficient and level bounds. This will be used both for congruence counts and for oscillatory sums over rough integers. The construction is a version of the Brun–Hooley sieve, using a product of block upper bounds and a one-block correction. We prove the form needed here, including its coefficient and level bounds.

Lemma 9.9 (Block sieve)**.** *Consider a finite set of objects with nonnegative weights. For some primes $p \le z$, let a bad condition be specified. If d is a squarefree product of these primes, suppose the weight of the objects on which all conditions at $p \mid d$ hold is*

$$Xg(d) + r(d),$$

including $d = 1$, where $X \ge 0$, g is multiplicative, and

$$0 \le g(p) \le 1 - \eta_0, \qquad \sum_{w<p\le w^2} g(p) \le C \quad (w > 1) \tag{9.4}$$

for fixed $\eta_0 > 0$ and C. For every sufficiently large even integer h, the weight S of objects avoiding all bad conditions satisfies

$$S = X \prod_{p\le z}(1 - g(p))(1 + O(e^{-h})) + O\left(\sum_{d\le z^{4h+2}} |r(d)|\right). \tag{9.5}$$

Only the indicated primes and their squarefree products are used. The implied constants depend on η_0, C, uniformly also when h grows. For the fixed even choice $h = 2$, there is an upper bound by a constant times $X\prod(1 - g(p))$ plus the same remainder sum.

More precisely, the proof supplies polynomials U and V in the indicators of the bad conditions such that

$$V \le \mathbf{1}_{\textit{no bad condition}} \le U, \qquad U \ge 0.$$

Both have coefficients of absolute value at most one, supported on squarefree $d \le z^{4h+2}$. The nonnegative overcount $U - \mathbf{1}_{\textit{no bad condition}}$ is dominated by the pointwise nonnegative polynomial $U - V$, whose absolute coefficients are also at most one and which has the same level bound.

Proof. Partition the indicated primes into blocks

$$\mathcal{B}_j = \{p : z^{2^{-j-1}} < p \le z^{2^{-j}}\}, \qquad j \ge 0.$$

Only finitely many blocks are nonempty. Set $h_j = (j + 1)h$, which is even. In block j, write I_j for avoidance of all its bad conditions, U_j for the inclusion–exclusion polynomial through degree h_j, and E_j for the sum of all monomials of degree $h_j + 1$. If exactly t bad conditions hold in the block, then

$$U_j = \sum_{a=0}^{h_j} (-1)^a \binom{t}{a} = \begin{cases} 1, & t = 0, \\ \binom{t-1}{h_j}, & t \ge 1. \end{cases}$$

Here a binomial coefficient is zero when its lower index exceeds its nonnegative upper index. It follows that $U_j \ge I_j \ge 0$ and $U_j - I_j \le E_j$. Telescoping a product of nonnegative factors gives

$$0 \le \prod_j U_j - \prod_j I_j \le \sum_j E_j \prod_{k\ne j} U_k.$$

The upper and lower polynomials are therefore

$$U = \prod_j U_j, \qquad V = U - \sum_j E_j \prod_{k \neq j} U_k. \tag{9.6}$$

In U, every block has degree at most h_j. A monomial from correction j has degree exactly $h_j + 1$ in block j and at most h_k elsewhere. These supports are disjoint across j and disjoint from the support of U. Thus the absolute coefficients in U, V are at most one. The same disjointness gives this bound for $U - V$.

The logarithm of a product in U, divided by $\log z$, is at most

$$\sum_{j \geq 0} h(j+1)2^{-j} = 4h.$$

A correction adds at most one more unit to this upper bound. Hence the stated level z^{4h+2} is valid for all the polynomials above.

Now give the bad conditions independent probabilities $g(p)$. Their block avoidance probability $a_j = \mathbb{E} I_j$ obeys

$$a_j = \prod_{p \in \mathcal{B}_j} (1 - g(p)) \geq \exp(-C/\eta_0) =: a_* > 0,$$

because $-\log(1-u) \leq u/\eta_0$ for $0 \leq u \leq 1 - \eta_0$. Also

$$e_j := \mathbb{E} E_j \leq \frac{C^{h_j+1}}{(h_j+1)!}, \qquad a_j \leq \mathbb{E} U_j \leq a_j + e_j.$$

For all large even h,

$$R_h := \sum_j e_j / a_j \ll e^{-h}.$$

Indeed the factorial tails eventually decrease geometrically in their index, and $h_j = (j+1)h$; the same sum is bounded by an absolute constant when $h = 2$. Independence between blocks yields

$$\prod_j a_j \leq \mathbb{E} U \leq \prod_j a_j \, e^{R_h}, \qquad \mathbb{E} \sum_j E_j \prod_{k \neq j} U_k \leq \prod_j a_j \, R_h e^{R_h}.$$

Consequently the expectations of both bounding polynomials differ from $\prod a_j$ by $O(e^{-h}) \prod a_j$ for large h. For $h = 2$, the upper expectation is at most a constant times $\prod a_j$.

Evaluate the two polynomials in the original counting problem. The main terms of their monomials are exactly their independent expectations multiplied by X. The absolute remainder for either polynomial is at most $\sum_{d \leq z^{4h+2}} |r(d)|$, by the coefficient bound. Squeezing S between them proves the result. □

Corollary 9.10 (A polynomial for rough integers)**.** *For the ordinary bad conditions $p \mid n$ at primes $p \leq z$, let $U(n) = \sum_{d \mid n} \lambda_d$ be the upper polynomial in Equation* (9.6)*, and put $D = z^{4h+2}$. Then*

$$U(n) \geq \mathbf{1}_{P^-(n) > z} \geq 0, \quad |\lambda_d| \leq 1, \quad \lambda_d = 0 \quad \textit{unless } d \leq D \textit{ is squarefree,}$$

and

$$\sum_{n \in I \cap \mathbb{N}} \left(U(n) - \mathbf{1}_{P^-(n) > z} \right) \ll |I| e^{-h} + D \tag{9.7}$$

for every finite interval $I \subset [1, \infty)$ and all sufficiently large even h. The bound is uniform in h, z, I. Moreover,

$$\sum_d \frac{|\lambda_d|}{d} \leq \prod_{p \leq z} (1 + p^{-1}) \ll \log(2z). \tag{9.8}$$

Proof. The pointwise claims and harmonic sum follow from the coefficient and support bounds. For the overcount, use its pointwise upper bound $U - V = \sum_j E_j \prod_{k \neq j} U_k$, which is nonnegative pointwise and has absolute coefficients at most one. In the interval I, the count of multiples of d is $|I|/d + O(1)$. The independent expectation of $U - V$ was bounded in the proof by $O(e^{-h}) \prod_{p \leq z} (1 - 1/p)$; the sum of absolute remainders is $O(D)$. Dropping the avoidance product proves Equation (9.7). Finally, $\prod (1 + 1/p) \leq \prod (1 - 1/p)^{-1} \ll \log(2z)$ by Mertens. □

10 Transference for dilation graphs

We transfer a norm estimate for independent prime labels to an averaged moment of a dilation graph whose labels must divide the integer at their vertex. Averaging a required divisibility condition supplies a factor $1/p$, which explains the reciprocal-prime law in the independent model. A prime used at several vertices pays this cost only once. The main problem is to retain that dependence while using the independent-model norm on most edges. The resulting moment will control pairings in which one endpoint is constant on all mark lists at its integer position. The memory identity and transference estimate below are proved for the present model.

10.1 Prime groups and the two operators

Definition 10.1 (Prime groups)**.** Fix constants

$$0 < a < b < c < d < 0.47, \qquad c_0, C_0, v_-, v_+ > 0, \qquad 0 < q < 1, \qquad \ell \in \mathbb{N}.$$

For each sufficiently large x, let the finite, pairwise disjoint sets of primes $\mathcal{P}_g$ be indexed by the disjoint sets $\mathcal{S}$ and $\mathcal{B}$, where

$$c_0 T \le |\mathcal{S}|, |\mathcal{B}| \le C_0 T.$$

The small and big groups, respectively, satisfy

$$\begin{aligned} &\exp(L^a) \le p \le \exp(L^b) \quad (p \in \mathcal{P}_g,\ g \in \mathcal{S}), \\ &\exp(L^c) \le p \le \exp(L^d) \quad (p \in \mathcal{P}_g,\ g \in \mathcal{B}), \\ &v_- \le V_g := \sum_{p \in \mathcal{P}_g} \frac{1}{p} \le v_+, \qquad \mu_g(p) := \frac{1}{V_g p}. \end{aligned} \tag{10.1}$$

Write $s = |\mathcal{S}| + |\mathcal{B}|$ and

$$\omega_g(n) = \sum_{p \in \mathcal{P}_g} \mathbf{1}_{p|n}, \qquad \omega(n) = \sum_g \omega_g(n) \quad (n \in \mathbb{Z}).$$

Thus divisibility at $n = 0$ has its usual meaning. We call primes in $\bigcup_g \mathcal{P}_g$ group primes.

Let J be a sufficiently large fixed integer and put $M = J + \ell$. An ordered label list has M slots in each group. A pattern ν specifies sets $C_g \subseteq \{1, \dots, J\}$, with

$$C_g = \varnothing \quad (g \in \mathcal{S}), \qquad |C_g| \le m, \qquad t_g = \ell + |C_g|.$$

The slots in $\{1, \dots, J\} \setminus C_g$ are shared; their labels are copied from the source list to the target list. The other t_g slots in each endpoint list are unshared. To form D, use the shared labels and, in the C_g slots, independent labels of law μ_g. Thus D has J prime factors from each group, counted with multiplicity. Write $D = D_{\mathcal{S}} D_{\mathcal{B}}$.

For each pattern there is a complex coefficient K_ν, depending only on the ordered big-group source labels, target labels, and the auxiliary free labels used in $D_{\mathcal{B}}$. The finite pattern family may depend on x and J, but

$$\sum_\nu \sup |K_\nu| \le L^{C_1}, \tag{10.2}$$

where m, C_1 are fixed independently of J. In every operator below, symmetrization means averaging all $M!$ slot permutations independently in each relevant group. This is an orthogonal projection because the measures are invariant under these permutations. Our convention is that a row operator integrates a function at the target and returns a function at the source.

Definition 10.2 (Ideal operator)**.** On the space

$$\mathcal{H}_{\mathrm{id}} = L^2\left(\prod_{g\in\mathcal{B}} \mathcal{P}_g^M,\ \bigotimes_{g\in\mathcal{B}} \mu_g^{\otimes M}\right),$$

repetitions of prime values are allowed. For a real ζ, the unsymmetrized ideal row operation for ν retains shared labels, samples all unshared target labels independently with laws μ_g, and samples the auxiliary labels of $D_{\mathcal{B}}$ independently with the same laws. Its multiplier is

$$K_\nu D_{\mathcal{B}}^{i\zeta} e(\Theta D_{\mathcal{B}})\,.$$

The sum over ν, composed on both sides with big-group symmetrization, is denoted $\mathcal{T}(\Theta)$.

Definition 10.3 (Physical operator)**.** A physical state $(n, \mathbf{p})$ consists of $n \in \mathbb{Z}$ and an ordered list $\mathbf{p} = (p_{g,i})_{g,1\le i\le M}$ such that, within each group, the labels are distinct and divide n. Give every state at n mass $\prod_g V_g^{-M}$ and use counting measure in n. This defines $\mathcal{H}_{\mathrm{ph}}$.

Fix an integer k with $0 < |k| \le L^{C_2}$. For pattern ν, sample the auxiliary labels forming D as above and move from n to $n' = n + kD$. Sum over physical target states at n' with the prescribed shared labels, with coefficient $\prod_g V_g^{-t_g}$ for each target. Forbid every auxiliary free label of D from both endpoint lists. Auxiliary free labels may equal one another. Multiply this row action by

$$K_\nu D^{i\zeta} q^{(\omega(n)-Ms)/2} q^{(\omega(n')-Ms)/2}. \tag{10.3}$$

The sum over patterns, symmetrized in all groups on both sides, is $\mathcal{A}$.

For clarity, the adjoint moves by $-kD$, interchanges source and target labels in the coefficient, and conjugates the multiplier. Indeed, after both lists and the auxiliary labels have been fixed, the shared product and hence D are unchanged on reversal. The joint measure of the two lists in group g has normalization

$$V_g^{-M-t_g} \tag{10.4}$$

in either direction. The auxiliary-label probability is also unchanged. This proves the assertion before symmetrization, and the symmetrizing projections are self-adjoint.

Lemma 10.4 (Absolute physical bounds)**.** *Replace the coefficients of each row transition of $\mathcal{A}$ by their absolute values before adding patterns or averaging permutations. The resulting operator and its adjoint have row sums at most $L^{C_{\mathrm{abs}}}$, where C_{abs} may depend on m, C_1 and the fixed data in Definition 10.1, but is independent of J. The same assertion holds if the position is replaced by independent uniform residues at all group primes.*

Proof. If the target has $y \ge M$ divisors from group g, then, after the $M - t_g$ distinct shared labels have been fixed, there are $(y - M + t_g)_{t_g}$ possible ordered unshared target lists. Hence their sum, with their endpoint damping, is at most

$$\sup_{z\ge 0} V_g^{-t_g}(z + t_g)_{t_g} q^{z/2} \le C. \tag{10.5}$$

Here $t_g \le \ell + m$, so C is independent of J. The source damping is at most one. Free-label probabilities have total mass one, and restrictions can only decrease an absolute sum. Taking the product over $s = O(T)$ groups and using Equation (10.2) gives the row bound. The reversed normalization in Equation (10.4) gives the identical argument for columns. Neither argument used any property of integer positions beyond divisibility and translation. □

In particular all these operators are bounded on their stated Hilbert spaces, including the physical space with unrestricted integer position.

Theorem 10.5 (Local transference)**.** *For every fixed $E > 0$ there is $E_{\mathrm{id}} > 0$, depending only on E and the fixed data in Definition 10.1, with the following property. Assume*

$$\sup_{\Theta\in\mathbb{R}} \|\mathcal{T}(\Theta)\| \le L^{-E_{\mathrm{id}}}. \tag{10.6}$$

Then, for every sufficiently large fixed J, where the lower bound on J may also depend on m, C_1, put

$$R = \lceil L^{0.52} \rceil, \qquad N = 2R.$$

For every fixed $\gamma > 0$ and $I = [X, 2X)$ with $x^\gamma \le X \le x^{1/\gamma}$, one has

$$\frac{1}{|I \cap \mathbb{Z}|} \sum_{n\in I\cap\mathbb{Z}} \langle u_n, (\mathcal{A}\mathcal{A}^*)^R u_n\rangle \le L^{-EN}, \tag{10.7}$$

for all sufficiently large x. Here u_n is one on every physical state at position n and zero elsewhere. The threshold for x may depend on all fixed parameters, including J, C_2, γ, and the bound is uniform subject to these parameters and Equation (10.6)*. No bound on ζ is needed apart from the assumed ideal estimate.*

The vector u_n is the constant function on the mark lists at n; it is not normalized. Thus the moment in Equation (10.7) sums paths whose final integer position returns to n, with arbitrary initial and final mark lists. Corollary 10.11 will turn precisely this estimate into cancellation against a mark-independent first endpoint. Those are the endpoints supplied in Section 17.

Here is the structure of the proof. First replace the integer root by independent residues and expand each big prime's contribution over intervals containing all its active visits. A memory records its congruence between active uses. The small-prime residues remain physical: the many choices of omitted small marks make transfers between memory and active lists rare. On an edge with no such transfer, Fourier transformation of the residue coordinates leaves the ideal big-label operator. We first allow different lifespans to use the same prime, and then exclude such coincidences by inclusion–exclusion organized by the number of independent equalities. Only the rare-transfer estimate needs the complexity-dependent choice of J; the ideal accuracy is fixed beforehand.

10.2 Replacing the root by independent residues

Expand the left side of Equation (10.7) into paths with N alternating forward and backward edges. Fix the patterns, all endpoint permutations, the auxiliary labels, and the active lists at visits $0, \dots, N$. These choices fix offsets $h_0 = 0, h_1, \dots, h_N$, with

$$h_{i+1} - h_i = \sigma_i k D_i, \qquad \sigma_i \in \{1, -1\}, \qquad h_N = 0.$$

Only return to the position is imposed; the last ordered list need not be the first one. Let Λ_i be the set of active labels at visit i. Put $q_0 = q_N = \sqrt{q}$ and $q_i = q$ for $0 < i < N$. The dependence on the initial integer n is precisely the nonnegative factor

$$\prod_{i=0}^{N} \left(\prod_{p\in\Lambda_i} \mathbf{1}_{n+h_i\equiv 0 \pmod p} \right) q_i^{\#\{p \text{ a group prime}: p\notin\Lambda_i,\ n+h_i\equiv 0 \pmod p\}}. \tag{10.8}$$

The label normalization is initially $\prod_g V_g^{-M}$ and at each edge $\prod_g V_g^{-t_g}$, in addition to the free-label probabilities. All remaining factors are independent of n.

Lemma 10.6 (Uniform residue replacement)**.** *For each compatible fixed path and every fixed $A > 0$, the average of Equation* (10.8) *over $I \cap \mathbb{Z}$ equals its expectation under independent uniform residues at all group primes, multiplied by $1 + O(\exp(-ANT))$. Incompatible active congruences give zero in both models.*

Proof. Let Q_{act} be the product of the distinct active primes over all visits. There are at most $(N+1)Ms$ of them, so

$$\log Q_{\mathrm{act}} = O_J(NTL^d) = o(L). \tag{10.9}$$

Compatibility fixes one congruence modulo Q_{act}. All damping factors belonging to these primes are then deterministic; factor them out in both models. This step permits a relative estimate even when these deterministic factors are very small.

For a remaining prime, collect repeated offsets into distinct residues. Its factor has the form

$$1 - X_p(n), \qquad X_p(n) = \sum_r c_{p,r} \mathbf{1}_{n \equiv r \pmod p}, \quad 0 \le c_{p,r} \le 1,$$

with at most $N+1$ residues and $0 \le X_p \le 1$. In the independent model,

$$\lambda := \sum_p \mathbb{E}X_p \le (N+1) \sum_p \frac{1}{p} = O(NT), \qquad \prod_p (1 - \mathbb{E}X_p) \ge \exp(-O(NT)),$$

since $\max_p \mathbb{E}X_p = o(1)$. Bonferroni inequalities for numbers in $[0,1]$ bracket $\prod_p(1 - X_p)$ by consecutive truncations of its elementary-symmetric expansion. At degree h, the independent expectation of the difference is at most $\lambda^h/h!$. Choose h of order NT, with its fixed constant sufficiently large after A. Stirling's lower bound $h! \ge (h/e)^h$ makes the difference smaller than $\exp(-(A+C)NT)$ for any required fixed C.

Each truncated term specifies one residue at each of at most h additional primes. Its CRT count on I differs from its independent density by $O(|I \cap \mathbb{Z}|^{-1})$, also when the active congruence is imposed. The number of such terms is at most

$$\exp\left(O(h(L^d + \log N + \log s))\right) = \exp(o(L)),$$

and their moduli, including Q_{act}, have product $\exp(o(L))$. Here $0.52 + d < 1$. Thus the total counting error is $\exp(-\gamma L + o(L))$. By Equation (10.9), this is negligible relative to $Q_{\mathrm{act}}^{-1} \exp(-O(NT))$. Restoring the factored deterministic damping proves the stated relative estimate. □

This replacement may be summed over paths with absolute coefficients. Indeed retain the full vector of residues as a position, with its Haar probability measure, and translate it by $\sigma_i k D_i$ on an edge. The expected total mass of all initial lists is at most one: in group g it is

$$V_g^{-M} \sum_{\substack{p_1,\dots,p_M \in \mathcal{P}_g \\ \text{distinct}}} \frac{1}{p_1 \cdots p_M} \le 1. \tag{10.10}$$

By Lemma 10.4, the absolute mass of all length N paths, even without a return condition, is $L^{O(N)}$. Taking A sufficiently large in Lemma 10.6 makes the summed error smaller than $L^{-A'N}$ for any desired fixed A'. We may therefore use the independent residue model. The return condition is now imposed by the exact identity

$$\mathbf{1}_{h_N = 0} = \int_0^1 e(\theta h_N)\, d\theta. \tag{10.11}$$

It suffices to bound the unrestricted signed path expression uniformly in θ. Small-prime residues will remain actual Haar coordinates throughout the argument.

10.3 The exact expansion for one big prime

Put $\eta_i = 1 - q_i$, $P_* := \exp(L^c)$, and, for each big prime,

$$b_p = 1 - \frac{1}{p} \sum_{i=0}^{N} \eta_i > 0, \qquad \widetilde{\mu}_g(p) = \frac{1}{V_g p b_p}. \tag{10.12}$$

The ratio $\widetilde{\mu}_g(p)/\mu_g(p)$ is $1+O(N/P_*)$, uniformly in big groups. We extract the common baseline $\prod_{p \text{ big}} b_p \le 1$ from every path sum.

Suppose first that p is active at some visits. Incompatible active offsets give zero. Otherwise let their common residue be c_p, let a_p, z_p be the first and last active visits, and put $H_i = \mathbf{1}_{h_i \equiv c_p \pmod p}$. Its residue expectation is

$$\frac{1}{p} \prod_{i:\ p \notin \Lambda_i} (1-\eta_i H_i).$$

The two exact telescoping identities

$$\begin{aligned} \prod_{i<a_p} (1-\eta_i H_i) &= 1 - \sum_{j<a_p} \eta_j H_j \prod_{j<i<a_p} (1-\eta_i H_i), \\ \prod_{i>z_p} (1-\eta_i H_i) &= 1 - \sum_{j>z_p} \eta_j H_j \prod_{z_p<i<j} (1-\eta_i H_i) \end{aligned} \tag{10.13}$$

follow by subtracting consecutive partial products. Thus, in addition to its active uses, the prime either starts at a_p or has an earlier ghost birth at a hit $j < a_p$, with coefficient $-\eta_j$; it either ends at z_p or has a later ghost termination at a hit $j > z_p$, again with coefficient $-\eta_j$. At strictly internal unmarked hits it receives q_i.

If p is never active, expand its full product and group each term of degree at least two by its earliest and latest indices. This gives

$$b_p + \frac{1}{p} \sum_{j<i} \eta_j \eta_i \mathbf{1}_{h_j \equiv h_i \pmod p} \prod_{j<t<i} \left(1 - \eta_t \mathbf{1}_{h_t \equiv h_j \pmod p}\right). \tag{10.14}$$

The empty and singleton terms give b_p; the interior product is the sum over every possible further chosen index. Relative to the baseline, the prime is either absent or has an orphan interval from j to i, $j<i$, of cost $(pb_p)^{-1}(-\eta_j)(-\eta_i)$ and the indicated internal damping.

Consequently each represented big prime has exactly one *lifespan*: an interval containing all its active visits, possibly containing none. Its endpoints and all its active visits have equal offsets modulo p. It pays $(pb_p)^{-1}$ once, the ghost endpoint coefficients if present, and q_i at each strictly internal unmarked hit. No offset outside the lifespan occurs in its weight.

There are two useful consequences of physical compatibility. For large x every group prime exceeds $|k|$. If a prime is active at both ends of an edge, then it divides kD, hence D. It cannot be an auxiliary free label, by the ban, so it must be a prescribed shared label. Conversely, a prime at an unshared entry or exit cannot divide D: it would either duplicate a shared label in that state or violate the free-label ban. Thus the original mark normalization assigns V_g^{-1} to each maximal run of activity, and no additional factor at shared continuation. In particular, a lifespan with r active runs requires total label weight

$$\frac{V_g^{-r}}{pb_p}.$$

The memory construction must therefore charge the reciprocal prime once, at the first appearance, and charge V_g^{-1} once for each active run. Later congruence tests must not introduce further reciprocal-prime costs. These requirements guide the choice of measures and transfer coefficients below; the memory identity will verify them.

10.4 A Hilbert space which remembers lifespans

Let $Y_{\mathcal{S}} = \prod_{p \text{ small}} \mathbb{Z}/p\mathbb{Z}$, with Haar probability measure. A small state at $y \in Y_{\mathcal{S}}$ is an ordered list of M distinct zero coordinates in each small group, with mass $\prod_{g \in \mathcal{S}} V_g^{-M}$. Write $\mathcal{H}_{\mathcal{S}}$ for the resulting L^2 space. For a big group set

$$\mathcal{Z}_g = \{(p,r) : p \in \mathcal{P}_g, r \in \mathbb{Z}/p\mathbb{Z}\}, \qquad d\lambda_g(p,r) = \widetilde{\mu}_g(p)\, d\,\mathrm{count}(r).$$

Define its memory space by

$$\mathcal{F}_g = \bigoplus_{j\geq 0} L^2_{\mathrm{sym}}\left(\mathcal{Z}_g^j, \frac{1}{j!}\lambda_g^{\otimes j}\right), \qquad \mathcal{H} = \mathcal{H}_{\mathcal{S}} \otimes L^2\left(\prod_{g\in\mathcal{B}} \mathcal{P}_g^M, \bigotimes_{g\in\mathcal{B}} \widetilde{\mu}_g^{\otimes M}\right) \otimes \bigotimes_{g\in\mathcal{B}} \mathcal{F}_g. \tag{10.15}$$

Here "symmetric" means invariant under permutations of the j particle indices, including when their numerical values coincide. The $j = 0$ summand is $\mathbb{C}$. A pending particle (p, r) records displacement from its required hit, modulo p. The residue measure in λ_g is *counting measure*: requiring $r = 0$ selects one point and incurs no factor $1/p$. The boundary vector $\mathbf{1}_\varnothing$ is one when every memory is empty and zero otherwise; it is independent of all other coordinates.

We next specify row actions. The factorial measure in Equation (10.15) is used for inner products and adjoints, not inserted anew in every row transition. All operations are first understood on finite memory truncations; the absolute bounds below justify the unrestricted sums.

Separate the evolution into an operation G_i at each visit $0 \leq i \leq N$ and an operation $\mathcal{E}_i$ across the following edge for $i < N$. The visit operation handles ghost endpoints and unmarked hits. The edge operation moves residues and transfers particles between active lists and memory: a store keeps a departing active label pending, and a promotion returns a pending particle to an active slot. Thus G_i acts after arrival and promotion at visit i, and before storage and departure.

To make both operations bounded, choose $\rho \in (0, 1)$ with $\rho^2 > \sqrt{q}$. At a strictly internal unmarked hit, split the required damping as

$$q_i = \rho\,(q_i/\rho^2)\,\rho.$$

The adjacent edges supply the two factors ρ, and the visit operation supplies the middle factor. The edge factors will control the number of possible promotions, while $q_i/\rho^2 < 1$ damps the visit operation. The ghost endpoint factor is similarly split as $-\eta_i = (-\eta_i/\rho)\rho$; the birth normalization is specified below. The definitions implement these local factorizations, including the initial and terminal visits.

For a group memory $\mathbf{z} = ((p_h, r_h))_{h=1}^j$, write $Z(\mathbf{z}) = \{h : r_h = 0\}$. At visit i the ghost operation G_i is the product over big groups of the following row action, leaving every other coordinate fixed:

$$\begin{aligned}(G_{i,g}F)(\mathbf{z}) = \sum_{A\subseteq Z(\mathbf{z})} \left(-\frac{\eta_i}{\rho}\right)^{|A|} \left(\frac{q_i}{\rho^2}\right)^{|Z(\mathbf{z})|-|A|} \sum_{b\geq 0} \frac{1}{b!} \left(-\frac{\eta_i V_g}{\rho}\right)^b \\ \cdot \int_{\mathcal{P}_g^b} F(\mathbf{z}\setminus A, (p_1', 0), \ldots, (p_b', 0)) \prod_{h=1}^b d\widetilde{\mu}_g(p_h').\end{aligned} \tag{10.16}$$

Thus selected old zeros terminate, continuing zeros are damped, and a new unordered batch is born at zero. Termination precedes birth; an orphan cannot be born and terminate at the same visit.

The edge operation $\mathcal{E}_i$, $0 \leq i < N$, sums the given patterns in orientation σ_i, with independent source and target symmetrizations in all active lists. Between these permutations its row action is as follows.

1. Form D from shared source labels and the independent free draws. Retain the ban of free labels from both endpoint active lists. Multiply by the oriented coefficient $K_\nu D^{i\zeta}$, or its conjugate with roles interchanged, and by $e(\theta\sigma_i kD)$.

2. Multiply by ρ for every old pending zero, before storing any active particle. In each big group select any subset of its t_g unshared source slots to store, appending their particles at residue zero; drop the other unshared source particles. Shared particles remain active.

3. Translate y and all pending residues, including the newly stored ones, by $\Delta = \sigma_i kD$. Choose any subset of the t_g unshared big target slots and an injection of those ordered slots into distinct *old* pending particle indices of that group whose translated residue is zero. Promote

the selected particles into those slots, remove them from memory, and multiply by V_g^{-1} per promotion. Newly stored particles cannot be promoted across this edge. Require $p \nmid D$ for every stored or promoted particle. Fill the other unshared target slots by independent fresh integrations against $\widetilde{\mu}_g$.

4. Multiply by ρ for every pending zero remaining after promotions. In the small groups use the physical target sum, with normalization $V_g^{-\ell}$, and the two small endpoint factors $q^{(\omega_g - M)/2}$. These counts concern small states at y and $y + \Delta$.

Selections of promotion slots and injections are summed without a factorial divisor. All choices of source or target slots refer to their canonical order between the sampled endpoint permutations. No big endpoint damping is included apart from the pending-particle factors specified above. Auxiliary free draws of D are not particle births.

For the moment impose, in the complete path sum, the rule that all prime values assigned at *different births* are distinct. Births comprise initial active slots, fresh active target slots, and ghost births. This is a global rule, even if two births occur far apart. Apart from it, place no distinctness restriction on big active lists or memories.

Lemma 10.7 (Exact memory identity)**.** *With the distinct-birth rule, the independent-residue path expression with the phase in Equation* (10.11) *equals*

$$\left(\prod_{p \ big} b_p \right) \langle \mathbf{1}_\varnothing, G_0 \mathcal{E}_0 G_1 \cdots \mathcal{E}_{N-1} G_N \mathbf{1}_\varnothing \rangle_{\text{distinct}}. \tag{10.17}$$

The subscript means that the indicator is inserted in the expanded path integral, not that each local operator separately enforces it.

Proof. A particle born active begins with its required hit at that visit. When it leaves an active run, storing it at zero and then translating records subsequent displacement from that hit. Promotion checks displacement zero. A later store resets the anchor at a congruent offset, so it changes no congruence requirement. A ghost birth sets the anchor at an unmarked hit, and a ghost termination also checks zero. Shared continuation automatically respects the congruence.

The ρ factors cancel locally. At an internal visit a continuing pending zero pays $\rho(q_i/\rho^2)\rho = q_i$. A ghost birth pays $(-\eta_i/\rho)\rho = -\eta_i$, and a ghost termination pays $\rho(-\eta_i/\rho) = -\eta_i$. An active particle pays none of these factors: promotion occurs before output damping and storage after input damping. At visit zero memory starts empty, and at visit N it must end empty, so there is no missing boundary factor. The values $q_0 = q_N = \sqrt{q}$ are therefore treated exactly.

For a prime with r active runs, a first active birth costs $1/(V_g p b_p)$ and each of its $r - 1$ later promotions costs $1/V_g$. A first ghost birth costs $1/(pb_p)$ and its r promotions cost V_g^{-r}. Both give

$$\frac{V_g^{-r}}{pb_p}.$$

An orphan has $r = 0$. These are precisely the baseline-relative costs in Equations (10.13) and (10.14), including all original mark normalizations.

Conversely, fix an original compatible path and a chosen lifespan term for every represented big prime. Its active visits determine exactly when it is shared, stored, promoted, or dropped; its ghost endpoints determine its birth and termination. Between these uses it must remain pending. The exclusions $p \nmid D$ follow from unshared entries and exits as proved above, and immediate promotion of a newly stored particle is impossible for the same reason. A numerical prime occupies at most one active slot or pending particle, because it has only one birth. Distinct ghost births in an unordered batch are counted once by its factorial divisor. Thus these constructions are mutually inverse and preserve all coefficients. This proves Equation (10.17). □

Figure 1 illustrates an admissible lifespan with two runs of activity and ghost endpoints.

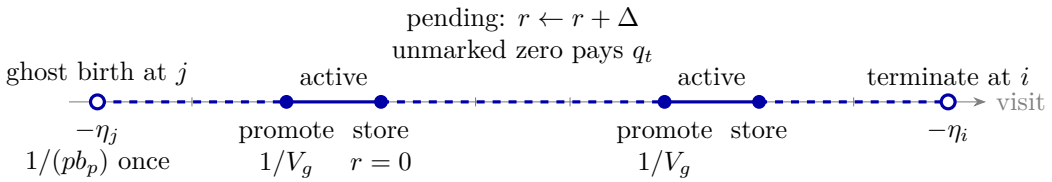


Figure 1: One possible lifespan. Solid segments denote shared activity; dashed segments denote memory. Every marked use and ghost endpoint has the same offset modulo p. The reciprocal prime factor is charged only at birth; $\eta_t = 1 - q_t$ at visit t.

10.5 Absolute bounds, including the adjoint measures

The memory identity is exact with the global distinct-birth rule. We now allow different births to have equal prime values, so that each operation acts locally on $\mathcal{H}$. We will restore the global rule after estimating this enlarged evolution. The absolute bounds proved here justify the memory truncation and remain available when equality constraints later modify individual operations. Their column bounds require the factorial adjoint measures explicitly.

For a positive row kernel K, if a positive function w satisfies $Kw \le Rw$ and $K^*w \le Cw$, then

$$\|K\|_{2\to 2} \le \sqrt{RC}. \tag{10.18}$$

Indeed, Cauchy's inequality in each row bounds $|Kf|^2$ by $(Kw)K(|f|^2/w)$; integration and the column inequality prove Equation (10.18). We apply this to absolute kernels, with

$$w(\text{state}) = v^{j_{\text{tot}}}, \qquad j_{\text{tot}} = \sum_{g\in\mathcal{B}} j_g,$$

where $v > 1$ is a fixed constant chosen below.

Here is a direct verification of the measures in the ghost adjoint. In one group abbreviate

$$A_i = \frac{q_i}{\rho^2}, \qquad D_i = -\frac{\eta_i}{\rho}, \qquad C_i = V_g D_i, \qquad \nu = \widetilde{\mu}_g,$$

and let $z(\mathbf{u})$ count zero residues in a memory list. For test functions f, h, the joint integral from Equation (10.16) is

$$\begin{aligned}\langle f, G_{i,g} h\rangle = \sum_{k,t,b\ge 0} \frac{1}{k!t!b!} \int A_i^{z(\mathbf{u})} D_i^t C_i^b \overline{f(\mathbf{u}, \boldsymbol{\alpha}^0)} h(\mathbf{u}, \boldsymbol{\beta}^0) \\ \cdot d\lambda_g^{\otimes k}(\mathbf{u}) d\nu^{\otimes t}(\boldsymbol{\alpha}) d\nu^{\otimes b}(\boldsymbol{\beta}).\end{aligned} \tag{10.19}$$

Here $\boldsymbol{\alpha}^0$ means that all its residues are zero. The identity follows by selecting t old zero indices from a list of size $k + t$:

$$\frac{1}{(k+t)!}\binom{k+t}{t} = \frac{1}{k!t!}.$$

Swapping $(t, \boldsymbol{\alpha})$ with $(b, \boldsymbol{\beta})$ and conjugating shows that the adjoint has the same row formula with birth coefficient D_i and termination coefficient C_i. In particular, reversal moves the factor V_g from births to terminations; it does not create any factor p or $1/p$.

For the absolute ghost kernel the weighted row ratio is exactly

$$\exp\big(|C_i| v \nu(\mathcal{P}_g)\big) \left(A_i + \frac{|D_i|}{v}\right)^{z(\mathbf{z})}, \tag{10.20}$$

and its column ratio has C_i, D_i interchanged. Choose v large enough that, for both possible q_i,

$$\frac{q_i}{\rho^2} + \frac{2\max(1, v_+)\eta_i}{\rho v} < 1. \tag{10.21}$$

This is possible since $q_i \le \sqrt{q} < \rho^2$. The same choice works when every original ghost birth receives an extra factor two. Since $\nu(\mathcal{P}_g) = 1 + o(1)$, the exponential factor in Equation (10.20) is bounded by a fixed constant per group. Thus for some fixed C_v,

$$|G_i|w \le L^{C_v} w, \qquad |G_i|^* w \le L^{C_v} w, \qquad \|G_i\| \le L^{C_v}. \tag{10.22}$$

The exponent C_v is independent of m, C_1, J. All these assertions include the version with doubled ghost births.

We verify the edge adjoint just as explicitly. Fix one big group, a pattern, endpoint permutations, free labels, and the chosen transfer subsets. Suppose a target slots are promotions and b source slots are stores, so $a, b \le t_g$. Let the source memory have size j, and write u for its $j - a$ surviving particles. Let $p_1, \dots, p_a$ be the promoted prime labels and $s_1, \dots, s_b$ the stored labels. For step Δ, the endpoint memories are

$$X = u \sqcup ((p_h, -\Delta \bmod p_h))_{h \le a}, \qquad Y = T_\Delta u \sqcup ((s_h, \Delta \bmod s_h))_{h \le b}, \tag{10.23}$$

where T_Δ translates every residue. The joint measure on these variables is

$$\frac{d\lambda_g^{\otimes(j-a)}(u)}{(j-a)!} \prod_{h \le a} d\widetilde{\mu}_g(p_h) \prod_{h \le b} d\widetilde{\mu}_g(s_h), \tag{10.24}$$

along with one $\widetilde{\mu}_g$ integration for every shared active label, dropped source label, and fresh target label. The transfer coefficient is V_g^{-a}, and the damping is $\rho^{z(X)+z(Y)}$. The factorial accounting is

$$\frac{(j)_a}{j!} = \frac{1}{(j-a)!} = \frac{(j-a+b)_b}{(j-a+b)!}. \tag{10.25}$$

The first equality selects old memory indices into the a specified ordered target slots. The second selects b indices of the output memory into the original source slots on reversal. Each required residue in Equation (10.23) selects exactly one point of counting measure. Translation preserves that measure.

It follows that the reversed row operation retrieves the original stores and stores the original promotions; the coefficient V_g^{-a} remains unchanged. If it is expressed using the forward convention that charges V_g^{-b} for its reversed promotions, its additional factor is V_g^{b-a}. This is bounded uniformly for $a, b \le \ell + m$. Fixing shared and free D-labels fixes Δ independently of which memory indices are retrieved, as required for this change of variables. In small groups Haar translation and the joint normalization $V_g^{-M-\ell}$ are invariant on reversal.

The identities above continue to hold with numerical coincidences. For example, a memory multiset with multiplicities n_z has mass $\prod_z \lambda_g(z)^{n_z}/n_z!$. Subset deletions and ordered retrievals retain their index multiplicities. No passage from indices to distinct numerical values was made in Equation (10.19) or Equation (10.25).

Lemma 10.8 (Absolute memory bounds)**.** *There is C_a, allowed to depend on m, C_1 but independent of J, such that every absolute edge has weighted row and column bounds L^{C_a}. These bounds and Equation* (10.22) *persist under arbitrary multipliers of modulus at most one on individual local transition choices, and under memory-size projections.*

Proof. For a row, fix a pattern, free labels, permutations, and transfer-slot subsets of sizes $a_g, b_g \le t_g$. If Z' old pending particles have zero translated residue in a group, their ordered promotions and the remaining output damping cost at most

$$(Z')_{a_g} \rho^{Z' - a_g} \le \sup_{z \ge a_g} (z)_{a_g} \rho^{z - a_g} < \infty. \tag{10.26}$$

Damping of stored particles can be discarded for this upper bound. The Schur weight contributes $v^{b_g - a_g}$, promotion coefficients contribute $V_g^{-a_g}$, and fresh integrations have mass $\widetilde{\mu}_g(\mathcal{P}_g)^{t_g - a_g} \le 2^{t_g}$

for large x. There are at most 4^{t_g} choices of transfer subsets. Each cost is bounded by a constant per group, independent of J. Small transitions satisfy Equation (10.5). Multiplying over groups, averaging permutations and free labels, and summing Equation (10.2) proves the row bound.

For the column, use Equations (10.23) and (10.25) in reverse. Original stores are now ordered retrievals. Their count is controlled by the original input ρ factors after reversed translation, in exactly the form in Equation (10.26). The remaining factor $V_g^{b_g-a_g}$ is bounded per group. This proves the column bound. Restrictions and local multipliers of modulus at most one are dominated by these absolute kernels. □

We now restrict the total memory size to

$$B = \lceil L^2 \rceil \tag{10.27}$$

before and after every operation, writing the projections implicitly. At most $(\ell+m)|\mathcal{B}|N$ particles can be supplied by stores. Thus a path violating this restriction has at least $B - O((\ell+m)sN)$ ghost births. Multiply every ghost birth by two in the absolute path sum and use the preceding bounds. The total omitted mass, even with the distinct-birth indicator, is at most

$$2^{-B+O((\ell+m)sN)} L^{O(N)} = \exp(-\Omega(L^2)) = o(L^{-AN}) \tag{10.28}$$

for every fixed A. The empty-memory boundary vector has bounded norm: the small part has mass at most one by Equation (10.10), and the big part has mass

$$\prod_{g\in\mathcal{B}} \widetilde{\mu}_g(\mathcal{P}_g)^M = 1 + O_J(sN/P_*) = 1 + o(1).$$

These estimates also prove absolute summability before truncation, for instance by first restricting all batch sizes and then applying monotone convergence to absolute kernels.

10.6 Rare transfers and the role of the small groups

It remains to obtain a small signed edge norm and then restore the omitted distinct-birth rule. Decompose each edge as $\mathcal{E}_i = \mathcal{E}_{i,\mathrm{clean}} + \mathcal{E}_{i,\mathrm{dirty}}$: a term is *dirty* if it has at least one store or promotion, and *clean* otherwise. Dirty terms connect memory to the active lists; we suppress them by averaging the omitted small labels. Clean terms only translate memory residues and refresh unshared active labels; the next subsection will reduce their norm to the ideal estimate. Define

$$\mathcal{D} = \binom{M}{\ell}^{|\mathcal{S}|}. \tag{10.29}$$

Lemma 10.9 (Rare transfer)**.** *With the memory truncation in Equation* (10.27)*, the aggregate of all dirty terms of any edge has norm at most*

$$2L^{C_a} \left(\frac{B}{\mathcal{D}}\right)^{1/2}, \tag{10.30}$$

after enlarging C_a independently of J if necessary.

Proof. First consider the positive sum of terms with a promotion. Fix its source, pattern, big-group permutations and shared choices, and free D-labels. For some one of its at most B old pending particles (p, r), promotion requires

$$r + \sigma_i k D_{\mathcal{B}} D_{\mathcal{S}} \equiv 0 \pmod{p}, \qquad p \nmid D. \tag{10.31}$$

In each small group the source symmetrization omits an independent uniformly chosen ℓ-subset of its M distinct labels. If F is the full product of all small source labels and U the product of the

omitted labels, then $D_{\mathcal{S}} = F/U$. Unique factorization and disjointness of groups show that the $\mathcal{D}$ choices give distinct positive integers U. Moreover,

$$U \le \exp(\ell|\mathcal{S}|L^b) < \exp(L^c) \le p$$

for sufficiently large x. They are therefore distinct modulo p. The quantities $F, k, D_{\mathcal{B}}$ are invertible modulo every p eligible in Equation (10.31). Hence each pending particle allows at most one omission choice. At most B of the $\mathcal{D}$ equally likely choices allow any promotion.

Conditional on every omission choice, the remaining weighted row costs are uniformly bounded by the proof of Lemma 10.8. Thus the promotion part improves its row bound to $L^{C_a}B/\mathcal{D}$, while its column bound remains L^{C_a}. Schur's test gives the first half of Equation (10.30). Terms with stores and no promotion have the same improvement in their column bound by reversing the edge: a stored particle becomes a retrieval from output memory. The exclusion $p \nmid D$ is retained under reversal, even when numerical coincidences have been allowed. Add these two bounds. □

The coefficient restriction to big labels is used here: conditioning on all such labels does not bias the small omission choices. In particular,

$$2L^{C_a}\sqrt{B/\mathcal{D}} \ll L^{C_a+1-(c_0/2)\log\binom{J+\ell}{\ell}}. \tag{10.32}$$

An arbitrarily strong fixed saving is obtained by increasing J, without changing the ideal norm target.

10.7 Clean edges and Fourier reduction

Lemma 10.10 (Clean norm)**.** *There is C_s, depending only on the fixed data in Definition 10.1, such that the clean part of every edge satisfies*

$$\|\mathcal{E}_{i,\mathrm{clean}}\| \le 2L^{-E_{\mathrm{id}}+C_s} \tag{10.33}$$

for all sufficiently large x, under Equation (10.6)*. In particular C_s is independent of m, C_1, J.*

Proof. A clean edge leaves all memory prime lists and sizes unchanged, translates their residues, drops every unshared big source slot, and fills every unshared big target slot freshly. Its input and output ρ factors and its memory projections are contractions. They commute with active-list symmetrizations. Absorb these factors and the symmetrizations into the two test vectors w_1, w_2. These vectors remain symmetric in the big active lists.

Fix an ordered tuple $\mathbf{d}$ of J shared labels in each small group and let d' denote their joint product. Define a map to unrestricted small residue functions by

$$(P_{\mathbf{d}}w)(y) = \left(\prod_{g\in\mathcal{S}} V_g^{-J/2-\ell}\right) q^{\frac{1}{2}\sum_{g\in\mathcal{S}}(\omega_g(y)-M)} \sum_{\mathbf{R}:\ (\mathbf{d},\mathbf{R}) \text{ a small state at } y} w(y,(\mathbf{d},\mathbf{R})). \tag{10.34}$$

All other coordinates are left untouched. The map is zero when the displayed list is not a small state, so the damping is used only where $\omega_g(y) \ge M$. The two endpoint factors in Equation (10.34) give $V_g^{-J-2\ell} = V_g^{-M-\ell}$, exactly the physical joint normalization.

By Cauchy's inequality in $\mathbf{R}$, followed by summation over $\mathbf{d}$, one has

$$\sum_{\mathbf{d}} \|P_{\mathbf{d}}w\|^2 \le L^{C_s}\|w\|^2. \tag{10.35}$$

To see the independence of J, the multiplier relative to the original small state measure in group g is bounded by

$$\sup_{z\ge M} q^{z-M}\frac{(z-M+\ell)_\ell}{V_g^\ell},$$

a constant depending only on q, ℓ, v_-. Taking products over $O(T)$ groups proves Equation (10.35).

The clean bilinear form is now exactly

$$\sum_{\mathbf{d}} \langle P_{\mathbf{d}} w_1, \mathcal{T}'_{\mathbf{d}} P_{\mathbf{d}} w_2 \rangle. \tag{10.36}$$

The operator between these maps uses the big-list transitions and translates all unrestricted small and memory residues by $\sigma_i k d' D_{\mathcal{B}}$. Fix the memory sizes and their prime lists. Fourier transformation on the finite abelian group consisting of the small residue torus and the pending residue coordinates diagonalizes these translations. Counting measure on pending residues and Haar measure on the small torus each have the usual unitary finite Fourier transform. One can first use ordered memory representatives; restriction to symmetric functions preserves the bound. Big active-list symmetry is unaffected.

At any fixed Fourier frequency, translation and the closure factor contribute

$$e(\sigma_i \Theta D_{\mathcal{B}})$$

for a real Θ depending on $\mathbf{d}$, the memory prime list, the frequency, and θ, but independent of the variable big active labels. The scalar factor $(d')^{\sigma_i i\zeta}$ has modulus one. Consequently the remaining operator, on symmetric big active tests, is $\mathcal{T}(\Theta)$ or its adjoint, with two changes only: the measures of active labels and fresh target integrations are $\widetilde{\mu}_g$ in place of μ_g, and the free/active collision ban remains in force.

Both changes are negligible in operator norm. Across the $M|\mathcal{B}|$ active coordinates, the product density changes by

$$1 + O_J(sN/P_*).$$

The corresponding square-root density identification is unitary between the two L^2 spaces and commutes with symmetrization. Fresh integration densities obey the same bound. The ideal absolute row and column sums are at most L^{C_1} before these changes, since all its unrestricted fresh laws are probabilities. Schur's test therefore bounds the density perturbation by $O_J(L^{C_1} sN/P_*)$.

For the ban, in either row direction condition on the fixed active state. Each free auxiliary draw has maximal atom $O(1/P_*)$; its chance to match a fixed active label is bounded by that quantity. A match to a fresh opposite unshared label has the same bound by independence. There are $O_{J,m}(s)$ possible comparisons. Reversal has the same estimate. Thus removing the banned transitions costs $O_{J,m}(L^{C_1} s/P_*)$ in norm. These estimates are $o(L^{-A})$ for every fixed A. They do not require exclusion of coincidences with pending labels: such labels are fixed in the Fourier decomposition and are already accounted for by translation.

It follows uniformly in $\mathbf{d}$ that the between-map norm is at most $2L^{-E_{\mathrm{id}}}$ for large x. Apply Cauchy's inequality to Equation (10.36) and then Equation (10.35); this proves Equation (10.33). □

Choose, in this order,

$$G > E + C_v + 3, \qquad E_{\mathrm{id}} > G + C_s + 2, \tag{10.37}$$

and finally choose the fixed integer J so large that Equation (10.32) is smaller than $\frac{1}{2} L^{-G}$ for large x. For example it suffices that

$$\frac{c_0}{2} \log \binom{J+\ell}{\ell} > C_a + G + 3.$$

The clean estimate supplies the other half. Hence the whole signed truncated edge, with birth distinctness still omitted, satisfies

$$\|\mathcal{E}_i\| \le L^{-G}. \tag{10.38}$$

Only this final choice of J needs to depend on m, C_1. This establishes the required quantifier order, but a further argument is necessary to impose distinct births without losing the signed contraction on every edge.

10.8 Restoring distinct births by equality rank

A single absolute collision estimate cannot finish the proof: one reciprocal-prime saving P_*^{-1} does not absorb a length-N absolute path cost $L^{O(N)}$. Instead, expand the global distinctness condition by equalities between birth values. Many independent equalities yield many point-mass savings. A small number of independent equalities can be imposed by phases at their birth operations, preserving the signed contraction on all the other edges.

In the truncated evolution allocate potential birth addresses as follows: initial big slots; every possible big target slot at every edge, in its canonical order before final symmetrization; and indices $1, \dots, B$ in each group's ordered ghost batch at each visit. A target slot is performed as an address precisely when it is a fresh birth rather than shared or promoted. A ghost address is performed precisely when its batch is at least that large. These are conditions on the choices made in their single local operation. The number A_* of potential addresses satisfies

$$A_* \le M|\mathcal{B}|(N+1) + (N+1)|\mathcal{B}|B = L^{O(1)}. \tag{10.39}$$

The implied exponent can be taken independent of J once x is sufficiently large for fixed J.

We use the following elementary form of distinctness inclusion-exclusion. Sum over collections π of disjoint blocks of potential addresses, every block having size at least two. Let $\mathcal{C}_\pi$ require that all addresses in those blocks be performed and that the prime values within each block be equal. Set

$$\mu(\pi) = \prod_{B' \in \pi} (-1)^{|B'|-1}(|B'|-1)!, \qquad r(\pi) = \sum_{B' \in \pi} (|B'|-1). \tag{10.40}$$

For a fixed realized path,

$$\mathbf{1}_{\text{distinct performed birth values}} = \sum_{\pi} \mu(\pi)\mathbf{1}_{\mathcal{C}_\pi}. \tag{10.41}$$

To verify this, interpret a block of size h as a cycle on its h addresses: there are $(h-1)!$ such cycles, each with sign $(-1)^{h-1}$. The right side is the sum of signs of permutations of the performed addresses which preserve their numerical prime values. It factors over equal-value classes. The sign sum is one for a singleton class and zero for any class of size at least two, by pairing permutations with their product by a fixed transposition.

A rank-r collection involves at most $2r$ addresses. More precisely its total absolute coefficient, summed over all collections of rank r, obeys

$$\sum_{r(\pi)=r} |\mu(\pi)| \le A_*^{2r} \le L^{C'r}. \tag{10.42}$$

For instance the unsigned cycle generating polynomial on A_* addresses is $\prod_{j=0}^{A_*-1}(1+jt)$, where the power of t is the sum of cycle lengths minus the number of cycles. Its coefficient of t^r is at most $(\sum_j j)^r/r! \le A_*^{2r}$.

Large rank: chronological fresh integrations. Fix π. Order birth addresses chronologically, using canonical slot order inside an edge or batch, and call the first address in each block its pivot. Every other address is a dependent birth. When such an address is performed, its fresh prime integration is constrained to the already determined pivot value. Since

$$\sup_{g \in \mathcal{B}, p \in \mathcal{P}_g} \widetilde{\mu}_g(p) \ll P_*^{-1}, \tag{10.43}$$

this improves the weighted absolute row bound by $O(P_*^{-1})$ per dependent birth.

Here the chronological assertion uses genuinely fresh integrations. At an edge choose stores, promotions, and the step first, and then fill fresh target slots in canonical order. The step, promotion count, and memory weight do not depend on these new prime values; the coefficient is

bounded by its supremum and exclusions may be discarded. In the proof of Lemma 10.8, replace the total fresh mass in each constrained slot by its single-point bound. For a ghost operation keep its ordered batch integral and factorial divisor; constraining indicated fresh coordinates replaces their mass factors by the same single-point bound. The birth coefficient V_g and the memory weight only give fixed constants per constrained coordinate. Initial active slots satisfy the same product estimate. Pivots and dependents within one operation are handled in their prescribed integration order.

For completeness, these conditional row bounds can be iterated although a pivot may have died before its dependent birth. Retain all pivot values in the conditioning. Bound the remaining terminal sum by backward induction using the uniform weighted row bounds for every possible conditioning, beginning with $\mathbf{1}_{\varnothing} \le w$. Whenever a dependent birth is reached, the preceding point-mass improvement is uniform in its retained pivot value. At the initial boundary $w = 1$, and the remaining integrated boundary mass is bounded. Thus the total absolute contribution for this fixed collection is at most

$$L^{C_h(N+1)}(C/P_*)^{r(\pi)}, \tag{10.44}$$

for fixed C_h, C; both may depend on m, C_1.

Choose a sufficiently large fixed C'', after these constants and C', and set

$$r_0 = \left\lceil \frac{C''NT}{\log P_*} \right\rceil. \tag{10.45}$$

By Equation (10.42) and Equation (10.44), the sum over $r \ge r_0$ is bounded by

$$L^{C_h(N+1)} \sum_{r \ge r_0} (CL^{C'}/P_*)^r = o(L^{-EN}).$$

Indeed $\log P_* = L^c \gg T$, and increasing C'' beats the fixed exponent $C_h + E$.

Small rank: phases only at the affected births. For a block with pivot β_0 and other addresses $\beta_1, \dots, \beta_{h-1}$, equality of the integer prime values has the exact representation

$$\mathbf{1}_{p_{\beta_j} = p_{\beta_0}\ (1 \le j < h)} = \int_{[0,1]^{h-1}} e\left(\sum_{j=1}^{h-1} \tau_j (p_{\beta_j} - p_{\beta_0}) \right) d\boldsymbol{\tau}, \tag{10.46}$$

on paths performing all these addresses. For fixed Fourier parameters, collect the factors by their birth operation. Each affected operation is multiplied, on its individual local choices, by

$$\mathbf{1}_{\text{designated local births are performed}} e\left(\sum_{\beta \text{ local}} c_\beta p_\beta \right),$$

of modulus at most one. Initial-slot factors merely modify the initial boundary vector by such a multiplier. An affected edge can lose its signed norm saving, but Lemma 10.8 bounds its new norm by L^{C_a}. Unaffected edges retain their complete signed pattern sum and both symmetrizations, so Equation (10.38) still applies.

The ordered indices in a ghost batch do not require ordered-memory functions. For a batch of size b, insert its local multiplier $\phi_b(p_1, \dots, p_b)$ inside the existing $1/b!$ product integral in Equation (10.16). The rest of that integral is symmetric in its fresh variables, so its value is unchanged on replacing ϕ_b by

$$\frac{1}{b!} \sum_{\sigma \in S_b} \phi_b(p_{\sigma(1)}, \dots, p_{\sigma(b)}).$$

This average has modulus at most one, also on numerical diagonals. The resulting operator acts on symmetric memories, and its reversed multiplier is the conjugate on the corresponding deleted batch in Equation (10.19). Its absolute row and column bounds remain Equation (10.22). In

particular no phase has to follow a particle through later storage, propagation, or promotion: equality was encoded entirely at its two birth addresses.

At most $2r$ edges are affected by a rank-r collection. The boundary norms are bounded, and all $N+1$ ghost operations cost at most L^{C_v}. Therefore its Fourier-integrated contribution is at most

$$O\left(L^{(-G+C_v)N+C_v+2r(G+C_a)}\right). \tag{10.47}$$

Since

$$\frac{r_0}{N} = O(T/L^c) + O(N^{-1}) = o(1),$$

the sum over $r < r_0$, with the coefficient bound Equation (10.42), is

$$L^{(-G+C_v+o(1))N} = o(L^{-EN})$$

by Equation (10.37).

Thus the high-rank terms are controlled by their fresh-label costs, while the low-rank terms retain signed contraction on $N - o(N)$ edges. Together they bound the expansion with distinct births, which is the expression required by Lemma 10.7.

Completion of Theorem 10.5. Apply Equation (10.41) in the truncated memory identity. The large- and small-rank estimates bound it by $o(L^{-EN})$, uniformly in θ. Restore the tail Equation (10.28); multiply by the baseline, which is at most one; and integrate Equation (10.11). Finally restore the uniform residue-replacement error, choosing its precision after the absolute path bound. The result is $o(L^{-EN})$, and therefore Equation (10.7) for sufficiently large x. The choice of E_{id} in Equation (10.37) depends only on E, C_v, C_s, and the dependence of these constants was stated above. This completes the proof with the asserted quantifiers. □

10.9 The endpoint pairing consequence

We now convert the root moment into the pairing estimate used in Section 17. Constancy on a mark fiber is needed only at the first endpoint; the second endpoint can be any vector with the stated norm.

Corollary 10.11. *Under the hypotheses and choices of Theorem 10.5, let $f \in \mathcal{H}_{\mathrm{ph}}$ be supported on positions $I = [X, 2X)$ and independent of the chosen marks, with value f_n at position n. Suppose*

$$\sup_{n\in I} |f_n| \le \exp(O(\sqrt{L})), \qquad \|f\|, \|g\| \le X^{1/2} L^{C_3}.$$

Then

$$|\langle f, \mathcal{A}g\rangle| \ll X L^{-E+2C_3}. \tag{10.48}$$

Only the first endpoint must be independent of its mark list.

Proof. Let $H \ge 1$ bound every displacement $|kD|$. Since D has Js factors, all at most $\exp(L^d)$,

$$\log H = O_J(TL^d). \tag{10.49}$$

Partition I into consecutive intervals I_j of length $\lceil H \rceil$, except possibly the last. Put $f_j = \mathbf{1}_{I_j} f$ and restrict g to the H-neighborhood of I_j to obtain g_j. These neighborhoods have bounded overlap, and

$$\langle f, \mathcal{A}g\rangle = \sum_j \langle f_j, \mathcal{A}g_j\rangle, \qquad \sum_j \|g_j\|^2 \ll \|g\|^2.$$

For $B_0 = \mathcal{A}\mathcal{A}^* \ge 0$, spectral Hölder gives

$$|\langle f_j, \mathcal{A}g_j\rangle| \le \|g_j\| \|f_j\|^{1-1/R} \langle f_j, B_0^R f_j\rangle^{1/(2R)}.$$

The finite matrix $(\langle u_n, B_0^R u_m\rangle)_{n,m\in I_j\cap\mathbb{Z}}$ is positive semidefinite. Its largest eigenvalue is at most its trace, so mark independence implies

$$\langle f_j, B_0^R f_j\rangle \le \left(\sum_{n\in I_j\cap\mathbb{Z}} |f_n|^2\right) d_j, \qquad d_j := \sum_{n\in I_j\cap\mathbb{Z}} \langle u_n, B_0^R u_n\rangle.$$

There is no assertion here that the u_n have equal norm, or that ordered lists return to their starting values. By Equation (10.49),

$$\left(O(H)\sup|f_n|^2\right)^{1/(2R)} = \exp\left(O_J\left(\frac{TL^d+\sqrt{L}}{L^{0.52}}\right)\right) = O(1).$$

Sum the resulting block bounds by Hölder with exponents 2, $2R/(R-1)$, and $2R$. This gives

$$|\langle f, \mathcal{A}g\rangle| \ll \|g\|\,\|f\|^{1-1/R}\left(\sum_j d_j\right)^{1/(2R)}.$$

The theorem bounds the last sum by $|I\cap\mathbb{Z}|L^{-2RE} \ll XL^{-2RE}$. Substitution of the two endpoint norm bounds proves Equation (10.48); the extra factor $L^{-C_3/R}$ is bounded for fixed C_3. □

11 Small ideal kernels at every frequency

We use the groups and probability measures of Section 10. In this section each big group $\mathcal{P}_g$, $g\in\mathcal{B}$, is the set of primes in an interval contained in $[\exp(L^c),\exp(L^d)]$. In particular, intersecting a group with a further interval again gives an interval of primes. This additional assumption allows us to use Theorem 9.2 with one label varying and all other labels fixed.

Theorem 11.1 (Construction of the residual ideal operator)**.** *Fix $E_{\mathrm{id}} > 0$ and a fixed frequency exponent $C_4 \ge 0$. There are fixed integers m, J_0 and a fixed constant C_1, depending only on these parameters, ℓ, and the prime-group bounds, with the following property. For every fixed $J \ge J_0$, there is a family consisting of the raw pattern*

$$C_g = \varnothing \quad \text{for every } g, \qquad K_0 = 1,$$

and signed comparison patterns satisfying

$$C_g = \varnothing\ (g\in\mathcal{S}), \qquad |C_g| \le m\ (g\in\mathcal{B}), \qquad \sum_\nu \sup|K_\nu| \le L^{C_1},$$

such that the symmetrized ideal operator of Section 10 satisfies

$$\sup_{\Theta\in\mathbb{R},\ |\zeta|\le L^{C_4}} \|\mathcal{T}(\Theta)\|_{2\to2} \le L^{-E_{\mathrm{id}}} \tag{11.1}$$

for sufficiently large x. The coefficients and the family are independent of Θ and of the individual value of ζ. Their defining parameters are independent of J; J only supplies the available shared slots. The threshold for x may depend on J.

More precisely, designate m of the first J slots in each big group as probes, and split the big groups into two blocks. Each comparison frees nonempty sets A and B of designated probes in the respective blocks, and its coefficient has the form

$$-(-1)^{|A|+|B|}\mathcal{K}_A(D_A,X_A)\mathcal{K}_B(D_B,X_B). \tag{11.2}$$

Here D_A, D_B are the products of the free labels used to form D, X_A is the product of the target labels in A, and X_B the product of the input labels in B. These coefficients use no shared labels and none of the last ℓ labels. Each kernel is a finite sum of products of matched log-cell indicators and pairs of Dirichlet characters of conductor bounded by a fixed power of L.

The construction separates an approximation from a cancellation. We first build kernels that replace selected shared products by independent products in a bilinear pairing, for each prescribed fixed error exponent. The tests may depend on the entire ordered label tuples, as they will when we condition on the labels outside the freed slots.

The cancellation then decomposes the input test in the first-block probes and the target test in the second-block probes. Each decomposition is orthogonal: its pieces are mean zero in a specified set of these probe coordinates and independent of the others. Freeing A averages the source coordinates in A, and freeing B averages the target coordinates in B; the asymmetric coefficient above leaves these coordinates absent from the multiplier. Thus only components independent of the freed coordinates survive. Alternating subset signs cancel every pair for which the input component is independent of at least one first-block probe and the target component is independent of at least one second-block probe. Every remaining pair contains a component that is mean zero in every probe of one block. Such components have small norms by permutation symmetry and the averaging of the last ℓ coordinates. We prove this cancellation after the comparison estimate, so that its use on conditional tuple tests is explicit.

All label tuples in the next three subsections have their product probability measure, with repetitions allowed.

11.1 Products of labels and an elementary bilinear bound

Let Λ be a nonempty set of slots, using $k_g \le m$ slots from group g, and let P_Λ denote the product of its labels. Disjointness of the prime groups and unique factorization give, on the support of this product,

$$\begin{aligned} \rho_\Lambda(n) := \mathbb{P}(P_\Lambda = n) = \frac{1}{n} \prod_g \frac{k_g!}{V_g^{k_g} \prod_{p \in \mathcal{P}_g} a_p!} \le \frac{L^\kappa}{n}, \\ \kappa = C_0\big(\log(m!) + m \log \max(1, v_-^{-1})\big), \end{aligned} \tag{11.3}$$

where a_p is the multiplicity of p in n. The probability is zero if the prescribed group multiplicities do not hold. Indeed, each factor apart from $1/n$ is at most $m! \max(1, v_-^{-1})^m$, and there are at most $C_0 T$ big groups. In particular, this exponent does not depend on J.

This estimate applies to tests on the entire ordered label tuple. If f is any such L^2 test, supported where $c_1 U \le P_\Lambda \le c_2 U$ for fixed $c_1, c_2 > 0$, put $a_n = \mathbb{E}[f \mathbf{1}_{P_\Lambda = n}]$. Cauchy–Schwarz on each fiber gives

$$\sum_n |a_n|^2 \le \sum_n \rho_\Lambda(n) \mathbb{E}[|f|^2 \mathbf{1}_{P_\Lambda = n}] \le \frac{L^\kappa}{c_1 U} \|f\|_2^2. \tag{11.4}$$

No assumption that f is a function of the product has been made.

Lemma 11.2 (Integer bilinear estimate)**.** *Let $U, V \ge 1$. Suppose a_n, b_t are supported in integer intervals of lengths at most CU, CV, respectively, where C is fixed. If*

$$\alpha = u/r + \beta, \qquad (u, r) = 1, \quad r \ge 1, \qquad |\beta| \le r^{-2},$$

then

$$\left| \sum_{n,t} a_n b_t e(\alpha n t) \right| \ll_C \|a\|_{\ell^2} \|b\|_{\ell^2} \left[U + (1 + V/r)\{U + r \log(2r)\} \right]^{1/2}. \tag{11.5}$$

The containing intervals need not begin at 1, and the coefficients may vanish on arbitrary subsets of them.

Proof. Extend the coefficients by zero to their containing intervals. Cauchy–Schwarz in n, followed

by expansion of the square and the geometric-sum bound on the full interval of n, gives

$$\left|\sum_{n,t} a_n b_t e(\alpha nt)\right|^2 \ll_C \|a\|_{\ell^2}^2 \left(U\sum_t |b_t|^2 + \sum_{1\le|h|\le CV} \min\{U, \|\alpha h\|_{\mathbb{R}/\mathbb{Z}}^{-1}\} \sum_t |b_t b_{t+h}|\right)$$
$$\ll_C \|a\|_{\ell^2}^2 \|b\|_{\ell^2}^2 \left(U + \sum_{1\le h\le CV} \min\{U, \|\alpha h\|_{\mathbb{R}/\mathbb{Z}}^{-1}\}\right).$$

At a zero denominator the minimum is interpreted as U. For $r \ge 2$, split the h-range into $O_C(1 + V/r)$ consecutive blocks, each of diameter at most $r/2$. If $h \ne h'$ belong to a block, their difference is nonzero and has magnitude less than r, so

$$\|\alpha(h - h')\|_{\mathbb{R}/\mathbb{Z}} \ge \|u(h - h')/r\|_{\mathbb{R}/\mathbb{Z}} - \frac{|h - h'|}{r^2} \ge \frac{1}{2r}.$$

Thus the phases in a block are $1/(2r)$-separated on the circle. Ordering them by distance from zero bounds their contribution by

$$O_C\left(U + \sum_{1\le j\le 2r} \frac{r}{j}\right) = O_C(U + r\log(2r)).$$

For $r = 1$, the trivial bound $O_C(UV)$ for the whole sum gives the asserted estimate. This proves the lemma. □

For two independent selected products of sizes U, V, combining Equations (11.4) and (11.5) shows that the probability-space bilinear form with phase $e(\alpha P_A P_B)$ has norm at most

$$CL^{\kappa}\left(\frac{1}{V} + \frac{1}{r} + \frac{\log(2r)}{U} + \frac{r\log(2r)}{UV}\right)^{1/2}. \tag{11.6}$$

The same assertion holds with the phase multiplied by $(P_A P_B)^{i\zeta}$ for any real ζ, since this factor separates between the two tests and preserves their norms.

11.2 Log cells and conditional character operators

For a fixed nonempty slot set Λ, let D, X be independent products of label tuples of this type. Given $Q \ge 1$, $0 < h \le 1$, and $0 < \xi \le 1$, put

$$I_j = [jh, (j+1)h), \qquad \nu_{\Lambda j} = \mathbb{P}(\log D \in I_j), \qquad \mathcal{C}_Q = \{\chi : \chi \text{ primitive, } \operatorname{cond}\chi \le Q\}.$$

The conductor-1 character is included. Define

$$\mathcal{K}_\Lambda(D, X) = \sum_{j:\nu_{\Lambda j}\ge\xi} \frac{\mathbf{1}_{\log D\in I_j}\mathbf{1}_{\log X\in I_j}}{\nu_{\Lambda j}} \sum_{\chi\in\mathcal{C}_Q} \overline{\chi(D)}\chi(X). \tag{11.7}$$

We use $\mathcal{K}_A, \mathcal{K}_B$ for the corresponding slot sets. All powers of L chosen below are fixed before J.

There are at most Q^2 characters in $\mathcal{C}_Q$. The number N_Λ of cells of positive mass satisfies

$$N_\Lambda \ll_m 1 + TL^d/h, \qquad \sup|\mathcal{K}_\Lambda| \le Q^2\xi^{-1}, \qquad \sup_X \mathbb{E}_D|\mathcal{K}_\Lambda(D, X)| \le Q^2. \tag{11.8}$$

The last inequality follows because, for fixed retained X, the mass $\nu_{\Lambda j}$ of its cell cancels the denominator. The kernel is zero for discarded X and has support only where $|\log D - \log X| < h$.

We record explicitly the character estimate under a cell restriction. If Q is a fixed power of L and χ, χ' are distinct members of $\mathcal{C}_Q$, then, for every fixed $D_* > 0$,

$$\left|\mathbb{E}\left[\mathbf{1}_{\log D\in I_j}\overline{\chi(D)}\chi'(D)\right]\right| \ll_{D_*} L^{-D_*}, \tag{11.9}$$

uniformly in j and in the nonempty slot set with at most m slots per group. To see this, hold all but one label fixed. The cell and the selected prime group restrict the remaining prime to an interval, possibly empty. The character $\overline{\chi}\chi'$ on the common modulus is nonprincipal: otherwise the uniqueness of primitive induction would imply $\chi = \chi'$. Its modulus is at most Q^2. For every fixed K, Theorem 9.2, summed against the character over reduced residue classes and applied with a larger accuracy exponent, gives

$$A_\rho(t) := \sum_{p \le t} \rho(p) \ll_K t(\log t)^{-K}$$

for these nonprincipal characters, uniformly for $t \ge y := \exp(L^c)$. The allowed modulus is a fixed power of $\log y$, as required there. Partial summation on any subinterval of $[y, \exp(L^d)]$ bounds its reciprocal-prime sum by

$$O_K\left((\log y)^{-K} + \int_y^\infty \frac{dt}{t(\log t)^K}\right) = O_K\big((\log y)^{1-K}\big).$$

Divide by $V_g \ge v_-$ and choose K sufficiently large in terms of D_*. Averaging the other labels proves Equation (11.9); its precision is independent of the width or location of the interval.

For a retained cell of mass ν, the integral operator defined by Equation (11.7) is the character frame operator on the conditional probability space:

$$f \longmapsto \sum_{\chi \in \mathcal{C}_Q} \chi \, \langle \chi, f \rangle_{L^2(\mathbb{P}(\cdot|I_j))}. \tag{11.10}$$

Its nonzero eigenvalues are those of the character Gram matrix. Every diagonal entry is 1, since all selected primes exceed Q^2 for large x. By Equation (11.9), every off-diagonal entry is $O_{D_*}(\xi^{-1}L^{-D_*})$. The Gram matrix therefore has norm at most

$$1 + O_{D_*}(Q^2\xi^{-1}L^{-D_*}) \le 2 \tag{11.11}$$

once the character precision has been chosen sufficiently large. Passing from the conditional to the original measure multiplies both squared norms by ν, so the operator norm is unchanged. Different cells are orthogonal blocks, and discarded cells have zero operator. Consequently kernel integration on the full label space has norm at most 2. The adjoint and transpose have the same bound, so the integration may be moved onto either test in a bilinear form. In particular neither a cell-count factor nor a factor ξ^{-1} is lost in this operator norm.

11.3 Uniform comparison for arbitrary tuple tests

Lemma 11.3 (Comparison kernel)**.** *Fix m, $A_0 > 0$, and $C_4 \ge 0$. There are fixed powers Q, h^{-1}, ξ^{-1} of L, independent of J, such that the kernels in Equation* (11.7) *have the following property. Let A, B be nonempty sets of slots, using at most m slots per big group. Take independent label tuples for D_A, X_A, D_B, X_B with the prescribed slot laws. If f and g are arbitrary L^2 tests on the tuples making up X_B and X_A, respectively, then for every $\alpha \in \mathbb{R}$ and $|\zeta| \le L^{C_4}$,*

$$\begin{aligned}\Big|\mathbb{E}\,\overline{f(X_B)}g(X_A)\big[&(X_AX_B)^{i\zeta}e(\alpha X_AX_B)\\ &- \mathcal{K}_A(D_A,X_A)\mathcal{K}_B(D_B,X_B)(D_AD_B)^{i\zeta}e(\alpha D_AD_B)\,\big]\Big| \le L^{-A_0}\|f\|_2\|g\|_2.\end{aligned} \tag{11.12}$$

Here notation such as $f(X_B)$ denotes a tuple test, not an assumption of dependence only on the product. The kernels are independent of α and ζ.

Proof. Split the tests according to dyads $X_A \in [U, 2U)$ and $X_B \in [V, 2V)$. Each selected product lies between $\exp(L^c)$ and $\exp(mC_0TL^d)$; thus for sufficiently large x there are at most L^{d+1} dyads for each product. We prove a bound L^{-B_0} on each pair of dyads, with $B_0 > A_0 + 2(d+1) + 10$.

Summing the bounds proves the lemma, even using the crude bound of the original test norm for every dyadic restriction.

Let $Z = UV$ and introduce a further fixed power S of L. Dirichlet approximation with maximum denominator $\lfloor Z/S \rfloor$ gives a reduced fraction satisfying

$$\alpha = u/r + \beta, \qquad 1 \le r \le Z/S, \qquad |\beta| \le r^{-2}, \qquad |\beta| Z \le 2S/r. \tag{11.13}$$

Here $Z/S \to \infty$ and the factor 2 accounts for the integer part. This approximation is used only in the proof: the kernel does not depend on the choice of the rational approximation.

Denominators larger than Q. Suppose $r > Q$. Apply Equation (11.6) to the raw term. For the comparison term, move the two separate kernel integrations onto the two tests. Their norms increase by at most 2 each by Equation (11.11). The resulting tests on D_A, D_B are supported in $[U/2, 4U]$ and $[V/2, 4V]$ for large x, by the log localization. The same bilinear bound therefore applies with an absolute constant change. In both uses we collapse arbitrary tuple tests by Equation (11.4) before applying the integer lemma.

The two scales exceed every fixed power of L, and $\log(2r) \le L^{d+1}$ for large x. Since $r > Q$ and $r \le UV/S$, the bound for either term, apart from an absolute constant and the two test norms, is

$$L^{\kappa} \left(o(L^{-A}) + Q^{-1} + L^{d+1} S^{-1} \right)^{1/2} \tag{11.14}$$

for every fixed A. Choosing Q, S sufficiently large gives the required dyadic precision. This part is uniform in all real ζ, because the Mellin factor separates.

Denominators at most Q. Suppose $r \le Q$. All products under consideration are units modulo r. Fourier expansion on the finite group of units gives

$$e(uv/r) = \sum_{\psi \bmod r} c_\psi \psi(v), \qquad \sum_{\psi} |c_\psi| \le \sqrt{\varphi(r)} \le \sqrt{Q}. \tag{11.15}$$

Indeed, in normalized counting measure the left side has squared norm 1, so Parseval gives $\sum |c_\psi|^2 = 1$ and Cauchy–Schwarz proves the stated bound. The unique primitive character inducing ψ has conductor at most r and is included in $\mathcal{C}_Q$.

First remove discarded cells. Their total probability for either product is at most

$$\mathbb{P}(\mathcal{E}) \le (N_A + N_B)\xi, \qquad \mathcal{E} = \{X_A \text{ or } X_B \text{ lies in a discarded cell}\}. \tag{11.16}$$

The comparison term is zero on this event. The raw term there has absolute value at most

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{\mathcal{E}} |f(X_B) g(X_A)|] &\le \mathbb{P}(\mathcal{E})^{1/2} (\mathbb{E}|f(X_B) g(X_A)|^2)^{1/2} \\ &= \mathbb{P}(\mathcal{E})^{1/2} \|f\|_2 \|g\|_2. \end{aligned}$$

The last equality uses independence of the two tuple spaces. This argument also applies to tests concentrated in a discarded cell; it does not assert that restriction to a small event is small as an L^2 operator.

On retained cells, write the remaining scalar multiplier in the log variable as

$$M(v) = \exp(i\zeta v) e(\beta e^v).$$

On the enlarged product range its derivative has magnitude $O(L^{C_4} + S)$, by Equation (11.13). Since both matched log products differ by less than h, we may replace $M(\log(D_A D_B))$ by $M(\log(X_A X_B))$ while leaving the rational phase unchanged. For each fixed retained pair (X_A, X_B), Equation (11.8) bounds the error after integration in D_A, D_B by

$$O(Q^4 h (S + L^{C_4})). \tag{11.17}$$

This is a pointwise error. Pairing it with the tests costs at most the same bound times $\|f\|_2\|g\|_2$, since their spaces are independent.

For a character ψ in Equation (11.15), the Gram calculation also gives the following pointwise reproduction formula for retained X:

$$\mathbb{E}_D\mathcal{K}_\Lambda(D,X)\psi(D) = \psi(X) + O_{D_*}(Q^2\xi^{-1}L^{-D_*}). \tag{11.18}$$

In fact, the summand corresponding to its inducing primitive character is exactly $\psi(X)$: the diagonal conditional expectation is 1. Each other summand is bounded by Equation (11.9) divided by the retained-cell mass. There are at most Q^2 such summands. Choose D_* so that the displayed error is at most 1. Applying the formula on both sides and then Equation (11.15) reproduces $e(uX_AX_B/r)$ with pointwise error

$$O_{D_*}(Q^{5/2}\xi^{-1}L^{-D_*}). \tag{11.19}$$

Multiplication by the fixed scalar $M(\log(X_AX_B))$ has modulus one and does not change this error. Together, Equations (11.16), (11.17) and (11.19) establish the desired comparison on a pair of dyads.

A noncircular choice of parameters. For completeness the following sufficient inequalities fix all the precisions just used. Write

$$Q = L^{q_*}, \quad S = L^{s_*}, \quad h = L^{-h_*}, \quad \xi = L^{-x_*}.$$

After m, A_0, C_4 have been fixed, choose successively

$$\begin{aligned} B_0 &> A_0 + 2(d+1) + 10, \\ q_*, s_* &> 2(B_0 + \kappa + 10) + d + 1, \\ h_* &> 4q_* + \max(s_*, C_4) + B_0 + 10, \\ x_* &> 2B_0 + d + 1 + h_* + 10, \\ D_* &> B_0 + x_* + \tfrac{5}{2}q_* + 10. \end{aligned} \tag{11.20}$$

Indeed $N_A, N_B \le L^{d+1+h_*}$ for large x. The successive lines control, respectively, dyadic summation, Equation (11.14), Equation (11.17), the square root of Equation (11.16), and both Equations (11.11) and (11.19). Fixed implied constants are absorbed by the margins in these inequalities. Finally use Theorem 9.2 at the precision needed for this D_*. None of these choices involves J. □

11.4 Symmetric probes and exact cancellation

Proof of Theorem 11.1. Split $\mathcal{B}$ into blocks $\mathcal{B}_1, \mathcal{B}_2$ of sizes s_1, s_2 differing by at most one. For sufficiently large x each size is at least $c_0T/3$. In each group designate the first m of the first J slots as probes; their sets in the two blocks are $\mathcal{I}_1, \mathcal{I}_2$. Let $P_i = |\mathcal{I}_i| = ms_i$ and $P = P_1 + P_2$. For every pair of nonempty subsets $A \subseteq \mathcal{I}_1$, $B \subseteq \mathcal{I}_2$, free exactly $A \cup B$ and use the coefficient in Equation (11.2), with the kernels just constructed. The other comparison target slots, including the last ℓ, have exactly the independent laws in the ideal operator's definition.

It suffices to bound pairings against symmetric vectors F, H, since the operator has the orthogonal symmetrizing projection on both sides. In every pairing, average out the last ℓ input and target slots in each group. No multiplier uses these coordinates. Denote the resulting functions on the first J coordinates by F', H'. Both operations are contractions. The raw pairing is

$$R(F', H') = \mathbb{E}\left[\overline{F'}H'D_{\mathcal{B}}^{i\zeta}e(\Theta D_{\mathcal{B}})\right], \tag{11.21}$$

where input and target use the same first J labels. Its bilinear norm is at most 1.

We use the orthogonal product-space decomposition and then count the components retained by the symmetry and probe constraints. For a coordinate i, write E_i for averaging its label. Decompose F' by the probes in the first block, and H' by those in the second:

$$F' = \sum_{S \subseteq \mathcal{I}_1} F_S, \qquad F_S = \prod_{i \in S}(1 - E_i) \prod_{i \in \mathcal{I}_1 \setminus S} E_i F',$$
$$H' = \sum_{T' \subseteq \mathcal{I}_2} H_{T'}, \qquad H_{T'} = \prod_{i \in T'}(1 - E_i) \prod_{i \in \mathcal{I}_2 \setminus T'} E_i H'.$$

These are orthogonal decompositions. The indicated components are separately mean zero in each indicated probe and independent of the other probes of their block. Put $F_{\text{full}} = F_{\mathcal{I}_1}$ and $H_{\text{full}} = H_{\mathcal{I}_2}$.

We need the precise effect of having first averaged out the last ℓ slots. In one group, decompose the original symmetric vector on all $M = J + \ell$ coordinates by the same coordinate averaging projections. Among its components on subsets of cardinality j, permutation symmetry makes their squared norms equal. Requiring all m probes to occur in the subset and all last ℓ slots to be absent retains exactly the fraction

$$\frac{\binom{M-m-\ell}{j-m}}{\binom{M}{j}} = \frac{(j)_m (M-j)_\ell}{(M)_{m+\ell}}. \tag{11.22}$$

The fraction is zero when a required cardinality is impossible. If $M \geq (m+\ell)(m+\ell-1)$, then

$$(M)_{m+\ell} = M^{m+\ell} \prod_{i=0}^{m+\ell-1} (1 - i/M) \geq \tfrac{1}{2} M^{m+\ell},$$

using $\prod(1 - a_i) \geq 1 - \sum a_i$ for $0 \leq a_i \leq 1$. Consequently Equation (11.22) is at most

$$2(j/M)^m (1 - j/M)^\ell \leq 2 \left(\frac{m}{m+\ell}\right)^m \left(\frac{\ell}{m+\ell}\right)^\ell \leq 2\ell^\ell m^{-\ell} =: \theta_m. \tag{11.23}$$

This bound tensorizes even when the vectors are not products across groups. Indeed, first resolve the orthogonal decomposition by the tuple of component cardinalities in the groups of the block. Independent within-group permutations give equal squared norms for all tuples of subsets with those cardinalities. The fraction surviving the stated inclusions and exclusions is the product of Equation (11.22) over those groups. Sum the resulting inequality over the cardinality tuples. Averaging the last ℓ coordinates in other groups is a further contraction. We obtain

$$\|F_{\text{full}}\|_2 \leq \theta_m^{s_1/2} \|F\|_2, \qquad \|H_{\text{full}}\|_2 \leq \theta_m^{s_2/2} \|H\|_2. \tag{11.24}$$

Choose m so large that $\theta_m < 1$ and $-(c_0/6) \log \theta_m > E_{\text{id}} + 3$. The right sides are then at most $L^{-E_{\text{id}}-3}\|F\|_2$ and $L^{-E_{\text{id}}-3}\|H\|_2$. This is the only use of large m, and it is possible because $\ell \geq 1$.

Now fix a component pair $(F_S, H_{T'})$ and let

$$U = \mathcal{I}_1 \setminus S, \qquad V = \mathcal{I}_2 \setminus T'$$

be its missing probes. A comparison term with free sets A, B vanishes unless $A \subseteq U$ and $B \subseteq V$. For if $i \in A \cap S$, the input label in coordinate i is unshared, is absent from the multiplier, and is absent from the target. Integrating it annihilates the mean-zero input component. The same reasoning applies to a target label in $B \cap T'$. This uses the deliberate asymmetry in Equation (11.2): its A labels come from the target, and its B labels from the input.

Suppose the inclusions hold. Condition on all shared first-J labels outside $A \cup B$, and write C for their product. The input component is independent of the A labels, and the target component is independent of the B labels. Thus in the raw term the remaining tests are an arbitrary tuple test on X_B and one on X_A, respectively, on independent product spaces. In the comparison term,

the unused input A labels and target B labels integrate out, leaving exactly the same two tests and independent free products D_A, D_B. The fixed shared product contributes $C^{i\zeta}$, of modulus one, and changes the additive frequency to $\alpha = \Theta C$. Lemma 11.3 therefore matches the comparison term before its sign to $R(F_S, H_{T'})$ with error at most

$$L^{-A_0}\|F_S\|_2\|H_{T'}\|_2. \tag{11.25}$$

To justify this bound after conditioning, apply the lemma to the two conditional tests, average its error over the shared labels, and use Cauchy–Schwarz. The averages of their squared conditional norms are precisely $\|F_S\|_2^2$ and $\|H_{T'}\|_2^2$. Uniformity in every real α is essential at this step.

When U, V are nonempty, the signs give exact cancellation:

$$\sum_{\substack{\varnothing\neq A\subseteq U\\ \varnothing\neq B\subseteq V}} (-1)^{|A|+|B|} = \left(\sum_{\varnothing\neq A\subseteq U} (-1)^{|A|}\right)\left(\sum_{\varnothing\neq B\subseteq V} (-1)^{|B|}\right) = (-1)(-1) = 1. \tag{11.26}$$

The initial minus sign in the comparison coefficient therefore cancels the raw pairing for this component pair, up to the errors just estimated. If either missing set is empty, no comparison survives. All these exceptional raw component pairs together are exactly

$$R(F_{\mathrm{full}}, H') + R(F' - F_{\mathrm{full}}, H_{\mathrm{full}}).$$

We sum them before taking absolute values. By Equations (11.21) and (11.24), their total is at most $2L^{-E_{\mathrm{id}}-3}\|F\|_2\|H\|_2$. In particular there is no subset-count loss in this bound.

For the approximation errors, there are at most 3^P admissible triples consisting of a component pair and its free subsets: each probe is either indicated, missing but not freed, or freed. Since $P \le mC_0T$, choose

$$A_0 > E_{\mathrm{id}} + mC_0\log 3 + 10.$$

Then the sum of Equation (11.25), even bounding each component norm by the original norm, is at most $L^{-E_{\mathrm{id}}-10}\|F\|_2\|H\|_2$. Together with the exceptional contribution this proves Equation (11.1) for sufficiently large x.

Finally, the number of comparisons is at most $2^P \le L^{mC_0\log 2}$, and Equation (11.8) bounds each coefficient by $Q^4\xi^{-2}$. For example, after making the choices in Equation (11.20), any fixed

$$C_1 > mC_0\log 2 + 4q_* + 2x_* + 1$$

bounds the total coefficient cost, including the raw pattern. Expanding the two cell sums and two character sums also has only a fixed power of L terms. Both free products contain a prime, so $D_A, D_B \ge \exp(L^c)$. The coefficient uses only the four products specified in Equation (11.2); all assertions about labels and subsequent separation follow directly from Equation (11.7).

The order of choices is now explicit: choose m from Equation (11.24), then A_0 to pay for 3^P, then the dyadic and kernel precisions in Equation (11.20), and hence C_1. Only afterward require $J \ge m$ and $J + \ell \ge (m+\ell)(m+\ell-1)$. These requirements define J_0. A still larger fixed J can therefore be chosen to meet Theorem 10.5 without changing any kernel parameter or the exponent C_1. □

12 Conventions and analytic inputs

Throughout the proof x is a real parameter tending to infinity, and

$$L = \log x, \qquad W = \exp(L^{0.24}), \qquad V(W) = \prod_{p\le W}\left(1 - \frac{1}{p}\right), \qquad \mathrm{e}(t) = \exp(2\pi i t).$$

A positive integer is *rough* if none of its prime factors is at most W; we set $P^-(1) = \infty$. Prime variables in explicitly indicated prime sums range only over primes. The functions $\tau(n)$ and $\tau_j(n)$

count divisors and ordered factorizations into j positive integers, respectively. Dirichlet characters are zero on nonunits. We use $n \asymp H$ for a fixed bounded enlargement of $[H, 2H]$.

Every unspecified exponent of L is fixed before x tends to infinity. Constants may depend on previously fixed data. Whenever an exponent must be independent of a marking parameter, this is stated explicitly. The number of determinant pads in Sections 14–16 grows slowly with L; the padding parameter in Section 17 is instead a sufficiently large fixed integer. These are separate constructions.

The graph transfer and ideal kernels have been proved above. The long prime-polynomial and logarithmic-phase bounds are proved in the section entitled Prime polynomials and logarithmic phases. Here the roughness cutoff is $W = \exp(L^{0.24})$.

12.1 Prime estimates and mean squares

Lemma 12.1 (Prime estimates)**.** *The prime number theorem holds with an error smaller than every fixed negative power of the logarithm. In particular, as $y \to \infty$,*

$$\sum_{p \le y} \frac{1}{p} = \log\log y + c_M + o(1), \qquad \prod_{p \le y} \Bigl(1 - \frac{1}{p}\Bigr) \sim \frac{e^{-\gamma_E}}{\log y},$$

where c_M and γ_E are constants. For fixed $A, C > 0$, uniformly for $r \le (\log y)^C$ and $(a, r) = 1$,

$$\#\{p \le y : p \equiv a \pmod r\} = \frac{\mathrm{Li}(y)}{\varphi(r)} + O_{A,C}\Bigl(\frac{y}{(\log y)^A}\Bigr).$$

The constants need not be effective.

These classical estimates follow from the quantitative prime number theorem, Siegel–Walfisz, and Mertens' theorems; see the Siegel–Walfisz theorem and Mertens' theorems. The harmonic asymptotic also follows by partial summation of the prime number theorem. Thus, for every fixed $a > 0$,

$$\sum_{\exp(L^a) \le p \le \exp(2L^a)} \frac{1}{p} = \log 2 + o(1). \tag{12.1}$$

We use prime asymptotics only on fixed positive-power ranges or on intervals for which taking differences of the stated estimates gives the required absolute error.

For the classical, stronger coefficient-independent mean-value theorem, see the Dirichlet polynomial mean-value theorem. We prove the weaker bounds needed here directly.

Lemma 12.2 (Divisor moments and Dirichlet mean squares)**.** *For every fixed nonnegative integer k there is C_k such that*

$$\sum_{n \le Y} \tau(n)^k \ll_k Y(\log(2Y))^{C_k}, \qquad \sum_{n \le Y} \frac{\tau(n)^k}{n} \ll_k (\log(2Y))^{C_k}.$$

Let $P(t) = \sum_{n \le Y} c_n n^{it}$. On every real interval I of length $T \ge 0$,

$$\int_I |P(t)|^2 \, \mathrm{d}t \ll (T + Y\log(2Y)) \sum_n |c_n|^2, \tag{12.2}$$

$$\sum_{t \in \mathcal{T}} |P(t)|^2 \ll (T + 1 + Y\log(2Y))(\log(2Y))^2 \sum_n |c_n|^2, \tag{12.3}$$

whenever $\mathcal{T} \subset I$ has mutual spacing at least one. The constants in these two inequalities are absolute and do not depend on how the coefficients were formed.

Proof. Unique factorization and positivity bound the harmonic divisor sum by

$$\prod_{p\le Y}\sum_{a\ge 0}\frac{(a+1)^k}{p^a}=\prod_{p\le Y}\left(1+\frac{2^k}{p}+O_k(p^{-2})\right)\ll_k (\log(2Y))^{C_k}.$$

Multiplication by Y gives the counting bound. For the continuous mean square, expand the square and integrate. The diagonal is $T\sum_n |c_n|^2$. For $m\neq n$, the integral has absolute value at most $2/|\log(n/m)|\ll Y/|n-m|$. The inequality $2|c_nc_m|\le |c_n|^2+|c_m|^2$ and the harmonic sum over $|n-m|$ give (12.2). On the unit interval centered at each point of $\mathcal{T}$, the one-dimensional Sobolev inequality bounds $|P(t)|^2$ by a constant times the integral of $|P|^2+|P'|^2$. These intervals have bounded overlap and lie in an interval of length $T+1$. Apply the continuous estimate to P and P', whose coefficients are $i(\log n)c_n$, to obtain (12.3). □

The coefficient-independent formulation in (12.2)–(12.3) is important: later a small-prime polynomial is raised to a growing power, whose coefficient norm is bounded separately by a factorial estimate. Fixed divisor moments alone are not applied at that growing order.

12.2 Prime polynomials and logarithmic phases

Lemma 12.3 (Long prime polynomial)**.** *Fix $0<\tau<\eta<1$ and $C,A>0$. There is $B_0=B_0(\tau,\eta,C,A)$ such that, uniformly for*

$$\frac{x^\tau}{2}\le N\le x^\eta,\qquad q\le L^C,\qquad L^{B_0}\le |t|\le x^2,$$

every Dirichlet character χ modulo q and every interval $I\subset[N,2N]$ satisfy

$$\left|\sum_{p\in I}\chi(p)p^{-1+it}\right|\ll_{\tau,\eta,C,A} L^{-A}. \tag{12.4}$$

The complete proof is Lemma 13.1 in Section 13, with exactly the displayed ranges. Partial summation converts the reciprocal weight in (12.4) to normalization by N on a dyadic interval.

Lemma 12.4 (Logarithmic phases on progressions)**.** *Fix $C>0$. There is an absolute constant $C_3>0$ such that, for sufficiently large x in terms of C, the following holds uniformly:*

$$\exp\Big(\frac{L}{(\log L)^2}\Big)\le N\le 2x^5,\quad q\le L^C,\quad \exp\Big(\frac{L}{2(\log L)^2}\Big)\le |t|\le 4x^3.$$

For every residue $a \pmod q$ and every interval $I\subset[N,2N]$,

$$\left|\sum_{\substack{n\in I\\ n\equiv a \pmod q}} n^{it}\right|\ll \frac{N}{q}\exp\Big(-\frac{L}{(\log L)^{C_3}}\Big). \tag{12.5}$$

The proof is Lemma 13.5 in Section 13. The estimate applies to arbitrary residue classes, not only units. At lower frequencies we use the elementary sum–integral comparison on a progression. For $N\ge x^\delta$, $q\le L^C$ and $1\le |t|\le \exp(L/(\log L)^2)$, it gives

$$\frac{q}{N}\left|\sum_{\substack{n\in I\\ n\equiv a \pmod q}} n^{it}\right|\ll \frac{1}{|t|}+x^{-c_\delta}, \tag{12.6}$$

after decreasing $c_\delta>0$ if necessary. Indeed the integral is $O(N/(q|t|))$, and the endpoint/total-variation error is $O(1+|t|)$. The same estimate holds after partial summation against a smooth factor with fixed logarithmic derivative costs, with those costs included in the implied logarithmic power.

12.3 Elementary rough counts

Lemma 12.5 (Rough integers in long intervals)**.** *Fix $\delta > 0$, $C > 0$, $0 < a < 1$ and $A > 0$. Suppose $x^\delta \le H \le x^C$ and $I \subset [H, 2H]$ is an interval. Then*

$$\#\{n \in I : P^-(n) > W\} = V(W)|I| + O(HL^{-A}). \tag{12.7}$$

If $d \le \exp(L^a)$ and every prime divisor of d exceeds W, then for every residue $b \pmod d$,

$$\#\{n \in I : n \equiv b \pmod d,\ P^-(n) > W\} = \frac{V(W)|I|}{d} + O\Big(\frac{H}{d}L^{-A}\Big). \tag{12.8}$$

For every character χ modulo $q \le L^C$,

$$\sum_{\substack{n\in I\\ P^-(n)>W}} \chi(n) = \mathbf{1}_{\chi \text{ principal}} V(W)|I| + O(HL^{-A}). \tag{12.9}$$

All statements are uniform in the indicated intervals and residues.

Proof. Use consecutive even and odd Bonferroni truncations for the events $p \mid n$, $p \le W$, at depths h and $h+1$, with $h \asymp C_A \log L$. Their divisors are at most

$$W^{h+1} = \exp(O_A(L^{0.24}\log L)) = x^{o(1)}.$$

The number of divisor terms and their total absolute integer coefficient mass are also $x^{o(1)}$. Counting each progression by its length divided by the modulus costs at most one per term. Writing $S_W = \sum_{p\le W} 1/p = O(\log L)$, the difference of the two model bounds is at most

$$\frac{S_W^{h+1}}{(h+1)!},$$

which is smaller than any prescribed power of L^{-1} on choosing the fixed constant C_A sufficiently large. The full model product is $V(W)$.

For (12.8), each sieve divisor is coprime to d, so the Chinese remainder theorem gives main term $|I|/(de)$ for the divisor e. The same omitted-degree bound is multiplied by H/d. The total endpoint error remains $x^{o(1)}$, which is smaller than $(H/d)L^{-A}$, since $d = x^{o(1)}$ and $H \ge x^\delta$. This proves both counting formulas.

For (12.9), first count a fixed unit residue class modulo q. Primes dividing q are automatically absent; the model density in that progression is

$$\frac{1}{q}\prod_{\substack{p\le W\\ p\nmid q}}\Big(1-\frac{1}{p}\Big) = \frac{V(W)}{\varphi(q)}.$$

The endpoint and Bonferroni errors have arbitrary logarithmic precision, uniformly for $q \le L^C$. Sum with the character over the unit residue classes, choosing the precision first to absorb their number. Character orthogonality gives the displayed answer. □

12.4 A sieve for nonnegative weighted objects

Lemma 12.6 (Block sieve)**.** *Consider finitely many objects with nonnegative weights and a bad condition at each of some designated primes $p \le z$. Suppose that the total weight on which all conditions indexed by the primes dividing a squarefree d hold is*

$$Xg(d) + r(d),$$

including $d = 1$, where $X \geq 0$, g is multiplicative, and fixed $\eta_0 > 0, C_0$ satisfy

$$0 \leq g(p) \leq 1 - \eta_0, \qquad \sum_{w<p\leq w^2} g(p) \leq C_0 \quad (w > 1).$$

For every sufficiently large even integer h, the weight avoiding all designated bad conditions is

$$X \prod_{p\leq z}(1 - g(p))(1 + O(e^{-h})) + O\left(\sum_{\substack{d\leq z^{4h+2}\\ d \text{ squarefree}}} |r(d)|\right). \tag{12.10}$$

Products and sums use only designated primes. Constants depend only on the density bounds and are uniform in h. For $h = 2$, an upper bound holds with a fixed constant times the main term and the same remainder sum.

This is the block sieve of S Lemma 9.9. Its upper and lower polynomials have coefficients of absolute value at most one, supported on squarefree $d \leq z^{4h+2}$; the upper polynomial is nonnegative. We will supply all needed remainder estimates for our particular weights. No assertion detecting primes from congruence information alone is included in this input.

13 Prime polynomials and logarithmic phases

This section proves the two estimates used in Section 12: cancellation in prime polynomials up to height x^2, and cancellation of logarithmic phases in arithmetic progressions. We retain the quantitative dependence on the degree in the classical Vinogradov mean-value iteration. The forms below suffice and will be proved in full. Throughout this section, $L = \log x$ and $T = \log L$.

Lemma 13.1 (Long prime polynomial)**.** *Fix $0 < \tau < \eta < 1$, $C > 0$, and $A > 0$. There is a constant $B_0 = B_0(\tau, \eta, C, A)$ such that, uniformly for*

$$\frac{x^\tau}{2} \leq N \leq x^\eta, \qquad q \leq L^C, \qquad L^{B_0} \leq |t| \leq x^2,$$

every Dirichlet character χ modulo q and every interval $I \subset [N, 2N]$ satisfy

$$\left|\sum_{p\in I} \chi(p) p^{-1+it}\right| \ll_{\tau,\eta,C,A} L^{-A}. \tag{13.1}$$

The proof proceeds from a quantitative power-sum estimate to cancellation for a logarithmic phase, then to a high-height zero-free strip, and finally to primes by Mellin inversion.

13.1 Elementary estimates and quantitative power sums

We use two elementary estimates before the mean-value iteration. The first is an immediate consequence of the prime number theorem already recorded in Lemma 12.1.

Lemma 13.2 (Primes for the congruence iteration)**.** *There is an absolute constant $A_1 > 0$ such that, for every integer $k \geq 2$ and every real $y \geq A_1 k^6$, the interval $[y, 2y]$ contains at least $2k^3 + 5$ primes, all greater than k.*

Proof. The prime number theorem gives an absolute y_0 such that $[y, 2y]$ contains at least $y/(2 \log y)$ primes for $y \geq y_0$. For a sufficiently large absolute A_1, this lower bound exceeds $2k^3 + 5$ whenever $y \geq A_1 k^6$, uniformly for $k \geq 2$. Increasing A_1 also ensures $y > k$. □

Lemma 13.3 (Monotone first derivative estimate)**.** *Let f be real-valued and continuously differentiable on an interval. Suppose f' is monotone and takes values in $[j+\lambda, j+1-\lambda]$ for an integer j and $0 < \lambda \le 1/2$. Then, on every subinterval,*

$$\left|\sum_n \mathrm{e}(f(n))\right| \ll \lambda^{-1},$$

where the sum is over the integers in that subinterval and the implied constant is absolute.

Proof. The assertion is immediate for at most one integer. Otherwise write $z_n = \mathrm{e}(f(n))$ and $\delta_n = f(n+1) - f(n) = \int_n^{n+1} f'(u)\,du$ between consecutive summation indices. The numbers δ_n are monotone and lie in $[j+\lambda, j+1-\lambda]$. Put

$$w_n = (\mathrm{e}(\delta_n) - 1)^{-1} = -\frac{1}{2} - \frac{i}{2}\cot(\pi\delta_n).$$

Then $z_n = w_n(z_{n+1} - z_n)$. Summation by parts bounds the sum, including its final term, by $1 + 2\sup_n |w_n| + \sum_n |w_{n+1} - w_n|$. The supremum is $O(\lambda^{-1})$. Since the cotangent is real and monotone on the indicated interval modulo integers, the variation has the same bound. □

We first establish a power-sum estimate with sufficient uniformity in its degree. For integers $k \ge 2$, $s \ge 1$, and $M \ge 1$, let $J_{s,k}(M)$ count the solutions of

$$\sum_{i=1}^{s} u_i^j = \sum_{i=1}^{s} v_i^j \quad (1 \le j \le k), \qquad 1 \le u_i, v_i \le M.$$

Writing $K = k(k+1)/2$ and

$$f(\boldsymbol{\alpha}) = \sum_{n=1}^{M} \mathrm{e}\left(\sum_{j=1}^{k} \alpha_j n^j\right),$$

orthogonality gives $J_{s,k}(M) = \int_{[0,1]^k} |f(\boldsymbol{\alpha})|^{2s}\,d\boldsymbol{\alpha}$.

The argument uses the classical Vinogradov mean-value method in Linnik's p-adic form. We derive the required degree-uniform quantitative estimate below.

Lemma 13.4 (A quantitative power-sum bound)**.** *There are absolute constants $C_1, C_2 > 0$ such that, for every integer $k \ge 2$, some integer s with $k \le s \le C_1 k^4$ satisfies*

$$J_{s,k}(M) \le \exp(C_1 k^{C_2}) M^{2s-K+1/100} \qquad (M \ge 1). \tag{13.2}$$

Proof. We iterate estimates

$$J_{s,k}(M) \le C_s M^{E_s}, \qquad E_s = 2s - K + \epsilon_s, \tag{13.3}$$

starting with $s = k$, $C_k = 1$, and $\epsilon_k = K$. The iteration sends s to $s+k$ and ϵ_s to $(1-1/k)\epsilon_s$.

Choose a sufficiently large absolute constant A_1. If $M^{1/k} < A_1 k^6$, then $\log M \ll k\log(2k)$, and the trivial bound $J_{u,k}(M) \le M^{2u}$ costs at most

$$M^K \le \exp\bigl(O(k^3 \log(2k))\bigr) \tag{13.4}$$

relative to every proposed estimate with exponent $2u - K + \epsilon$, $\epsilon \ge 0$. Thus it suffices to treat $M^{1/k} \ge A_1 k^6$. Lemma 13.2, after enlarging A_1, supplies a fixed list of $2k^3 + 5$ primes r in $[M^{1/k}, 2M^{1/k}]$, all greater than k.

At moment $s+k$, call a tuple degenerate if it has fewer than k distinct coordinates. There are at most $k^{s+k} M^{k-1}$ such tuples. If G is their exponential sum, its contribution on one side of the equations is at most

$$\int_{[0,1]^k} |G|\,|f|^{s+k} \le k^{s+k} M^{k-1} J_{s+k,k}(M)^{1/2}.$$

The contribution with a degenerate tuple on either side is therefore at most twice this quantity.

For a nondegenerate solution, select k distinct coordinates on each side and permute them to the first k positions. This costs at most $(s+k)^{2k}$. The product of the two Vandermonde products is a nonzero integer of absolute value at most $M^{k(k-1)}$. Fewer than $k^2(k-1)+1$ primes of size at least $M^{1/k}$ can divide it. Hence some prime r in our list makes these first k coordinates distinct modulo r on each side.

For such a prime put

$$F_r(\boldsymbol{\alpha}) = \sum_{\substack{1\le z_1,\dots,z_k\le M \\ z_i\not\equiv z_j \pmod r\ (i\neq j)}} \mathrm{e}\left(\sum_{j=1}^{k}\alpha_j\sum_{i=1}^{k} z_i^j\right).$$

The relevant number of solutions is bounded by

$$(s+k)^{2k}\sum_r \int_{[0,1]^k} |F_r|^2 |f|^{2s}.$$

The integrands are nonnegative. Write $f = \sum_{a \bmod r} f_a$, where f_a is restricted to $n \equiv a \pmod r$. Hölder's inequality gives

$$|f|^{2s} \le r^{2s-1} \sum_{a \bmod r} |f_a|^{2s}.$$

Fix a. In the system counted by $\int |F_r|^2 |f_a|^{2s}$, translate all variables by $-a$. Translation preserves the equations for the first k powers by the binomial formula. The remaining s variables on either side are multiples of r, so the first lists $\boldsymbol{z}, \boldsymbol{z}'$ obey

$$\sum_{i=1}^{k} z_i^j \equiv \sum_{i=1}^{k} (z_i')^j \pmod{r^j} \qquad (1 \le j \le k). \tag{13.5}$$

There are at most M^k choices for the first list. For each such list, there are at most $k! r^{K-k}$ choices for the second. To see this, lift the prescribed jth sum modulo r^j to a residue modulo r^k. The number of choices for all lifts is $\prod_{j=1}^{k} r^{k-j} = r^{K-k}$. For each full vector of sums modulo r^k, Newton's identities determine the multiset of roots modulo r, since $r > k$. There are at most $k!$ orderings. The Jacobian of the power sums has determinant

$$k! \prod_{i<j} (z_j' - z_i'),$$

up to sign, and is invertible modulo r. Each ordering therefore lifts uniquely from modulus r to modulus r^k: at each stage the next digits are the unique solution of the corresponding linear system modulo r. Finally, an interval of M integers contains at most one representative of any residue modulo r^k, because $M \le r^k$.

After both first lists are fixed, divide the remaining variables by r. They range over a common interval of at most $\lceil M/r \rceil$ integers and have prescribed differences of power sums. Translation to an initial interval changes only these prescribed differences. The count is a Fourier coefficient of the nonnegative function $|f|^{2s}$ at that shorter length, and consequently is at most $J_{s,k}(\lceil M/r \rceil)$. Summing over a accounts for the final factor r, and gives

$$\begin{aligned} J_{s+k,k}(M) \le\ & 2k^{s+k} M^{k-1} J_{s+k,k}(M)^{1/2} \\ & + (s+k)^{2k}(2k^3+5)k! \max_r r^{2s+K-k} M^k J_{s,k}(\lceil M/r \rceil). \end{aligned} \tag{13.6}$$

Insert (13.3). Since $E_s \ge 0$, rounding costs at most 2^{E_s}, and the exponent of r becomes

$$2s + K - k - E_s = k^2 - \epsilon_s \ge 0.$$

Replacing r by at most $2M^{1/k}$ therefore gives the exponent

$$k + E_s + \frac{k^2 - \epsilon_s}{k} = 2(s+k) - K + (1 - 1/k)\epsilon_s.$$

The inequality $J \le a\sqrt{J} + b$ implies $J \le 2a^2 + 2b$. The exponent $2k-2$ from a^2 is admissible: the target exponent starts at $2k$, and at each step its increase is $2k - \epsilon_s/k \ge 2k - K/k > 0$. Thus the asserted iteration holds. Its constants can be chosen with

$$\log C_{s+k} \le \log(1 + C_s) + O\Big((s+k)\log(2k) + k\log(s+k) + k^3\log(2k)\Big),$$

including (13.4).

After $O(k\log(2k))$ steps, $\epsilon_s = K(1-1/k)^{(s-k)/k} \le 1/100$. Then $s = O(k^2\log(2k)) \le C_1 k^4$, and summing the displayed costs gives $\log C_s \le C_1 k^{C_2}$ for absolute constants. Increasing the exponent from ϵ_s to $1/100$ proves the Lemma. □

13.2 Logarithmic phases on progressions

The quantitative moment bound now gives cancellation for a logarithmic phase even when the Taylor degree grows with x. This step yields the second estimate from Section 12 and will also control Dirichlet series near the line of absolute convergence.

Lemma 13.5 (A logarithmic phase on progressions)**.** *Fix $C > 0$. There is an absolute constant $C_3 > 0$ such that, for sufficiently large x in terms of C, the following holds uniformly:*

$$\exp(L/T^2) \le N' \le 2x^5, \qquad q \le L^C, \qquad \exp(L/(2T^2)) \le |v| \le 4x^3.$$

For every residue $a \pmod q$ and every interval $I \subset [N', 2N']$,

$$\left| \sum_{\substack{n \in I \\ n \equiv a \pmod q}} n^{iv} \right| \ll \frac{N'}{q} \exp(-L/T^{C_3}). \tag{13.7}$$

Proof. Put $H = N'/q$ and $y = \log|v|/\log H$. Then

$$\log H \ge L/T^2 - CT \gg L/T^2, \qquad y \ll T^2. \tag{13.8}$$

Writing $n = q(b + a/q)$, with $0 \le a < q$, reduces the sum, up to a factor of modulus one, to $\sum_{b \in \mathcal{I}} (b + a/q)^{iv}$, where $\mathcal{I}$ is an interval of integers and $H \le b + a/q \le 2H$.

If $y < 4/5$, the derivative of $v\log(b + a/q)/(2\pi)$ is monotone, has magnitude comparable to $|v|/H$, and has magnitude less than $1/2$. The monotone first derivative estimate in Lemma 13.3 therefore bounds the sum by $O(H/|v|)$. The lower bound on $|v|$ makes this smaller than (13.7).

Suppose now that $y \ge 4/5$. Set

$$M = \lfloor H^{3/4} \rfloor, \qquad k = \lceil 4y + 8 \rceil.$$

Averaging the sum over forward shifts $1 \le h \le M$ changes it by $O(M)$: a shift changes an interval at only $O(h)$ endpoints. For $b \in \mathcal{I}$, Taylor expansion gives

$$\frac{v}{2\pi}\log(b + h + a/q) = \frac{v}{2\pi}\log(b + a/q) + \sum_{j=1}^{k} \alpha_j(b) h^j + O(H^{-2}), \qquad \alpha_j(b) = \frac{(-1)^{j-1} v}{2\pi j (b + a/q)^j}.$$

Indeed the remainder is $O(|v|(M/H)^{k+1}) = O(H^{y-(k+1)/4}) = O(H^{-9/4})$, uniformly even as k grows. Thus, with

$$S(\boldsymbol{\alpha}) = \sum_{h=1}^{M} \mathrm{e}\left(\sum_{j=1}^{k} \alpha_j h^j \right),$$

the original sum is bounded by

$$\frac{1}{M}\sum_{b\in\mathcal{I}}|S(\boldsymbol{\alpha}(b))| + O(M + H^{-1}). \tag{13.9}$$

Partition the coefficient torus $[0,1)^k$ into boxes with side length M^{-j} in coordinate j. Each box contains at most

$$\exp(O(k))H^{4/5} \tag{13.10}$$

of the vectors $\boldsymbol{\alpha}(b) \pmod 1$. For this it suffices to use the coordinate $j = \lceil y + 3/20 \rceil$, which lies between 1 and k. Before reduction modulo one, this coordinate has magnitude $O(H^{-3/20})$, is monotone, and has derivative of magnitude at least

$$\frac{|v|}{2\pi(2H)^{j+1}} \geq \exp(-O(k))H^{y-j-1}.$$

A coordinate interval of length M^{-j} on the torus pulls back to at most two intervals in b. Their total length is at most $\exp(O(k))H^{1+j/4-y}$. Here the floor in M costs $\exp(O(k))$, while

$$y - \frac{j}{4} \geq \frac{3y}{4} - \frac{23}{80} \geq \frac{5}{16} > \frac{1}{5}.$$

Counting integer points proves (13.10).

Let s be supplied by Lemma 13.4. We also need the following bound for suprema over boxes Q:

$$\sum_{Q}\sup_{\boldsymbol{\alpha}\in Q}|S(\boldsymbol{\alpha})|^{2s} \leq \exp(O(sk+k))M^K J_{s,k}(M). \tag{13.11}$$

To prove it, rescale each box to $[0,1]^k$. Iterating the one-dimensional fundamental theorem of calculus gives, for any smooth function F on that cube,

$$\sup|F| \leq \sum_{\boldsymbol{\epsilon}\in\{0,1\}^k}\int_{[0,1]^k}|\partial^{\boldsymbol{\epsilon}}F|.$$

Hölder's inequality bounds the $2s$th power of the right-hand side by $2^{k(2s-1)}$ times the sum of the corresponding $2s$th moments. For the rescaled S, every mixed derivative has coefficients bounded in modulus by $(2\pi)^k$: each differentiation in coordinate j introduces the factor $2\pi i h^j/M^j$. On summing over the boxes, change of variables contributes M^K. Orthogonality then bounds the full-torus moment of every such derivative by $(2\pi)^{2sk}J_{s,k}(M)$. This proves (13.11), with constants controlled at the growing degree.

Apply Hölder's inequality to the sum in (13.9), and use (13.10), (13.11), and (13.2). Since $|\mathcal{I}| \ll H$, the result is

$$\frac{1}{M}\sum_{b\in\mathcal{I}}|S(\boldsymbol{\alpha}(b))| \ll H\left(\exp(O(k^{C_4}))H^{-1/5}M^{1/100}\right)^{1/(2s)} \tag{13.12}$$

for an absolute constant C_4. We have $k \ll T^2$ and $s \ll T^8$, whereas

$$-\frac{1}{5}\log H + \frac{1}{100}\log M \leq -\frac{77}{400}\log H.$$

Every fixed power of T is $o(L/T^2)$. Thus the right-hand side of (13.12) is at most $H\exp(-c_1 L/T^{10})$ for some absolute $c_1 > 0$, once x is sufficiently large. The errors in (13.9) are smaller. Increasing a fixed exponent $C_3 > 10$ absorbs c_1 and proves the Lemma. □

13.3 A zero-free strip from the phase estimate

We next convert progression cancellation into a bound for a Dirichlet L-function near $\operatorname{Re} s = 1$. A local logarithmic-derivative formula and the classical positive trigonometric polynomial then exclude zeros in the narrower strip needed for Mellin inversion.

We write $D(s,\chi)$ for the Dirichlet L-function, to distinguish it from $L = \log x$. Characters need not be primitive. Periodicity gives $\sum_{n\le u}\chi(n) = c_\chi u + O(q)$, where $c_\chi = q^{-1}\sum_{a \bmod q}\chi(a)$. Partial summation therefore continues $D(s,\chi)$ meromorphically to $\operatorname{Re} s > 0$, with only the possible simple pole at $s = 1$, and gives, for $\sigma = \operatorname{Re} s$ in a fixed compact subinterval of $(0,\infty)$,

$$D(s,\chi) = \sum_{n\le Y}\frac{\chi(n)}{n^s} + \frac{c_\chi Y^{1-s}}{s-1} + O(q(1+|s|)Y^{-\sigma}). \tag{13.13}$$

Lemma 13.6 (A bound near the line $\operatorname{Re} s = 1$)**.** *Fix $C > 0$ and put $r_* = T^4/L$. Uniformly for $q \le L^C$,*

$$\log|D(\sigma+iv,\chi)| \ll_C T^2 \quad \left(|\sigma-1| \le 10r_*, \quad \frac{1}{2} \le |v| \le 3x^3\right). \tag{13.14}$$

Proof. First suppose $|v| \le \exp(L/(2T^2))$, and use $Y = \exp(2L/T^2)$ in (13.13). Absolute summation gives

$$\sum_{n\le Y} n^{-\sigma} \ll (1+\log Y)\max(1, Y^{1-\sigma}) \le \exp(O(T^2)).$$

Since $|s-1| \ge 1/2$, the pole term has the same bound. The error is at most

$$\exp\left(-\frac{3L}{2T^2} + O(T^2 + CT)\right),$$

and hence is negligible.

For $\exp(L/(2T^2)) < |v| \le 3x^3$, take $Y = x^5$. The terms with $n \le \exp(L/T^2)$ again contribute $\exp(O(T^2))$ in absolute value. On a dyadic interval above this threshold, sum (13.7), with phase $-v$, over the residue classes modulo q, including their character coefficients. The factors q and $1/q$ cancel. Partial summation with $n^{-\sigma}$ bounds that dyad by

$$\exp(-L/T^{C_3})\exp(O(T^4)).$$

There are $O(L)$ dyads, and the same bound applies to a final partial dyad. Their total is negligible, since L/T^{C_3} exceeds every fixed power of T. Finally, the pole term and remainder in (13.13) are bounded respectively by

$$\exp\left(-\frac{L}{2T^2} + O(T^4)\right), \qquad \exp(-2L + O(T^4 + CT)).$$

This proves (13.14). □

Lemma 13.7 (Local logarithmic derivative)**.** *Fix $C > 0$, let $q \le L^C$, and put $s_0 = 1 + r_* + iv$, where $1 \le |v| \le \frac{5}{2}x^3$. There is a radius R with $4r_* \le R \le 5r_*$, with no zero on its boundary, for which*

$$\frac{D'}{D}(s,\chi) = \sum_{|\rho-s_0|<R}\frac{1}{s-\rho} + O_C(T^2/r_*) \qquad (|s-s_0| \le 2r_*), \tag{13.15}$$

away from zeros. The sum counts zeros with multiplicity and has $O_C(T^2)$ terms.

Proof. For large x, the disk $|s-s_0| \le 8r_*$ avoids the possible pole at $s = 1$ and lies in the region of Lemma 13.6. At its center the Euler product gives

$$|D(s_0,\chi)| \ge \prod_p (1+p^{-1-r_*})^{-1} = \frac{\zeta(2+2r_*)}{\zeta(1+r_*)} \gg r_*.$$

Thus $\log|D(s_0,\chi)| \geq -O(T)$. Jensen's formula, using the radius $8r_*$ and the upper bound $O_C(T^2)$, shows that the disk of radius $5r_*$ contains $O_C(T^2)$ zeros. Choose $R \in [4r_*, 5r_*]$ so that none lies on its boundary.

Use centered coordinates $z = s - s_0$, and write $a = \rho - s_0$ for each zero in $|z| < R$. Divide $D(s_0+z,\chi)$ by the disk Blaschke factors

$$B_a(z) = \frac{R(z-a)}{R^2 - \overline{a}z},$$

repeated with multiplicity. The quotient G is holomorphic and nonvanishing on the closed disk, and has the same boundary modulus as D. Maximum modulus gives $\log|G| \leq M_0$ throughout the disk for some $M_0 = O_C(T^2)$. Since $|B_a(0)| < 1$, $\log|G(0)| \geq -O(T)$.

Let h be an analytic logarithm of G. The positive harmonic function $M_0 - \operatorname{Re} h$ has value $O_C(T^2)$ at the center. The Poisson formula, or its derivative together with Harnack's inequality on concentric disks, gives

$$|h'(z)| \ll_C T^2/r_* \qquad (|z| \leq 2r_*).$$

Restoring the factors uses

$$\frac{B_a'}{B_a}(z) = \frac{1}{z-a} + \frac{\overline{a}}{R^2 - \overline{a}z}.$$

The second term is $O(1/r_*)$ on $|z| \leq 2r_*$, uniformly in $|a| < R$. There are $O_C(T^2)$ such terms, giving exactly (13.15). □

Lemma 13.8 (A zero-free strip at large height)**.** *For each fixed $C > 0$ there is $c = c(C) > 0$ such that every character of modulus at most L^C has no zero in*

$$\operatorname{Re} s \geq 1 - cT^2/L, \qquad 1 \leq |\operatorname{Im} s| \leq x^3. \tag{13.16}$$

Moreover,

$$\left|\frac{D'}{D}(s,\chi)\right| \ll_C L \quad \left(1 - \frac{cT^2}{2L} \leq \operatorname{Re} s \leq 1 + \frac{1}{L}, \quad 2 \leq |\operatorname{Im} s| \leq \frac{x^3}{2}\right). \tag{13.17}$$

Proof. For $\sigma > 1$, the Euler products and the inequality $3 + 4\cos\theta + \cos(2\theta) = 2(1+\cos\theta)^2 \geq 0$ give

$$0 \leq -3\frac{\zeta'}{\zeta}(\sigma) - 4\operatorname{Re}\frac{D'}{D}(\sigma+iv,\chi) - \operatorname{Re}\frac{D'}{D}(\sigma+2iv,\chi^2). \tag{13.18}$$

Indeed this follows term by term in the absolutely convergent prime-power expansions; primes dividing the modulus contribute only the positive zeta term. Also $-\zeta'/\zeta(\sigma) = 1/(\sigma-1) + O(1)$ near 1.

Suppose $\rho = \beta + iv$ were a zero in (13.16), and set $\sigma = 1 + 20cT^2/L$. Apply Lemma 13.7 at heights v and $2v$. The evaluation points lie in the respective inner disks, and ρ lies in the first zero sum, because $T^2/L = o(r_*)$. No zero has real part greater than 1, by the absolutely convergent Euler product. All terms in the zero sums therefore contribute nonpositively to the negative real logarithmic derivatives. Retaining the term at ρ, and using $0 < \sigma - \beta \leq 21cT^2/L$, bounds the right-hand side of (13.18) by

$$\left(\frac{3}{20c} - \frac{4}{21c} + O_C(1)\right)\frac{L}{T^2} = \left(-\frac{17}{420c} + O_C(1)\right)\frac{L}{T^2}.$$

Here the errors are $O_C(T^2/r_*) = O_C(L/T^2)$. Choosing a sufficiently small fixed $c > 0$ gives a contradiction.

For s in the region of (13.17), apply the local formula centered at $1 + r_* + i\operatorname{Im} s$. Every zero in its sum has absolute imaginary part between 1 and x^3, since $2 \leq |\operatorname{Im} s| \leq x^3/2$ and $r_* = o(1)$. By (13.16), its horizontal distance from s is at least $cT^2/(2L)$. The $O_C(T^2)$ zero terms and the local error therefore total $O_C(L)$, proving (13.17). □

13.4 Mellin inversion and the prime polynomial

The zero-free strip permits a short contour shift while keeping the imaginary part away from zero. Smoothing the interval first makes the horizontal edges and discarded Mellin tails uniformly negligible.

Proof of Lemma 13.1. We first establish the analogous bound with the von Mangoldt weight. Let $\delta = L^{-A-4}$. Smooth the indicator of $I/N \subset [1,2]$ by convolution with a nonnegative smooth kernel of width δ. This gives $0 \le g \le 1$, supported in $[1/2, 3]$, whose difference from the indicator is supported within $O(\delta)$ of its endpoints, and with $\|g^{(j)}\|_\infty \ll_j \delta^{-j}$. The construction also applies when I is shorter than δN. Since $\Lambda(n) \le \log n$, the error in replacing the interval by $g(n/N)$ is

$$O\Big((\delta N + 1)\frac{\log(3N)}{N}\Big) = O(L^{-A-2}), \tag{13.19}$$

uniformly in I.

Define the Mellin transform by

$$\widehat{g}(z) = \int_0^\infty g(w) w^z \, \frac{dw}{w}.$$

On every fixed bounded real-part strip, integration by parts gives

$$|\widehat{g}(u+iv)| \ll_j L^{(j+1)(A+5)}(1+|v|)^{-j}. \tag{13.20}$$

Also $|\widehat{g}(u+iv)| \ll 1$ there, by absolute integration. Mellin inversion and the absolutely convergent logarithmic derivative on $\operatorname{Re} z = 1/L$ yield

$$\sum_n \Lambda(n)\chi(n)n^{-1+it}g(n/N) = \frac{1}{2\pi i}\int_{1/L-i\infty}^{1/L+i\infty} \widehat{g}(z)N^z \left(-\frac{D'}{D}(1-it+z,\chi)\right) dz. \tag{13.21}$$

On this full line, absolute convergence gives

$$\left|\frac{D'}{D}(1+1/L+iu,\chi)\right| \le \sum_n \frac{\Lambda(n)}{n^{1+1/L}} = -\frac{\zeta'}{\zeta}(1+1/L) \ll L,$$

at every height u.

Choose a fixed $C_5 > A+6$ and put $H_0 = L^{C_5}$. Then choose a fixed integration-by-parts order j large enough that (13.20) makes the two tails with $|\operatorname{Im} z| > H_0$ in (13.21) $O(L^{-A-2})$. Explicitly their bound is

$$O_j\Big(L^{1+(j+1)(A+5)}H_0^{1-j}\Big),$$

since $N^{1/L} \ll 1$. Fix $B_0 > C_5 + 2$. For $L^{B_0} \le |t| \le x^2$, every point in the rectangle

$$-\frac{cT^2}{2L} \le \operatorname{Re} z \le \frac{1}{L}, \qquad |\operatorname{Im} z| \le H_0$$

has

$$2 \le |-t + \operatorname{Im} z| \le x^2 + H_0 < \frac{x^3}{2} \tag{13.22}$$

for large x. Thus Lemma 13.8 applies throughout the rectangle. There are no zeros or poles of the logarithmic derivative inside it; in particular, the possible principal-character pole at $z = it$ is outside it.

Shift the truncated contour to $\operatorname{Re} z = -cT^2/(2L)$. The horizontal edges are $O(L^{-A-2})$, by (13.17) and (13.20), increasing the already fixed order j if necessary. On the new vertical segment,

$$|N^z| = \exp\left(-\frac{cT^2\log N}{2L}\right) \ll \exp(-c\tau T^2/2).$$

Its remaining factors and length cost at most a fixed power of L. Since $\exp(-c\tau T^2/2)$ is smaller than every prescribed fixed power of L^{-1}, the shifted integral is $O(L^{-A-2})$. Together with (13.19), this proves, uniformly for all subintervals $I \subset [N, 2N]$,

$$\sum_{n\in I} \Lambda(n)\chi(n)n^{-1+it} \ll L^{-A-2}. \tag{13.23}$$

Prime powers of exponent at least two contribute in absolute value at most $O(N^{-1/2}(\log(3N))^2)$: there are $O(\sqrt{N}\log(3N))$ possible bases and exponents with $N \le p^a \le 2N$, and every summand has size $O(\log(3N)/N)$. This is smaller than every fixed power of L^{-1} in the present range. Removing them from (13.23) gives the same uniform bound for $\sum_{p\in I}(\log p)\chi(p)p^{-1+it}$. Finally, partial summation against $1/\log u$, whose value and total variation on $[N, 2N]$ are $O_\tau(1/L)$, removes the logarithmic weight and proves (13.1). □

14 A determinant estimate with marks on one side

The first correlation estimate removes prime conditions from factors of the unmarked endpoint. Throughout this section and the next two sections, put

$$L = \log x, \qquad W = \exp(L^{0.24}), \qquad V(W) = \prod_{p\le W}(1 - p^{-1}). \tag{14.1}$$

An integer is *rough* if it has no prime factor at most W. Fix disjoint groups consisting of all the primes in the bands

$$\mathcal{P}_i = \{p : \exp(L^{a_i}) \le p \le \exp(2L^{a_i})\}, \qquad 0.1 < a_1 < \cdots < a_K < 0.2. \tag{14.2}$$

Here K and the a_i are fixed before x tends to infinity. Write

$$\begin{aligned} V_i &= \sum_{p\in\mathcal{P}_i} \frac{1}{p}, \qquad \mu_i(p) = \frac{1}{pV_i}, \\ \omega_i(h) &= \sum_{p\in\mathcal{P}_i} \mathbf{1}_{p|h}, \quad \omega(h) = \sum_{i=1}^{K} \omega_i(h). \end{aligned} \tag{14.3}$$

Lemma 12.1 and partial summation give $V_i = \log 2 + o(1)$. For a fixed $q \in (0, 1)$, the marked weight is

$$\mathcal{W}(h) = q^{\omega(h)-K} \prod_{i=1}^{K} \frac{\omega_i(h)}{V_i}. \tag{14.4}$$

The weight is zero if some group supplies no divisor, and is bounded by a constant depending only on K, q. Indeed, $q^{j-1}j$ is bounded for positive integers j.

Theorem 14.1 (One-sided marked Type II estimate)**.** *Fix $\delta, C, D_* > 0$ and $q \in (0, 1)$. Suppose $H_m, H_n \ge x^\delta$ and $X = H_mH_n \asymp x$. Let F satisfy $|F(h)| \le 1$ and $F(ph) = F(h)$ for every prime in the groups* (14.2)*. Let α_m be supported on an arbitrary interval in $[H_m, 2H_m]$, where it has the value*

$$\alpha_m = m^{iv}\left(\mathbf{1}_{m \text{ prime}} - \frac{\mathbf{1}_{m \text{ rough}}}{V(W)\log m}\right), \qquad |v| \le L^C.$$

Let β_n be supported on rough integers in $[H_n, 2H_n]$, with $|\beta_n| \le L^C$. If K is sufficiently large in terms of δ, C, D_*, q, then

$$\left|\sum_{m,n} \alpha_m\beta_n F(mn-1)\mathcal{W}(mn-1)\right| \ll XL^{-D_*}. \tag{14.5}$$

The required lower bound on K is independent of the particular a_i. The implicit constant and the threshold for x may depend on all the fixed parameters, including the a_i.

We first give the reduction and geometric construction. The signed moment estimate is proved in Proposition 15.1; Proposition 16.1 estimates the resulting major term and completes the proof of the theorem. Exponents denoted $O_C(1)$ below can be chosen independently of K. A factor depending on a fixed K is harmless, but a loss L^{cK} would not be harmless at the final choice of K; we keep this distinction explicit.

14.1 Cauchy's inequality and the normalization of the square

Expand the product of the ω_i by selecting one prime from each group, and denote their product by b_0. For $mn-1=b_0h$, invariance gives $F(mn-1)=F(h)$. Except when some selected prime has square dividing $mn-1$, one also has $\omega(h)=\omega(mn-1)-K$. The replacement of the original damping by $q^{\omega(h)}$ costs

$$O_A(XL^{-A}) \qquad \text{for every fixed } A>0. \tag{14.6}$$

To verify this, the sum over labels of either nonnegative weight is $O_{K,q}(1)$: removing K distinct prime factors can lower ω by at most K, so the new weight is at most the old one. For fixed m and a group prime p, the congruence $mn\equiv 1 \pmod{p^2}$ either has no solution or specifies one residue class of n. Since $H_n\ge x^{\delta}\gg p^2$, its count is $O(H_n/p^2)$. Finally $\sum_{i,p\in\mathcal{P}_i}p^{-2}\ll_K \exp(-L^{0.1})$. The coefficient bounds absorb only a fixed power of L, proving (14.6).

Choose a real smooth dyadic partition with factors $\eta(b_0/Y)$, where η is supported in $[1,4]$ and has bounded derivatives. There are $O_K(L)$ relevant Y, and

$$cL^{0.1}\le \log Y\le C_KL^{0.2}. \tag{14.7}$$

For a fixed factor of this partition the sum becomes

$$\sum_{h\ll X/Y}F(h)q^{\omega(h)}\sum_{\substack{mn-1=b_0h\\ b_0=\prod_i p_i,\ p_i\in\mathcal{P}_i}}\left(\prod_i V_i^{-1}\right)\eta(b_0/Y)\alpha_m\beta_n.$$

Cauchy's inequality bounds its square by $O(X/Y)$ times the nonnegative expanded square

$$\mathcal{Q}_Y=\left(\prod_i V_i^{-2}\right)\sum_{\substack{a,b,m,n,r,s\\ mn-1=ah,\ rs-1=bh \text{ for some } h}}\eta(a/Y)\eta(b/Y)\alpha_m\overline{\alpha_r}\beta_n\overline{\beta_s}. \tag{14.8}$$

Both a and b select one prime per group. In particular, proving $\mathcal{Q}_Y\ll XYL^{-D_1}$ for arbitrarily large fixed D_1 is sufficient; the required D_1 depends on C,D_* but not on K.

We may first discard pairs a,b sharing a label. The elementary bound used here, uniform for $l\asymp Y$, is

$$\sum_{h\ll X/Y}\tau(1+lh)^2\ll (X/Y)L^3. \tag{14.9}$$

Indeed, $\tau(n)^2\le\tau_4(n)$. In an ordered four-factor decomposition of an integer at most $O(X)$, deleting a largest factor leaves a product $d\ll X^{3/4}$. There are at most $4\tau_3(d)$ choices for the three retained factors and their positions. If $d\mid 1+lh$, then $(d,l)=1$, and h lies in one progression modulo d. Consequently the left side of (14.9) is at most a constant times

$$\sum_{d\ll X^{3/4}}\tau_3(d)\left(\frac{X}{Yd}+1\right)\ll \frac{X}{Y}L^3+X^{3/4}L^2\ll\frac{X}{Y}L^3.$$

These divisor estimates also follow from Lemma 12.2. For fixed a,b, Cauchy's inequality and (14.9) bound the absolute h-sum of the two representation counts by $(X/Y)L^{O_C(1)}$. The number of pairs sharing a prime is

$$\ll Y^2\sum_{i,p\in\mathcal{P}_i}p^{-2}\ll_K Y^2\exp(-L^{0.1});$$

here the number of integers in $[Y, 4Y]$ divisible by p is $O(Y/p)$. Thus the discarded part is $O_A(XYL^{-A})$.

For the remaining pairs $(a, b) = 1$. Their two equations in (14.8) are equivalent to

$$t = bmn - ars = b - a. \tag{14.10}$$

In fact, the displayed equality gives $a \mid mn - 1$ and $b \mid rs - 1$, and the quotients agree and are positive.

Fix a real compactly supported smooth ψ, equal to one on $[-4, 4]$. For a fixed exponent $A_0 > 0$, let $\mathfrak{M}$ be the union, on $\mathbb{R}/\mathbb{Z}$, of the arcs

$$\left|\theta - \frac{h}{k}\right| \le \frac{2L^{A_0}}{Y}, \qquad 1 \le k \le L^{A_0}, \quad (h, k) = 1.$$

They are disjoint for large x. The replacement for (14.10) is

$$\mathcal{H}_{\mathfrak{M}}(t; a, b) = \psi(t/Y) \int_{\mathfrak{M}} \mathrm{e}\big(\theta(t - b + a)\big)\, \mathrm{d}\theta. \tag{14.11}$$

In the ultimate major sum a, b need not be disjoint. The signed error in this replacement is controlled by the lift constructed next.

14.2 Physical states, good positions, and endpoint vectors

Use the growing parameters

$$J = \lfloor L^{0.01} \rfloor, \quad M = J + 1, \quad R = \lceil L^{0.5}/2 \rceil, \quad N = 2R. \tag{14.12}$$

An auxiliary list has J primes per group; its product is denoted by D. Partition all possible D into ordinary dyads $[d_0, 2d_0)$. There are $O_K(L)$ such dyads and $\log(2d_0) \le C_K L^{0.21}$. Fix one and put

$$H = d_0 Y, \qquad U = HH_m, \qquad V = H_n, \qquad \Omega = [U, 16U] \times [V, 2V]. \tag{14.13}$$

A *physical state* is a primitive integer vector $P = (P_1, P_2) \in \Omega$ together with an ordered list $\boldsymbol{p}_i = (p_{i,1}, \dots, p_{i,M})$ of distinct primes from $\mathcal{P}_i$ for each i, all dividing P_1. Let $\mathcal{H}$ be the Hilbert space on these states with measure

$$\mathrm{d}\sigma(P, \boldsymbol{p}) = \prod_{i=1}^{K} V_i^{-M} \quad \text{times counting measure.} \tag{14.14}$$

Averaging independently over permutations of each list defines an orthogonal projection $\mathcal{S}$ on $\mathcal{H}$.

We now specify the row operation before symmetrization. Copy slots $1, \dots, J$ in every group from source to target; their product is D. The product of the source's last slots is b. The target's last slots are new labels with product a. Each new slot is summed with weight V_i^{-1}; target positions are summed with counting measure. The labels shared across the edge are exactly the copied slots. Thus all unshared labels are distinct from one another and from every shared label, including across the two endpoints. This restriction will be called the *cross-edge ban*. Require $D \in [d_0, 2d_0)$ and $t = \det(P, Q)/D \in \mathbb{Z}$, and use the multiplier

$$\begin{aligned} \mathcal{K}(P, Q; D, a, b) = \frac{d_0}{D} &\eta(b/Y)\eta(a/Y)\psi(t/Y) \\ &\times \left\{ \mathbf{1}_{t=b-a} - \int_{\mathfrak{M}} \mathrm{e}\big(\theta(t - b + a)\big)\, \mathrm{d}\theta \right\} \\ &\times q^{(\omega(P_1) - KM)/2 + (\omega(Q_1) - KM)/2}. \end{aligned} \tag{14.15}$$

Let $\mathcal{T}$ be this row operation. All sums are finite. The adjoint reverses the row rule and conjugates its multiplier. Indeed, the joint measure of the two lists in an edge pairing is $\prod_i V_i^{-(M+1)}$ in either direction: one factor V_i^{-M} comes from (14.14), and one V_i^{-1} from the new slot.

After slot symmetrization, the copied products D range over all choices obtained by omitting one prime from each full list and lying in the chosen pad dyad $[d_0, 2d_0)$. Each of the M^K omission tuples has weight M^{-K} under the source symmetrization.

We next restrict the positions at which this row operation is used. The first restriction will keep the lattice determined by the shared labels from becoming too thin. The second separates phases associated with the omission choices; in the repeated-prime argument it will leave at most one compatible omission tuple.

For a primitive P, complete it to an integral matrix of determinant one, and let r_P be the first coordinate of the second column divided by P_1, considered modulo one. Changing the completion adds an integer, so r_P is well defined. A position with its full unordered lists is *good* if the following hold.

(i) For every $D \in [d_0, 2d_0)$ obtained by omitting one label per group, there is no integer $1 \le l \le Y^{0.2}$ such that $\|lDr_P\|_{\mathbb{R}/\mathbb{Z}} \le Y^{-0.7}$.

(ii) Let I be any nonempty subset of the groups and $i_* = \max I$. Draw one fresh prime in each group outside I, independently according to μ_i, and let T_f be their product. Except for a set of these draws of probability at most $\exp(-\frac{1}{4}L^{a_{i_*}})$, the points $DZT_f r_P$ corresponding to all distinct integers DZ are pairwise separated in circle distance by $100\exp(-L^{a_{i_*}})$. Here D ranges over the omission products just described, and Z selects one prime from each group in $I \setminus \{i_*\}$.

Let $\mathcal{G}$ be the projection onto good states. The definition is symmetric in the ordered slots, so $\mathcal{G}$ commutes with $\mathcal{S}$. We use the restricted, symmetrized operator

$$\mathcal{A} = \mathcal{G}\mathcal{S}\mathcal{T}\mathcal{S}\mathcal{G}.$$

The first test is used in the lattice construction leading to (15.27); the second is used to prove the unique omission choice in the argument following (15.36).

Lemma 14.2. *For any fixed lists, the set of $r \in \mathbb{R}/\mathbb{Z}$ failing goodness has Lebesgue measure at most $\exp(-cL^{0.1})$, for some $c > 0$ and all sufficiently large x.*

Proof. There are at most M^K omission products. The union bound for the first test is at most $2M^K Y^{-0.5}$, which has the required size by (14.7). For the second test fix I, T_f. The number of trial integers DZ is

$$M^K \prod_{i \in I \setminus \{i_*\}} |\mathcal{P}_i| = \exp(o(L^{a_{i_*}})).$$

For distinct $DZ, D'Z'$, the integer $(DZ - D'Z')T_f$ is nonzero. Multiplication by this integer preserves uniform measure on the circle. The measure on which this pair violates the required separation is therefore at most $200\exp(-L^{a_{i_*}})$. Summing over pairs and then averaging over T_f gives $\exp(-L^{a_{i_*}} + o(L^{a_{i_*}}))$. Markov's inequality at the threshold $\exp(-\frac{1}{4}L^{a_{i_*}})$ gives $\exp(-\frac{3}{4}L^{a_{i_*}} + o(L^{a_{i_*}}))$ for the exceptional set of r. There are only $2^K - 1$ possible I. □

We record precisely the vectors that recover the Cauchy square. At an endpoint require the complete part of P_1 supported on the group primes to be squarefree, with exactly M primes from each group. Denote that part by $G(P_1)$. Put

$$\overline{f(P)} = \alpha_{P_1/G(P_1)}\overline{\beta_{P_2}}, \qquad g(P) = \overline{\alpha_{P_1/G(P_1)}}\beta_{P_2}, \tag{14.16}$$

and set these vectors to zero when the stated conditions or the coefficient supports fail. The vectors are constant on lists. In an edge contributing to their pairing, the coordinates are

$$P = (Dbm, s), \qquad Q = (Dar, n), \qquad \det(P, Q)/D = bmn - ars. \tag{14.17}$$

No new restriction is imposed on the original coefficients: every m, r, n, s with nonzero coefficient is rough, and every group prime is below W. The damping in (14.15) equals one at these endpoints.

Their norms have the uniform bounds

$$\|f\|, \|g\| \ll (UV)^{1/2} L^{O_C(1)}, \qquad \sup|f| + \sup|g| \ll L^{O_C(1)}. \tag{14.18}$$

For example, at a fixed ordered list with product G, the number of possible positions is $O(UV/G)$, since $G = x^{o(1)} \ll U$. Summing its reciprocal product with the state normalization gives

$$\prod_i V_i^{-M} \sum_{\substack{p_{i,1},\dots,p_{i,M} \in \mathcal{P}_i \\ \text{distinct in each group}}} \prod_{i,j} \frac{1}{p_{i,j}} \le 1.$$

This proves (14.18) without a logarithmic exponent depending on K or J.

Conversely, positions in (14.17) on the support of (14.15) are automatically primitive. A prime dividing both Dbm and s cannot divide Db, by roughness. It would therefore divide m, s and be larger than W. It then divides $t = bmn - ars$. The cutoff gives $|t| \ll Y < W$, so $t = 0$. But $bmn = ars$ is impossible: any prime in b divides none of a, r, s. The same proof applies to Q. The ranges in (14.17) lie in (14.13) because $D \in [d_0, 2d_0)$ and $a, b \in [Y, 4Y]$.

14.3 From the moment to endpoint pairings

For a primitive $P \in \Omega$, let $\mathbf{u}_P$ be one on every list at P and zero at other positions. These vectors are not normalized. The estimate proved in Proposition 15.1 is, for any prescribed $E_0 > 0$,

$$\frac{1}{UV} \sum_{P \in \Omega \text{ primitive}} \langle \mathbf{u}_P, (\mathcal{A}\mathcal{A}^*)^R \mathbf{u}_P \rangle \le L^{-E_0 N}. \tag{14.19}$$

First A_0 and then K are chosen sufficiently large; thresholds in absolute errors may subsequently depend on the fixed group exponents. We prove here the consequence

$$|\langle f, \mathcal{A}g \rangle| \ll UV L^{-E_0 + O_C(1)}. \tag{14.20}$$

Partition positions into half-open intervals of the slope P_2/P_1 of length H/U^2. A nonzero edge has $|\det(P, Q)| \ll H$, so only a bounded number of neighboring intervals can interact. Each interval contains $O(H)$ primitive positions. To see this, choose a primitive pivot in it. Its determinant with any other position in the interval is an integer of size $O(H)$. With that determinant fixed, all integer solutions differ by integral multiples of the pivot; the first coordinate ranges over $[U, 16U]$, permitting $O(1)$ multiples.

Let f_j be f restricted to one interval, and g_j the restriction of g to its interacting neighbors. Set $B_0 = \mathcal{A}\mathcal{A}^*$ and

$$d_j = \sum_{P \text{ in interval } j} \langle \mathbf{u}_P, B_0^R \mathbf{u}_P \rangle.$$

The matrix $C^{(j)}_{PQ} = \langle \mathbf{u}_P, B_0^R \mathbf{u}_Q \rangle$ is positive semidefinite on the ordinary coefficient space. Its largest eigenvalue is at most its trace d_j, irrespective of the norms or orthogonality of the testing vectors. Since $f_j = \sum_P f(P)\mathbf{u}_P$, this gives

$$\langle f_j, B_0^R f_j \rangle \le d_j \sum_P |f(P)|^2 \ll H \sup|f|^2 d_j. \tag{14.21}$$

Spectral Hölder for the positive operator B_0, followed by Cauchy's inequality, now gives

$$|\langle f_j, \mathcal{A}g_j \rangle| \le \|g_j\| \, \|f_j\|^{1-1/R} (CH \sup|f|^2 d_j)^{1/(2R)}.$$

Hölder's inequality in j, with exponents $2, 2R/(R-1), 2R$, and bounded overlap of the g_j imply

$$|\langle f, \mathcal{A}g \rangle| \ll \|g\| \, \|f\|^{1-1/R} (CH \sup|f|^2)^{1/(2R)} \left(\sum_j d_j \right)^{1/(2R)}.$$

By (14.7) and (14.13), $\log H = O_K(L^{0.21})$, so $H^{1/N} = O(1)$. Equations (14.19) and (14.18) prove (14.20). This argument uses the unnormalized trace in (14.19); no normalization by the number of lists has been introduced.

14.4 Uniform residues at a primitive root

The root average in (14.19) will be replaced by independent projective residue lines. We give an elementary count that works even when the two coordinate scales are comparable. For a primitive (u, v) in Ω there is a unique completion

$$g = \begin{pmatrix} u & c \\ v & d \end{pmatrix}, \qquad ud - vc = 1, \qquad 0 \le c < u. \quad \text{Put } r = c/u. \tag{14.22}$$

Lemma 14.3 (Root residues)**.** *There is $c_\delta > 0$ such that the following holds. Let S be squarefree with $S \le \exp(C_K L^{0.98})$, and prescribe $g_0 \in \mathrm{SL}_2(\mathbb{Z}/S\mathbb{Z})$. If I_1, I_2, I_3 are intervals in $[1, 16], [1, 2], [0, 1]$, respectively, the number of primitive roots satisfying*

$$u/U \in I_1, \qquad v/V \in I_2, \qquad r \in I_3, \qquad g \equiv g_0 \pmod{S}$$

is

$$\frac{UV\,|I_1||I_2||I_3|}{\zeta(2)|\mathrm{SL}_2(\mathbb{Z}/S\mathbb{Z})|} + O(UVx^{-c_\delta}). \tag{14.23}$$

The error is uniform in the intervals and in g_0.

Proof. We first prove the elementary exponential-sum estimate needed for the count. For $m \ge 1$ put

$$K_m(h, k) = \sum_{y \in (\mathbb{Z}/m\mathbb{Z})^\times} \mathrm{e}\left(\frac{hy + ky^{-1}}{m}\right).$$

Orthogonality shows that $\sum_{h,k \bmod m} |K_m(h, k)|^4 = m^2 T_m$, where T_m counts unit quadruples (y_1, y_2, z_1, z_2) with equal sums and equal inverse sums. The first relation determines $z_2 = y_1 + y_2 - z_1$. After multiplying the inverse-sum relation by the product of the four units, one obtains

$$m \mid (y_1 + y_2)(z_1 - y_1)(z_1 - y_2).$$

The map from the three free residues to these three linear forms has determinant of absolute value two, and its kernel modulo m has size at most two. Distribute, prime by prime, the valuation of m among the three factors. There are $\tau_3(m)$ distributions $d_1 d_2 d_3 = m$; for each, the number of triples of forms with the prescribed divisibilities is $m^3/(d_1 d_2 d_3) = m^2$. Consequently $T_m \le 2m^2 \tau_3(m)$.

The value of $K_m(h, k)$ is constant on the unit-scaling orbit $(h, k) \mapsto (hs, ks^{-1})$. If $g = (h, k, m)$, its stabilizer consists of the units $s \equiv 1 \pmod{m/g}$, so the orbit has $\varphi(m/g) \ge (m/g)m^{-o(1)}$ elements. Dividing the fourth moment bound by this orbit size gives

$$|K_m(h, k)| \ll m^{3/4 + o(1)}(h, k, m)^{1/4}. \tag{14.24}$$

Only the elementary bounds $\tau_3(m) = m^{o(1)}$ and $\varphi(m) \ge m^{1-o(1)}$ enter here; both follow by separating the finitely many small primes from the remaining prime factors.

Suppose first $U \le V$. Write the entries of the prescribed matrix as u_0, c_0, v_0, d_0'; the prime on d_0' distinguishes it from the pad scale. Fix $u \equiv u_0 \pmod{S}$. Modulo $m_0 = uS$, the required pairs (c, v) satisfy

$$c \equiv c_0 \pmod{S}, \quad v \equiv v_0 \pmod{S}, \quad cv \equiv ud_0' - 1 \pmod{uS}. \tag{14.25}$$

There are exactly

$$M(u) = u \prod_{\substack{p \mid u \\ p \nmid S}} (1 - p^{-1}) \tag{14.26}$$

such pairs. Indeed, the admissible c must be coprime to u. At primes common to u, S this is forced by $u_0 d_0' - c_0 v_0 = 1$ modulo S; at other primes the indicated Euler factors count the admissible

c in its progression. For any such c, writing $v = v_0 + Sj$ reduces the last congruence to a linear congruence modulo u with invertible coefficient c, and hence determines exactly one $j \pmod u$.

These $M(u)$ pairs have discrepancy

$$O(u^{0.99}S^2) \tag{14.27}$$

for rectangles in the torus $(c/(uS), v/(uS))$. Here are details of the uniformity in S. Put $S_1 = \prod_{p|(S,u)} p$ and $S_2 = S/S_1$. The Chinese remainder theorem fixes (c, v) modulo S_2 and leaves an inverse graph modulo $m_1 = uS_1$. Write $a' = ud_0' - 1$ and $\gamma = S_2^{-1} \pmod{m_1}$; both are units modulo m_1. On the inverse graph $cv = a'$ the condition $c \equiv c_0 \pmod{S_1}$ already forces $v \equiv v_0 \pmod{S_1}$, since c_0 is a unit there. Additive orthogonality thus gives the Fourier sum, up to a constant of absolute value one, as

$$\frac{1}{S_1} \sum_{j \bmod S_1} \mathrm{e}(-jc_0/S_1) K_{m_1}(\gamma h + ju, \gamma k a'). \tag{14.28}$$

For $0 < \max(|h|, |k|) \le u^{0.02}$,

$$(\gamma h + ju, \gamma k a', m_1) \le S_1(h, k, u) \le S_1 u^{0.02}.$$

Using the exponent $3/4 + 1/200$ in (14.24), each nonzero sum (14.28) is therefore $O(u^{0.76}S^2)$.

For completeness, sandwich each interval indicator between smooth upper and lower functions obtained by enlarging or contracting the interval by $2u^{-0.01}$ and convolving with a probability bump of width $u^{-0.01}$. Their zeroth coefficients differ from the interval length by $O(u^{-0.01})$. Their Fourier coefficients are bounded. There are $O(u^{0.04})$ retained frequency pairs, whose total cost is $O(u^{0.80}S^2)$. For explicit control of the tail, if $\Delta = u^{-0.01}$ and $T_1 = u^{0.02}$, ten integrations by parts give a double Fourier tail outside $\max(|h|, |k|) \le T_1$ bounded by $O(\Delta^{-11}T_1^{-9}) = O(u^{-0.07})$. The trivial bound $M(u) \le u$ makes its contribution $O(u^{0.93})$. The error from the zeroth coefficient is $O(u^{0.99})$. This proves (14.27).

The c interval has length $u|I_3|$, and the v interval has length $V|I_2|$. If the latter passes through several periods of length uS, apply the rectangle count to each; the number of pieces is $O(V/u+1)$. Equations (14.26)–(14.27) give the fixed-u main term

$$\frac{V|I_2||I_3|}{S^2} \prod_{\substack{p|u \\ p\nmid S}} (1 - p^{-1})$$

and error $O((V/u + 1)u^{0.99}S^2)$. For $u \asymp U$ the sum of these errors is $O(UVU^{-0.01}S^2)$.

It remains to average the Euler factor in the progression of u. Expanding it as $\sum_{l|u,(l,S)=1} \mu(l)/l$ gives

$$\sum_{\substack{u/U \in I_1 \\ u \equiv u_0 \ (S)}} \prod_{\substack{p|u \\ p\nmid S}} (1 - p^{-1}) = \frac{U|I_1|}{S} \sum_{(l,S)=1} \frac{\mu(l)}{l^2} + O(\log(2U)).$$

The tail beyond $16U$ contributes $O(1/S)$ and is covered by the error. The series equals $\zeta(2)^{-1} \prod_{p|S}(1 - p^{-2})^{-1}$. Counting first columns and their completions over each field shows

$$|\mathrm{SL}_2(\mathbb{Z}/S\mathbb{Z})| = S^3 \prod_{p|S} (1 - p^{-2}). \tag{14.29}$$

This proves the stated main term. Since $U \ge x^\delta$ and $S^2 = x^{o(1)}$, all the errors admit a saving x^{-c_δ}; for example, $c_\delta = \delta/1000$ is sufficient.

If $V < U$, fix v instead and apply the same argument to the congruence $ud \equiv 1 + vc_0 \pmod{vS}$, with the roles of the coordinates reversed and determinant sign reversed. Use d/v as the third coordinate. For $u, v > 1$ both c/u and d/v belong to $[0, 1)$, and

$$\frac{d}{v} - \frac{c}{u} = \frac{1}{uv}.$$

Enlarging and contracting the prescribed interval by $O((UV)^{-1})$ changes its main mass by $O(1)$ and its count by the already available discrepancy. The same formula follows. □

14.5 Independent residue lines and the archimedean error budget

Expand (14.19) as a path with N edges, returning to its initial position but with no return condition on the lists. In the basis (14.22), each position is gz with primitive $z \in \mathbb{Z}^2$, and the initial and final vectors are $e_1 = (1, 0)$. Put $\tau(z) = z_1 + rz_2$. The cutoff and the physical box imply

$$|z_2| \ll NH, \qquad |z_1| \ll NH, \qquad \tau(z) \asymp 1. \tag{14.30}$$

Indeed, one edge changes slope by $O(H/U^2)$; summing at most N changes gives $z_2 = \det((u, v), gz) = O(NH)$. Also $(gz)_1 = u\tau(z) \asymp U$, which proves the remaining claims. For a group prime p, the condition $p \mid (gz)_1$ means that the projective line $[z]_p$ equals the kernel line of the first row of g modulo p.

Lemma 14.4 (Replacement of the root average)**.** *In the normalized path expansion of* (14.19)*, one may replace the root average by*

$$\frac{1}{\zeta(2)} \int_{[1,16]\times[1,2]\times[0,1]} \mathrm{d}(u/U)\ \mathrm{d}(v/V)\ \mathrm{d}r$$

and by independent uniformly distributed lines in $\mathbb{P}^1(\mathbb{F}_p)$ for all the group primes. The real matrix at an archimedean root is

$$g(u, v, r) = \begin{pmatrix} u & ur \\ v & vr + 1/u \end{pmatrix}.$$

All box and good-state conditions are retained. The total error is $O_A(L^{-AN})$ for every fixed $A > 0$. The same assertion, with polynomial logarithmic coefficient bounds, holds for one edge and for failure of either endpoint's goodness.

Proof. We specify both the combinatorial and the archimedean budgets. There are $\exp(O_K(L^{0.73}))$ choices for paths z satisfying the size bounds in (14.30), labels of all visits, and slot permutations. For example, the logarithm of the number of position paths is $O(N \log(NH)) = O_K(L^{0.71})$; the label choices have logarithm at most $O_K(NML^{0.2}) = O_K(L^{0.71})$; and the permutations cost $O_K(NM \log M)$. These also bound the total absolute coefficients after dropping congruences and bounded damping. The product of the primes active anywhere on a path has logarithm $O_K(L^{0.72})$.

Keep the exact factors of the primes active at some visit. At the other primes the joint damping can be written $1 - X_p$, where $0 \le X_p \le 1$. Here the damping exponent at an endpoint is $q_j = \sqrt{q}$ and at an interior visit is $q_j = q$. In the independent-line model,

$$\sum_p \mathbb{E}X_p \le \sum_p \sum_{j=0}^{N} \frac{1 - q_j}{p + 1} \ll_K N + 1.$$

Upper and lower Bonferroni polynomials of consecutive degrees near $T = \lceil L^{0.76} \rceil$ sandwich $\prod_p (1 - X_p)$ pointwise. Their expected difference is at most

$$\frac{(C_K(N + 1))^T}{T!} = \exp(-\Omega(L^{0.76} \log L)).$$

The product inequality is ordinary inclusion–exclusion, applied to numbers in $[0, 1]$; it requires no independence for the pointwise sandwich. Independence between different residue lines is used only to estimate its gap. This gap absorbs the $\exp(O_K(L^{0.73}))$ path and coefficient budget.

Each monomial in the truncated expressions requires residues at the active primes and at most T further primes. Its squarefree modulus satisfies $\log S = O_K(L^{0.96})$, and hence lies in the range of Lemma 14.3. The total number of monomials, prime choices, and residue matrices is $\exp(o(L))$.

The uniform distribution on $\mathrm{SL}_2(\mathbb{Z}/S\mathbb{Z})$ factors over the primes by the Chinese remainder theorem. At each prime its first row is uniform among nonzero rows, and its kernel is uniform among the $p+1$ projective lines. This is exactly the asserted independent model, with primitive-root density $1/\zeta(2)$.

We next make the archimedean conditions compatible with interval counts. Choose an integral complement w to each primitive z, with $\det(z,w)=1$ and $|w| \ll 1+|z|$. The ratio for the position gz is

$$r_{gz} = \frac{\tau(w)}{\tau(z)} \pmod 1. \tag{14.31}$$

On $\tau(z) \asymp 1$ its derivative in r is $1/\tau(z)^2$. The good tests are therefore constant except at $\exp(o(L))$ values of r: enumerate every product and fresh draw in the tests, every pair of distinct integers DZ, and the integer translates in each circle inequality. All their numbers and sizes are $\exp(o(L))$. The exceptional-probability condition in the second good test is also constant between these endpoints, because it is a finite weighted sum of such indicators.

The physical coordinates are

$$(gz)_1 = u\tau(z), \qquad (gz)_2 = v\tau(z) + z_2/u. \tag{14.32}$$

The normalized box faces consequently have derivatives bounded by $\exp(o(L))$. Near a first-coordinate face the derivative in u/U is bounded below, and near a second-coordinate face the derivative in v/V is bounded below, because $\tau(z) \asymp 1$. One may first exclude $\tau(z)$ near zero, which is incompatible with the first-coordinate box condition. Thus the union of cubes meeting any face or good-test boundary has volume at most $x^{-\epsilon_\delta+o(1)}$ on a grid of side $x^{-\epsilon_\delta}$ in $(u/U, v/V, r)$.

Choose the grid exponent only after the saving in Lemma 14.3; concretely take $\epsilon_\delta = c_\delta/8$. The grid has $O(x^{3\epsilon_\delta})$ cubes, not a subpower number. Summing all root-count errors, including the $\exp(o(L))$ discrete choices, costs at most

$$UVx^{-c_\delta+3\epsilon_\delta+o(1)} = UVx^{-5c_\delta/8+o(1)}. \tag{14.33}$$

The main mass of boundary cubes is at most

$$UVx^{-\epsilon_\delta+o(1)} = UVx^{-c_\delta/8+o(1)}. \tag{14.34}$$

Their discrepancy is already included in (14.33). On each remaining cube all archimedean indicator conditions are fixed. The edge factors in (14.15) depend, after fixing labels, on $\det(z,z')/D$ and hence do not vary with the root; the finite residue factors have already been handled. This proves the replacement with a fixed-power error after normalization. Such an error is $O_A(L^{-AN})$, since $N \asymp L^{0.5}$. The one-edge versions have smaller enumeration budgets and use the same proof. □

It remains to bound the signed independent-line path expansion. The next section keeps the precise correlations caused by a prime that leaves an active list and later returns, proves (14.19), and then removes the good-state projections and the auxiliary pads. This yields the replacement of (14.8) by (14.11) with error $O_A(XYL^{-A})$.

15 Signed memory and the determinant moment

We retain the notation, physical state space, good-state tests, and operator $\mathcal{A}$ of the preceding section. In particular, the number $J = \lfloor L^{.01} \rfloor$ of pads per group grows with x, $M = J+1$, and $N = 2R \asymp L^{.5}$. All constants depending on the fixed number K of groups are allowed to affect the threshold for x. We will explicitly distinguish those constants from powers of L.

Proposition 15.1 (The minor-arc moment)**.** *For every fixed $E_0 > 0$, one can choose A_0 sufficiently large and then K sufficiently large, in terms of E_0, q and the fixed parameters of Theorem 14.1, so that the operator with the good-state projections satisfies*

$$\frac{1}{UV} \sum_{\substack{P \in \Omega \\ P \text{ primitive}}} \langle \mathbf{u}_P, (\mathcal{A}\mathcal{A}^*)^R \mathbf{u}_P \rangle \le L^{-E_0 N}. \tag{15.1}$$

The choices of A_0, K do not depend on the particular fixed band exponents $a_1 < \cdots < a_K$; the threshold for x may depend on them. Consequently, for any prescribed fixed $A > 0$, the determinant indicator in the expanded square of the preceding section can be replaced by (14.11)*, with error $O_A(XYL^{-A})$. In the resulting major term the disjointness restriction on the two unshared products a, b can be omitted. Equivalently, with the notation of* (14.8) *and* (16.1)*,*

$$|\mathcal{Q}_Y - \mathcal{Q}_Y^{\mathrm{maj}}| \ll_A XYL^{-A}. \tag{15.2}$$

The precision needed in the pairing estimate (14.20) *for this consequence is independent of K.*

The proof constructs operators which remember a prime between two of its uses. The primewise identity and every operator estimate needed for the determinant graph are proved below.

15.1 The exact primewise identity

Apply Lemma 14.4 to the expansion of the left side of (15.1). We work for now at a fixed archimedean root and with independent uniform kernel lines at the group primes. Write the path as $z_0, \dots, z_N$, where $z_0 = z_N = e_1$; all the z_j are primitive and satisfy the tube and box conditions of (14.30). Put

$$q_j = \begin{cases} \sqrt{q}, & j = 0, N, \\ q, & 0 < j < N, \end{cases} \qquad \eta_j' = 1 - q_j, \qquad b_p' = 1 - \frac{\sum_{j=0}^{N} \eta_j'}{p+1}, \qquad \nu_i(p) = \frac{1}{V_i(p+1)b_p'} \quad (p \in \mathcal{P}_i). \tag{15.3}$$

Thus $b_p' > 0$ for large x. With $p_{\min} = \min \bigcup_i \mathcal{P}_i$, we have

$$\frac{\nu_i(p)}{\mu_i(p)} = 1 + O\Big(\frac{N+1}{p}\Big), \qquad s_i := \sum_{p \in \mathcal{P}_i} \nu_i(p) = 1 + O\Big(\frac{N+1}{p_{\min}}\Big), \qquad \sup_{i,p} \nu_i(p) \le C_K e^{-L^{.1}}. \tag{15.4}$$

The estimates are uniform in the path. Here and below C_K can also depend on q and on fixed smooth cutoffs.

For one prime p, let A_p be its set of active visits, namely the visits where p occurs in the active list. For a line $\ell \in \mathbb{P}^1(\mathbb{F}_p)$ set $H_j(\ell) = \mathbf{1}_{\ell = [z_j]_p}$. Before active-list normalizations, its factor in the independent-line expectation is exactly

$$\frac{1}{p+1} \sum_{\ell \in \mathbb{P}^1(\mathbb{F}_p)} \prod_{j \in A_p} H_j(\ell) \prod_{j \notin A_p} (1 - \eta_j' H_j(\ell)). \tag{15.5}$$

Indeed the physical damping charges q_j precisely when the prime divides the first coordinate at visit j without belonging to its active list.

First suppose $A_p = \varnothing$. Expand the product in (15.5). The empty and singleton subsets contribute b_p'. In every larger subset fix its first and last indices $a < c$; the sum over all choices of internal indices gives

$$b_p' + \frac{1}{p+1} \sum_{0 \le a < c \le N} \eta_a' \eta_c' \mathbf{1}_{[z_a]_p = [z_c]_p} \prod_{a < j < c} (1 - \eta_j' \mathbf{1}_{[z_j]_p = [z_a]_p}). \tag{15.6}$$

After division by b_p', a summand is a lifespan beginning with a ghost at a and ending with a ghost at c. It has one factor $((p+1)b_p')^{-1}$, one factor $-\eta_j'$ at each endpoint, and a factor q_j at each

strictly internal visit that hits its line. There is no lifespan with just one ghost: all singleton terms have already been included in the baseline.

Next suppose $A_p \neq \varnothing$. If its active lines disagree, the factor is zero. Otherwise let ℓ be their common line and put $a = \min A_p$, $c = \max A_p$. This line is chosen with probability $1/(p+1)$. The inactive visits between a and c retain their damping factors. The products outside this interval have the exact identities

$$\prod_{j<a}(1-\eta'_j H_j(\ell)) = 1 - \sum_{h<a} \eta'_h H_h(\ell) \prod_{h<j<a} (1-\eta'_j H_j(\ell)), \tag{15.7}$$

$$\prod_{j>c}(1-\eta'_j H_j(\ell)) = 1 - \sum_{h>c} \eta'_h H_h(\ell) \prod_{c<j<h} (1-\eta'_j H_j(\ell)). \tag{15.8}$$

These follow by telescoping a finite product, in opposite orders at the two ends. They attach either no ghost or one ghost extension to each end of the active interval. Every inactive internal hit then pays q_j, and each chosen ghost endpoint pays $-\eta'_j$.

Extract $\prod_p b'_p \leq 1$ over all group primes. Equations (15.6)–(15.8) show that every term of the expansion assigns at most one lifespan to each prime. All active visits and ghost endpoints of a lifespan must have the same line. Its probability $1/(p+1)$, together with division by b'_p, contributes the factor $((p+1)b'_p)^{-1}$ exactly once. The remaining factors are the damping at inactive internal visits and the active-list normalization. An *active run* is a maximal interval of consecutive active visits. The ban in the physical edge ensures that a prime active at two consecutive visits is continued in a shared slot. Thus a lifespan in group i with r active runs has factor

$$\frac{V_i^{-r}}{(p+1)b'_p} \tag{15.9}$$

before its ghost signs and internal damping. The power of V_i^{-1} counts active runs; the line probability is paid only once.

A lifespan retains its prime and line while the prime is outside the active list; call such a retained prime *pending*. At a strictly internal inactive visit it is a hit if $[z_j]_p$ equals that line, and a miss otherwise. The prime keeps the same line across both kinds of visit. The same lifespan may join two active runs across pending hits and misses; the stored prime and line remain unchanged.

15.2 The symmetric memory space and its operations

We realize the preceding expansion by storing each pending prime together with its line. A later active run retrieves that stored pair, instead of paying for a new line. The following measures and transfer factors are chosen to reproduce (15.9).

Let $\mathcal{Z}$ be the finite set of primitive integer vectors in the tube which satisfy the fixed-root box condition; it has counting measure. An active list is an ordered M-tuple in each group, endowed with measure $\bigotimes_i \nu_i^{\otimes M}$. The underlying list space allows repetitions. Distinctness within physical lists and the other local restrictions will be imposed in the operators.

A memory particle in group i consists of a prime and an anchor,

$$\mathcal{X}_i = \{(p,\ell) : p \in \mathcal{P}_i,\ \ell \in \mathbb{P}^1(\mathbb{F}_p)\}, \qquad \lambda_i(p,\ell) = \nu_i(p). \tag{15.10}$$

The line coordinate has counting measure, not uniform probability measure. In particular, anchoring a particle at $[z]_p$ leaves the prime measure $\nu_i(p)$, without another factor $1/(p+1)$. A particle *hits* z when its anchor equals $[z]_p$.

For a vector of memory sizes $\boldsymbol{k} = (k_1, \ldots, k_K)$, use symmetric functions of the k_i particles in each group and the measure $\lambda_i^{\otimes k_i}/k_i!$. The Hilbert space is

$$\mathscr{H} = \ell^2(\mathcal{Z}) \otimes L^2\left(\bigotimes_i \nu_i^{\otimes M}\right) \otimes \bigotimes_{i=1}^{K} \left(\bigoplus_{k_i \geq 0} L^2_{\mathrm{sym}}\left(\mathcal{X}_i^{k_i}, \frac{\lambda_i^{\otimes k_i}}{k_i!}\right)\right). \tag{15.11}$$

Products and sums of the operators below use the row convention: the input state is fixed, choices leading to output states are summed, and the resulting operator acts on a test function at the output. Let $\mathbf{b}$ be one when $z = e_1$ and memory is empty, and zero otherwise; it is independent of the active lists. By (15.4),

$$\|\mathbf{b}\|^2 = \prod_i s_i^M = \exp\Big(O_K\Big(\frac{M(N+1)}{p_{\min}}\Big)\Big) = 1 + o(1). \tag{15.12}$$

Fix once and for all

$$0 < \rho < 1, \qquad \rho^2 > \sqrt{q}. \tag{15.13}$$

The ghost operation $\mathcal{G}_j$ leaves the position and active list unchanged. Independently in each group it first chooses a subset of the indexed pending particles which hit the current position and deletes them, paying $-\eta_j'/\rho$ per deletion. Each surviving hit pays q_j/ρ^2. It then appends an ordered batch of $l \geq 0$ primes, integrated with measure $\nu_i^{\otimes l}/l!$, anchored at the current lines, and with coefficient $(-\eta_j' V_i/\rho)^l$. The divisor $l!$ makes the batch an unordered birth set when the primes are distinct; the ordered integral will also be useful after that restriction is removed.

The edge $\mathcal{E}_j$ retains the geometric multiplier in (14.15), or its adjoint according to the alternating physical path, but removes its ω-damping and its active divisibility tests. It replaces target unshared counting and normalization by the following operations between the same endpoint slot symmetrizations.

(i) Pay ρ for every old pending particle which hits the source z. Choose a subset of the unshared source slots to store in memory, anchored at $[z]_p$, and drop the other unshared source slots. Copy the shared J slots in each group.

(ii) Choose a target position z' and its unshared label in each group. A subset of these slots is filled by promoting indexed old particles which hit z', removing them from memory and paying $1/V_i$ per promotion in group i. Particles just stored on this edge cannot be promoted on the same edge. Every other unshared target slot is filled by an independent ν_i draw.

(iii) Pay ρ for every particle remaining in memory which hits z', including newly stored particles. Impose the physical endpoint list restrictions, the cross-edge ban, the pad dyad, all good-state tests and geometric cutoffs, using the actual source and target label products.

Positions are always summed with counting measure. Since the real root matrix has determinant one, its edge determinant is $\det(z, z')$.

For the moment impose also that all *performed births* have globally distinct prime values. Births comprise initial active entries, fresh target entries, and ghost creations. This is a restriction on the fully expanded history, and is not claimed to be a local projection on $\mathscr{H}$. Then the independent-line path sum, divided by its extracted baseline, is exactly

$$\langle \mathbf{b}, \mathcal{G}_0\mathcal{E}_0\mathcal{G}_1 \cdots \mathcal{E}_{N-1}\mathcal{G}_N \mathbf{b}\rangle, \qquad \text{with global birth distinctness imposed.} \tag{15.14}$$

Here equality refers to the choice-by-choice expansion of the right side. To verify it, a shared prime continues with the same line: $p \mid D$ implies $\det(z, z') = 0 \pmod p$, and primitivity implies $[z]_p = [z']_p$. A prime leaving activity either terminates there or is stored. If it is used again, global birth distinctness forces its promotion from that stored particle. Ghost creations and deletions are exactly the optional endpoints of (15.6)–(15.8).

At an internal pending hit, the incoming edge, ghost survival, and outgoing edge give

$$\rho\,(q_j/\rho^2)\,\rho = q_j. \tag{15.15}$$

At a ghost birth the creation coefficient and outgoing hit factor give $-\eta_j' V_i$; at termination the incoming hit factor and deletion give $-\eta_j'$. There are no unmatched factors at $0, N$, because memory starts and ends empty. An active birth has measure $\nu_i(p) = ((p+1)b_p')^{-1} V_i^{-1}$, a ghost birth has

prime measure $V_i\nu_i(p) = ((p+1)b'_p)^{-1}$ after this cancellation, and each promotion pays V_i^{-1}. These are exactly the factors in (15.9): each new active run contributes V_i^{-1}, while the lifespan has only one line-probability factor. Conversely a compatible primewise lifespan determines all these choices: store on leaving any nonfinal active run, promote on entering the next, and create or terminate at its specified ghost endpoints. The factorial birth integral counts its unordered ghost birth set once. This proves the identity, including primes which leave and later re-enter activity.

15.3 Truncation and adjoints

Let $B = \lceil L^2 \rceil$ and let Π_B restrict total memory size to at most B. We may insert Π_B before and after every operation in (15.14), with error

$$O_A(L^{-AN}) \qquad \text{for every fixed } A > 0. \tag{15.16}$$

This assertion is made while births are still globally distinct. There are at most KN stores in a history. A history reaching memory size greater than B has therefore had at least $B - KN$ ghost births. Insert a factor two at each ghost birth in an absolute majorant. For a prime active somewhere, there are at most $(N+2)^2$ possible ghost endpoint pairs. There are $O_K(M+N)$ such primes along a fixed active path. For a never-active prime, fixing its earlier ghost endpoint and summing its possible later endpoints costs $O_q(1)$: successive later endpoints with the required line acquire successive internal factors at most $\sqrt{q} < 1$. Summing the earlier endpoint gives a total $O_q((N+1)/p)$ for the absolute ghost-only contribution, including the inserted factor two. Consequently the product of all never-active prime costs is $\exp(O_K(N+1))$.

The extracted baseline and its reciprocal have logarithms $O_K(N+1)$, by (15.3) and the bounded reciprocal prime sums. The path and label enumeration used in the root replacement costs $\exp(O_K(L^{.73}))$; multiplying by the active endpoint choices and the preceding absolute costs gives, with room to spare, $\exp(O_K(L^{.74}))$. Removing the inserted factors on the omitted histories gives the upper bound

$$2^{-B+KN} \exp(O_K(L^{.74})),$$

which proves (15.16).

We next omit global birth distinctness and work on $\mathscr{H}_B = \Pi_B \mathscr{H}$. The resulting signed norm estimates apply only to the unrestricted history; we will restore distinctness by an exact expansion in equality constraints. Repeated pending values are now permitted. All deletion and promotion choices remain choices of indices, so that the following adjoint identities hold also in their presence.

Suppose a particles are deleted from a memory list of size $k+a$. For an unordered ghost deletion batch its indexed choices and the factorial memory measure give

$$\frac{\binom{k+a}{a}}{(k+a)!} = \frac{1}{k!a!}. \tag{15.17}$$

Its anchors are forced by the hit condition, leaving exactly the ν_i integrations. A fresh ghost batch has the same factorial integral. Interchanging the deleted and appended batches therefore gives the adjoint ghost operation: its deletion coefficient is $-\eta'_j V_i/\rho$ and its creation coefficient is $-\eta'_j/\rho$; the surviving-hit multiplier stays q_j/ρ^2.

For retrieval into a prescribed ordered active slots the corresponding identity is

$$\frac{\binom{k+a}{a}a!}{(k+a)!} = \frac{1}{k!}. \tag{15.18}$$

The retrieved anchors again force one line and leave ν_i prime integrations. Newly stored particles get those same prime measures from their source active entries. Thus at two fixed positions the joint measure of all source and target active entries and all memory coordinates is unchanged on exchanging stores and promotions. The factor $1/V_i$ on a forward promotion stays on that transfer, becoming a reversed store factor; this is a bounded factor. The two endpoint ρ factors exchange

roles. Summing over the two positions preserves the equality because their measure is counting measure. Slot symmetrizations are self-adjoint averages, and all endpoint and edge restrictions transpose with the kernel. These observations prove the adjoint rules even when list restrictions depend jointly on both positions.

For later use, a local choice multiplier of modulus at most one may be inserted provided it is equivariant under simultaneous permutations of the old memory coordinates and the selected deletion or promotion indices. The indexed choice sum then preserves symmetry in the old particles. In a ghost birth integral, a multiplier depending on the ordered fresh batch is averaged over permutations of that batch when acting on symmetric test functions. This leaves the integral unchanged, keeps its modulus at most one, and retains the factor $1/l!$. These modified operations therefore act on the same symmetric spaces. In both adjoint directions they are dominated in absolute value by the positive kernels obtained by summing the absolute values of the unmodified operation-choice coefficients. An arbitrary multiplier depending on an old particle's index alone need not preserve symmetry and is not allowed here.

15.4 Absolute bounds for ghosts and edges

Call an edge *clean* if it stores and promotes no particle, and *dirty* otherwise. A clean edge draws all unshared target labels freshly, and its signed minor-arc kernel will give cancellation. For a dirty edge the good-state tests instead make an absolute row or column sum small. We first bound the ghosts and record crude edge bounds, then prove these two kinds of small edge estimate.

For a state with total pending size k_{tot} use the positive Schur weight

$$w = v_0^{k_{\mathrm{tot}}}, \tag{15.19}$$

where v_0 is a sufficiently large fixed constant. A weighted row bound R means that the absolute row sum against output weight is at most R times input weight; the analogous bound for the adjoint is the weighted column bound C. Weighted Schur gives norm at most $\sqrt{RC}$. Restrictions by Π_B can only decrease these absolute sums.

In one group with h pending hits, the ghost row bound, divided by the input weight, is

$$\exp\left(\frac{v_0\eta_j' V_i s_i}{\rho}\right)\left(\frac{q_j}{\rho^2}+\frac{\eta_j'}{\rho v_0}\right)^h. \tag{15.20}$$

The exponential sums fresh births; the power sums deletion or survival of each old hit. The adjoint bound is

$$\exp\left(\frac{v_0\eta_j' s_i}{\rho}\right)\left(\frac{q_j}{\rho^2}+\frac{\eta_j' V_i}{\rho v_0}\right)^h. \tag{15.21}$$

Since $q_j \le \sqrt{q} < \rho^2$ and the V_i are bounded above and below, one fixed v_0 makes both parentheses at most one, for all i, j and large x. Multiplication over the fixed groups proves

$$\|\mathcal{G}_j\|_{\mathscr{H}_B\to\mathscr{H}_B} \le C_K, \tag{15.22}$$

with the same absolute row and column bounds under the bounded choice modifications just described.

For the crude edge bounds fix the source z and choose an integral complement w_z with $\det(z, w_z) = 1$. Write

$$z' = j_1 w_z + h z, \qquad j_1 = \det(z, z'), \qquad \left|h + j_1\frac{\tau(w_z)}{\tau(z)}\right| \le 16. \tag{15.23}$$

The last inequality follows from $\tau(z')/\tau(z)$ lying in the fixed box range. Enlarging its absolute constant would have no effect on any argument below. For fixed source, labels, and determinant j_1,

this permits $O(1)$ targets. In the raw determinant part $j_1 = D(b-a)$ is fixed. In the comparison part $j_1 = Dt$ has $O(Y)$ possibilities, and its multiplier is at most

$$C|\mathfrak{M}| \ll \frac{L^{3A_0}}{Y}. \tag{15.24}$$

The masses of all fresh label draws are bounded by C_K. There are at most $C_K B^K$ indexed promotion choices. Store subsets, Schur weight ratios and V_i factors cost C_K. Source and target symmetrizations are probability averages. Using the adjoint calculation for columns, we obtain a constant $C = C(K, A_0, q)$ such that

$$R(\mathcal{E}_j),\ C(\mathcal{E}_j) \le C_K B^K (1 + L^{3A_0}) \le L^C. \tag{15.25}$$

This holds for arbitrary multipliers of modulus at most one on the choices. It also holds after fixing any or all fresh prime values, with their measures left outside the bound. This uniformity will preserve the cost of a forced prime atom in the birth-distinctness argument.

15.5 The lattice box and the clean signed norm

We first extract a geometric consequence of the first good-state test. Partition $\mathcal{Z}$ by intervals of length $H = d_0 Y$ for $z_2/\tau(z)$. The identity

$$\frac{z_2'}{\tau(z')} - \frac{z_2}{\tau(z)} = \frac{\det(z, z')}{\tau(z)\tau(z')}$$

shows that an edge connects only cells whose indices differ by a bounded amount. At fixed shared list, hence fixed squarefree product D, partition further by the tuple $([z]_p)_{p|D}$. An edge preserves this tuple. The resulting operator is a sum over a bounded number of cell-index shifts of direct sums of blocks. It suffices to bound each block uniformly.

Choose a relevant good source z in one such pair of cells and a complement w_z as in (15.23). All vectors in either cell with the specified line tuple belong to the lattice

$$\Lambda = \mathbb{Z}z + \mathbb{Z}Dw_z.$$

Indeed, on writing a vector as $hz + jw_z$, equality of the projective lines modulo every $p \mid D$ implies $p \mid j$; squarefreeness gives $D \mid j$. Its coordinates (h, l) in this lattice satisfy

$$|l| \ll Y, \qquad |h + l\alpha| \ll 1, \qquad \alpha = D\frac{\tau(w_z)}{\tau(z)}. \tag{15.26}$$

The l bound follows from the cell separation and $D \asymp d_0$; the other follows from the box condition.

The image of this integer lattice under $(h, l) \mapsto (h + l\alpha, l/Y)$ has covolume $1/Y$. Let λ be the length of its shortest nonzero vector. If $\lambda < Y^{-.8}$, its second coordinate gives $0 < |l| < Y^{.2}$, and its first gives $\|l\alpha\| < Y^{-.8} < Y^{-.7}$; $l = 0$ is impossible for a nonzero vector this short. This contradicts the first good-state test, since $\tau(w_z)/\tau(z)$ is a lift of r_{gz}. Thus $\lambda \ge Y^{-.8}$. Dirichlet's elementary pigeonhole approximation, using denominators at most $\lceil\sqrt{Y}\rceil$, gives $\lambda \ll Y^{-.5}$.

A shortest vector is primitive in the lattice, so complete it to a basis. Subtract a multiple of the first vector from the second to make its parallel component at most $\lambda/2$. Its perpendicular component is $1/(Y\lambda)$; since $\lambda^2 \ll 1/Y$, its length is $O(1/(Y\lambda))$. Inverting this basis on the bounded region in (15.26) puts all relevant vectors in a centered coordinate box with side parameters $O(1/\lambda)$ and $O(Y\lambda)$. Choose its orientation positively. Enlarging constants, the box has parameters A', B' satisfying

$$A'B' \ll Y, \qquad Y^{.1} \le A', B' \le Y^{.9}. \tag{15.27}$$

The change of integer basis has determinant one, so that $\det(z, z')/D$ is the ordinary determinant of their new integer coordinates. This proof applies unchanged for a continuous root: the tested quantity r_{gz} is still $\tau(w_z)/\tau(z) \bmod 1$.

For a clean edge, before the slot symmetrizations, condition on the shared list and all pending particles. They are unchanged by the central operation. Its remaining pending-hit factors are diagonal contractions at the two endpoints. We may omit the extra cross-edge ban at a norm cost smaller than any power of L^{-1}. In fact, keeping distinctness of each full active list, a forbidden coincidence requires one of the fresh labels to equal a fixed source label, costing at most $C_K M \sup_{i,p} \nu_i(p)$. After the labels are fixed, (15.23) and (15.24) give an absolute row bound $C_K L^{3A_0}$; the same argument for the adjoint gives the column bound. Equation (15.4) proves the claimed negligible norm cost.

Inside a lattice block, the unshared ordered list, containing one prime from each disjoint band, is uniquely determined by its product b. Its measure is

$$m(b) = \prod_{i=1}^{K} \nu_i(p_i) \le \frac{C_K}{Y} \quad \text{when } \eta(b/Y) \neq 0. \tag{15.28}$$

Conjugation from $L^2(m)$ to counting measure inserts $\sqrt{m(b)m(a)}$. Consequently the clean norm is at most C_K times the norm of the following operator on the integer box in (15.27) and an unrestricted integer product variable:

$$\frac{1}{Y}\psi\Big(\frac{\det(\boldsymbol{h},\boldsymbol{l})}{Y}\Big)\int_{\mathbb{T}\setminus\mathfrak{M}} \mathrm{e}(\theta(\det(\boldsymbol{h},\boldsymbol{l}) - b + a))\ \mathrm{d}\theta. \tag{15.29}$$

Here $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Extending the product variable means composing with extension by zero and restriction. All actual product supports, full-list restrictions, goodness tests, cutoff factors, and the bounded square-root factors from (15.28) are endpoint diagonal multipliers. The factor d_0/D is bounded. Thus none of these operations increases the bound except by C_K. We use the transposed conjugate kernel on a reversed edge.

Fourier transformation in the unrestricted integer product variable diagonalizes its convolution: for each $\theta \in \mathbb{T} \setminus \mathfrak{M}$ the fiber is the determinant exponential matrix with its smooth cutoff, divided by Y. If $\widehat{\psi}(u) = \int_{\mathbb{R}} \psi(t)\mathrm{e}(-ut)\ \mathrm{d}t$, then

$$\psi\Big(\frac{\det(\boldsymbol{h},\boldsymbol{l})}{Y}\Big)\,\mathrm{e}(\theta\det(\boldsymbol{h},\boldsymbol{l})) = \int_{\mathbb{R}} \widehat{\psi}(u)\mathrm{e}((\theta + u/Y)\det(\boldsymbol{h},\boldsymbol{l}))\ \mathrm{d}u. \tag{15.30}$$

The density $\widehat{\psi}$ decreases faster than any inverse power.

For $|u| \le L^{A_0}/2$, put $\vartheta = \theta + u/Y$. Dirichlet approximation with $Q = \lceil Y/L^{A_0} \rceil$ gives a reduced c/d such that

$$L^{A_0} < d \le Q, \qquad \Big|\vartheta - \frac{c}{d}\Big| \le \frac{1}{dQ} \le \frac{1}{d^2}. \tag{15.31}$$

Indeed $d \le L^{A_0}$ would put θ within $\frac{3}{2}L^{A_0}/Y$ of a rational of allowed denominator, contrary to the definition of $\mathfrak{M}$ with radius $2L^{A_0}/Y$.

For completeness, let $T_{A',B'}(\vartheta)$ have entries $\mathrm{e}(\vartheta hk)$ for integer $|h| \ll A'$, $|k| \ll B'$. Its Gram matrix and the geometric-sum estimate give

$$\|T_{A',B'}(\vartheta)\|^2 \ll \sum_{|h|\ll A'} \min\Big(B', \frac{1}{\|h\vartheta\|}\Big) \ll (A'/d + 1)(B' + d\log(2d)). \tag{15.32}$$

In the term $h = 0$ the minimum is interpreted as B'. To justify the last inequality, split the h interval into blocks of length at most $d/2$. For two different indices in a block, reduction of c/d and the error in (15.31) separate their fractional parts by at least $1/(2d)$. Ordering their distances to the nearest integer bounds each block by $O(B' + d\sum_{1\le j\le d} j^{-1})$.

Up to a permutation of columns, the determinant matrix $\mathrm{e}(\vartheta(h_1l_2 - h_2l_1))$ is the tensor product of $T_{A',B'}(\vartheta)$ and its transposed conjugate. Its norm is therefore the squared norm in (15.32). After division by Y, (15.27) implies the bound

$$\frac{\|(\mathrm{e}(\vartheta\det(\boldsymbol{h},\boldsymbol{l})))\|}{Y} \ll \frac{1}{d} + Y^{-.1}\log(2Y) + \frac{d\log(2d)}{Y} \ll_K L^{.2-A_0} + Y^{-.1}\log(2Y). \tag{15.33}$$

For the complementary Fourier tail in (15.30), the trivial matrix norm is $O(A'B') = O(Y)$, so rapid decay of $\widehat{\psi}$ gives any required negative power of L. Combining the direct sums of cell blocks and the endpoint symmetrizations, we conclude that, for any fixed $G > 0$, choosing A_0 sufficiently large in terms of G gives

$$\|\mathcal{E}_j^{\mathrm{clean}}\| \le L^{-G} \tag{15.34}$$

for large x, with an arbitrarily fixed margin in the exponent. Constants depending on the later fixed value of K are absorbed by that margin and the threshold for x.

15.6 Dirty raw edges: the two Schur sides

Fix the set S of stored groups and the set I of promoted groups. There are at most 4^K choices, a constant. All store and promotion coefficients and the ratio of Schur weights cost C_K.

If $I = \varnothing$, the weighted raw row bound is C_K, independently of B. For each source omission tuple, sample the fresh target label in every group and use (15.23) with its fixed determinant. There are no pending-index choices. The source symmetrization averages the omission tuples and thus does not multiply their number. This argument allows arbitrary S.

Suppose $I \neq \varnothing$ and put $i_* = \max I$. Fix the source state and sample the fresh target labels in the groups outside I, with product T_f. Their product law ν is boundedly comparable to the law μ in the second good-state test, by (15.4). Outside an exceptional mass $C_K \exp(-\frac{1}{4}L^{a_{i_*}})$, that test supplies its asserted phase separation. Fix one old index for promotion in group i_*; there are at most B such indices, and write its prime and anchor as (p_*, ℓ_*). Let Z be the product of the other promoted numeric primes and let P_{full} be the product of the complete source active list. The raw determinant equation is

$$j_1 = P_{\mathrm{full}} - Dp_*ZT_f. \tag{15.35}$$

The ban excludes p_* from the entire source list. Thus $j_1 \equiv P_{\mathrm{full}} \neq 0 \pmod{p_*}$, independently of the omission tuple and of Z.

In (15.23) the line condition $[j_1 w_z + hz]_{p_*} = \ell_*$ is either impossible or fixes a single residue $h = h_* \pmod{p_*}$. To see uniqueness, in the basis (z, w_z) the second coordinate j_1 is nonzero modulo p_*; therefore its projective line determines the first coordinate uniquely. This residue depends on the fixed index and P_{full}, but not on D, Z. Substitution in (15.23), using the real lift $r_z^* = \tau(w_z)/\tau(z)$, yields

$$\left\| DZT_f r_z^* - \frac{P_{\mathrm{full}} r_z^* + h_*}{p_*} \right\| \le \frac{16}{p_*}. \tag{15.36}$$

The phase separation is $100e^{-L^{a_{i_*}}}$, whereas the diameter of the interval in (15.36) is at most $32/p_* \le 32e^{-L^{a_{i_*}}}$. Hence at most one distinct integer DZ is possible.

This also determines D and Z separately. All prime factors of D are source labels and all prime factors of Z are excluded from the source list by the ban. Their prime factorizations therefore separate unambiguously. Because the source list is distinct, D determines which single label was omitted in every group. Under source symmetrization this one numeric omission tuple has mass exactly M^{-K}. It is an average over tuples, including over their internal orders, rather than an unnormalized sum. The disjoint prime bands also make Z determine each other promoted numeric prime.

The labels now fix j_1, so there are $O(1)$ target positions. At any one of them, multiplicities among the remaining promoted indices are absorbed by the outgoing pending damping. If a group has h old particles hitting that target, the choice of one eligible index and the damping of all other old hits cost at most

$$h\rho^{h-1} \le \sup_{n \ge 1} n\rho^{n-1} < \infty. \tag{15.37}$$

This remains true for repeated numeric values and repeated anchors. Additional newly stored hits only decrease the factor. Applying this in each of the other promoted groups costs C_K. For the

exceptional fresh draws use the crude B^K count and (15.23). We obtain the weighted raw row bound

$$\varepsilon_K = C_K \left(\frac{B}{M^K} + B^K e^{-cL^{.1}} \right) \qquad (I \neq \varnothing). \tag{15.38}$$

Reversal exchanges S and I by (15.18), changes only bounded V_i factors, and retains both good-state tests and the ban. Its raw determinant equation, with the reversed sign, again has source full product minus D times the target unshared product. The preceding argument therefore gives the following complete table. Every entry includes the fixed C_K factors.

Stored groups S	Promoted groups I	Weighted row	Weighted column
$\varnothing$	$\varnothing$	C_K	C_K
$\varnothing$	nonempty	ε_K	C_K
nonempty	$\varnothing$	C_K	ε_K
nonempty	nonempty	ε_K	ε_K

The first row is only an absolute estimate; its signed estimate is (15.34). Once K is large enough that $\varepsilon_K \leq 1$, weighted Schur bounds the sum of the three dirty raw cases by

$$C_K \sqrt{\varepsilon_K} \leq C_K L^{1-.005K+o(1)} + O_A(L^{-A}) \quad \text{for every fixed } A. \tag{15.39}$$

The bounded entry opposite a small Schur entry is essential here: it contains no factor B^K. Thus making $.005K$ larger than a prescribed constant purchases that constant in log decay. All remaining K-dependence in this estimate is a fixed multiplicative constant.

15.7 Dirty comparison edges

Use the supremum (15.24), discarding its dependence on the target unshared labels. Fix a source omission tuple, hence D, and suppose $I \neq \varnothing$. A promoted old index with prime p requires one specified projective line at the target. By the ban, $p \nmid D$. The lattice basis used to obtain (15.27) has determinant D in the original integer coordinates and is therefore invertible modulo p. The prescribed line is consequently one nonzero homogeneous linear congruence in the two box coordinates.

For a box of sides A', B', the number of solutions of such a congruence is

$$O\Big(\frac{A'B'}{p} + \max(A', B') \Big) \ll Y \left(\frac{1}{p} + \frac{1}{\min(A', B')} \right). \tag{15.40}$$

If the coefficient of the shorter coordinate is nonzero, fix the other coordinate and count its solutions in that shorter interval. If that coefficient vanishes, the congruence restricts the longer coordinate instead; the same displayed upper bound follows. We have included vectors divisible by p in this count, which only increases it.

For each source only a bounded number of neighboring cell blocks occurs. Take the union over at most B old indices in one promoted group. The other index choices can be bounded by B^{K-1}, or by (15.37). Sum the fresh prime masses and the source omission average. Equations (15.24), (15.27), and (15.40) give a row bound

$$\zeta_K := C_K L^{3A_0} B^K \left(p_{\min}^{-1} + Y^{-.1} \right), \qquad \zeta_K = O_A(L^{-A}) \quad \text{for every fixed } A. \tag{15.41}$$

All parameters K, A_0 are fixed in the last assertion. The opposite Schur side is at most L^C by (15.25). For a store-only piece apply the same argument to the adjoint; with both stores and promotions it applies to both sides. For clarity the comparison counterpart of the dirty table is

Stored groups S	Promoted groups I	Weighted row	Weighted column
$\varnothing$	nonempty	ζ_K	L^C
nonempty	$\varnothing$	L^C	ζ_K
nonempty	nonempty	ζ_K	ζ_K

Weighted Schur makes every dirty comparison piece smaller than every fixed negative power of L.

Choose $G > E_0 + 3$. First choose A_0 so that the clean argument gives more than G powers of decay, and then choose K so that $.005K > G + 2$. Combining the clean, dirty raw, and dirty comparison estimates, with their fixed finite sums, proves

$$\|\mathcal{E}_j\|_{\mathscr{H}_B \to \mathscr{H}_B} \le L^{-G} \tag{15.42}$$

before global birth distinctness is restored. If desired all strict inequalities here can be enlarged by one to absorb the constants. The crude exponent $C(K, A_0, q)$ in (15.25) has not entered the choice needed in (15.39).

15.8 Restoring global birth distinctness

The contractions just proved did not require globally distinct births. We now restore that condition exactly. A union bound over one repeated prime would not suffice against the absolute cost of a long path; we instead separate equality constraints by their rank.

Allocate a finite ordered set $\mathcal{B}$ of potential birth addresses as follows: the KM initial slots; the K canonical target slots at each edge, before its final symmetrization; and B ordered fresh-batch indices per group at each ghost operation. The truncation ensures that a ghost batch has length at most B. Each address has a local *performed flag*: at an edge it is performed exactly if that slot is filled freshly, and in a ghost batch exactly if its index does not exceed the batch length. Initial addresses are always performed. Thus

$$Q_{\text{birth}} := |\mathcal{B}| \le KM + KN + K(N+1)B = L^{O(1)}. \tag{15.43}$$

Order these addresses chronologically, with a fixed order within each operation.

Let $\mathcal{C}$ be a collection of disjoint nonsingleton subsets of $\mathcal{B}$. For each block $C \in \mathcal{C}$ require that every address is performed and that their prime values are equal. Denote this event by E_C. Then the exact distinctness indicator is

$$\mathbf{1}_{\text{distinct performed birth values}} = \sum_{\mathcal{C}} \prod_{C \in \mathcal{C}} (-1)^{|C|-1}(|C|-1)! \prod_{C \in \mathcal{C}} \mathbf{1}_{E_C}. \tag{15.44}$$

One proof expands the right side as the sum of signs of all permutations of the performed addresses which preserve their values: a nontrivial cycle on C has sign $(-1)^{|C|-1}$ and there are $(|C|-1)!$ such cycles. In each equal-value class of size at least two the signs of permutations sum to zero, and for a singleton they sum to one. Unperformed addresses cannot occur in a nontrivial cycle, which is exactly enforced by the flags.

Define the rank

$$r(\mathcal{C}) = \sum_{C \in \mathcal{C}} (|C| - 1). \tag{15.45}$$

A rank-r collection involves at most $2r$ addresses. Its total absolute coefficient mass, summed over all collections of that rank, is at most

$$Q_{\text{birth}}^{2r} = L^{O(r)}. \tag{15.46}$$

Indeed the coefficients count permutations with that cycle rank, and each such permutation has a fixed canonical decomposition into r transpositions. Its ordered list of transpositions, drawn from at most Q_{birth}^2 possibilities, determines the permutation.

We first bound a fixed system of equality constraints absolutely. In each block choose its earliest address as pivot, and condition chronologically on the entire earlier history and all known pivot values. Each nonpivot performed birth must draw one prescribed prime. By (15.4) this costs an atom at most

$$\Delta := \sup_{i,p} \nu_i(p) \le C_K e^{-L^{\cdot 1}}. \tag{15.47}$$

If its prescribed value belongs to a wrong group, the term is zero. No later endpoint conditioning is imposed in this absolute bound: we drop the final return requirement and use weighted row iteration. Thus dependence of intermediate positions on the earlier pivot values cannot change the atom cost.

Here is the precise uniform estimate justifying that iteration. For an edge with s prescribed nonpivot fresh values, fix all fresh values first. The possible target positions still number $O(1)$ in the raw part or $O(Y)$ in the comparison part. The latter count is compensated by (15.24). Consequently its weighted absolute row is at most

$$C_K B^K (1 + L^{3A_0}) \Delta^s, \tag{15.48}$$

uniformly in the input state and all fixed values. The remaining fresh values have bounded total mass, and all local restrictions may be discarded for this upper bound.

For a ghost in group i, set $c_i = v_0 \eta_j' V_i / \rho$. If s specified ordered coordinates of its new batch are prescribed, its birth row sum is at most

$$\sum_{l \geq s} \frac{c_i^l \Delta^s s_i^{l-s}}{l!} \leq (c_i \Delta)^s \exp(c_i s_i). \tag{15.49}$$

The actual requirement that the batch reaches its largest named index can only reduce this sum. Within a batch the earlier pivot coordinates are integrated before the prescribed later ones, which gives the same estimate. Deletion and survival have weighted factor at most one by (15.20). Anchors add no factor $p+1$, as their measure is counting measure. Initial slots have the same single-atom estimate. Thus for each fixed rank-r constraint system, the absolute boundary contribution is at most

$$L^{C(N+1)} (C_K e^{-L^{\cdot 1}})^r, \tag{15.50}$$

for a fixed $C = C(K, A_0, q)$. The boundary weight is one because memory is empty; after dropping return, its terminal indicator is at most the positive Schur weight, so the chronological row bounds apply.

Choose, after C is fixed,

$$r_0 = \left\lceil C' \frac{N \log L}{L^{\cdot 1}} \right\rceil, \tag{15.51}$$

with C' sufficiently large. Combining (15.46) and (15.50), and using $\log Q_{\mathrm{birth}} = O(\log L)$, bounds all ranks $r \geq r_0$ by

$$L^{C(N+1)} \sum_{r \geq r_0} \left(C_K Q_{\mathrm{birth}}^2 e^{-L^{\cdot 1}} \right)^r \leq L^{-(E_0+2)N}. \tag{15.52}$$

The series is geometric for large x. For example its ratio is at most $e^{-L^{\cdot 1}/2}$, so choosing $C'/2 > C + E_0 + 3$ suffices after enlarging the threshold for x.

For ranks $r < r_0$, use signed contraction on most edges. If $C = \{a_0, a_1, \ldots, a_s\}$ is one block, with earliest address a_0, write its equality event by ordinary circle orthogonality as

$$\mathbf{1}_{E_C} = \left(\prod_{a \in C} \mathbf{1}_{a \text{ performed}} \right) \int_{\mathbb{T}^s} \prod_{h=1}^{s} \mathrm{e}(\theta_h (p_{a_h} - p_{a_0}))\, \mathrm{d}\theta_1 \cdots \mathrm{d}\theta_s. \tag{15.53}$$

When an address is unperformed the integrand is interpreted as zero; no value is assigned to an unperformed birth. After the circle variables are fixed, every phase is attached only to the operation where its prime was created. In particular the pivot receives the single local multiplier $\mathrm{e}(-p_{a_0} \sum_h \theta_h)$. Its value need not be remembered through later operations. Flags are local choice restrictions of modulus at most one. An edge flag tests whether its slot is freshly drawn rather than promoted; it does not distinguish old promotion indices. A ghost flag tests whether its fresh batch reaches the specified address. Thus these flags and birth-value phases satisfy the old-memory equivariance condition stated after (15.18). For fixed circle variables, backward induction through

the later operations gives a symmetric continuation function. Averaging the ghost-batch multiplier over its new coordinates consequently leaves the full integral unchanged, without increasing its absolute kernel, changing its factor $1/l!$, or modifying any later operation.

It follows that a rank-r system modifies at most $2r$ operations, and hence at most $2r$ edges. Initial phases change $\mathbf{b}$ by a bounded multiplier and preserve (15.12). Every ghost, modified or not, has norm at most C_K; a modified edge has norm at most L^C by (15.25); all other edges retain (15.42). For each fixed set of phases the boundary matrix element is therefore at most

$$C_K^{N+1}\|\mathbf{b}\|^2 L^{-G(N-2r)} L^{2Cr}. \tag{15.54}$$

Integration of phases does not increase this bound. As $r_0/N = O(\log L/L^{\cdot 1}) = o(1)$, summing (15.54) with (15.46) gives

$$\sum_{r<r_0} Q_{\mathrm{birth}}^{2r} C_K^{N+1}\|\mathbf{b}\|^2 L^{-G(N-2r)+2Cr} \le L^{(-G+o(1))N}. \tag{15.55}$$

This remains true even though C depends on K: K was fixed before letting x grow, and its crude exponent is paid on only $o(N)$ edges.

Equations (15.52) and (15.55) bound (15.14) with exact global birth distinctness, uniformly in all archimedean root variables. Multiply back the baseline, whose modulus is at most one. The root integral has bounded mass. The error (15.16) and the root-replacement error are $O_A(L^{-AN})$ for arbitrary fixed A. Because $G > E_0 + 3$, these margins absorb all constants and prove (15.1). The passage from this moment to (14.20) was proved in the preceding section.

15.9 Removing goodness and stripping the pads

We complete the second assertion of Proposition 15.1. First remove the good-state projections from the endpoint pairing. The part where the source is bad, and separately the part where the target is bad, may be majorized absolutely; bound the endpoint coefficients by their log-power suprema. Apply the single-edge version of the root replacement. In the independent-line model, a shared prime requires the same line at both positions, or contributes zero. Every distinct prime in the union of the two active lists is charged exactly once by $1/(p+1)$. The ban makes the union consist of $M+1$ distinct primes per group. Its list normalization is $V_i^{-(M+1)}$, so summing these probabilities over ordered union lists is at most

$$\prod_{i=1}^{K}\left(\sum_{p\in\mathcal{P}_i}\frac{1}{V_i(p+1)}\right)^{M+1} \le 1. \tag{15.56}$$

We have dropped any incompatibilities for an upper bound. Damping is also at most one on the physical states and may be discarded.

For fixed labels, (15.23) counts $O(1)$ raw targets; the comparison target count is $O(Y)$ and is compensated by (15.24). For every fixed source list, the set of root ratios failing either good-state test has measure $O(e^{-cL^{\cdot 1}})$ by Lemma 14.2. This estimate is uniform in the other root coordinates. The preceding target counts hold for every ratio, so they may be integrated over that bad set. Reversal proves the identical assertion for failure of target goodness. Equation (15.56) then gives the normalized pairing error

$$O\Big(UVL^{C_1+3A_0}e^{-cL^{\cdot 1}}\Big) + O_A(UVL^{-A}), \tag{15.57}$$

where C_1 comes from the endpoint suprema. The root-replacement error can here be made smaller than every log power by its single-edge estimate; this also handles the indicators of goodness failure. Therefore removing the projections costs $O_A(UVL^{-A})$ for every fixed A.

The endpoint vectors are invariant under permutations of their lists. Hence the two slot symmetrizations disappear when the pairing is expanded. Write the ordered pads as $p_{i,1},\dots,p_{i,J}$ in

group i, with total product D. The remaining source and target slots have products b and a. The common state normalization and the fresh-target normalization are $\prod_i V_i^{-(J+2)}$. Divide the pairing for the dyad d_0 by d_0. Its scalar multiplier becomes $1/D$, and the exact measure identity is

$$\frac{1}{D}\prod_{i=1}^{K} V_i^{-(J+2)} = \left(\prod_{i=1}^{K}\prod_{h=1}^{J} \mu_i(p_{i,h})\right)\prod_{i=1}^{K} V_i^{-2}. \tag{15.58}$$

Thus the pads are independent ordered harmonic draws, and the remaining factors are exactly the two unshared-list normalizations in the expanded square. There is no factorial or M-dependent normalization left over: the pads occupy prescribed ordered shared slots and the unshared label occupies its prescribed remaining slot.

The change of variables is $P = (Dbm, s)$, $Q = (Dar, n)$, so $\det(P, Q)/D = bmn - ars$. The support and roughness properties of the endpoint vectors give the original coefficients and make the physical damping equal to one. Primitivity and the box conditions are automatic on their nonzero support, as established in the preceding section. On summing the ordinary pad dyads every ordered pad tuple occurs once. Therefore, apart from the pad collision restrictions, (15.58) gives exactly the signed difference between the determinant condition and $\mathcal{H}_{\mathfrak{M}}$ in the expanded square.

For fixed disjoint unshared products a, b, independent pad draws violate pad distinctness or the ban with probability at most

$$C_K M^2 \sup_{i,p} \mu_i(p) \ll_K M^2 e^{-L^{\cdot 1}}. \tag{15.59}$$

This follows by a union bound over pairs of pad addresses and over coincidences with the two fixed unshared primes in each group; for two independent draws their equality probability is at most the largest atom. To remove this probability loss from a signed sum, we need absolute bounds for each of its two parts.

The absolute raw expanded square is

$$\ll_K XYL^{C_2}, \tag{15.60}$$

with C_2 depending on the coefficient bounds and fixed divisor moments, independently of K. Indeed for each pair a, b its original h representation has $h \ll X/Y$ and contributes at most a log-power coefficient bound times $\sum_{h \ll X/Y} \tau(1 + ah)\tau(1 + bh)$. Cauchy and the divisor moment estimate from the preceding section bound this by $(X/Y)L^{C_2}$. There are $O(Y^2)$ possible numeric products a, b, and their normalization costs only C_K.

For the absolute comparison sum, fix $a, b \asymp Y$. If $k = mn$ and $l = rs$, the cutoff imposes

$$|bk - al| \ll Y, \qquad k, l \ll X.$$

For each k there are $O(1)$ choices of l, and conversely, because $a, b \asymp Y$. Thus Cauchy on this bounded-degree relation and the ordinary divisor second moment give

$$\sum_{\substack{k,l \ll X \\ |bk-al| \ll Y}} \tau(k)\tau(l) \ll \sum_{k \ll X} \tau(k)^2 \ll XL^{C_3}. \tag{15.61}$$

The endpoint coefficients cost a fixed log power, and (15.24) contributes $O(L^{3A_0}/Y)$. Summing the $O(Y^2)$ possible a, b proves

$$\ll_K XYL^{C_4+3A_0}, \tag{15.62}$$

where C_3, C_4 are again independent of K. Multiplying (15.60) and (15.62) by (15.59) gives $O_A(XYL^{-A})$ for every fixed A. This strips all pads.

The comparison estimate also allows the two unshared products to have a common prime. The number of pairs $a, b \asymp Y$ sharing any group prime is at most

$$C_K Y^2 \sum_{p \in \bigcup_i \mathcal{P}_i} \frac{1}{p^2} \ll_K Y^2 e^{-L^{.1}}. \tag{15.63}$$

For fixed p there are at most $O(Y/p)$ possible numeric products divisible by p, and each product determines its prime list. Apply (15.61) to each such pair and then (15.24); their total is again smaller than XY times every fixed negative log power.

Finally, if C_5 denotes the log-power loss in (14.20), its endpoint calculation gives C_5 depending only on the fixed coefficient bounds, independently of K. Since $UV = d_0 YX$, division by d_0 bounds each pad dyad by $XYL^{-E_0+C_5}$. There are $O_K(L)$ pad dyads, so their sum is

$$O_K(XYL^{-E_0+C_5+1}) + O_A(XYL^{-A}). \tag{15.64}$$

Given a desired replacement precision A, choose $E_0 > A + C_5 + 2$ first, then choose A_0, then K as above. All exponentially small errors have already been bounded after these parameters are fixed and impose no further condition on E_0. This proves the replacement assertion of Proposition 15.1 and leaves the unrestricted major term for the next section.

16 The major term of the determinant estimate

We retain the notation of Theorem 14.1, in particular $X = H_m H_n \asymp x$ and $W = \exp(L^{0.24})$. The exponent A_0 defining $\mathfrak{M}$ is fixed throughout this section. Constants may depend on K and on its fixed prime bands; the powers of L used before choosing K will not depend on K.

Proposition 16.1 (The determinant major term)**.** *For every fixed $D > 0$, the unrestricted major sum*

$$\mathcal{Q}_Y^{\mathrm{maj}} = \left(\prod_{i=1}^{K} V_i^{-2} \right) \sum_{a,b,m,n,r,s} \eta(a/Y)\eta(b/Y)\alpha_m \overline{\alpha_r} \beta_n \overline{\beta_s}\, \mathcal{H}_{\mathfrak{M}}(bmn - ars; a, b) \tag{16.1}$$

satisfies

$$|\mathcal{Q}_Y^{\mathrm{maj}}| \ll_D XYL^{-D}. \tag{16.2}$$

Here a, b run independently over products of one prime from each $\mathcal{P}_i$, with no disjointness condition, and $\mathcal{H}_{\mathfrak{M}}$ denotes the kernel in (14.11)*. The estimate is uniform in the coefficient intervals, in $|v| \le L^C$, and in the admissible scale Y.*

The major-arc decomposition will separate three factors: the centered prime-minus-rough coefficient α, the arbitrary rough coefficient β, and the product of small prime labels. Uniform cancellation of the first factor alone does not control an integral over a frequency range of length comparable to X. We isolate one small prime label. Where its polynomial is small, a mean-square bound for the product of the two long polynomials suffices; where it is large, the frequencies are sparse enough to control the energy of the β polynomial. The next two lemmas supply the uniform cancellation and the sparse-frequency bound, respectively. All characters below are extended by zero away from the units.

Lemma 16.2 (The long coefficient)**.** *Fix $A_0, B, A > 0$. For every character χ of modulus at most L^{A_0}, put*

$$M_\alpha(t) = \frac{1}{H_m} \sum_m \alpha_m \chi(m) m^{it}. \tag{16.3}$$

Then, uniformly for $|t| \le 2XL^B$,

$$|M_\alpha(t)| \ll_A L^{-A}. \tag{16.4}$$

The assertion also holds with conjugated coefficients and characters.

Proof. Write $u = t+v$ and let $I \subset [H_m, 2H_m]$ be the coefficient interval. We give the approximation of rough numbers explicitly. Set $\mathcal{P}(W) = \{p : p \le W\}$ and $S_W = \sum_{p\le W} p^{-1} = 0.24 \log L + O(1)$. For an integer $r \asymp c \log L$, define

$$T_r(m) = \sum_{\substack{d\mid m\,,\ d \text{ squarefree} \\ p\mid d \Rightarrow p\le W\,,\ \#\{p:p\mid d\}\le r}} \mu(d).$$

The two consecutive Bonferroni truncations bracket $\mathbf{1}_{P^-(m)>W}$. Their difference is supported on products of $r+1$ distinct primes. Consequently, on every subinterval $I' \subset I$,

$$\sum_{m\in I'} |\mathbf{1}_{P^-(m)>W} - T_r(m)| \ll H_m \frac{S_W^{r+1}}{(r+1)!} + W^{r+1}, \tag{16.5}$$

where changing r by one, if necessary, is harmless. This follows by summing $\mathbf{1}_{d\mid m}$ over the first omitted degree, using $\#\{m \in I' : d \mid m\} = |I'|/d + O(1)$. Since $r! \ge (r/e)^r$, choosing c sufficiently large makes the first term $O(H_m L^{-D'})$ for any prescribed fixed D'. The second is $x^{o(1)}$, since $r \log W = O(L^{0.24} \log L)$. Moreover, the absolute reciprocal coefficient mass of every truncation satisfies

$$\sum_{d \text{ used}} \frac{1}{d} \le \prod_{p\le W} (1+p^{-1}) \ll L^{0.24}. \tag{16.6}$$

In particular, increasing the depth to improve the error does not increase the exponent in this bound.

First consider large $|u|$. For each surviving divisor d with $(d,k) = 1$, where k is the character modulus, write $m = dz$ and split z into residues modulo k. Its scale is $N' = H_m/d = x^{\Omega(1)}$. For $1 \le |u| \le T_* = \exp(L/(\log L)^2)$, summation against an integral on a progression gives

$$\sum_{\substack{z\in J\subset[N',2N'] \\ z\equiv a \pmod k}} z^{iu} = \frac{1}{k}\int_J y^{iu}\,\mathrm{d}y + O(1+|u|), \qquad \left|\int_J y^{iu}\,\mathrm{d}y\right| \ll \frac{N'}{|u|}. \tag{16.7}$$

Indeed the total variation of y^{iu} on this dyad is $O(|u|)$; the integral formula follows by evaluating $y^{1+iu}/(1+iu)$ at the endpoints. The error, after summing the at most k residues and all $d \le W^r$, is x^{-c_δ} after division by H_m, uniformly up to T_*. At $T_* \le |u| < x^2$, apply Lemma 12.4: its lower scale condition holds because $N' \ge x^{\delta/2}$ for sufficiently large x, and its lower frequency condition is weaker than $|u| \ge T_*$. Together with (16.6), these bounds and partial summation for $1/\log m$ imply

$$\left|\frac{1}{H_m V(W)} \sum_{\substack{m\in I \\ P^-(m)>W}} \frac{\chi(m) m^{iu}}{\log m}\right| \ll L^{-D'} + \begin{cases} L^{C_0}(|u|^{-1} + x^{-c_\delta}), & 1 \le |u| \le T_*, \\ L^{C_0}\exp(-L/(\log L)^{C_3}), & T_* \le |u| < x^2. \end{cases} \tag{16.8}$$

Here C_0 is fixed after A_0, δ; it need not grow with the chosen Bonferroni depth. The factor $1/V(W) \ll L^{0.24}$ and the bounded variation of $1/\log m$ have been included in C_0.

For the prime part, choose fixed $0 < \tau < \delta$ and $1-\delta < \eta < 1$. The scale conditions imply $x^\tau/2 \le H_m \le x^\eta$ for large x. Lemma 12.3, followed by partial summation to replace p^{-1} by H_m^{-1}, gives an arbitrary negative power of L when $L^{B_0'} \le |u| \le x^2$. Choose $B_2 > C+2$ sufficiently large after A, A_0, B and the exponent C_0. Then $L^{B_2} \le |t| \le 2XL^B$ implies $|u| \asymp |t|$, $|u| \ge L^{B_0'}$, and $|u| < x^2$. The prime estimate and (16.8) prove (16.4) on this range.

It remains to treat $|t| \le L^{B_2}$, hence $|u| \le 2L^{B_2}$. Choose the counting precision D' after B_2. Lemma 12.5 gives, on every subinterval,

$$\sum_{m\in I'} \chi(m)\mathbf{1}_{P^-(m)>W} = \mathbf{1}_{\chi \text{ principal}} V(W)|I'| + O(H_m L^{-D'}). \tag{16.9}$$

The principal character equals one on rough numbers because all primes dividing k are at most W; for nonprincipal characters the count has zero main term by complete-period cancellation. For primes, Lemma 12.1 and character orthogonality give, to any prescribed precision,

$$\sum_{p\in I'}\chi(p)=\mathbf{1}_{\chi\ \text{principal}}\int_{I'}\frac{\mathrm{d}y}{\log y}+O(H_mL^{-D'}).$$

Partial summation introduces at most a fixed power of L on this low-frequency range. Dividing the rough main term by $V(W)\log y$ produces exactly $\mathrm{d}y/\log y$; its twisted main term therefore cancels the prime main term. Increasing D' gives (16.4). Complex conjugation changes only the signs of the real frequencies and the character, so the proof is unchanged. □

Lemma 16.3 (A small prime factor and sparse frequencies)**.** *Let P satisfy $cL^{0.1}\le\log P\le C_1L^{0.2}$, and let $\mathcal{P}$ be any set of primes in $[P,2P]$. For any real ξ and any character χ, put*

$$P_s(t)=P^{-1}\sum_{p\in\mathcal{P}}\chi(p)p^{i(t+\xi)}.$$

Let $N_\beta(t)=H_n^{-1}\sum_n\beta_n\chi_n(n)n^{it}$, with $|\beta_n|\le L^C$ on $[H_n,2H_n]$ and $x^\delta\le H_n\ll x$. For every fixed $B,A_s>0$, the unit intervals meeting

$$\{t:|t|\le 2XL^B,\ |P_s(t)|>L^{-A_s}\}$$

form a family $\mathcal{I}$ satisfying

$$|\mathcal{I}|\le\exp(O(L^{0.91})),\qquad\sum_{I\in\mathcal{I}}\sup_{t\in I}|N_\beta(t)|^2\ll L^{2C+1}.\tag{16.10}$$

Both assertions are uniform in ξ and in the prime set.

Proof. Put $Z=4XL^B$ and $j=\lfloor\log Z/\log(2P)\rfloor$. Write $P_s(t)^j=\sum_{n\le(2P)^j}c_nn^{it}$. For each product n, unique factorization bounds the number of ordered prime tuples producing it by $j!$. Hence

$$\sum_n|c_n|^2\le j!P^{-2j}|\mathcal{P}|^j\le j!(C_2/P)^j.\tag{16.11}$$

This is a coefficient estimate for the actual growing degree j; no fixed-divisor-moment constant is used for it.

Choose an exceeding point from each member of $\mathcal{I}$, and split these points into three classes according to the integer left endpoint modulo three. Each class is separated by at least one. The separated mean-square estimate of Lemma 12.2, whose constant is independent of the coefficients, gives

$$|\mathcal{I}|\ll ZL^{O(1)}L^{2A_sj}j!(C_2/P)^j.\tag{16.12}$$

Since $\log Z-j\log P\ll\log P+j$ and $j\ll L^{0.9}$, the logarithm of its right-hand side is at most

$$O(\log P+j\log(j+1)+A_sj\log L+\log L)=O(L^{0.9}\log L+L^{0.2})=O(L^{0.91}).$$

The factor $p^{i\xi}$ affects no coefficient absolute value, so this estimate is uniform for all real ξ.

For the second assertion choose a maximizing point t_I on the closure of each unit interval, and again use three separated classes. In one class consider the evaluation matrix on the full integer interval $[H_n,2H_n]$. Its Gram entries are

$$G_{I,J}=\sum_{H_n\le n\le 2H_n}n^{i(t_I-t_J)}.$$

Set $T_*=\exp(L/(\log L)^2)$. The diagonal is $O(H_n)$. For $1\le|\Delta|\le T_*$, (16.7) with modulus one gives

$$H_n^{-1}|G_{I,J}|\ll|\Delta|^{-1}+x^{-c_\delta}.\tag{16.13}$$

For $T_* < |\Delta| \le 4XL^B + 2 < 4x^3$, Lemma 12.4 gives

$$H_n^{-1}|G_{I,J}| \ll \exp(-L/(\log L)^{C_3}). \tag{16.14}$$

All its scale hypotheses hold because $H_n \ge x^\delta$ and $H_n \ll x$. Separation implies that the sum of $1/|\Delta|$ in any row is $O(\log(2Z)) = O(L)$. The other row sums tend to zero, since

$$|\mathcal{I}|x^{-c_\delta} = o(1), \qquad |\mathcal{I}|\exp(-L/(\log L)^{C_3}) = o(1).$$

Thus the absolute Gram row sums are $O(H_nL)$, and the same is true of its operator norm. The coefficient vector of N_β has squared norm $O(L^{2C}/H_n)$. Applying the Gram bound and adding the three classes proves (16.10). □

Proof of Proposition 16.1. For large x the arcs are disjoint: distinct reduced rationals of denominator at most L^{A_0} are at distance at least L^{-2A_0}, whereas their radii are $2L^{A_0}/Y$ and $Y \ge \exp(cL^{0.1})$. On an arc write $\theta = h/k + \lambda/Y$, with $|\lambda| \le 2L^{A_0}$ and $\mathrm{d}\theta = \mathrm{d}\lambda/Y$. Every variable in (16.1) is a unit modulo k: the group primes and all prime factors of the rough variables exceed k. The function

$$(a,b,m,n,r,s) \longmapsto \mathrm{e}\big(h(bmn - ars - b + a)/k\big)$$

on $((\mathbb{Z}/k\mathbb{Z})^\times)^6$ has an exact Fourier expansion in products of six multiplicative characters. Each Fourier coefficient has absolute value at most one, so their total absolute mass is at most $\varphi(k)^6 \le k^6$. There are $O(L^{2A_0})$ arcs. It is therefore enough to estimate each separated character term, uniformly in λ, with arbitrarily large logarithmic saving.

Set $w = bmn/(YX)$ and $w' = ars/(YX)$. Both lie in a fixed compact subinterval of $(0,\infty)$. Apart from the factors $\mathrm{e}(-\lambda b/Y)\mathrm{e}(\lambda a/Y)$, the remaining common kernel is

$$\psi_\lambda(X(w-w')), \qquad \psi_\lambda(y) = \psi(y)\mathrm{e}(\lambda y).$$

Here is the Mellin separation with its normalization. In coordinates $z = \log w'$ and $\zeta = X\log(w/w')$, it is

$$\psi_\lambda\big(Xe^z(e^{\zeta/X} - 1)\big).$$

Insert fixed smooth cutoffs equal to one on the possible values of w, w'. The resulting function is supported on a fixed compact set of (ζ, z): the support of ψ forces $|\zeta| \ll 1$. Its derivatives of order $r+s$ are $O_{r,s}(L^{B_0(r+s)})$ for some fixed B_0 chosen after A_0. Fourier inversion, with inverse kernel $e^{i(v\zeta + s'z)}$, and the change of variable $t = Xv$ consequently give the exact identity

$$\psi_\lambda(X(w-w')) = \frac{1}{X}\iint_{\mathbb{R}^2} F_\lambda(t/X, s')w^{it}(w')^{i(s'-t)}\,\mathrm{d}t\,\mathrm{d}s' \tag{16.15}$$

on the coefficient support. Repeated integration by parts yields, for every fixed integer $j \ge 1$,

$$|F_\lambda(v,s')| \ll_j (1 + |v|/L^{B_0})^{-j}(1 + |s'|/L^{B_0})^{-j}. \tag{16.16}$$

Define the normalized endpoint polynomials

$$\begin{aligned} B_\lambda(t) &= \frac{1}{Y}\sum_b \eta(b/Y)\mathrm{e}(-\lambda b/Y)\left(\prod_i V_i^{-1}\right)\chi_b(b)b^{it}, \\ N_\beta(t) &= \frac{1}{H_n}\sum_n \beta_n\chi_n(n)n^{it}, \qquad Q(t) = B_\lambda(t)M_\alpha(t)N_\beta(t). \end{aligned} \tag{16.17}$$

The other endpoint gives a polynomial Q' of the same type with conjugations and sign changes. Each unnormalized endpoint sum is YX times its polynomial. Combining these two factors with the X^{-1} in (16.15) and the arc measure Y^{-1} gives

$$(YX)^2 \cdot X^{-1} \cdot Y^{-1} = XY. \tag{16.18}$$

The additional factor $(YX)^{-is'}$ has absolute value one. Thus, after division by XY, the separated expression is an integral of $F_\lambda(t/X, s')Q(t)Q'(s'-t)$, followed by the λ integral.

The bound $|B_\lambda(t)| \ll 1$ follows directly from $b \asymp Y$ and

$$\sum_b \frac{1}{b} \prod_i V_i^{-1} \le \prod_i \left(V_i^{-1} \sum_{p \in \mathcal{P}_i} p^{-1} \right) = 1.$$

The coefficient of index $u = mn$ in $M_\alpha N_\beta$ is $O(L^C \tau(u)/X)$ and is supported on $u \asymp X$. Lemma 12.2 therefore gives a squared coefficient norm $O(L^{C_4}/X)$ and, on every real interval I of length $Z \ge 1$,

$$\int_I |M_\alpha(t) N_\beta(t)|^2 \, \mathrm{d}t \ll L^{C_4}(1 + Z/X). \tag{16.19}$$

The fixed exponent C_4 depends only on the original coefficient bounds. The estimate is uniform in translates of I, by twisting the coefficients, and also bounds the mean square of Q and Q'.

Fix $B_1 > B_0 + 1$. Equations (16.16) and (16.19) permit us, to arbitrary logarithmic precision, to restrict to

$$|s'| \le L^{B_1}, \qquad |t| \le XL^{B_1}. \tag{16.20}$$

For completeness, on a dyadic t interval of length Z, Cauchy's inequality bounds the integral of $|Q(t)Q'(s'-t)|$ by $L^{C_4}(1+Z/X)$, uniformly in s'. For $Z \ge XL^{B_1}$ multiply this by $(1+Z/(XL^{B_0}))^{-j}$ and sum the geometric tail. For the s' tail, the full weighted t integral is $O(L^{C_4+B_0+1})$, uniformly in s', and

$$\int_{|s'|>L^{B_1}} (1 + |s'|/L^{B_0})^{-j} \, \mathrm{d}s' \ll_j L^{B_0-(B_1-B_0)(j-1)}.$$

Taking j large makes both errors as small as required, including the character and arc costs. This argument also covers translated frequencies $s' - t$ without a pointwise tail assumption.

On (16.20), Cauchy's inequality in t reduces the claim to proving, for every prescribed $A' > 0$,

$$\int_{|t| \le 2XL^{B_1}} |B_\lambda(t) M_\alpha(t) N_\beta(t)|^2 \, \mathrm{d}t \ll_{A'} L^{-A'}. \tag{16.21}$$

Indeed the retained s' integral has length $2L^{B_1}$ and all remaining character and λ costs are fixed powers of L.

Write $b = pb'$ with $p \in \mathcal{P}_1$ and split this prime band into $O(L)$ ordinary dyads $[P, 2P)$. The cutoff on b permits restriction to $Y/(2P) \le b' \le 4Y/P$. Mellin inversion of $f_\lambda(y) = \eta(y)\mathrm{e}(-\lambda y)$ gives

$$f_\lambda(y) = \int_{\mathbb{R}} \widehat{f}_\lambda(\xi) y^{i\xi} \, \mathrm{d}\xi, \qquad \int_{\mathbb{R}} |\widehat{f}_\lambda(\xi)| \, \mathrm{d}\xi \ll L^{C_5},$$

where C_5 is fixed after A_0. This last bound follows, for example, by twice integrating by parts outside $|\xi| \le 1$; the first two derivatives of $f_\lambda(e^z)$ have fixed compact support and size $O((1+|\lambda|)^2)$. The remaining factor

$$Q_{b'}(u) = \frac{P}{Y} \sum_{Y/(2P) \le b' \le 4Y/P} \left(\prod_{i=2}^{K} V_i^{-1} \right) \chi_b(b')(b')^{iu}$$

has absolute value $O(1)$ by its reciprocal coefficient mass. Precisely, the dyad contribution to $B_\lambda(t)$ is

$$V_1^{-1} \int_{\mathbb{R}} \widehat{f}_\lambda(\xi) Y^{-i\xi} \left(P^{-1} \sum_{p \in \mathcal{P}_1 \cap [P,2P)} \chi_b(p) p^{i(t+\xi)} \right) Q_{b'}(t+\xi) \, \mathrm{d}\xi.$$

Minkowski's integral inequality shows that it suffices to prove

$$\int_{|t| \le 2XL^{B_1}} |P_s(t) M_\alpha(t) N_\beta(t)|^2 \, \mathrm{d}t \ll L^{-A''} \tag{16.22}$$

with arbitrary fixed A'', uniformly in every real shift ξ. The $O(L)$ dyads and the Mellin L^1 norm are then paid by increasing A''. In particular there is no unaccounted Fourier tail in this factor separation.

On $|P_s(t)| \le L^{-A_s}$, Equation (16.19) bounds (16.22) by $O(L^{-2A_s+C_4+B_1})$. Choose A_s large after A''. On the remaining set use the unit intervals of Lemma 16.3. We have $|P_s(t)| \ll 1$, and Lemma 16.2 gives $|M_\alpha(t)| \ll L^{-A}$ everywhere in the retained range, including its low frequencies. Therefore its contribution is

$$\ll L^{-2A} \sum_{I \in \mathcal{I}} \sup_{t \in I} |N_\beta(t)|^2 \ll L^{-2A+2C+1}.$$

Taking A sufficiently large proves (16.22), then (16.21), and finally (16.2). □

Completion of Theorem 14.1. Fix the requested precision D_* and choose the expanded-square precision D_1 sufficiently large in terms of D_* and the original coefficient bound C. These choices precede K. The squarefree-label and coincident-label errors from the Cauchy reduction are smaller than every fixed logarithmic power. Proposition 15.1, including its restoration of good states and harmonic pads, replaces the remaining determinant equality by (14.11) with error $O(XYL^{-D_1})$.

To record the parameter order, first choose E_0 after D_1, C, then the arc exponent A_0, and finally K, as in that proposition. For a pad dyad, division of its pairing bound by d_0 uses $UV = d_0YX$ and gives $XYL^{-E_0+O_C(1)}$. There are $O_K(L)$ pad dyads; thus E_0 can absorb this loss with an exponent independent of K. Constants involving the now fixed K and its band exponents only affect the threshold for x. The analytic accuracies in Proposition 16.1 are chosen last, after A_0. That proposition bounds the unrestricted major term by $O(XYL^{-D_1})$.

Consequently the Cauchy square in each Y dyad is $O(XYL^{-D_1})$. Multiplication by its Cauchy factor $O(X/Y)$ and taking square roots gives $O(XL^{-D_1/2})$ for that dyad. The $O_K(L)$ dyads of Y, and all preceding fixed coefficient losses, are absorbed by the original choice of D_1. This proves (14.5), with K depending on the stated fixed data and not on the particular exponents a_i. □

17 Two-sided marked correlations

We now allow marks on both endpoints of the shift. The integer J used in this section is sufficiently large and *fixed* as $x \to \infty$; it is different from the growing determinant parameter in the preceding sections. Likewise the active damping parameter q_0 below need not equal the parameter q in the fixed-band weight $\mathcal{W}$.

Retain the fixed K bands defining $\mathcal{W}$. Independently choose pairwise disjoint active groups $\mathcal{P}_g$, indexed by $\mathcal{S} \sqcup \mathcal{B}$, such that, for fixed positive constants c_0, C_0, v_-, v_+,

$$c_0 \log L \le |\mathcal{S}|, |\mathcal{B}| \le C_0 \log L, \qquad v_- \le V_g := \sum_{p \in \mathcal{P}_g} p^{-1} \le v_+, \tag{17.1}$$

$$\mathcal{P}_g \subset [\exp(L^{.27}), \exp(L^{.36})] \quad (g \in \mathcal{S}), \qquad \mathcal{P}_g \subset [\exp(L^{.39}), \exp(L^{.46})] \quad (g \in \mathcal{B}).$$

Every big group is the set of all primes in an interval. Write $\mathcal{P} = \bigcup_g \mathcal{P}_g$, $s_P = |\mathcal{S}| + |\mathcal{B}|$, and define

$$n_* := \prod_{p \notin \mathcal{P}} p^{v_p(n)}, \qquad \omega_g(n) := \sum_{p \in \mathcal{P}_g} \mathbf{1}_{p|n}, \qquad \omega_P(n) := \sum_g \omega_g(n), \qquad \mu_g(p) := \frac{1}{V_g p}.$$

Thus n_* removes the entire active part, including its multiplicities. For a vector $\mathbf{t} = (t_g)$ of nonnegative integers, a represented list at n consists of t_g ordered distinct prime divisors of n from each group. For a function $\mathbf{b}$ of these lists put

$$W_{\mathbf{t}}^{\mathbf{b}}(n) := q_0^{\omega_P(n) - \sum_g t_g} \prod_g V_g^{-t_g} \sum_{\text{represented lists at } n} \mathbf{b}(\mathbf{p}), \qquad 0 < q_0 < 1. \tag{17.2}$$

The weight is zero if no list exists. Write $W_{\mathbf{t}}$ when $\mathbf{b} = 1$, and $W_1 = W_{(1,\dots,1)}$. For every fixed m',

$$|W_{\mathbf{t}}^{\mathbf{b}}(n)| \le W_{\mathbf{t}}(n) \le L^{C(m')} \quad (t_g \le m',\ |\mathbf{b}| \le 1). \tag{17.3}$$

Write $(u)_t = u(u-1)\cdots(u-t+1)$, with $(u)_0 = 1$. Indeed, each factor $q_0^{u-t}(u)_t/V_g^t$, for $u \ge t$, is bounded by a constant depending only on m', q_0, v_-, and there are $O(\log L)$ factors.

Theorem 17.1 (Two-sided correlation)**.** *Fix $d \ge 1$, $\varepsilon > 0$, and intervals $I_1, \dots, I_d$ whose lower endpoints are at least x^ε and whose upper endpoints have product at most $x^{1-\varepsilon}$. Let $\mathbf{l} = (l_1, \dots, l_d)$ run over ordered tuples of distinct primes with $l_i \in I_i$, and put*

$$l := l_1 \cdots l_d, \qquad C_x := \sum_{\mathbf{l}} \frac{1}{l}, \qquad F_x(n) := \mathcal{W}(n) \left(\sum_{\mathbf{l}} \mathbf{1}_{l|n} - C_x \right). \tag{17.4}$$

Suppose $G(n) = G(n_*)$ and $|G(n)| \le L^C \tau(n_*)^C$, for fixed C. For every fixed smooth compactly supported $\Psi : (0,\infty) \to \mathbb{C}$ and every fixed $A > 0$,

$$\sum_{w \ge 1} \frac{\Psi(w/x)}{w} F_x(w) G(w+1) W_1(w) W_1(w+1) \ll_A L^{-A}. \tag{17.5}$$

The constants and sufficiently-large-x threshold may depend on all the fixed data, including the fixed inert bands, but the estimate is uniform over the active groups and slot intervals satisfying the stated bounds. The slots have no joint restriction other than distinctness.

The harmonic prime estimates give $C_x = O_\varepsilon(1)$. On every fixed positive-power range for n, there are only $O_\varepsilon(1)$ prime divisors in the slot ranges. Since $\mathcal{W}$ is bounded for fixed K, q, F_x is bounded there. Also $F_x(pn) = F_x(n)$ for every active prime p: the active groups are disjoint both from the inert bands and from the long-prime slots. These two facts will give the stronger endpoint hypotheses required by the graph argument.

The proof has two stages. First, the graph estimates established earlier give arbitrary logarithmic savings for a signed combination of correlations. After common prime labels have been divided out of both endpoints, one term, called the raw term, is the unit-shift correlation in (17.5); the other terms have shifts that are products of independently sampled big-group primes. We call these other terms the comparison correlations. We then bound each comparison separately, using the centered first endpoint, and recover the unit-shift term by subtraction. We state the graph inputs with their exact measures before making this reduction. The new endpoint estimates begin once the comparison correlations have been identified.

17.1 The exact graph inputs

For clarity we state the general graph statements, rather than using a correlation theorem with more restrictive endpoint hypotheses. In this subsection only, replace $.27, .36, .39, .46$ in (17.1) by arbitrary fixed $0 < a < b < c < d_0 < .47$, and fix an integer $\ell \ge 1$. Extend the divisibility definitions to all $n \in \mathbb{Z}$, with every group prime dividing zero. Put $M = J + \ell$. A pattern ν chooses sets

$$C_g \subset \{1, \dots, J\}, \qquad C_g = \varnothing\ (g \in \mathcal{S}), \qquad |C_g| \le m_0\ (g \in \mathcal{B}), \qquad t_g = \ell + |C_g|.$$

In each of the first J slots outside C_g, the source and target lists have the same prime label. The product of these shared labels is D_R. In each slot of C_g take an independent auxiliary label of law μ_g; their product is D_C. The dilation $D = D_R D_C$ therefore contains J labels from each group, counted with multiplicity. A coefficient K_ν may depend only on the ordered big-group source lists, target lists, and auxiliary free labels.

The physical Hilbert space consists of states $(n, \mathbf{p})$ with $n \in \mathbb{Z}$ and M ordered distinct divisors from every group. Each state has mass $\prod_g V_g^{-M}$, with counting measure in n. Fix an integer k

with $0 < |k| \le L^{C_2}$. A row operation samples the auxiliary labels, sets $n' = n + kD$, and retains the shared labels while summing over physical target lists with coefficient $\prod_g V_g^{-t_g}$. Every auxiliary free label is forbidden from both endpoint lists; auxiliary labels may coincide with one another. The row multiplier is

$$K_\nu D^{i\zeta} q_0^{(\omega_P(n)-Ms_P)/2} q_0^{(\omega_P(n')-Ms_P)/2}. \tag{17.6}$$

Add the patterns and average independent permutations of the M slots in every group at both ends. This defines $\mathcal{A}_\zeta$. In particular, the joint measure of the two lists in group g is $V_g^{-M-t_g}$ in either direction. A row operator integrates the target function and returns a function at the source.

The ideal Hilbert space is

$$\mathcal{H}_{\mathrm{id}} = L^2\left(\prod_{g\in\mathcal{B}} \mathcal{P}_g^M, \bigotimes_{g\in\mathcal{B}} \mu_g^{\otimes M}\right).$$

Repetitions are allowed here. Retain the shared coordinates and sample the unshared target and auxiliary coordinates independently. If $D_{\mathcal{B}}$ is the big-group part of D, use multiplier $K_\nu D_{\mathcal{B}}^{i\zeta} \mathrm{e}(\Theta D_{\mathcal{B}})$. The sum, symmetrized at both ends, is $\mathcal{T}_\zeta(\Theta)$.

Proposition 17.2 (General graph transference input)**.** *Assume $\sum_\nu \sup|K_\nu| \le L^{C_1}$, where m_0, C_1 are fixed independently of J. For every fixed $E > 0$ there is $E_{\mathrm{id}} > 0$, depending only on E and the fixed group data, such that*

$$\sup_{\Theta\in\mathbb{R}} \|\mathcal{T}_\zeta(\Theta)\|_{2\to 2} \le L^{-E_{\mathrm{id}}}$$

has the following consequence for all sufficiently large fixed J. Let $I = [X, 2X)$, with $x^\gamma \le X \le x^{1/\gamma}$ for fixed $\gamma > 0$. If f is supported on positions in I, is independent of the chosen marks, and satisfies

$$\sup|f| \le \exp(O(\sqrt{L})), \qquad \|f\|, \|g\| \le X^{1/2} L^{C_3},$$

then

$$|\langle f, \mathcal{A}_\zeta g\rangle| \ll X L^{-E+2C_3}. \tag{17.7}$$

The lower bound on J may also depend on m_0, C_1. The threshold for x may depend on J, C_2, γ. No further restriction on ζ is imposed beyond the ideal norm hypothesis. Taking absolute values of the individual transition coefficients gives row and column sums at most $L^{C_{\mathrm{abs}}}$, where C_{abs} is independent of J.

This is the combination of the local transference theorem, endpoint pairing corollary, and physical Schur bound of Theorem 10.5, Corollary 10.11, and Lemma 10.4. Its parameter q is our q_0; its laws, physical state masses, two-sided symmetrization, and ban on free endpoint labels are exactly those defined above. In the absolute bound, if a target has $y \ge M$ divisors in a group, its unshared list sum is bounded by

$$\sup_{z\ge 0} V_g^{-t_g} (z + t_g)_{t_g} q_0^{z/2} = O_{m_0,\ell}(1).$$

The source damping is at most one. Products over $O(\log L)$ groups and the pattern mass give the asserted exponent independent of J; the symmetric joint normalization gives the column bound as well.

Proposition 17.3 (Residual ideal family input)**.** *For every $E_{\mathrm{id}} > 0$ and fixed $C_4 \ge 0$, there are fixed m_0, J_0, C_1 such that, for each fixed $J \ge J_0$, a family consisting of the raw pattern $C_g = \varnothing$, $K_0 = 1$, and signed comparison patterns satisfies*

$$\sum_\nu \sup|K_\nu| \le L^{C_1}, \qquad \sup_{\substack{\Theta\in\mathbb{R} \\ |\zeta|\le L^{C_4}}} \|\mathcal{T}_\zeta(\Theta)\|_{2\to 2} \le L^{-E_{\mathrm{id}}}. \tag{17.8}$$

The family is independent of Θ, ζ; its defining parameters are independent of J.

Split the big groups into two blocks. Every comparison frees nonempty sets of probes A, B in the respective blocks, with coefficient

$$-(-1)^{|A|+|B|}\mathcal{K}_A(D_A, X_A)\mathcal{K}_B(D_B, X_B). \tag{17.9}$$

Here D_A, D_B are the free products, X_A is the corresponding target-mark product, and X_B the corresponding source-mark product. No shared or final ℓ label enters these kernels. For a nonempty slot set Λ with at most m_0 slots in each group,

$$\mathcal{K}_\Lambda(D, X) = \sum_{j:\nu_{\Lambda j}\ge\xi} \frac{\mathbf{1}_{\log D\in I_j}\mathbf{1}_{\log X\in I_j}}{\nu_{\Lambda j}} \sum_{\substack{\chi \text{ primitive}\\ \mathrm{cond}(\chi)\le Q}} \overline{\chi(D)}\chi(X), \qquad I_j = [jh, (j+1)h). \tag{17.10}$$

The number $\nu_{\Lambda j}$ is the cell probability for the independent slot laws. The parameters Q, h^{-1}, ξ^{-1} are fixed powers of L, with $h \le 1$. There are at most Q^2 characters and $O_{m_0}(1 + L^{d_0}\log L/h)$ nonempty cells, and

$$\sup|\mathcal{K}_\Lambda| \le Q^2\xi^{-1}, \qquad \sup_X \mathbb{E}_D|\mathcal{K}_\Lambda(D, X)| \le Q^2.$$

More precisely, for any prescribed fixed $A_0 > 0$, the parameters may be chosen, independently of J, so that for arbitrary L^2 tuple tests $f(X_B), g(X_A)$ and all $\theta \in \mathbb{R}$, $|\zeta| \le L^{C_4}$,

$$\begin{aligned}\big|\mathbb{E}\,\overline{f(X_B)}g(X_A)\big[&(X_AX_B)^{i\zeta}\mathrm{e}(\theta X_AX_B)\\ &- \mathcal{K}_A(D_A, X_A)\mathcal{K}_B(D_B, X_B)(D_AD_B)^{i\zeta}\mathrm{e}(\theta D_AD_B)\big]\big| \le L^{-A_0}\|f\|_2\|g\|_2.\end{aligned} \tag{17.11}$$

The four label tuples in this expectation are independent; the tests need not depend only on their products. Expanding the two kernels, including all patterns, has total coefficient mass and number of terms bounded by fixed powers of L.

This is Theorem 11.1. The kernels in (17.10) are two global cell and character expansions; there is not a separate character expansion for every group. We use these two propositions with $\ell = k = 1$ and the exponents in (17.1).

17.2 Endpoint bounds and removal of shared labels

On a physical state $\omega_P(n) \ge Ms_P$. Use endpoint vectors with values $\overline{F_x(n)}$ and $G(n')$, respectively, and with one extra half-damping factor at each endpoint. The conjugation compensates for that in the Hilbert-space pairing. These extra factors are at most one and are independent of the lists. For a fixed ordered physical list with squarefree product P, $n_* \mid n/P$. The divisor moment bound of Lemma 12.2 gives

$$\sum_{\substack{n\precsim X\\ P\mid n}} |G(n)|^2 \le L^{2C} \sum_{v\ll X/P} \tau(v)^{2C} \ll (X/P)L^{C_*}.$$

The same statement holds for F_x, which is bounded. Here $P \le \exp(O_J(L^{.46}\log L)) = x^{o(1)}$, and the normalized list sum satisfies

$$\prod_g V_g^{-M} \sum_{\text{physical lists}} \frac{1}{P} \le \prod_g \left(\sum_{p\in\mathcal{P}_g} \frac{1}{V_g p}\right)^M = 1. \tag{17.12}$$

Consequently both endpoint norms on an enlarged dyad are at most $X^{1/2}L^{C_3}$, where C_3 is independent of J and m_0. The first vector has bounded supremum. It is essential here to keep $q_0^{(\omega_P(n)-Ms_P)/2} \le 1$ intact; separating its two powers would introduce an unnecessary exponent depending on J.

Insert $\Psi(n/(xD))/n$ into the physical edge pairing. Its support has $n \asymp n' \asymp xD = x^{1+o(1)}$. Partition n into $O(L)$ dyads $[X, 2X)$, restricting n' to fixed enlargements. Let $\phi(u) = \Psi(e^u)$ and use the convention $\widehat{\phi}(\zeta) = \int \phi(u)e^{-i\zeta u}\,\mathrm{d}u$. Then

$$\Psi\left(\frac{n}{xD}\right) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{\phi}(\zeta)(n/x)^{i\zeta}D^{-i\zeta}\,\mathrm{d}\zeta. \tag{17.13}$$

Put X/n into the first vector and $1/X$ outside. The endpoint bounds just proved are unchanged. Fix the cutoff $|\zeta| \le L$ now, so $C_4 = 1$ suffices in Proposition 17.3.

The choices have the following order. After the desired saving A and the original data, fix C_3, then a transfer saving E large enough to cover A, $2C_3$, and the fixed dyadic costs. Choose E_{id} from Proposition 17.2, obtain m_0, J_0, C_1 from Proposition 17.3, and finally choose a fixed J satisfying both lower bounds. For the tail of (17.13), the absolute Schur estimate and the endpoint norms cost at most a fixed power of L. Since

$$\int_{|\zeta|>L} |\widehat{\phi}(\zeta)|\,\mathrm{d}\zeta \ll_N L^{-N}$$

for every fixed N, its differentiation order can be chosen after the family is fixed. The cutoff exponent does not have to change. Thus the total residual edge pairing is $O_A(L^{-A})$, with as much extra fixed saving as will be needed below.

For a single pattern put $n = D_R w$, $n' = D_R w'$. The shift becomes $w' = w + D_C$. Away from shared-label collisions,

$$\omega_g(D_R w) - M = \omega_g(w) - t_g, \qquad V_g^{-M-t_g} = V_g^{-(M-t_g)}V_g^{-2t_g}, \qquad \frac{1}{n} = \frac{1}{D_R w}. \tag{17.14}$$

The factor D_R^{-1} changes each shared sum into its probability law μ_g, and the remaining list factors and damping are exactly $W_{\mathbf{t}}(w)W_{\mathbf{t}}(w')$. The endpoint cores are invariant under these labels. The comparison coefficients involve no shared labels. Consequently the stripped expression is

$$\mathbb{E}_{D_C}\sum_{w\ge 1}\frac{\Psi(w/(xD_C))}{w}F_x(w)G(w+D_C)W_{\mathbf{t}}(w)W_{\mathbf{t}}(w+D_C)\mathbb{E}_{\mathbf{p}(w),\mathbf{p}(w+D_C)}K_\nu, \tag{17.15}$$

where the last expectation is uniform on each endpoint's ordered unshared lists. The independent permutations have total mass one and only rename the ordered slots, so they introduce no factorial factor.

We justify restoring all the restrictions suppressed in this formula. There are $O_J(\log L)$ shared or free slots. Every group atom has size $O(\exp(-L^{.27}))$. Conditional on w, w' and the free labels, the harmonic mass of a shared/shared coincidence, a shared/free coincidence, or a shared prime dividing ww' is therefore

$$O_J(L^{C_J}\exp(-L^{.27})). \tag{17.16}$$

For the last assertion use that an integer of size $x^{O(1)}$ has at most $O(L)$ distinct prime factors. On actual overlap exceptions the unshared labels still divide w, w'. Changing the damping loses at most $q_0^{-2Ms_P} = L^{O(J)}$. The marked bounds and Cauchy with fixed divisor moments on the translated dyads bound the remaining harmonic sum by a fixed log power. Thus (17.16) also controls these actual exceptions.

If a free prime occurs in either unshared endpoint list, it divides D_C and one of $w, w + D_C$, hence both. It is at least $\exp(L^{.39})$. On $w \asymp xD_C$ the count of its multiples is $O(xD_C/p)$; Cauchy and a fixed higher divisor moment bound the corresponding weighted harmonic mass by $L^{O(1)}p^{-1/2}$. Summing over the $O_{m_0}(\log L)$ free slots costs $L^{O(1)}\exp(-cL^{.39})$. This also restores any coincidence between the two unshared endpoint lists, since such a label must divide their difference D_C. All errors remain smaller than every negative log power after the fixed pattern cost. They are incurred after J is fixed and do not change the exponent C_3 used in transference. For the raw pattern, $D_C = 1$ and $t_g = 1$, so (17.15) is precisely the left side of (17.5).

The transferred estimate therefore controls the desired correlation plus the signed comparison correlations (17.15). It remains to bound the latter separately. Their shifts contain two nonempty products of free big-group labels; these products will give a bilinear Fourier multiplier. All cancellation required at an endpoint will come from the centered function F_x.

17.3 Comparison multipliers and local energy

Expand the two kernels in (17.9). Their contract supplies only a fixed log-power total cost. In each term the two nonempty free products are independent and lie in log cells of width at most one, say $D_A \asymp U_1$, $D_B \asymp U_2$. Put

$$H' = U_1U_2, \qquad Y = xH', \qquad U_i \ge \exp(L^{.39} - O(1)), \qquad H' \le \exp(O_{m_0}(L^{.46}\log L)). \tag{17.17}$$

The cell and character expansion factors into separate bounded tests on the two free tuples and on each endpoint's big marks. Insert fixed smooth cutoffs at scale Y at both endpoints. Fourier inversion in logarithmic size separates $\Psi(w/(xD_AD_B))/w$ as in (17.13). After a factor $1/Y$, it is enough to estimate combinations of

$$\mathbb{E}\, b_1(D_A)b_2(D_B)\sum_w \mathcal{U}(w)\mathcal{V}(w + D_AD_B), \qquad |b_i| \le 1. \tag{17.18}$$

Tuple tests may be used in place of product tests; conditioning on the product produces the notation here. The endpoint sequences have smooth cutoffs at Y, fixed log-power twists, and the form of their respective invariant cores times $W_{\mathbf{t}}^{\mathbf{b}}$, with $|\mathbf{b}| \le 1$ depending only on big marks. Fixed divisor moments give

$$\sum_w |\mathcal{U}(w)|^2 + \sum_w |\mathcal{V}(w)|^2 \ll YL^{C_5}. \tag{17.19}$$

These estimates also bound the separated Fourier tails absolutely: on any translated time interval, the corresponding collapsed Dirichlet polynomials obey the coefficient mean-square estimate. Smooth Fourier decay can therefore give any required precision. All exponents chosen from now on may depend on the fixed ideal family.

The additive multiplier for (17.18) is

$$\mathfrak{m}(\theta) := \mathbb{E}\, b_1(D_A)b_2(D_B)\mathrm{e}(\theta D_AD_B), \qquad |\mathfrak{m}(\theta)| \le 1.$$

The product probability of a tuple using $k_g \le m_0$ slots per group is, by unique factorization,

$$\rho(u) = \frac{1}{u}\prod_g \frac{k_g!}{V_g^{k_g}\prod_p a_p!} \le \frac{L^{C_7}}{u}.$$

Since its total mass is at most one, each collapsed sequence on $u \asymp U_i$ has squared coefficient norm at most L^{C_7}/U_i. The elementary bilinear Fourier estimate (Cauchy followed by the spacing estimate for $\|\theta u\|^{-1}$) consequently gives, if $(h,r) = 1$ and $|\theta - h/r| \le r^{-2}$,

$$|\mathfrak{m}(\theta)| \ll L^{C_7}\left(\frac{(U_1/r+1)(U_2 + r\log(2r))}{H'}\right)^{1/2}. \tag{17.20}$$

For completeness, Cauchy in the variable on the interval of length $O(U_2)$, expansion, and the geometric-sum bound give an unnormalized square at most

$$\|a\|_2^2\|b\|_2^2\left(U_2 + \sum_{1\le v\ll U_1}\min\{U_2, \|\theta v\|_{\mathbb{R}/\mathbb{Z}}^{-1}\}\right).$$

For $r \ge 2$, split the v-range into $O(1 + U_1/r)$ blocks of diameter at most $r/2$. Distinct points in one block have residues θv separated by at least $1/(2r)$, because $|\theta - h/r| \le r^{-2}$. The sum in a

block is at most $O(U_2 + r\log(2r))$. The case $r = 1$ is immediate from the trivial bound U_2 on each summand. This proves (17.20) with the two coefficient norms above.

Given any required saving, choose C_6 sufficiently large and set

$$H = H'/L^{C_6}, \qquad \mathfrak{M}_H = \bigcup_{\substack{k\le L^{C_6}\\(h,k)=1}} \{\theta : |\theta - h/k| \le H^{-1}\}. \tag{17.21}$$

Dirichlet approximation with denominator bound $\lfloor H'/L^{C_6-1}\rfloor$ gives a fraction h/r with $|\theta-h/r| \le r^{-2}$. If $r \le L^{C_6}$, this places θ in $\mathfrak{M}_H$. Otherwise $L^{C_6} < r \ll H'/L^{C_6-1}$, and the expression under the square root in (17.20) is bounded by

$$\frac{1}{r} + \frac{\log(2r)}{U_2} + \frac{1}{U_1} + \frac{r\log(2r)}{H'}.$$

The middle terms save every log power by (17.17), and the other terms give any desired fixed saving by increasing C_6. Thus $\mathfrak{m}$ is arbitrarily log-power small off $\mathfrak{M}_H$.

Use the Fourier convention $\widehat{a}(\theta) = \sum_w a_w \mathrm{e}(\theta w)$. Parseval and (17.19) handle the complement of $\mathfrak{M}_H$. On its $O(L^{2C_6})$ arcs, Cauchy and the global energy of $\mathcal{V}$ reduce the problem to

$$\int_{|\beta|\le H^{-1}} |\widehat{\mathcal{U}}(h/k+\beta)|^2 \,\mathrm{d}\beta \ll_{A'} YL^{-A'} \tag{17.22}$$

for every fixed A'. The scale conversion we need is as follows.

Lemma 17.4 (Local Mellin energy)**.** *Let a_w be supported on $w \asymp Y$, where $Y = x^{1+o(1)}$, and suppose $H = x^{o(1)} \to \infty$. For every fixed integer $j \ge 1$,*

$$\int_{|\beta|\le H^{-1}} |\widehat{a}(\beta)|^2 \,\mathrm{d}\beta \ll_j \frac{1}{Y}\int_{\mathbb{R}} (1+H|t|/Y)^{-j} \left|\sum_w a_w w^{it}\right|^2 \,\mathrm{d}t + \left(\frac{H}{Y}\right)^2 \sum_w |a_w|^2. \tag{17.23}$$

Proof. Choose a smooth bump K of integral one, supported close enough to zero that its additive Fourier transform has modulus at least $1/2$ on $[-1,1]$. Plancherel gives

$$\int_{|\beta|\le H^{-1}} |\widehat{a}(\beta)|^2 \,\mathrm{d}\beta \ll H^{-2}\int_{\mathbb{R}} \left|\sum_w a_w K((w-v)/H)\right|^2 \,\mathrm{d}v.$$

Only $v \asymp Y$ contributes. On the union of the relevant supports, replace $K((w-v)/H)$ by $K(v\log(w/v)/H)$: their arguments differ by $O(H/Y)$. Cauchy over the $O(H)$ available w, and integration over the $O(H)$ centers for each w, give normalized squared error $O((H/Y)^2\sum|a_w|^2)$.

Write $v = Ye^u$, $h_0 = H/Y$, and $P_a(t) = \sum_w a_w w^{it}$. Fourier inversion for the new kernel reads

$$\sum_w a_w K(v\log(w/v)/H) = \frac{h_0}{2\pi}\int_{\mathbb{R}} e^{-u}\widehat{K}(h_0 e^{-u}t/(2\pi))v^{-it}P_a(t)\,\mathrm{d}t.$$

Insert a fixed smooth cutoff in u for $v \asymp Y$. Two integrations by parts in u and the Schwartz bounds for $\widehat{K}$ bound the Fourier transform of this amplitude by $C_j(1+|s|)^{-2}(1+h_0|t|)^{-j}$. Expand in s, apply Minkowski and Plancherel in u, and use $\mathrm{d}v \ll Y\,\mathrm{d}u$. The outside normalization is $H^{-2}Yh_0^2 = Y^{-1}$. Increasing the decay order proves (17.23). □

Apply the lemma to $a_w = \mathcal{U}(w)\mathrm{e}(hw/k)$. The additive error is negligible by (17.19). The Dirichlet mean-square bound in Lemma 12.2 gives, on every dyadic time interval of length R,

$$Y^{-2}\int_{R\le|t|\le 2R} |P_a(t)|^2\,\mathrm{d}t \ll L^{C_5+1}(1+R/Y).$$

For $T_0 = LY/H$, the tail in (17.23) is thus at most

$$YL^{C_5+1}\sum_{r\ge 0}(2^rL)^{-j}(1+2^rL/H),$$

which is as small as required on choosing j last. It remains to prove, for every fixed $A' > 0$,

$$\int_{|t|\le T_0} \left| Y^{-1} \sum_w \mathcal{U}(w) \mathrm{e}(hw/k) w^{it} \right|^2 \, \mathrm{d}t \ll_{A'} L^{-A'}, \qquad T_0 = \frac{LY}{H} = xL^{C_6+1}. \tag{17.24}$$

Notice that the first term in (17.23) is Y times the integral in (17.24), as required by (17.22).

We now prove this one endpoint estimate. At low Mellin frequencies, a finite divisor model makes the centering in F_x cancel the main term. At high frequencies, one small-prime mark supplies a polynomial that is small outside a sparse set of times. On that sparse set we treat the positive tuple sum and its subtracted constant separately: the tuple sum has a long-prime factor, whereas the constant term requires cancellation on the divisor progressions of the finite model.

17.4 A divisor model and low Mellin frequencies

Lemma 17.5 (Truncated divisor model)**.** *Let $H_{\mathrm{marks}}(n) = \mathcal{W}(n) W_{\mathbf{t}}^{\mathbf{b}}(n)$, where $0 \le t_g \le m'$ for fixed m' and $|\mathbf{b}| \le 1$. In particular, zero values of t_g are allowed; these occur when a small mark is removed below. There is a divisor sum*

$$H_0(n) = \sum_{d'|n} A_{d'}, \qquad d' \le \exp(O(L^{.47})), \qquad \sum_{d'} \frac{|A_{d'}|}{d'} \le L^{C_8}, \tag{17.25}$$

whose prime divisors all belong to the active or inert groups, such that, on every positive integer dyad $n \asymp R$,

$$\sum_{n \asymp R} |H_{\mathrm{marks}}(n) - H_0(n)|^2 \ll_N RL^{-N} \tag{17.26}$$

for every fixed N. The assertion is uniform in the bounded tuple test $\mathbf{b}$, with no regularity assumption on that test.

Proof. Expand both weights into their represented lists. The inert list has one prime per band, and the active list has t_g per group. For a fixed represented list, the damping is exactly the product of q or q_0 over the remaining distinct group primes that divide n. Expand this product as

$$\prod_{p \text{ not represented}} (1 - (1 - q_p)\mathbf{1}_{p|n}), \qquad q_p \in \{q, q_0\},$$

and truncate after $h_0 = \lfloor L^{.01} \rfloor$ additional primes. All represented primes are distinct within their groups, and the additional primes are disjoint from them. Each resulting divisor has at most $O_{m'}(\log L) + K + h_0$ prime factors, all at most $\exp(L^{.46})$. This proves the support assertion in (17.25).

In the reciprocal coefficient sum the normalized represented lists have total mass at most one, by their harmonic definitions. The additional subset sums are at most

$$\prod_{p \text{ in all groups}} (1 + (1 - q_p)/p) \le \exp(O(\log L)) = L^{O(1)}.$$

This proves the coefficient bound, even after absolute values.

Let ω_{tot} count hits in all active and inert groups. For fixed list sizes, their normalized counts are at most $L^{C_9} 2^{\omega_{\mathrm{tot}}}$: use $(u)_t = t!\binom{u}{t} \le t! 2^u$ in each group. The discarded binomial expansion is at most $2^{\omega_{\mathrm{tot}}} \mathbf{1}_{\omega_{\mathrm{tot}} \ge h_0}$. Hence the absolute pointwise error is bounded by $L^{C_9} 4^{\omega_{\mathrm{tot}}} \mathbf{1}_{\omega_{\mathrm{tot}} \ge h_0}$. For any fixed $B > 1$, expanding $B^{\omega_{\mathrm{tot}}}$ into squarefree divisors and counting their multiples gives

$$\sum_{n \asymp R} B^{\omega_{\mathrm{tot}}(n)} \ll R \prod_{p \text{ in all groups}} (1 + (B-1)/p) \ll RL^{C(B)}. \tag{17.27}$$

Indeed only divisors at most a constant times R can occur, and each has $O(R/d)$ multiples in the dyad. Since $16^u \mathbf{1}_{u \ge h_0} \le 2^{-h_0} 32^u$, the square of the error has total at most $RL^{O(1)} 2^{-h_0}$. This proves (17.26) for every fixed N. □

Write the first endpoint as

$$\mathcal{U}(w) = \psi_1(w/Y) w^{i\sigma} \left(\sum_{\mathbf{l}} \mathbf{1}_{l|w} - C_x \right) H_{\mathrm{marks}}(w), \tag{17.28}$$

where ψ_1 has fixed compact support and its rescaled derivatives, as well as $|\sigma|$, have fixed log-power bounds. For every fixed $B' > 0$, we claim uniformly for $|t| \le L^{B'}$ that the normalized sum in (17.24) saves any prescribed log power.

In a tuple term put $w = ln$. All long primes lie outside the marked groups, so $H_{\mathrm{marks}}(ln) = H_{\mathrm{marks}}(n)$. Apply Lemma 17.5 on the scales Y/l and Y in the tuple and constant terms respectively. Cauchy and (17.26) make the total normalized error at most $L^{-N}(1 + \sum_{\mathbf{l}} 1/l)$, after renaming N. For a fixed divisor d', all primes in ld' exceed k for large x, so $\gcd(ld', k) = 1$. Smooth sum–integral comparison on the progressions modulo k gives

$$\begin{aligned} &\frac{1}{Y} \sum_{v \ge 1} \psi_1(ld'v/Y)(ld'v)^{i(t+\sigma)} \mathrm{e}(hld'v/k) \\ &\quad = \frac{1}{ld'} \left(\frac{1}{k} \sum_{a \bmod k} \mathrm{e}(ha/k) \right) \int_0^\infty \psi_1(z)(Yz)^{i(t+\sigma)} \, \mathrm{d}z + O(Y^{-1} L^{C(B')}). \end{aligned} \tag{17.29}$$

The sum has scale $Y/(ld') \ge x^{\varepsilon - o(1)}$. The error follows by summing the endpoint and total-variation errors over k residue classes; all derivative and modulus costs are fixed log powers. Moreover

$$\sum_{\mathbf{l}} 1 \le x^{1-\varepsilon} C_x, \qquad \sum_{d'} |A_{d'}| \le \exp(O(L^{.47})) L^{C_8} = x^{o(1)}.$$

The summed errors in (17.29) are therefore $O(x^{-\varepsilon + o(1)})$, since $Y \ge x^{1-o(1)}$. The main term depends on l only through $1/l$. Its tuple sum is exactly C_x times the main term for the subtracted constant. They cancel, including the complete residue average. Choosing N after B' proves (17.24) on every fixed log-power time interval.

17.5 High frequencies: factorization and exceptional times

Choose a small group g_s. It has $t_{g_s} = 1$ in every comparison, and the tuple test depends only on big marks. Except when a prime of this group divides w twice,

$$W^{\mathbf{b}}_{\mathbf{t}}(w) = \frac{1}{V_{g_s}} \sum_{\substack{p_s | w \\ p_s \in \mathcal{P}_{g_s}}} W^{\mathbf{b}}_{\mathbf{t} - \mathbf{e}_{g_s}}(w/p_s). \tag{17.30}$$

This follows directly by removing the represented small prime; the remaining damping exponent is unchanged. Both sides are bounded by fixed log powers. The square exceptions have relative count at most $L^{O(1)} \exp(-L^{.27})$ on the relevant dyads. In the positive tuple part we may also allow repeated slot primes: the extra terms have a square of a prime at least x^{ε} dividing w, a relative count $O(x^{-\varepsilon})$. Slot multiplicities at each $w \asymp Y$ remain bounded. Lemma 12.2, applied to these coefficient errors, makes their normalized integrated square over $|t| \le T_0$ smaller than every negative log power. Indeed their squared coefficient norm is at most $Y L^{O(1)}(\exp(-L^{.27}) + x^{-\varepsilon})$, and $T_0/Y = L/H = o(1)$.

In the tuple part factor $w = p_s p_l u$, where $p_l = l_1$ lies in the first long-prime slot. After allowing repeated slots the coefficient on u is exactly

$$W^{\mathbf{b}}_{\mathbf{t} - \mathbf{e}_{g_s}}(u) \mathcal{W}(u) \sum_{l_2, \dots, l_d} \mathbf{1}_{l_2 \cdots l_d | u}. \tag{17.31}$$

For $d = 1$ the final factor is one. This is bounded by a fixed log power times a fixed divisor power. In the constant part write $w = p_s n$; its coefficient is just the remaining marked weight times $\mathcal{W}(n)$.

Separate the rational phase by residue classes modulo k. In the tuple part, p_s, p_l are units modulo k; on fixed classes for p_s and u, expand the remaining test in characters of p_l by orthogonality on the units. In the constant part fix classes for p_s, n. Both operations have polynomial-in-k total cost and do not require u or n to be units. Divide all factors into dyads and Fourier-separate the smooth product cutoff in logarithmic size. There are only a fixed log-power number of boxes. Their factor scales have product comparable to Y, and every collapsed coefficient sequence of normalized scale R has

$$\sum_{v \asymp R} |c_v|^2 \ll L^{C_{11}}/R, \tag{17.32}$$

by fixed divisor moments. This applies also to the uncut product of all factors, so Dirichlet mean squares uniformly on translated time intervals justify discarding the Fourier tails with arbitrary log saving. Retain only shifts of fixed log-power size. Enlarge the low-time exponent B' beyond these shifts and any later prime thresholds. The low-time argument already proved is available for this enlarged B'. At the remaining times all retained shifts of t are comparable in magnitude to t.

It remains to treat products of normalized polynomials at a common time, with $L^{B'} \le |t| \le 2T_0$. In every such product there is a small-prime factor

$$P_s(t) = P^{-1} \sum_{\substack{p_s \asymp P \\ p_s \in \mathcal{P}_{g_s}}} b_s(p_s) p_s^{it}, \qquad cL^{.27} \le \log P \le L^{.36}, \qquad |b_s| \le L^{C_{12}}. \tag{17.33}$$

The coefficients include residue restrictions and retained twists. In the tuple part the remaining factors are a long-prime polynomial $P_l(t)$ and a polynomial $R(t)$ at scales N_l and

$$R_0 \asymp Y/(PN_l) \ge x^{\varepsilon - o(1)}. \tag{17.34}$$

In the constant part the remaining polynomial $N(t)$ has scale $R_0 \asymp Y/P$.

The threshold split and exceptional-frequency treatment follow the Matomäki–Radziwiłł strategy; see the Dirichlet polynomial mean-value method. The residual-scale sparse Gram estimate below is adapted here using Section 13, while the factorial coefficient estimate retains the Soundararajan credit stated at its use.

On the set $|P_s(t)| \le L^{-A_s}$, collapse all remaining factors. Their scale is Y/P, and

$$\frac{Y/P}{2T_0} = \frac{H}{2LP} \longrightarrow \infty, \tag{17.35}$$

because $H \ge \exp(2L^{.39} - O(\log L))$ and $P \le \exp(L^{.36})$. Equations (17.32) and the mean-square bound give a fixed log-power integrated square for this collapsed polynomial. Taking A_s sufficiently large handles this part.

Let $\mathcal{J}$ be the integer unit intervals meeting $\{|t| \le 2T_0 : |P_s(t)| > L^{-A_s}\}$. Then

$$|\mathcal{J}| \le \exp\big(O(L^{.36} + L^{.73} \log L)\big) \le \exp(O(L^{.9})). \tag{17.36}$$

For this step we use the factorial prime-polynomial coefficient estimate, as in the factorial moment bound, together with a coefficient-independent mean square. Put $Z = 8T_0 + 8$ and $j_0 = \lfloor \log Z / \log(2P) \rfloor$. Unique factorization gives squared coefficient norm for $P_s^{j_0}$ at most

$$j_0! \left(P^{-2} \sum_{p_s \asymp P} |b_s(p_s)|^2 \right)^{j_0} \le j_0! (L^{C_{13}}/P)^{j_0}.$$

Select a point exceeding the threshold from each occupied interval, and split their integer indices modulo three to get separated sets. The indices of this powered polynomial are at most Z. The separated-point mean-square bound of Lemma 12.2 therefore gives

$$|\mathcal{J}| \ll Z L^{O(1)} j_0! (L^{C_{13}+2A_s}/P)^{j_0}.$$

Since $ZP^{-j_0} \le 2P\,2^{j_0}$ and $j_0 = O(L^{.73})$, taking logarithms proves (17.36).

We will also use the following sparse bound for any normalized polynomial $R(t) = \sum_{n \asymp R_0} c_n n^{it}$ with $x^{\varepsilon/2} \le R_0 \le x^2$ and $\sum |c_n|^2 \le L^{C_{14}}/R_0$:

$$\sum_{I \in \mathcal{J}} \sup_{t \in I} |R(t)|^2 \ll L^{C_{15}}. \tag{17.37}$$

Choose maximizing points and again split into three separated classes. For $1 \le |\Delta| \le T_{\mathrm{mod}} := \exp(L/(\log L)^2)$, sum–integral comparison on the full integer dyad gives

$$R_0^{-1} \left| \sum_{n \asymp R_0} n^{i\Delta} \right| \ll |\Delta|^{-1} + x^{-c_\varepsilon}.$$

For larger differences, Lemma 12.4 gives $O(\exp(-L/(\log L)^{C_{16}}))$ instead. Its hypotheses hold: $R_0 \ge x^{\varepsilon/2} > \exp(L/(\log L)^2)$, all relevant scales are at most $2x^5$, and the differences are between T_{mod} and $4T_0 + O(1) < 4x^3$. The Gram matrix of the vectors $(n^{it_i})_{n \asymp R_0}$ consequently has absolute row sum

$$\ll R_0 \left(1 + \log(2T_0) + |\mathcal{J}|\{x^{-c_\varepsilon} + \exp(-L/(\log L)^{C_{16}})\}\right) \ll R_0 L.$$

Here separation bounds the reciprocal-difference sum by $O(\log T_0)$, and (17.36) absorbs both small errors. The Gram operator bound multiplied by $\sum |c_n|^2$ proves (17.37).

17.6 The tuple and constant contributions at high times

In the tuple term the long-prime polynomial is, after partial summation, a normalized prime sum with a character modulo k and a fixed log-power twist. Its dyad lies between $x^\varepsilon/2$ and $x^{1-\varepsilon}$. Choose fixed $0 < \tau < \varepsilon$ and $1 - \varepsilon < \eta < 1$ to apply Lemma 12.3. For any prescribed A_l, it gives

$$|P_l(t)| \ll L^{-A_l} \quad (L^{B'} \le |t| \le 2T_0), \tag{17.38}$$

after increasing B' beyond its threshold and all retained shifts. The upper frequencies are $< x^2$. The remaining coefficient (17.31) has scale (17.34) and obeys (17.32), so (17.37) applies. The small polynomial has absolute bound $L^{O(1)}$. Hence on the exceptional intervals the integrated square of $P_s P_l R$ is

$$\ll L^{O(1)-2A_l} \sum_{I \in \mathcal{J}} \sup_{t \in I} |R(t)|^2,$$

which has arbitrary log saving on choosing A_l large enough. Together with the ordinary-time estimate this handles the tuple part.

For the constant part set $R_0 \asymp Y/P$, and apply Lemma 17.5 to the remaining marked coefficient at this scale. Keep its residue-class restriction separately. If $E(n)$ is the coefficient error, then for every fixed N, $\sum_{n \asymp R_0} |E(n)|^2 \ll R_0 L^{-N}$. The mean-square bound on the *entire* retained time interval gives

$$\begin{aligned} \int_{|t| \le 2T_0} \left| R_0^{-1} \sum_{n \asymp R_0} E(n) n^{it} \right|^2 \mathrm{d}t &\ll L^{-N} \left(\frac{T_0}{R_0} + \log R_0 \right) \\ &\ll L^{-N} \left(\frac{LP}{H} + O(L) \right). \end{aligned} \tag{17.39}$$

By (17.35), $LP/H = o(1)$. Thus the known small-prime suprema and all fixed decomposition costs can be paid by choosing N last. In particular the model error is controlled before restricting to the exceptional times.

For the truncated model, a divisor d' leaves a progression modulo k at scale $R' = R_0/d'$. Its normalization in $N(t)$ is $A_{d'}/d'$ times normalization at R'. Since $d' \le \exp(O(L^{.47}))$, one has $R' = x^{1-o(1)}$. The divisor is a unit modulo k, so the original residue restriction becomes a single

residue restriction at this scale. For $L^{B'} \le |t| \le T_{\mathrm{mod}}$, comparison with the integral of the logarithmic phase gives

$$|N_0(t)| \ll L^{C_{17}}(|t|^{-1} + x^{-c'}). \tag{17.40}$$

Indeed the normalized integral over a dyad is $O(1/|t|)$ and the progression endpoint and variation error is $O(L^{O(1)}(1+|t|)/R') = O(x^{-c'})$; sum using $\sum |A_{d'}|/d' \le L^{C_8}$. Retained twists may either be incorporated in t or absorbed by increasing B'. The square of (17.40) is integrable with

$$\int_{L^{B'} \le |t| \le T_{\mathrm{mod}}} |N_0(t)|^2 \, \mathrm{d}t \ll L^{2C_{17}-B'} + x^{-c''}. \tag{17.41}$$

This has arbitrary log saving after enlarging B', even after the small-prime absolute bound. We used square integration of $1/|t|$, not a pointwise log saving multiplied by the full interval length.

For $T_{\mathrm{mod}} < |t| \le 2T_0$, apply Lemma 12.4 to each such progression. Its modulus is at most L^{C_6}, its scale is $x^{1-o(1)}$, and, including retained shifts, its frequency is at least $\frac{1}{2}T_{\mathrm{mod}} > \exp(L/(2(\log L)^2))$ and at most $x^{1+o(1)} < 4x^3$. After summing the reciprocal divisor coefficients this yields

$$|N_0(t)| \ll L^{C_{18}} \exp(-L/(\log L)^{C_{19}}). \tag{17.42}$$

We now integrate only on the exceptional unit intervals. By (17.36) and the small-prime absolute bound, their total contribution is at most

$$L^{O(1)} \exp(O(L^{.9}) - 2L/(\log L)^{C_{19}}),$$

which saves every fixed log power. No length-dominated mean-square estimate was required on an individual divisor progression: R' may be smaller than T_0, and the bounds (17.40)–(17.42) are pointwise. Equations (17.39), (17.41), and (17.42) complete the constant-term estimate.

We have proved (17.24) to every fixed log precision. The local Mellin inequality gives (17.22); the comparison multiplier estimate and Parseval then bound each term in (17.18), divided by Y, by an arbitrarily large negative log power. Choose that precision after the known fixed kernel, dyad, arc, and separation costs. Their total is therefore negligible. Subtracting these comparison terms from the residual pairing leaves its raw pattern, already identified with (17.5). This proves Theorem 17.1.

18 Extraction of the prime statistic

Fix $d \ge 1$ and $\varepsilon > 0$. For each $i \le d$, let $\mathcal{I}_i$ be an interval of primes with lower endpoint at least x^{ε}, and suppose that the product of the upper endpoints is at most $x^{1-\varepsilon}$. Define

$$f_x(u) = \sum_{\substack{\ell_i \in \mathcal{I}_i \\ \ell_1,\dots,\ell_d \text{ distinct}}} \mathbf{1}_{\ell_1 \cdots \ell_d \mid u}, \qquad C_x = \sum_{\substack{\ell_i \in \mathcal{I}_i \\ \ell_1,\dots,\ell_d \text{ distinct}}} \frac{1}{\ell_1 \cdots \ell_d}. \tag{18.1}$$

Lemma 12.1 gives $C_x = O_{d,\varepsilon}(1)$. On any fixed range $u \asymp x$, the same bound holds for $f_x(u)$: there are only $O_\varepsilon(1)$ prime divisors of u exceeding x^{ε}. All cutoffs below are supported in a fixed compact subset of $(0,\infty)$.

Theorem 18.1 (Interior prime statistic)**.** *For every nonnegative $\Phi \in C_c^\infty((0,\infty))$,*

$$\sum_{p \text{ prime}} \Phi((p-1)/x)(f_x(p-1) - C_x) = o(x/\log x). \tag{18.2}$$

The limit holds through all real $x \to \infty$ for the slot intervals just described.

Throughout the proof write $L = \log x$, $W = \exp(L^{.24})$, and $V(W) = \prod_{p \le W}(1 - 1/p)$. Fix $q, q_0 \in (0,1)$, for example $q = q_0 = 1/2$. A *marking array* will mean K disjoint bands from Theorem 14.1, with their weight

$$\mathcal{W}(u) = \prod_{i=1}^{K} q^{\omega_i(u)-1}\frac{\omega_i(u)}{V_i}, \qquad V_i = \sum_{p \in \mathcal{P}_i}\frac{1}{p}. \tag{18.3}$$

Each band consists of all primes between $\exp(L^{a_i})$ and $\exp(2L^{a_i})$, with $.1 < a_i < .2$. The number K and the exponents are fixed in each limit. Since $V_i = \log 2 + o(1)$ and $\sup_{n \ge 0} nq^{n-1} < \infty$, these weights are bounded by a constant depending only on K, q, for large x.

Presieving $u+1$ leaves both primes and composites in the average of $\mathcal{W}(u)(f_x(u) - C_x)$. We will show that the composite contribution is negligible in two stages. The one-sided estimate first replaces its prime factors by rough variables; additional divisor marks then allow the two-sided estimate to bound the resulting centered sum. This isolates a prime average carrying $\mathcal{W}$. Finally, a mean-square bound for averages of disjoint marking arrays removes that weight.

18.1 Progressions and the independent divisor model

For finitely many disjoint arrays, define the independent divisor model by replacing every $\mathbf{1}_{p|u}$ at a band prime with an independent Bernoulli variable of mean $1/p$. For a function H of these indicators write $M_x(H)$ for its expectation in this model. In particular put $m_{\mathcal{W}} = M_x(\mathcal{W})$. There are constants $0 < c_K < C_K < \infty$ such that

$$c_K \le m_{\mathcal{W}} \le C_K, \qquad M_x(\mathcal{W}^2) \le C_K. \tag{18.4}$$

For the lower bound, in each band the event of exactly one hit has probability $\prod_p(1-1/p)\sum_p 1/(p-1)$, bounded below since the reciprocal sum tends to $\log 2$ and the largest $1/p$ tends to zero. The weights have a fixed positive value on the event of one hit per band. Independence between bands proves the lower bound; boundedness proves the other two assertions. These constants can be chosen independently of the particular fixed exponents.

Lemma 18.2 (Summed progression remainder)**.** *Choose $0 < c_0 < \varepsilon/2$. Let H be a weight from a fixed finite collection of disjoint arrays, a product of two such weights (allowing a repeated array), a constant, or a bounded linear combination of these functions. Then, for every fixed $A > 0$,*

$$\sum_{r \le x^{c_0}}\left|\sum_{\substack{u \ge 1\\ r|u+1}}\Phi(u/x)f_x(u)H(u) - \frac{x}{r}\left(\int_0^\infty \Phi(t)\,\mathrm{d}t\right)C_x M_x(H)\right| \ll_A xL^{-A}. \tag{18.5}$$

The same assertion holds with f_x, C_x replaced by $1, 1$. The constants may depend on the fixed arrays and on the bounded coefficients in the linear combination.

Proof. We give the expansion including the repeated-array case needed later. It suffices to treat one product of weights. In any band g which occurs, let $a_g \in \{1,2\}$ be its multiplicity in the product. Its factor is

$$V_g^{-a_g} q^{a_g(\omega_g(u)-1)}\omega_g(u)^{a_g}.$$

Expand $\omega_g^{a_g}$ as the sum over ordered a_g-tuples of divisor primes, permitting coincidences. For a chosen tuple let S_g be its set of distinct primes. On the event $S_g \mid u$, where this notation means that every prime in S_g divides u, the contribution is

$$V_g^{-a_g} q^{a_g(|S_g|-1)}\mathbf{1}_{S_g|u}\prod_{p \in \mathcal{P}_g \setminus S_g}(1 - (1-q^{a_g})\mathbf{1}_{p|u}). \tag{18.6}$$

Thus repetitions change the damping to q^2 on the unrepresented primes. For instance, the reciprocal mass of the represented unions when $a_g = 2$ is exactly

$$V_g^{-2}\left\{V_g + q^2\left(V_g^2 - \sum_{p\in\mathcal{P}_g} p^{-2}\right)\right\},$$

and is bounded. For $a_g = 1$ that mass is one. Multiplying over the fixed number of bands shows that all represented unions have total reciprocal mass $O(1)$, with their nonnegative coefficients and their multiplicities included.

Expand the remaining product in (18.6) through degree $h_0 = \lfloor L^{.01}\rfloor$ in the unrepresented primes. Bonferroni inequalities, valid for factors $1 - t_p$ with $0 \le t_p \le 1$, bound the error in absolute value by the nonnegative term of degree $h_0 + 1$. If C_* bounds the sum of the reciprocal masses of all bands involved, the total reciprocal coefficient mass at degree t is at most a fixed constant times $C_*^t/t!$. Hence the truncated expansion has reciprocal coefficient mass $O(1)$ and its error envelope has reciprocal mass at most

$$O\left(\frac{C_*^{h_0+1}}{(h_0+1)!}\right) = O_A(L^{-A}) \quad \text{for every fixed } A. \tag{18.7}$$

Every divisor in the truncation or the error envelope satisfies

$$d' \le \exp\bigl(O_K((h_0+1)L^{.2})\bigr) \le \exp(O_K(L^{.22})) = x^{o(1)}. \tag{18.8}$$

It follows also that the ordinary absolute coefficient mass is $x^{o(1)}$: multiply its reciprocal mass by the largest divisor. All these facts hold as well for the constant function.

Fix an ordered long-prime tuple, put $\ell = \prod_i \ell_i$, and write $u = \ell y$. Every band prime is coprime to ℓ. Moreover, every $\ell_i > x^{c_0}$, so $(\ell, r) = 1$ for $r \le x^{c_0}$. The congruence $r \mid \ell y + 1$ therefore fixes one reduced residue class of y modulo r. For a divisor d' of band primes the additional condition $d' \mid y$ is impossible if $(d', r) > 1$. Otherwise the Chinese remainder theorem and smooth summation in one progression give

$$\sum_{\substack{y\ge 1\\ \ell y\equiv -1\ (r)\\ d'\mid y}} \Phi(\ell y/x) = \frac{x}{\ell r d'}\int_0^\infty \Phi(t)\ \mathrm{d}t + O_\Phi(1). \tag{18.9}$$

The error follows, for example, by comparing each sampling interval with its integral and using the bounded total variation of Φ. It is uniform in ℓ, r, d'.

Inserting the expansion into (18.9) recovers the model in which the band primes dividing r are forced absent. The same calculation for the positive Bonferroni envelope bounds the truncation error by (18.7) times $x/(\ell r)$, plus $x^{o(1)}$ in rounding errors. There are at most $d!\,x^{1-\varepsilon}$ ordered long-prime tuples, since a squarefree product determines at most $d!$ orders. Consequently the sum of all rounding errors over tuples and $r \le x^{c_0}$ is

$$O(x^{1-\varepsilon+c_0+o(1)}) = O(x^{1-\varepsilon/3}), \tag{18.10}$$

after increasing the threshold for x. When $f_x = 1$, use $\ell = 1$ and the stronger bound $x^{c_0+o(1)}$ instead. The principal Bonferroni errors sum to at most $O(xC_xLC_*^{h_0+1}/(h_0+1)!)$ and save every fixed log power.

Finally, coupling the forced-absence model to the unconditional one and using the boundedness of H bounds their mean difference by

$$O\left(\sum_{\substack{p\mid r\\ p \text{ in the arrays}}} \frac{1}{p}\right) \ll L\exp(-L^{.1}).$$

Its contribution after summing $1/\ell$ and $1/r$ is smaller than every xL^{-A}. This proves (18.5); bounded linear combinations follow by the triangle inequality. □

18.2 Presieving and replacement of composite prime slots

Fix a large even integer h, and set

$$\gamma = \frac{c_0}{4h+3}, \qquad z = x^{\gamma}. \tag{18.11}$$

Apply the block sieve, Lemma 12.6, to the bad conditions $r' \mid u+1$ at primes $r' \le z$, with density $g(r') = 1/r'$. Use separately the two nonnegative weights $\Phi(u/x)\mathcal{W}(u)f_x(u)$ and $\Phi(u/x)\mathcal{W}(u)$. The density is at most $1/2$, and its reciprocal mass in every block $(b, b^2]$ is bounded by an absolute constant by Lemma 12.1. The sieve remainder is bounded by the sum in Lemma 18.2, because

$$z^{4h+2} = x^{c_0(4h+2)/(4h+3)} < x^{c_0}. \tag{18.12}$$

Subtract the second sieve formula times C_x from the first. Their main terms cancel, while each relative sieve error is $O(e^{-h})$. Since the main terms both contain $m_{\mathcal{W}}$ and $\prod_{r' \le z}(1-1/r') \asymp 1/(\gamma L)$, we obtain

$$\limsup_{x\to\infty} \frac{L}{x m_{\mathcal{W}}} \left| \sum_{\substack{u \ge 1 \\ P^-(u+1) > z}} \Phi(u/x)\mathcal{W}(u)(f_x(u) - C_x) \right| \ll \frac{e^{-h}}{\gamma}. \tag{18.13}$$

The implied constant depends on the slot and cutoff data but is independent of K, h and of the fixed array. The progression remainders may have constants depending on these choices; their normalized limits are zero. In particular, no reciprocal of a rare-mark mean occurs in the surviving error in (18.13).

For integers in this sum that are composite, write their ordered prime factorization as $u+1 = p_1 \cdots p_j$, with $p_i > z$. For large x, $2 \le j \le 2/\gamma$. Integers divisible by p^2 for a prime $p > z$ number

$$\ll \sum_{p > z} \frac{x}{p^2} \ll x/z.$$

The weights and the possible ordered factorization counts are bounded for fixed γ, d, K. We may therefore discard these integers at error $o(x/L)$, count the others with weight $1/j!$, and restore repeated factors when convenient at the same cost. For each fixed j we will prove

$$\sum_{p_1,\dots,p_j > z} \Phi((p_1 \cdots p_j - 1)/x)\mathcal{W}(p_1 \cdots p_j - 1)(f_x(p_1 \cdots p_j - 1) - C_x) = o(x/L). \tag{18.14}$$

Let

$$\kappa(m) = \frac{1}{V(W)\log m} \quad (m \ge 2), \qquad a(m) = \mathbf{1}_{m>z}\mathbf{1}_{P^-(m)>W}\kappa(m). \tag{18.15}$$

Set $\kappa(1) = 0$; values at one are always excluded by the positive-power threshold tests. We first replace every prime variable in (18.14) by this coefficient. Telescope one variable m at a time and write n for the product of the other $j-1$ variables. Dyadically decompose m, n on the support of $\Phi((mn-1)/x)$. There are $O(L^2)$ boxes, with $H_m H_n \asymp x$, and both scales are at least a fixed positive power of x. The restriction $m > z$ only shortens its dyadic interval, as permitted in Theorem 14.1. The other coefficient β_n is supported on W-rough integers and is a fixed convolution of prime indicators and proxy coefficients. It is bounded by $L^{C_j}\tau_{j-1}(n)$ for a fixed exponent C_j.

The divisor moment bound in Lemma 12.2 permits truncation of β_n at L^B, for a sufficiently large fixed B, with an arbitrarily large log saving in its ℓ^1 norm. More explicitly, for any fixed integer $t > 1$,

$$\sum_{n \asymp H_n} |\beta_n|\mathbf{1}_{|\beta_n| > L^B} \le L^{-B(t-1)} \sum_{n \asymp H_n} |\beta_n|^t \ll H_n L^{C_{j,t} - B(t-1)}.$$

Multiplying by the number and maximum size of the m coefficients and by the bounded endpoint factor shows that the discarded terms are $O(xL^{-A})$ for any prescribed A, on taking B large enough. This truncation does not change rough support.

Replacing $\Phi((mn-1)/x)$ by $\Phi(mn/x)$ costs $O(x^{-1})$ per coefficient, hence only a fixed log power in the whole sum; it is negligible compared with x/L. Fourier inversion of the smooth function in $\log(mn/x)$ separates the two variables into factors m^{iv} and n^{iv}, with a rapidly decreasing Fourier density. Its tails may be discarded past a fixed log power, with any desired saving, using the same coefficient bounds. On the remaining frequencies the discrepancy in the m slot is exactly the one in Theorem 14.1.

To meet its invariance hypothesis globally, clip $f_x(u) - C_x$ to a fixed interval containing all its values on the support under consideration, and divide by a fixed bound so that its modulus is at most one. Every long-slot prime lies outside the array bands, so $f_x(pu) = f_x(u)$ for each band prime p; clipping preserves this identity. Thus the Type II theorem applies with F equal to this normalized clipped function. Choose its log-saving target after all the fixed dyadic, Fourier, and truncation losses. We then choose K sufficiently large for that target, simultaneously for all $2 \le j \le 2/\gamma$. This choice is made after h, γ, and is independent of the particular fixed band exponents. Telescoping all slots gives

$$\sum_{p_1,\dots,p_j>z} \Phi((p_1\cdots p_j-1)/x)\mathcal{W}(p_1\cdots p_j-1)(f_x(p_1\cdots p_j-1)-C_x)$$
$$= \sum_{m_1,\dots,m_j} \Phi((m_1\cdots m_j-1)/x)\mathcal{W}(m_1\cdots m_j-1)(f_x(m_1\cdots m_j-1)-C_x)\prod_{i=1}^{j} a(m_i) + o(x/L). \tag{18.16}$$

18.3 Removing candidate prime parts

Fix j in the preceding finite range. Construct candidate small groups from all primes in consecutive bands $[\exp(y), \exp(2y))$, starting at $y = L^{.28}$ and doubling y while $2y \le L^{.35}$. Construct candidate big groups in the same way, from $y = L^{.40}$ while $2y \le L^{.45}$. There are between positive constant multiples of $\log L$ groups of each kind. They are disjoint, their reciprocal sums V_g are bounded above and below by positive constants, and they lie in the allowed small and big ranges of Theorem 17.1. The big groups are full prime intervals. All candidate primes exceed W.

We will group factorizations that differ only in the allocation of candidate primes among their j factors. Their coefficients must agree, so we first remove all candidate prime parts from the threshold and logarithm in $a(m)$. For any integer m, let $\bar{m}$ be obtained by removing the entire prime-power parts belonging to all candidate groups. Replace $a(m)$ by

$$\widetilde{a}(m) = \mathbf{1}_{P^-(m)>W}\mathbf{1}_{\bar{m}>x^\gamma}\kappa(\bar{m}). \tag{18.17}$$

We show that this changes the right-hand side of (18.16) by $o(x/L)$ in absolute value. In particular, the estimate will remain valid before any cancellation is used.

For these error estimates, decompose tuples into full product dyads $m_i \asymp H_i$, with $\prod_i H_i \asymp x$ and $H_i \ge x^\gamma/2$. There are $O(L^{j-1})$ such boxes. By Lemma 12.5, the number of W-rough integers in each dyad is $O(H_i V(W))$. On an unconditioned integer dyad,

$$\sum_{m\asymp H_i} \log(m/\bar{m}) \le \sum_{p \text{ candidate}} \sum_{a\ge 1} (\log p)\left\lfloor \frac{2H_i}{p^a}\right\rfloor \ll H_i \sum_{p\le \exp(L^{.45})} \frac{\log p}{p-1} \ll H_i L^{.45}. \tag{18.18}$$

Thus the number with $\log(m/\bar{m}) > L^{.9}$ is $O(H_i L^{-.45})$. On the support of either coefficient in the comparison its size is $O((LV(W))^{-1})$. Taking the bad count in one coordinate and the rough counts in all others bounds the bad-tuple mass in a box by

$$O\left(\frac{x}{L^j}\frac{L^{-.45}}{V(W)}\right) = O(xL^{-j-.21}), \tag{18.19}$$

since $V(W) \asymp L^{-.24}$. After summing boxes this is $O(xL^{-1-.21})$.

For the other tuples, a change in the threshold test requires $\gamma L < \log m_i \le \gamma L + L^{.9}$ for at least one coordinate. There are $O(L^{.9}L^{j-2})$ product dyads meeting such a strip, because $j \ge 2$. Their positive mass is $O(x/L^j)$ per box, so the total is $O(xL^{-1.1})$. Away from those strips the supports coincide, and

$$\frac{\kappa(\bar{m}_i)}{\kappa(m_i)} = \frac{\log m_i}{\log \bar{m}_i} = 1 + O_\gamma(L^{-.1}).$$

The total positive tuple mass is $O(x/L)$, by the same rough counts and box count. Together with the boundedness of $\mathcal{W}(f_x - C_x)$ this proves the required absolute $o(x/L)$ error.

18.4 Many successes before the signed expansion

Write $v = m_1 \cdots m_j$ and $u = v - 1$. For each candidate group g, independently conditional on the tuple, toss a coin whose success probability is

$$s_g(u,v) = c_1 q_0^{\omega_g(u)-1} \frac{\omega_g(u)}{V_g} (q_0/j)^{\omega_g(v)-1} \frac{\omega_g(v)}{V_g}. \tag{18.20}$$

The constant $c_1 > 0$ is fixed sufficiently small, depending on j, q_0 and the bounds for V_g, that these probabilities are at most one. The factors with no hit are interpreted as zero. The division by j in the product-side damping anticipates the j possible allocations of each candidate prime among the factors when the candidate part of v is squarefree. Summing those allocations will recover the q_0 damping of the two-sided estimate. We claim that for some fixed $c_2 > 0$, outcomes with fewer than $c_2 \log L$ successes in either family have total positive weighted mass $o(x/L)$ under the coefficient $\prod_i \widetilde{a}(m_i)$.

We prove the uniformity needed for this claim on full rough product dyads. Select each m_i independently and uniformly among W-rough integers in its full dyad. For any fixed number of distinct candidate primes, their product Q satisfies $\log Q = O(L^{.45})$. Each prescribed residue class modulo Q is relatively equidistributed among these rough integers, with error $o(1)$ uniformly in the classes, primes, and dyads. To verify this, apply Bonferroni to divisibility by primes $p \le W$ at depth $D \log L$, where D is a sufficiently large fixed constant. The reciprocal sum of those primes is $O(\log L)$, so taking D large makes the omitted reciprocal mass smaller than any prescribed log power times $V(W)$. The moduli in the retained terms are at most

$$QW^{O(\log L)} = \exp(O(L^{.45} + L^{.24} \log L)) = x^{o(1)}.$$

Since every factor dyad has positive-power length, CRT counting errors divided by its expected class count tend to zero uniformly. Normalizing by the similarly evaluated total rough count proves the assertion, with arbitrary fixed logarithmic precision if needed.

In the independent residue model, for a candidate prime p the events $p \mid u$ and $p \mid v$ are disjoint, with probabilities

$$a_p = \frac{(p-1)^{j-1}}{p^j}, \qquad b_p = 1 - (1 - 1/p)^j, \tag{18.21}$$

respectively. For the first formula all residues must be nonzero, and the first $j - 1$ determine the last. For the second, at least one of the j residues must be zero. Distinct primes are independent in this model. In each group, $\sum_p a_p = V_g + o(1)$ and $\sum_p b_p = jV_g + o(1)$, while the largest individual probability tends to zero. Summing the uniform residue estimates proves convergence of every fixed joint factorial moment of the two hit counts, for any one or two groups, uniformly over their choices and over product dyads. These moments are bounded by C^r at total order r, with C depending only on the fixed j and the reciprocal group bounds.

Here is why fixed moments suffice for the bounded probabilities in (18.20), without any growing-moment assumption. First cut each of the at most four counts at a fixed bound a. The bounded first moments give tail probability $O(1/a)$ uniformly. For exact counts below this bound, list the required hits and impose the absence of further hits by Bonferroni. At a fixed depth b, the limit

superior of the remainder is at most $C_a C^b/b!$, by the corresponding factorial moment estimate. Take $x \to \infty$, then $b \to \infty$, and then $a \to \infty$. It follows that the expectations of the bounded one-group tests and their two-group products agree with the model up to a uniform $o(1)$.

Explicitly, for a count vector $N = (N_1, \dots, N_r)$ and a fixed target k, list the k_i required hits of each type. Conditional on $N \geq k$ coordinatewise, inclusion–exclusion for no other hit is the alternating binomial sum in $N_{\mathrm{tot}} - |k|$, multiplied by $\prod_i \binom{N_i}{k_i}$. The first omitted term at depth b is bounded by

$$\frac{(N_{\mathrm{tot}})_{|k|+b+1}}{(b+1)! \prod_i k_i!},$$

where $(n)_r = n(n-1)\cdots(n-r+1)$. Its expected limit superior is at most $C^{|k|+b+1}/((b+1)! \prod_i k_i!)$. This supplies the stated uniform remainder for every exact count used after the fixed truncation.

In the model, the probability of exactly one hit of each type in a group is bounded below. Indeed it equals

$$\prod_{p \in g} (1 - a_p - b_p) \sum_{p \neq q \text{ in } g} \frac{a_p}{1 - a_p - b_p} \frac{b_q}{1 - a_q - b_q},$$

whose product and sum are bounded below using the displayed reciprocal bounds and the vanishing largest atom. On this event the conditional success probability is c_1/V_g^2, also bounded below. Distinct groups are independent in the model. Therefore, if I_g denotes the actual coin indicator, there are $\beta > 0$ and $\epsilon_x \to 0$, uniform in the dyads and groups, such that

$$\mathbb{E} I_g \geq \beta, \qquad |\operatorname{Cov}(I_g, I_{g'})| \leq \epsilon_x \quad (g \neq g'). \tag{18.22}$$

Conditional independence of the coins identifies the second quantity with the covariance of their success probabilities. For a family of $N \asymp \log L$ groups, Chebyshev gives

$$\mathbb{P}\left(\sum_g I_g < \beta N/2 \right) \leq \frac{4}{\beta^2 N} + \frac{4\epsilon_x}{\beta^2} = o(1). \tag{18.23}$$

Only two families occur. Notice that a uniform $o(1)$ covariance suffices; no rate relative to $1/\log L$ is required.

Finally transfer this full-dyad probability to the actual weighted tuples. A rough product box contains $O(xV(W)^j)$ tuples, and $\prod_i \widetilde{a}(m_i) \ll (LV(W))^{-j}$ on its support. Thus its failure mass is $o(x/L^j)$ uniformly. There are $O(L^{j-1})$ boxes, proving the claimed $o(x/L)$ total. The bounded factor $\Phi(u/x)\mathcal{W}(u)(f_x(u) - C_x)$ does not alter this conclusion. We discard these failed outcomes now, while their weights form a nonnegative probability partition.

18.5 Allocation and the two-sided correlation

Let $\mathcal{C}$ be the full candidate set of groups. The coin partition is

$$1 = \sum_{S \subseteq \mathcal{C}} \prod_{g \in S} s_g \prod_{g \notin S} (1 - s_g).$$

Retain only S containing at least $c_2 \log L$ groups of each kind, at the error just proved. Now expand the failure factors in each retained term. Each expanded term has the form

$$\pm \prod_{g \in P} s_g, \qquad P = S \cup T, \quad T \subseteq \mathcal{C} \setminus S. \tag{18.24}$$

There are at most $3^{|\mathcal{C}|} = L^{O(1)}$ terms; their coefficients, including $c_1^{|P|}$, have fixed log-power bounds. Every active set P has between fixed positive multiples of $\log L$ groups of each kind and satisfies all the active-group hypotheses of Theorem 17.1. This expansion is performed after the failed mass has been removed, so it never multiplies the preceding qualitative $o(x/L)$ error.

For a fixed active set let $s_P = |P|$, let v_* be obtained from v by removing the entire parts of its active primes, and use W_{1,q_0} to denote the active one-mark weight with damping q_0. We may first restrict to v not divisible by the square of any candidate prime. Such square exceptions have integer count

$$O\left(x \sum_{p \text{ candidate}} p^{-2}\right) \ll x \exp(-cL^{.28}). \tag{18.25}$$

Their weighted tuple sums remain negligible after all fixed log-power losses. Indeed the number of ordered factor tuples of v is at most $\tau_j(v)$; Cauchy–Schwarz and a fixed divisor moment bound give an exponential saving in a power of L for sums of any fixed divisor power over this exceptional set. All active marked factors are bounded by fixed log powers, because there are $O(\log L)$ groups with exponential damping in each. The same argument will allow these exceptions to be restored at the end of the calculation.

Define the core

$$G_P(v) = \sum_{e_1 \cdots e_j = v_*} \prod_{i=1}^{j} \left[\mathbf{1}_{P^-(e_i) > W} \mathbf{1}_{\bar{e}_i > x^\gamma} \kappa(\bar{e}_i)\right]. \tag{18.26}$$

Here the bar still removes *all* candidate prime parts, including the inactive ones. This definition implies $G_P(v) = G_P(v_*)$ and

$$|G_P(v)| \le (C/(LV(W)))^j \tau_j(v_*) \le L^{C_j} \tau(v_*)^{C_j}. \tag{18.27}$$

The second bound follows from the elementary inequality $\tau_j(n) \le \tau(n)^{j-1}$, obtained prime by prime.

Off the square exceptions, fix a factorization $e_1 \cdots e_j = v_*$. Every active prime dividing v can be assigned independently to any of the j slots. These are all the original factorizations with that stripped factorization, and all have the same coefficient in (18.17): active primes exceed W and the tests and logarithms depend only on the bars. Thus there are $j^{\omega_P(v)}$ equal contributions. The exact identity

$$j^{\omega_P(v)} (q_0/j)^{\omega_P(v) - s_P} = j^{s_P} q_0^{\omega_P(v) - s_P} \tag{18.28}$$

shows that summing the v-side factor in (18.24) over factorizations gives

$$j^{s_P} W_{1,q_0}(v) G_P(v).$$

This is an identity of weights, including the mark factors $\prod_g \omega_g(v)/V_g$.

Restoring the square exceptions by (18.25) and its divisor-moment bound, each expanded term is, up to a negligible error, a constant of fixed log-power size times

$$\sum_{u \ge 1} \Phi(u/x) \mathcal{W}(u) (f_x(u) - C_x) W_{1,q_0}(u) W_{1,q_0}(u+1) G_P(u+1). \tag{18.29}$$

Take $\Psi(y) = y\Phi(y)$. Since $\Phi(u/x) = x\Psi(u/x)/u$, Theorem 17.1 bounds (18.29) by $O_A(xL^{-A})$ for every fixed A. Its invariant core hypothesis is exactly (18.26), and its growth hypothesis is (18.27). Its other endpoint is exactly $F_x(u) = \mathcal{W}(u)(f_x(u) - C_x)$. Choose A after the fixed costs from $3^{|\mathcal{C}|}$, j^{s_P}, and the coefficients. They are all absorbed. Summing the expanded terms and adding back the earlier, unamplified, discarded masses proves that the proxy sum in (18.16) is $o(x/L)$. This proves (18.14).

18.6 Removal of the small-band marks

Subtract the composite contribution from (18.13). All primes in the support exceed z for large x, and we have proved

$$\limsup_{x \to \infty} \frac{L}{x} \left| \sum_{p \text{ prime}} \Phi((p-1)/x) \frac{\mathcal{W}(p-1)}{m_{\mathcal{W}}} (f_x(p-1) - C_x) \right| \ll e^{-h}/\gamma. \tag{18.30}$$

The constant here remains independent of K, h.

Fix this h and its sufficient fixed K. For any finite $b \geq 1$ choose b disjoint arrays, each with K distinct exponents, using distinct exponents across all arrays inside $(.1, .2)$. This is possible for every finite b, and the Type II choice of K applies to every one of them. Write $\mathcal{W}_a, m_a$ for their weights and model means, and set

$$Z_b(u) = \left(\frac{1}{b} \sum_{a=1}^{b} \frac{\mathcal{W}_a(u)}{m_a} - 1 \right)^2. \tag{18.31}$$

Disjoint arrays are independent in the divisor model. Their normalized weights have mean one and uniformly bounded second moments by (18.4). Hence

$$M_x(Z_b) = \frac{1}{b^2} \sum_{a=1}^{b} \mathrm{Var}_M(\mathcal{W}_a/m_a) \leq C_K/b. \tag{18.32}$$

The square and all its cross terms lie within Lemma 18.2, whose coefficient bounds are uniform for these normalized weights when b, K are fixed.

Apply the nonnegative upper bound of the block sieve at the fixed depth $h' = 2$, with $z' = x^{c_0/20}$, to the weight $\Phi(u/x)Z_b(u)$. Its remainder range is

$$(z')^{4h'+2} = x^{c_0/2} < x^{c_0},$$

so Lemma 18.2 without tuple factors applies. Every prime $u + 1$ in the support exceeds z'. The sieve main term is bounded by a fixed constant times $xM_x(Z_b)/\log z'$, and its remainder is $o(x/L)$. It follows that

$$\limsup_{x\to\infty} \frac{L}{x} \sum_{p \text{ prime}} \Phi((p-1)/x)Z_b(p-1) \ll C_K/b. \tag{18.33}$$

The implied constant here depends on the fixed c_0 and cutoff, but not on b, K; the latter dependence is recorded in C_K.

Average (18.30) over the b arrays. By Cauchy–Schwarz, the difference between this averaged weighted sum and the unweighted sum is at most

$$\left(\sum_p \Phi((p-1)/x)Z_b(p-1) \right)^{1/2} \cdot \left(\sum_p \Phi((p-1)/x)(f_x(p-1) - C_x)^2 \right)^{1/2}.$$

The second factor is $O((x/L)^{1/2})$ by the boundedness of $f_x - C_x$ and the prime number theorem. We conclude that

$$\limsup_{x\to\infty} \frac{L}{x} \left| \sum_p \Phi((p-1)/x)(f_x(p-1) - C_x) \right| \leq C\frac{e^{-h}}{\gamma} + C\sqrt{C_K/b}. \tag{18.34}$$

All constants in the preceding remainder estimates were allowed to depend on the finite b; they have disappeared in this real x limit. First let b be arbitrarily large with h, K fixed. Then let the even integer h be arbitrarily large, choosing its new sufficient fixed K each time. Since $\gamma^{-1} = (4h+3)/c_0$, the surviving bound tends to zero. This proves Theorem 18.1.

In particular, the order of choices is: fix the interior data and c_0; fix h, then γ and the finitely many composite lengths; choose the Type II precision and K; fix finitely many disjoint exponent arrays; take the limit in real x; then remove the arrays by $b \to \infty$ and the sieve error by $h \to \infty$. No number of arrays grows with x in an application of either correlation theorem or the progression lemma.

19 A coefficient criterion and fixed character twists

We need the same marked bound for a rough-factor discrepancy. The following lemma isolates a sufficient condition on a scalar coefficient: cancellation against every character of logarithmic modulus throughout the stated frequency range. Its proof extracts the coefficient dependence from the argument of Sections 14 to 16.

Lemma 19.1 (Coefficient criterion for the marked estimate)**.** *Fix $\delta, C, D_* > 0$ and $q \in (0,1)$. Retain the marked data and the hypotheses on F, H_m, H_n in Theorem 14.1, with fixed comparison constants in $X = H_m H_n \asymp x$. Let (α_m) and (β_n) be scalar sequences indexed by positive integers, supported on W-rough integers in their respective dyads, with α supported on an arbitrary interval $I \subseteq [H_m, 2H_m]$, and with $|\alpha_m|, |\beta_n| \le L^C$. Suppose that for every fixed $A_0, B, A > 0$, every Dirichlet character χ modulo $k \le L^{A_0}$, and every $|t| \le 2XL^B$,*

$$\left| \frac{1}{H_m} \sum_m \alpha_m \chi(m) m^{it} \right| \ll_A L^{-A}, \tag{19.1}$$

uniformly in the permitted interval, character, and frequency. The constant and threshold in this hypothesis may depend on A_0, B, A, the fixed setup, and all further fixed parameters defining the coefficients; these parameters are chosen before x grows.

Then Equation (14.5) *holds when K is sufficiently large in terms of the fixed data, independently of the particular band exponents a_i. Its implicit constant and threshold may depend on all the fixed data, including the a_i.*

Proof. We extract the assertion from the proof of Sections 14 to 16. In its marked expansion, write $mn - 1 = ah$, where a is the product of the selected primes. Invariance gives $F(mn-1) = F(h)$. Except when a selected prime has its square dividing $mn-1$, one also has $\omega(mn-1) - K = \omega(h)$. Replacing the damping by $q^{\omega(h)}$ therefore costs $O_A(XL^{-A})$ for every fixed A. The resulting factor $F(h)q^{\omega(h)}$ has modulus at most one and is discarded in the following Cauchy bound. There are $O_K(L)$ dyads Y for the product of the selected primes. For the contribution S_Y of one dyad, Cauchy's inequality gives

$$|S_Y|^2 \ll \frac{X}{Y} \mathcal{Q}_Y,$$

where the nonnegative expanded square is

$$\mathcal{Q}_Y = \left(\prod_i V_i^{-2} \right) \sum_{\substack{a,b,m,n,m',n' \\ mn-1=ah,\ m'n'-1=bh \text{ for some } h}} \rho(a/Y)\rho(b/Y) \alpha_m \overline{\alpha_{m'}} \beta_n \overline{\beta_{n'}}. \tag{19.2}$$

Here a, b each select one prime from every group, and ρ is the fixed real smooth dyadic cutoff supported in $[1,4]$. Pairs a, b sharing a selected prime cost $O_A(XYL^{-A})$. For the remaining coprime pairs, the common-h condition is equivalent to $t = b - a$, where $t = bmn - am'n'$. These reductions use only coefficient bounds and the invariance of F; see Sections 14 to 16.

For a fixed $A_0 > 0$, let $\mathfrak{M} \subset \mathbb{R}/\mathbb{Z}$ be the union of the arcs

$$\operatorname{dist}_{\mathbb{R}/\mathbb{Z}}(\theta, c/k) \le \frac{2L^{A_0}}{Y}, \qquad 1 \le k \le L^{A_0}, \quad c \bmod k, \quad (c,k) = 1.$$

The selected-product range has $\log Y \gg L^{0.1}$, so these arcs are disjoint for large x. Choose a fixed real $\psi \in C_c^\infty(\mathbb{R})$, equal to one on $[-4,4]$, and define

$$\mathcal{H}_{\mathfrak{M}}(t; a, b) = \psi(t/Y) \int_{\mathfrak{M}} \exp\big(2\pi i \theta (t - b + a)\big) \, d\theta.$$

The major form is the scalar sum

$$\begin{aligned} \mathcal{Q}_Y^{\mathrm{maj}} = \left(\prod_i V_i^{-2} \right) \sum_{a,b,m,n,m',n'} & \rho(a/Y)\rho(b/Y) \\ & \times \alpha_m \overline{\alpha_{m'}} \beta_n \overline{\beta_{n'}} \mathcal{H}_{\mathfrak{M}}(bmn - am'n'; a, b), \end{aligned} \tag{19.3}$$

where a, b now run independently over products of one prime from every group.

The proof of the replacement conclusion in Proposition 15.1 applies to these scalar coefficients. Its operator moment is independent of their values. The passage from that moment to the scalar endpoint pairing uses only W-rough support and norm and supremum bounds with logarithmic exponents depending on C, not on K Sections 14 to 16. This pairing initially has projections onto the good endpoint states of that argument. In removing either projection, only the error from the endpoint outside its good set is majorized absolutely: each whole scalar endpoint factor is replaced by its logarithmic supremum bound, discarding both its coefficient value and its support indicator before the one-edge root replacement. The surviving unprojected pairing retains its exact endpoint vectors. Expanding those vectors and controlling collisions uses only size, rough support, and fixed divisor moments Sections 14 to 16. Thus the replacement conclusion, including restoration of independent a, b, gives

$$|\mathcal{Q}_Y - \mathcal{Q}_Y^{\mathrm{maj}}| \ll_A XYL^{-A} \tag{19.4}$$

for every prescribed fixed accuracy, with the choices specified below.

The proof of the major-term estimate Proposition 16.1 separates characters only at moduli $k \le L^{A_0}$. The group primes and all prime divisors of the rough variables exceed these moduli, so every variable is a unit there. After Mellin separation its retained rectangle, for a fixed B chosen after A_0, has $|s'| \le L^B$ and $|t| \le XL^B$, so both t and $s' - t$ lie in the range of Equation (19.1). Transform decay and the product mean square on translated intervals control the discarded tails using only scalar coefficient bounds and fixed divisor moments. The further Mellin shift enters only the one-group prime polynomial and the bounded factor from the other groups; the sparse-frequency estimate is uniform in this shift Lemma 16.3. On the remaining sparse frequency set, the only further input about α is the supremum in Equation (19.1), in the role of Lemma 16.2. The same hypothesis covers the conjugated endpoint, by conjugating and replacing χ, t by $\overline{\chi}, -t$. The major-term proof therefore gives

$$|\mathcal{Q}_Y^{\mathrm{maj}}| \ll_D XYL^{-D} \tag{19.5}$$

for every fixed $D > 0$.

Choose the desired square accuracy D_1 sufficiently large in terms of D_*, C, before choosing K. For Equation (19.4), choose the moment accuracy E_0 after D_1, C, then the arc exponent A_0, and then K, as in Proposition 15.1. These choices do not depend on the particular a_i. The analytic accuracies in Equation (19.5) are chosen after A_0, as permitted by Equation (19.1) for every fixed accuracy. Taking $A = D = D_1$ gives $\mathcal{Q}_Y \ll XYL^{-D_1}$. Cauchy's inequality yields $|S_Y| \ll XL^{-D_1/2}$. The $O_K(L)$ dyads and all fixed coefficient losses are absorbed by the choice of D_1. The logarithmic exponents used in that choice are independent of K; constants depending on K affect only the implicit constant and threshold. This proves Equation (14.5). □

20 Fixed residue classes

Lemma 20.1 (Finite projection)**.** *Let $D \ge 1$ be an integer and set $\Delta_D(n) = \mathbf{1}_{D|n}$. For every finitely supported complex sequence (a_n) on the positive integers, every real θ, and every $T \ge 0$,*

$$\sum_n \Delta_D(n) a_n n^{it} \mathrm{e}(\theta n) = \frac{1}{D} \sum_{j=0}^{D-1} \sum_n a_n n^{it} \mathrm{e}((\theta + j/D)n), \tag{20.1}$$

$$\int_{-T}^{T} \left| \sum_n \Delta_D(n) a_n n^{it} \mathrm{e}(\theta n) \right|^2 \mathrm{d}t \le \frac{1}{D} \sum_{j=0}^{D-1} \int_{-T}^{T} \left| \sum_n a_n n^{it} \mathrm{e}((\theta + j/D)n) \right|^2 \mathrm{d}t. \tag{20.2}$$

More generally, for any integer c,

$$\mathbf{1}_{n \equiv c \ (D)} = \frac{1}{D} \sum_{j=0}^{D-1} \mathrm{e}(j(n-c)/D),$$

and the same squared-integral inequality holds with $\Delta_D(n)$ *replaced by* $\mathbf{1}_{n\equiv c\ (D)}$. *For the zero residue, if* $(s,D)=1$, *then* $\Delta_D(sn)=\Delta_D(n)$.

Proof. The finite geometric sum $D^{-1}\sum_{j=0}^{D-1}\mathrm{e}(jn/D)$ equals 1 when $D\mid n$ and 0 otherwise. This proves (20.1). The inequality $|D^{-1}\sum_j z_j|^2\le D^{-1}\sum_j|z_j|^2$, followed by integration, proves (20.2). Replacing n by $n-c$ in the geometric sum proves the general residue identity. The extra coefficients $\mathrm{e}(-jc/D)$ have modulus one, so the same inequality proves its integral bound. Finally, coprimality gives $D\mid sn$ if and only if $D\mid n$. □

Proposition 20.2. *For fixed* $H\ge 1$, $(b,H)=1$, $1/2<\alpha<\beta<1$, *and smooth compactly supported* $\Phi\ge 0$,

$$\sum_p \Phi((p-1)/x)\mathbf{1}_{p\equiv b\ (H)}(f_x(p-1)-C_x)=o(x/\log x). \tag{20.3}$$

Proof. Choose $\varepsilon>0$ with $\varepsilon<\alpha$ and $\beta<1-\varepsilon$. Then the interval $(x^\alpha,x^\beta]$ is an admissible one-slot system. We extend the full prime-extraction proof of Theorem 18.1 to the fixed class. Define the periodic two-variable function

$$\mathcal{B}(s,t)=\mathbf{1}_{(t,H)=1}\mathbf{1}_{bs\equiv(b-1)t\ (H)}.$$

Direct simplification gives

$$\mathcal{B}(u,u+1)=\mathbf{1}_{u+1\equiv b\ (H)}, \tag{20.4}$$

$$\mathcal{B}(Rs,Rt)=\mathcal{B}(s,t)\quad((R,H)=1), \tag{20.5}$$

$$\mathcal{B}(w,w+d)=\mathbf{1}_{w\equiv(b-1)d\ (H)}\quad((d,H)=1). \tag{20.6}$$

For the last identity, the congruence reduces to $w\equiv(b-1)d$, which also gives $(w+d,H)=1$.

Write $\mathcal{B}$ as a sum of at most H^2 products of residue indicators, one at each endpoint. Insert these into the bilinear graph transfer. The endpoint norms do not increase, and the same ideal kernel applies to every term. Recombine the finite sum before dividing out the shared labels. Their product R is coprime to H for large x, so (20.5) preserves the joint factor exactly.

In a comparison correlation the shift is the product of the two free label products. Partition each free tuple by its residue modulo H. These bounded tests preserve the endpoint hypotheses. On each part the shift $d \bmod H$ is fixed, and (20.6) leaves a fixed residue condition on the first endpoint. For such a condition,

$$\mathbf{1}_{n\equiv c\ (H)}=\frac{1}{H}\sum_{j=0}^{H-1}\mathrm{e}(j(n-c)/H).$$

The finite projection identity in Lemma 20.1 applies with coefficients $\mathrm{e}(-jc/H)$ of modulus one. The shifted denominators are at most Hk, and (17.24) at a larger fixed logarithmic exponent bounds their full time intervals. The finite splitting cost is absorbed by its arbitrary logarithmic precision. Thus the two-sided estimate holds with $\mathbf{1}_{n+1\equiv b\ (H)}$.

For the one-sided prime replacement, use characters extended by zero on nonunits:

$$\mathbf{1}_{mn\equiv b\ (H)}=\frac{1}{\varphi(H)}\sum_{\lambda \bmod H}\overline{\lambda(b)}\lambda(m)\lambda(n). \tag{20.7}$$

Keep F invariant and place these characters in the two coefficients. If χ has modulus k, then $\lambda\chi$ has modulus $\operatorname{lcm}(H,k)\le Hk$. Since H is fixed, the original coefficient estimate with its modulus exponent increased by one proves (19.1) for $\alpha_m\lambda(m)$, on the same full time interval. The coefficient form of the one-sided estimate therefore applies. Recombining the character sum restores the congruence on the actual product.

For the progression main term, put $n=Qd'u$. For large x, $(Qd',H)=1$ and $(Q,r)=1$. The conditions $n+1\equiv b \pmod H$ and $r\mid n+1$ are impossible if $(r,Hd')>1$. Otherwise u lies in one

class modulo Hr, and its smooth main term is $x\int\Phi/(HQd'r)$. The rounding and forced-absence errors have the same bounds as before. The local sieve density is zero at primes dividing H and $1/s$ elsewhere, so its main factor is $1/\varphi(H)$. The normalized presieve error is $O(e^{-h}/(\varphi(H)\gamma))$.

All discarded masses in the composite extraction are bounded absolutely before the signed expansion. Reallocation preserves the actual product $v=n+1$, hence $v\equiv b\pmod H$. The preceding one-sided and two-sided estimates remove the retained terms. In particular the weighted-prime estimate becomes

$$\limsup_{x\to\infty}\frac{L}{x}\left|\sum_{p\equiv b\ (H)}\Phi((p-1)/x)\frac{\mathcal{W}(p-1)}{m_{\mathcal{W}}}(f_x(p-1)-C_x)\right|\ll\frac{e^{-h}}{\varphi(H)\gamma}.$$

For B disjoint arrays let

$$Z_B(u)=\left(\frac{1}{B}\sum_{a=1}^{B}\frac{\mathcal{W}_a(u)}{m_a}-1\right)^2.$$

The nonnegative variance satisfies $\mathbf{1}_{p\equiv b\ (H)}Z_B(p-1)\le Z_B(p-1)$. Thus (18.33) and Cauchy–Schwarz give

$$\limsup_{x\to\infty}\frac{L}{x}\left|\sum_{p\equiv b\ (H)}\Phi((p-1)/x)(f_x(p-1)-C_x)\right|\le C\frac{e^{-h}}{\varphi(H)\gamma}+C\sqrt{C_K/B}.$$

For each fixed even h choose a sufficient fixed K and then a finite number B of arrays. After the real limit $x\to\infty$, let $B\to\infty$ and finally $h\to\infty$. Since $\gamma^{-1}=(4h+3)/c_0$, both terms vanish. This proves (20.3). □

Proof of Theorem 2.1. Apply Proposition 20.2 with the fixed data in the statement. It gives exactly (2.1), for all sufficiently large real x. The preceding extraction proof fixes the sieve depth, the number and exponents of the bands, and the finite number of arrays before taking $x\to\infty$; the final limits in the array count and sieve depth remove the two errors. Thus no growing modulus or unproved marked-correlation hypothesis remains in this deduction. □

Proof of Theorem 2.2

Fix a cyclotomic field F containing μ_{12} and a finite-order Hecke character η. The proof constructs a single scalar average and evaluates it twice. The first estimate comes from cubic theta reflection and a quadratic large sieve. The second comes from Poisson summation and an Euler product. Their comparison supplies a power saving for an inverse Mellin transform and excludes zeros. We use the established local and global structure theorems for cubic theta representations, including the exceptional representation, its spherical vector, and Whittaker nonvanishing and uniqueness. We also use the quadratic Hecke large-sieve theorem, class field theory, the norm formula for cubic Gauss sums, adelic Poisson summation, local zeta functional equations, and analytic continuation of Hecke L-functions. The required specializations and normalizations of these general results, and the estimates for the present character, are developed below.

21 S-integers, residue symbols, and Gauss sums

The analytic argument will sum over ideals while also using their generators in local fields. We first choose generators for which reciprocity has no supplementary factor. We then identify the two Hecke characters which account for the phases of the cubic and sextic Gauss sums.

Fix a cyclotomic field F containing μ_{12}, and put $[F:\mathbb{Q}] = 2r$. At a finite place v, normalize the absolute value by $|\varpi_v|_v = (\mathrm{N}v)^{-1}$; at a complex place use the square of the ordinary modulus. These are the absolute values in the product formula. Let

$$\psi = \prod_v \psi_v : \mathbb{A}_F/F \longrightarrow \mathbb{C}^\times$$

be the standard trace additive character. We also fix a finite-order Hecke character η. Choose a finite set S containing the complex places, all places above 6, the finite conductor of η, and every place at which $\mathcal{O}_v$ is not self-dual for the pairing $\psi_v(xy)$. The theta normalization in Proposition 23.1 and the local Euler-product estimate in Proposition 26.3 require two additional fixed finite sets of excluded places. Include both sets in S. Their existence is established at those points, independently of the summation variables and of the arithmetic choices below. We may include any further fixed finite set of places. Enlarging S by finitely many primes which generate the ideal class group, we arrange that the ring of S-integers

$$R = \mathcal{O}_S = \{a \in F : v_\mathfrak{p}(a) \geq 0 \text{ for every } \mathfrak{p} \notin S\}$$

is principal. This is the final choice of S, and all arithmetic data below are constructed for it before the coefficient family is defined. All assertions in this section concern this fixed F and S; their auxiliary constants may depend on them.

Write S_f for the finite places in S, $s_f = |S_f|$, and

$$U_S = R^\times, \qquad F_S = \prod_{v\in S} F_v, \qquad |y|_S = \prod_{v\in S} |y_v|_v, \qquad F_S^1 = \{y \in F_S^\times : |y|_S = 1\}.$$

An *exterior ideal* is a nonzero integral ideal of R, equivalently an ideal supported at primes outside S. Its norm is $q_\mathfrak{a} = \mathrm{N}\mathfrak{a} = |R/\mathfrak{a}|$, extended multiplicatively to fractional ideals. Roman letters will denote generators of the corresponding fraktur ideals. For $a \in F^\times$, put $q_a = |a|_S$. Finally, on the additive group F_S set

$$\mathrm{e}(x) = \prod_{v\in S} \psi_v(x_v), \qquad \widehat{f}(y) = \int_{F_S} f(x)\mathrm{e}(-xy)\,dx,$$

where dx is the product of the self-dual local additive measures.

Lemma 21.1 (The S-integer lattice)**.** *The diagonal copy of R is a discrete cocompact subgroup of F_S, is its own annihilator for the pairing* $\mathrm{e}(xy)$*, and has covolume* 1*. For every Bruhat–Schwartz function f on F_S,*

$$\sum_{m\in R} f(m) = \sum_{\nu\in R} \widehat{f}(\nu).$$

If $\mathfrak{a} = aR$ is an exterior fractional ideal, then

$$q_a = q_{\mathfrak{a}}, \qquad (aR)^{\perp} = a^{-1}R, \qquad \sum_{m\in aR} f(m) = q_{\mathfrak{a}}^{-1} \sum_{\nu\in a^{-1}R} \widehat{f}(\nu).$$

Proof. Put $K^S = \prod_{v\notin S} \mathcal{O}_v$, embedded in the adeles with zero S-components. At every place outside S, the defining conductor condition says that $\mathcal{O}_v$ is its own annihilator. Its self-dual measure is consequently 1, and $\widehat{\mathbf{1}_{K^S}} = \mathbf{1}_{K^S}$. Adelic additive duality identifies the annihilator of F in $\mathbb{A}_F$ with F, and the additive quotient $\mathbb{A}_F/F$ has volume 1. Strong approximation identifies

$$F_S/R \simeq \mathbb{A}_F/(F + K^S).$$

The quotient is compact because $\mathbb{A}_F/F$ is compact. To check discreteness, choose a compact neighborhood C of zero in F_S. The intersection $F \cap (C \times K^S)$ is finite because diagonal F is discrete and closed in the adeles. Shrinking C excludes the finitely many nonzero S-components in this intersection, so zero is isolated in R. Taking annihilators in the last quotient gives

$$(F + K^S)^{\perp} = F \cap (F_S \times K^S) = R,$$

with R diagonally embedded, so its annihilator in F_S is indeed R. The covolume of a lattice and that of its annihilator are reciprocal under self-dual measure; hence this covolume is 1.

More directly, apply adelic Poisson summation to $f \otimes \mathbf{1}_{K^S}$. Only the elements of R occur on either side, and the exterior factor is unchanged by Fourier transform. This gives the displayed Poisson identity with no scalar factor. Multiplication by a has additive modulus $|a|_S$, so scaling this identity gives the formula for aR. The product formula gives

$$|a|_S = \prod_{v\notin S} |a|_v^{-1} = \mathrm{N}(aR) = q_{\mathfrak{a}},$$

which finishes the proof. □

We use Hilbert symbols to remove the supplementary factor in sextic reciprocity. Choose the local reciprocity map Art_v in which a uniformizer acts as arithmetic Frobenius on an unramified extension. For $n \in \{2, 3, 6\}$, set

$$(a, b)_{n,v} = \frac{\mathrm{Art}_v(b)(a^{1/n})}{a^{1/n}}.$$

This orders the two arguments oppositely to the convention in classical class field theory. Thus, at a place of residue characteristic prime to n, for a unit u and a uniformizer ϖ,

$$(u, \varpi)_{n,v} \equiv u^{(\mathrm{N}v-1)/n} \pmod{v}.$$

The value is read in $\mu_n \subset F$. On

$$\mathcal{V} = F_S^{\times}/F_S^{\times 6}$$

define $H_S(x, y) = \prod_{v\in S}(x_v, y_v)_{6,v}$, and let E be the image of U_S in $\mathcal{V}$. For $T > 0$, let $\ell_T \in F_S^{\times}$ have finite components 1 and component $T^{1/(2r)}$ at every complex place. Then $|\ell_T|_S = T$.

Lemma 21.2 (Generators with fixed power classes)**.** *The pairing H_S on $\mathcal{V}$ is nondegenerate and alternating. The map $U_S/U_S^6 \to \mathcal{V}$ is injective, its image E satisfies $E^{\perp} = E$, and there is an isotropic subgroup B such that $\mathcal{V} = E \oplus B$.*

Every exterior fractional ideal $\mathfrak{a}$ has a generator a whose class $[a]$ in $\mathcal{V}$ belongs to B. Any two such generators differ by an element of U_S^6. They can be chosen so that

$$a/\ell_{q_{\mathfrak{a}}} \in \mathcal{K}_{\mathrm{bal}}$$

for one fixed compact subset $\mathcal{K}_{\mathrm{bal}} \subset F_S^1$, independent of $\mathfrak{a}$.

Proof. For $\ell = 2, 3$, put $\mathcal{V}_\ell = F_S^\times / F_S^{\times \ell}$. Local Kummer theory and local reciprocity give a nondegenerate Hilbert pairing on this group. At a finite place v,

$$\dim_{\mathbb{F}_\ell} F_v^\times / F_v^{\times \ell} = 2 + \mathbf{1}_{v|\ell}[F_v : \mathbb{Q}_\ell].$$

Indeed the valuation contributes one dimension. When $v \nmid \ell$, the residue units contribute one dimension and the principal units are ℓ-divisible. When $v \mid \ell$, the principal units have a $\mathbb{Z}_\ell$-part of rank $[F_v : \mathbb{Q}_\ell]$ and an ℓ-power root-of-unity part contributing one more dimension. The latter is present because F contains μ_ℓ. This is also the local power-class formula from local class field theory. Complex power-class groups are trivial. All places above ℓ lie in S, so their local degrees sum to $2r$ and

$$\dim_{\mathbb{F}_\ell} \mathcal{V}_\ell = 2s_f + 2r. \tag{21.1}$$

The S-unit theorem gives $U_S \simeq \mu(F) \times \mathbb{Z}^{r+s_f-1}$. Since $\mu(F)$ is cyclic and contains μ_ℓ,

$$\dim_{\mathbb{F}_\ell} U_S / U_S^\ell = r + s_f. \tag{21.2}$$

We next check that the latter group injects into $\mathcal{V}_\ell$. Suppose that $u \in U_S$ is an ℓ-th power in every F_v with $v \in S$. The Kummer extension $F(u^{1/\ell})/F$ is cyclic and split at S. At every exterior prime u is a unit and ℓ is invertible, so this extension is unramified there. It is consequently an everywhere unramified abelian extension in which every S-prime splits. The Hilbert class-field correspondence, together with the quotient description of the S-class group, identifies such extensions with quotients of the class group of R, which is trivial. Hence $u = t^\ell$ for $t \in F$. Its exterior valuations show $t \in U_S$, proving the injection.

For two S-units, every exterior Hilbert symbol is 1, since both arguments are units and the residue characteristic is prime to 6. Hilbert reciprocity therefore makes their S-product 1. The Hilbert pairing is alternating here: $-1 = \zeta_{12}^6$, and the diagonal identity $(x, x)_{n,v} = (x, -1)_{n,v}$ makes its diagonal trivial for $n = 2, 3, 6$. By Equation (21.1) and Equation (21.2), the injected unit image in $\mathcal{V}_\ell$ is an isotropic subspace of half the dimension. Nondegeneracy implies that it is self-orthogonal. Symplectic linear algebra supplies an isotropic complementary subspace for each ℓ. The Chinese remainder isomorphism $\mathcal{V} \simeq \mathcal{V}_2 \oplus \mathcal{V}_3$, under which the two pairings are the third and second powers of H_S, combines these complements into B. It also gives the asserted injection and self-orthogonality for exponent 6.

Let $\mathfrak{a} = a_0 R$. Decompose $[a_0] = eb$ in $E \oplus B$, and choose an S-unit representing e^{-1}. Its product with a_0 is a generator a with $[a] \in B$. If a' is another, then a'/a is an S-unit whose class lies in $E \cap B$. It is therefore locally a sixth power at S, and the injection already proved says $a'/a \in U_S^6$.

Finally $a/\ell_{q_\mathfrak{a}}$ belongs to F_S^1, by Lemma 21.1. To see the relevant compactness explicitly, the norm-one idele class group is compact. The subgroup $F_S^1 \times \prod_{v \notin S} \mathcal{O}_v^\times$ is open in the norm-one ideles and meets diagonal $F^\times$ in U_S. Its quotient by U_S is therefore an open, hence closed, subgroup of that compact idele class group. Its projection onto F_S^1/U_S is surjective, so F_S^1/U_S is compact. Since U_S^6 has finite index in U_S, F_S^1/U_S^6 is compact as well. Choosing a compact set which maps onto this quotient and multiplying a by a suitable element of U_S^6 proves the last assertion. □

Call a generator with class in B *allowed*, and call it *balanced* when it satisfies the compactness condition in Lemma 21.2. We use balanced representatives for estimates involving S-components. In multiplicative identities we may instead use the product of the allowed generators of the prime factors: the product remains allowed and differs from the selected representative by an element of U_S^6.

For an exterior prime $\mathfrak{p}$, the reduction of μ_6 into $(R/\mathfrak{p})^\times$ is injective. Define the sextic residue character by

$$\chi_\mathfrak{p}(x) \equiv x^{(q_\mathfrak{p}-1)/6} \pmod{\mathfrak{p}} \quad (x \not\equiv 0 \pmod{\mathfrak{p}}), \qquad \chi_\mathfrak{p}(0) = 0.$$

For every integer j, the notation $\chi_{\mathfrak{p}}(x)^j$ means the usual character power on units and means 0 on nonunits, also when $j \equiv 0 \pmod 6$. If $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{k_{\mathfrak{p}}}$, put

$$\chi_{\mathfrak{a}}(x) = \prod_{\mathfrak{p}|\mathfrak{a}} \chi_{\mathfrak{p}}(x)^{k_{\mathfrak{p}}}, \qquad \chi_R(x) = 1.$$

The same zero convention applies to powers of $\chi_{\mathfrak{a}}$.

For distinct exterior primes with allowed generators p, p', global reciprocity gives the exact equality

$$\chi_{\mathfrak{p}}(p') = \chi_{\mathfrak{p}'}(p). \tag{21.3}$$

Indeed the only possibly nontrivial exterior symbols in the product formula for $(p, p')_6$ are $(p, p')_{6,\mathfrak{p}} = \chi_{\mathfrak{p}}(p')^{-1}$ and $(p, p')_{6,\mathfrak{p}'} = \chi_{\mathfrak{p}'}(p)$. The S-product is 1 because B is isotropic. If $m \in R \setminus \{0\}$ and $\mathfrak{b} = (m)R$, write $m = \epsilon b$ with b allowed and $\epsilon \in U_S$. Multiplying Equation (21.3) over prime factors gives, for an allowed generator a of $\mathfrak{a}$,

$$\chi_{\mathfrak{a}}(m) = \chi_{\mathfrak{a}}(\epsilon)\chi_{\mathfrak{b}}(a). \tag{21.4}$$

This includes the case $(\mathfrak{a}, \mathfrak{b}) \neq R$: both sides are then zero. For an S-unit ϵ, the same product formula gives

$$\chi_{\mathfrak{a}}(\epsilon) = H_S(\epsilon, a)^{-1}. \tag{21.5}$$

Thus its dependence on $\mathfrak{a}$ is through the finite class $[a]$.

Lemma 21.3 (Kummer and class characters)**.** *For $u \in R \setminus \{0\}$, the ideal function*

$$\chi_\bullet(u)(\mathfrak{a}) = \chi_{\mathfrak{a}}(u)$$

is induced away from S and the prime divisors of $(u)R$ by a finite-order Hecke character. Its values at those prime divisors are set equal to zero. If $\mathfrak{f}_u$ is the conductor of the inducing primitive character, then

$$\mathrm{N}\mathfrak{f}_u \leq C_{F,S}\mathrm{N}\,\mathrm{rad}((u)R) \leq C_{F,S}q_u. \tag{21.6}$$

The same bound, with a changed constant, holds after twisting by the fixed η. The inducing characters associated to u and v agree if and only if $u/v \in F^{\times 6}$.

Every character of the finite group B, evaluated on allowed generators, extends to a finite-order Hecke character unramified outside S.

The following parametrization will index the nonzero Poisson frequencies; its character distinctness and conductor bound will be used in zero detection. Choose a complete set of representatives $\mathcal{U}$ modulo U_S^6 among the nonzero S-integers whose exterior valuations belong to $\{0, 1, \ldots, 5\}$, with $1 \in \mathcal{U}$. For allowed generators r, the map

$$(u, \mathfrak{r}) \longmapsto [ur^6] \quad \textit{from } \mathcal{U} \times \{\textit{exterior ideals}\} \quad \textit{to } (R \setminus \{0\})/U_S^6 \tag{21.7}$$

is a bijection. Among $u \in \mathcal{U}$, only $u = 1$ induces the principal character, and distinct u's induce distinct characters, also after the fixed twist by η.

Proof. Because $\mu_6 \subset F$, Kummer theory identifies $u \in F^\times$ with the character

$$\kappa_u(\sigma) = \frac{\sigma(u^{1/6})}{u^{1/6}} \quad \text{of } \mathrm{Gal}(\overline{F}/F),$$

whose kernel as a function of u is exactly $F^{\times 6}$. Global class-field theory identifies this finite abelian character with a finite-order Hecke character. At an exterior prime $\mathfrak{p} \nmid (u)R$, arithmetic Frobenius satisfies

$$\kappa_u(\mathrm{Frob}_{\mathfrak{p}}) \equiv (u^{1/6})^{q_{\mathfrak{p}}-1} = u^{(q_{\mathfrak{p}}-1)/6} = \chi_{\mathfrak{p}}(u) \pmod{\mathfrak{p}}.$$

Reduction is injective on μ_6, so these are equal as roots of unity. This proves the asserted identification on good primes.

At an exterior place, the principal units are sixth powers by Hensel's lemma. Hence the associated local Hilbert character is trivial on principal units, giving conductor exponent at most one. The tame Hilbert formula shows that the pairing of two units is 1, so if u is a local unit the character is unramified. At a place in S, it depends only on the class of u in the finite group $F_v^\times/F_v^{\times 6}$; the finitely many resulting local conductors have a fixed maximum. This proves Equation (21.6). A fixed twist supported in S only changes that maximum. Equality of two inducing Hecke characters is equality of their Kummer characters, since unramified Frobenius elements determine a finite abelian character. The Kummer kernel gives the stated criterion. Setting extra prime values to zero merely omits Euler factors and does not alter the inducing character.

For a character θ_B of B, extend it to $\mathcal{V} = E \oplus B$ by $\theta_S(eb) = \theta_B(b)$. This is a continuous finite-order character of $F_S^\times$ trivial on U_S. If $\mathfrak{a} = aR$, define $\theta(\mathfrak{a}) = \theta_S(a)$. It is independent of the generator and multiplicative. At every exterior place take the unramified local character sending a uniformizer to $\theta(\mathfrak{p})$, and at S take θ_S^{-1}. Their product is trivial on diagonal $F^\times$ by construction and is the required Hecke character.

To prove Equation (21.7), divide each exterior valuation of a nonzero h by 6. This determines an ideal $\mathfrak{r}$ and the exterior valuations of an S-integer $u_0 = h/r^6$ in $\{0, \dots, 5\}$. Replacing u_0 by its representative in $\mathcal{U}$ changes $u_0 r^6$ only by an element of U_S^6. Conversely, the exterior valuations of $[ur^6]$ determine $\mathfrak{r}$ and those of u; equality of two such classes then says that the two representatives u differ by U_S^6. Changing an allowed r by U_S^6 does not affect its image in the quotient. This proves bijectivity.

If $u \in \mathcal{U}$ induces the principal character, the first part gives $u = t^6$ in F. Its exterior valuations are divisible by 6 and between 0 and 5, hence are all zero; the same is true for the exterior valuations of t. Thus $u \in U_S^6$, giving $u = 1$. If $u, v \in \mathcal{U}$ induce the same character, the differences of their exterior valuations are divisible by 6 and lie in $[-5, 5]$, so vanish. Their quotient is then a sixth power of an S-unit, and $u = v$. The fixed twist by η cancels from an equality of twisted characters. □

We now determine the phases of the Gauss sums. If $\mathfrak{c}$ is a squarefree exterior ideal with allowed generator c, define

$$\gamma_j(\mathfrak{c}) = q_{\mathfrak{c}}^{-1/2} \sum_{x \bmod \mathfrak{c}} \chi_{\mathfrak{c}}(x)^j \mathrm{e}(x/c), \qquad \gamma_j(R) = 1. \tag{21.8}$$

The additive character on $R/\mathfrak{c}$ here is primitive by Lemma 21.1. A change $c \mapsto \epsilon^6 c$, with $\epsilon \in U_S$, changes the additive argument by a sixth power and is absorbed by a change of residue variable; hence the sum is independent of the allowed generator. At an exterior prime, the adelic product formula gives

$$\mathrm{e}(x/p) = \psi_{\mathfrak{p}}(-x/p).$$

The reduction of μ_{12} is injective there, so $q_{\mathfrak{p}} \equiv 1 \pmod{12}$ and $\chi_{\mathfrak{p}}(-1) = 1$. The minus sign can therefore be suppressed in all these prime Gauss sums.

Lemma 21.4 (The Gauss correction characters)**.** *There are a unitary Hecke character Λ, unramified outside S, and a finite-order Hecke character G, unramified outside S, such that for every exterior prime $\mathfrak{p}$,*

$$\gamma_2(\mathfrak{p})^3 = -\Lambda(\mathfrak{p}), \qquad G(\mathfrak{p}) = \overline{\chi_{\mathfrak{p}}(4)}\,\gamma_3(\mathfrak{p}), \qquad \gamma_1(\mathfrak{p})\gamma_2(\mathfrak{p}) = -\Lambda(\mathfrak{p})G(\mathfrak{p}). \tag{21.9}$$

At each complex place the infinity type of Λ is purely angular of exponent 1 or -1, in the ordinary angular coordinate. The values $\gamma_3(\mathfrak{p})$ are signs and depend multiplicatively on the class of an allowed p in B.

For coprime squarefree exterior ideals,

$$\gamma_j(\mathfrak{c}_1\mathfrak{c}_2) = \chi_{\mathfrak{c}_1}(c_2)^j \chi_{\mathfrak{c}_2}(c_1)^j \gamma_j(\mathfrak{c}_1)\gamma_j(\mathfrak{c}_2). \tag{21.10}$$

Proof. First work over a finite field k of size Q, with a nontrivial additive character ψ_k, a character χ of exact order 6, and $\rho = \chi^2$. Write

$$G_j = \sum_{x \in k} \chi(x)^j \psi_k(x), \qquad J(\alpha, \beta) = \sum_{x \in k} \alpha(x)\beta(1-x).$$

For nontrivial characters the usual Gauss inversion and Gauss–Jacobi identities give

$$G_2 G_4 = Q, \qquad G_2^2 = J(\rho, \rho) G_4, \qquad (Q^{-1/2} G_2)^3 = Q^{-1/2} J(\rho, \rho). \tag{21.11}$$

The first equality uses $\rho(-1) = 1$. Changing $\psi_k(x)$ to $\psi_k(ax)$ multiplies G_j by $\overline{\chi(a)}^j$, so the cube in Equation (21.11) is independent of the additive character.

Let $E_0 = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. Every ideal of $\mathbb{Z}[\omega]$ prime to 3 has a unique generator congruent to 1 modulo 3; denote it by $\pi_+(\mathfrak{a})$. This follows from class number one and the bijection of the six units with $(\mathbb{Z}[\omega]/3)^\times$, and is the convention used in the cubic Gauss-sum identities. For a prime $\mathfrak{q} \nmid 6$ with residue size Q, take $\rho(x) \equiv x^{(Q-1)/3} \pmod{\mathfrak{q}}$. Reduction of the Jacobi sum gives

$$J(\rho, \rho) \equiv \sum_{x \in \mathbb{F}_Q} x^{(Q-1)/3} (1-x)^{(Q-1)/3} = 0 \pmod{\mathfrak{q}},$$

since every monomial has degree strictly between 0 and $Q-1$. The Gauss identities also give $|J(\rho, \rho)|^2 = Q$. To determine the unit, put $t = 1 - \omega$; the ideals (t^2) and (3) are equal. Writing $\rho(x) = \omega^{k(x)}$, with $k(x) \in \{0, 1, 2\}$, and expanding modulo t^2 yields

$$J(\rho, \rho) \equiv Q - 2 - t \sum_{x \neq 0,1} \big(k(x) + k(1-x)\big) \equiv Q - 2 - 2(Q-1)t \equiv -1 \pmod{3}.$$

Here each value of ρ occurs $(Q-1)/3$ times on $\mathbb{F}_Q^\times$, and $Q \equiv 1 \pmod 3$. Divisibility by $\mathfrak{q}$, the norm equality, and this congruence show

$$J(\rho, \rho) = -\pi_+(\mathfrak{q}). \tag{21.12}$$

This is also the prime case of the classical cubic Gauss identity $g(n)^3 = \mu(n)\mathrm{N}(n)n$ in the cubic Gauss-sum identities.

Define on ideals of E_0 prime to 3

$$\Lambda_{E_0}(\mathfrak{a}) = \frac{\pi_+(\mathfrak{a})}{|\pi_+(\mathfrak{a})|}.$$

The generators congruent to 1 (mod 3) multiply, so this is a multiplicative ideal character. On a principal ideal generated by $a \equiv 1 \pmod 3$ its value is $a/|a|$; thus it is the unitary Hecke character of finite conductor dividing (3) and angular infinity type of exponent 1, in ideal-character convention. Set $\Lambda = \Lambda_{E_0} \circ \mathrm{N}_{F/E_0}$. If $\mathfrak{p}$ lies above $\mathfrak{q}$ with residue degree f, the power definition of the residue character shows that the cubic character at $\mathfrak{p}$ is the lift of that at $\mathfrak{q}$ by the residue-field norm. The Davenport–Hasse Jacobi lifting formula says

$$J_{\mathbb{F}_{Q^f}}(\rho \circ \mathrm{N}, \rho \circ \mathrm{N}) = (-1)^{f-1} J_{\mathbb{F}_Q}(\rho, \rho)^f;$$

its hypotheses hold because ρ, ρ, and ρ^2 are all nontrivial. By Equation (21.12), the right side is $-\pi_+(\mathfrak{q})^f$. Since $\mathrm{N}_{F/E_0}\mathfrak{p} = \mathfrak{q}^f$, Equation (21.11) proves $\gamma_2(\mathfrak{p})^3 = -\Lambda(\mathfrak{p})$. The local norm at infinity is a product of complex coordinates or their conjugates. Consequently its angular exponent at each complex place of F is 1 or -1, with no radial factor. Pullback by norm is unramified wherever Λ_{E_0} is unramified, so its finite conductor is supported in S.

For the quadratic sum we use one global additive character to relate the local signs. For $a \in F_v^\times$, define the normalized Weil index

$$\mathfrak{w}_v(a) = \frac{\mathrm{WeilIndex}_{\psi_v}(ax^2)}{\mathrm{WeilIndex}_{\psi_v}(x^2)}.$$

It depends only on the square class of a. Weil's product formula and the Hilbert-symbol identity are

$$\prod_v \mathfrak{w}_v(a) = 1 \quad (a \in F^\times), \qquad \mathfrak{w}_v(a)\mathfrak{w}_v(b) = (a,b)_{2,v}\mathfrak{w}_v(ab); \tag{21.13}$$

At an odd conductor-zero place, the local Gauss integral for the Weil index gives $\mathfrak{w}_v(u) = 1$ for a unit u and

$$\mathfrak{w}_\mathfrak{p}(p^{-1}) = q_\mathfrak{p}^{-1/2} \sum_{x \bmod \mathfrak{p}} \psi_\mathfrak{p}(x^2/p) = q_\mathfrak{p}^{-1/2} \sum_{x \bmod \mathfrak{p}} \chi_\mathfrak{p}(x)^3 \psi_\mathfrak{p}(x/p) = \gamma_3(\mathfrak{p}).$$

For the second equality, the number of square roots of x is $1 + \chi_\mathfrak{p}(x)^3$, and the sum of a nontrivial additive character is zero. The last equality uses $\chi_\mathfrak{p}(-1) = 1$.

Put $\mathfrak{w}_S = \prod_{v \in S} \mathfrak{w}_v$. Applying the first identity of Equation (21.13) to p^{-1}, all exterior places other than $\mathfrak{p}$ contribute 1, and hence

$$\gamma_3(\mathfrak{p}) = \mathfrak{w}_S(p^{-1})^{-1}.$$

On B the multiplicative defect in the second identity is

$$\prod_{v \in S} (a,b)_{2,v} = H_S(a,b)^3 = 1.$$

Thus $[a] \mapsto \mathfrak{w}_S(a^{-1})^{-1}$ is a character of B factoring through square classes. It takes values in $\{\pm 1\}$; equivalently Gauss inversion gives $\gamma_3(\mathfrak{p})^2 = \chi_\mathfrak{p}(-1)^3 = 1$. Extend this B-character by Lemma 21.3. Since 4 is an S-unit, $\chi_\bullet(4)$ is also a finite-order Hecke character unramified outside S. Their product, with $\chi_\bullet(4)$ conjugated, defines G and gives its displayed prime value.

It remains to check the sextic duplication factor. In the finite-field notation above put $\phi = \chi^3$. Completing the square and counting square roots gives

$$J(\chi,\chi) = \overline{\chi(4)} \sum_{t \in k} \chi(1 - t^2) = \overline{\chi(4)} J(\phi,\chi).$$

In the second equality the part without ϕ vanishes, since χ is nontrivial. Applying the Gauss–Jacobi identity to both sides yields

$$G_1 G_4 = \overline{\chi(4)} G_2 G_3.$$

Together with $G_2 G_4 = Q$, this gives

$$\gamma_1(\mathfrak{p})\gamma_2(\mathfrak{p}) = \overline{\chi_\mathfrak{p}(4)}\, \gamma_3(\mathfrak{p})\gamma_2(\mathfrak{p})^3 = -\Lambda(\mathfrak{p})G(\mathfrak{p}).$$

Finally use the allowed product $c_1 c_2$ to evaluate the sum at $\mathfrak{c}_1\mathfrak{c}_2$. Chinese remaindering writes its additive character as the product of the characters modulo $\mathfrak{c}_i$, with their arguments multiplied respectively by c_2^{-1} and c_1^{-1}. Changing the two residue variables contributes $\chi_{\mathfrak{c}_1}(c_2)^j \chi_{\mathfrak{c}_2}(c_1)^j$. This proves Equation (21.10), including $j \equiv 0 \pmod 6$ with the specified zero convention. □

For later use set

$$\vartheta = \overline{\Lambda}\,\overline{G}\,\eta.$$

Its finite conductor is supported in S. If ϑ_v denote its idele-class components, define

$$L_S(y) = \prod_{v \in S} \vartheta_v(y_v)^{-1}.$$

The global character is trivial on $F^\times$, so for every exterior ideal with allowed generator a,

$$\vartheta(\mathfrak{a}) = L_S(a), \qquad L_S(\epsilon) = 1 \quad (\epsilon \in U_S). \tag{21.14}$$

This is a smooth unitary character of $F_S^\times$. Finite-order characters are trivial on the connected group $\mathbb{C}^\times$; hence the infinity components contributed by G and η are trivial. Those of L_S are therefore purely angular of exponent 1 or -1 at each complex place.

22 The coefficient family and analytic comparison

The two analytic evaluations use the same family of coefficient sums. We define that family and state the second moment proved by the theta and reflection arguments, then explain how the moment enters the zero-free argument.

Retain the field F, the final set S, the ring $R = \mathcal{O}_S$, and the generator convention fixed in Section 21. Unless explicitly stated otherwise, all ideals in this section are integral exterior ideals. A roman letter representing an ideal is always an allowed B-generator; it need not be balanced unless this is stated. In particular, products of allowed prime generators are permitted. The row variable m below ranges over all of R.

For $\operatorname{Re} s > 1$, write $\zeta_F^S(s) = \sum_{\mathfrak{a}} q_{\mathfrak{a}}^{-s}$ for the Dedekind zeta function with its factors at S omitted, and set

$$L_F^S(s,\eta) = L^S(s,\eta) := \prod_{\mathfrak{p}\notin S} (1 - \eta(\mathfrak{p}) q_{\mathfrak{p}}^{-s})^{-1}.$$

The latter notation also denotes the usual meromorphic continuation of this incomplete Hecke L-function; a principal pole is permitted. All primes dividing the conductor of η already lie in S, so there are no further omitted exterior factors for this target character. For $t > 0$ put

$$P_G(t) = \frac{1}{2\sqrt{\pi}} \exp\left(-\frac{(\log t)^2}{4}\right), \qquad \int_0^\infty P_G(t) t^{\xi} \, \frac{\mathrm{d}t}{t} = e^{\xi^2} \quad (\xi \in \mathbb{C}). \tag{22.1}$$

The transform identity follows by putting $v = \log t$ and completing the square, first for real ξ and then by entire continuation. Write $\vartheta = \overline{\Lambda}\,\overline{G}\,\eta$. For $m \in R$ and $Z > 0$ define

$$B_m(Z) = \sum_{\substack{\mathfrak{c} \text{ squarefree} \\ \mathfrak{n}}} \frac{\gamma_2(\mathfrak{c})}{q_{\mathfrak{c}}^{1/2} q_{\mathfrak{n}}} \vartheta(\mathfrak{A}) \chi_{\mathfrak{A}}(m) P_G\left(\frac{q_{\mathfrak{A}}}{Z}\right), \qquad \mathfrak{A} = \mathfrak{c}\mathfrak{n}^3, \quad A = cn^3. \tag{22.2}$$

Here c, n, A are allowed generators, and $\mathfrak{n}$ is unrestricted, so it may share primes with $\mathfrak{c}$. Every residue-character power has the zero extension on nonunits fixed in Section 21, including powers with exponent zero modulo six. The sum is independent of the permitted generator choices. It converges absolutely: $|\gamma_2(\mathfrak{c})| = 1$ for squarefree $\mathfrak{c}$, and for any $\sigma > 1/2$ the bound $P_G(t) \ll_\sigma t^{-\sigma}$ reduces the absolute sum, after a factor $O_\sigma(Z^\sigma)$, to

$$\sum_{\mathfrak{c},\mathfrak{n}} q_{\mathfrak{c}}^{-1/2-\sigma} q_{\mathfrak{n}}^{-1-3\sigma} \le \zeta_F^S(\sigma + 1/2) \zeta_F^S(3\sigma + 1) < \infty. \tag{22.3}$$

At $m = 0$, the mask kills every term with $\mathfrak{A} \neq R$, so $B_0(Z) = P_G(1/Z)$. The factor $\gamma_2(\mathfrak{c})/(q_{\mathfrak{c}}^{1/2} q_{\mathfrak{n}})$ will be identified in Section 23 as the exterior coefficient supplied by the normalized local theta formula.

The comparison uses rows at scale XY, where

$$\begin{aligned} a = \frac{9}{10}, \qquad b = \frac{133}{1000}, \qquad M_0 = a + b = \frac{1033}{1000}, \\ h_0 = 1 - a = \frac{1}{10}, \qquad X = Z^a, \quad Y = Z^b. \end{aligned} \tag{22.4}$$

Recall the norm section ℓ_T fixed in Section 21, for which $|\ell_T|_S = T$. If $\mathscr{K} \subset F_S^\times$ is compact, set

$$\mathscr{B}_Z(\mathscr{K}) = R \cap \ell_{Z^{M_0}} \mathscr{K}.$$

Thus every local absolute value of $\ell_{Z^{M_0}}^{-1} m$ is bounded above and away from zero on this set. These are the compact balanced annular row sets used below. For $m \neq 0$ decompose its exterior ideal uniquely as

$$(m)_S = \mathfrak{h}(m)\mathfrak{d}(m), \qquad \mathfrak{h}(m) = \prod_{v_{\mathfrak{p}}(m) \ge 2} \mathfrak{p}^{v_{\mathfrak{p}}(m)}, \qquad \mathfrak{d}(m) = \prod_{v_{\mathfrak{p}}(m) = 1} \mathfrak{p}.$$

The ideal $\mathfrak{d}(m)$ is squarefree and coprime to $\mathfrak{h}(m)$; every prime exponent of $\mathfrak{h}(m)$ is at least two. We call such an ideal powerful.

The theta and reflection arguments in Sections 23 and 25 prove the following estimate.

Proposition 22.1 (Reflected second moment)**.** *Let F be a fixed cyclotomic field containing μ_{12}, let S and the arithmetic data be as above, and fix a compact subset $\mathscr{K} \subset F_S^\times$. For every $\epsilon > 0$, every $Z \geq 2$, and all dyadic lengths $P, D \geq 1$,*

$$\sum_{\substack{m \in \mathscr{B}_Z(\mathscr{K}) \\ P \leq q_{\mathfrak{h}(m)} < 2P \\ D \leq q_{\mathfrak{d}(m)} < 2D}} |B_m(Z)|^2 \ll_{F,S,\eta,\mathscr{K},\epsilon} Z^\epsilon P^{1/2} \left(D + \frac{D^2 P}{Z} \right). \tag{22.5}$$

In particular,

$$\sum_{m \in \mathscr{B}_Z(\mathscr{K})} |B_m(Z)|^2 \ll_{F,S,\eta,\mathscr{K},\epsilon} Z^{2M_0 - 1 + \epsilon}. \tag{22.6}$$

The estimate is uniform if $\mathscr{K}$ is multiplied by an element of any fixed compact subset of the norm-one group of $F_S^\times$.

Here is the analytic comparison proved in Sections 26 and 27. The first of those sections constructs a scalar $I_Z = I(Z^a, Z^b, Z)$ by pairing $B_m(Z)$, on a fixed compact annular row set of scale XY, with a normalized sum of $\chi_{\mathfrak{s}}(d)\mathrm{e}(-md/s)$, where $q_{\mathfrak{s}} \asymp Y$ and $d \in (R/\mathfrak{s})^\times$. It specifies the weights and the average over F_S^1/U_S^6. The whole additive factor has squared row mass $O(Y^{-1})$, uniformly in that average. Hence Equation (22.6) and Cauchy–Schwarz give, for every $\epsilon > 0$,

$$I_Z \ll Z^{(2M_0 - 1 - b)/2 + \epsilon} = Z^{933/2000 + \epsilon}.$$

The constants here and below may depend on the fixed F, S, η and the fixed weights, as well as on any displayed loss.

Poisson summation gives a second evaluation. Its nonzero frequencies are grouped by $u \in \mathcal{U}$ through Equation (21.7). Only the principal class $u = 1$, the class of sixth powers, contributes both numerator poles in the auxiliary Mellin variables: $w = 1$ and $z = 1/6$. The Euler calculation constructs a specific principal correction $\mathcal{H}_\eta(s)$ by specializing the correction for $u = 1$ at $(w, z) = (1, 1/6)$. It proves that $\mathcal{H}_\eta$ is holomorphic and nowhere zero on an open neighborhood of the closed half-plane $\operatorname{Re} s \geq .998$, and is bounded on that closed half-plane uniformly in height. The double residue equals $c_0 f_\eta(Z)$, where $c_0 \neq 0$ and

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{s - 8/15} e^{(s-5/6)^2} \frac{\mathcal{H}_\eta(s)}{L^S(s, \eta)}\, ds \qquad (Z > 0).$$

The exponent left by the two residues is $a/2 + s - 1 + h_0/6 = s - 8/15$, whose value at $s = 1$ is $7/15$.

The zero-detection argument proves, for sufficiently large Z,

$$I_Z = c_0 f_\eta(Z) + O(Z^{7/15 - .0004}).$$

Since $933/2000 = 7/15 - 1/6000$, this identity and the preceding bound, with the stated losses, give $f_\eta(Z) \ll Z^{7/15 - 1/20000}$. Moving the original s-line in the inverse Mellin integral to the right gives $f_\eta(Z) = O_A(Z^A)$ for every $A > 0$ and $0 < Z \leq 1$. Consequently the shifted Mellin transform $\int_0^\infty f_\eta(Z) Z^{-s + 8/15}\, dZ/Z$ is holomorphic for $\operatorname{Re} s > 1 - 1/20000$. Fourier inversion on $\operatorname{Re} s = 2$, followed by meromorphic continuation, identifies it throughout that half-plane with

$$e^{(s-5/6)^2} \frac{\mathcal{H}_\eta(s)}{L^S(s, \eta)}.$$

Its numerator is holomorphic and nowhere zero there, so $L^S(s, \eta)$ has no zero there; a principal pole only gives a zero of the reciprocal. The numerical saving $1/20000$ is absolute; the constants and thresholds may depend on F, S, η. The later sections supply the residue calculation and the contour and Fourier justifications.

23 The normalized cubic theta input and its radial profiles

To prove the reflected moment, we first realize the exterior coefficient $\gamma_2(\mathfrak{c})/(q_{\mathfrak{c}}^{1/2}q_{\mathfrak{n}})$ in the family B_m. The cubic theta representation used here has one-dimensional local Whittaker spaces. Its selected central datum, together with normalized induction, gives the factor $q_{\mathfrak{p}}^{-1}$ for a triple valuation step. Compatible rational, torus, root, and compact sections fix the unit factor at valuation e as $\chi_{\mathfrak{p}}(u)^{2e}$. We state this normalized input first and then identify its exterior coefficient. The verification of the Kazhdan–Patterson specialization is in Section 24.

Throughout this section F is the fixed cyclotomic field from the preceding section, and $\mathbb{A}_F$ is its adele ring. In particular, $\mu_{12} \subset F$. We use the standard trace character $\psi : F\backslash\mathbb{A}_F \longrightarrow \mathbb{C}^\times$, and choose the self-dual additive Haar measures, for which $F\backslash\mathbb{A}_F$ has volume one. Write

$$n(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \qquad \bar{n}(x) = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}, \qquad a(y) = \begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The same notation will denote the specified lifts below. A Whittaker functional with character ψ_v means a linear functional λ satisfying $\lambda(\pi(n(x))\varphi) = \psi_v(x)\lambda(\varphi)$.

23.1 The automorphic input and its exterior coefficients

We use the determinant-modified cover of Kazhdan and Patterson with parameter $c = 2$, with the ordinary torus section and the canonical root splittings. The embedding $\iota : \mu_3(F) \hookrightarrow \mathbb{C}^\times$ is oriented as in the exterior cocycle formula below. If $a_v(y)$ denotes the chosen torus lift, define the effective scalar $\mathcal{C}_v(x, y)$ by

$$\pi(a_v(x))\pi(a_v(y)) = \mathcal{C}_v(x, y)\pi(a_v(xy))$$

in a genuine representation π. Thus this notation includes the chosen embedding of the central μ_3.

Proposition 23.1 (Cubic theta input)**.** *For the fixed field F, there is a threefold central cover*

$$1 \longrightarrow \mu_3 \longrightarrow \widetilde{\mathrm{PGL}}_2(\mathbb{A}_F) \longrightarrow \mathrm{PGL}_2(\mathbb{A}_F) \longrightarrow 1$$

which splits over $\mathrm{PGL}_2(F)$, and an irreducible genuine automorphic representation $\Theta = \bigotimes'_v \Theta_v$ on this cover, with the following properties.

(i) Global representation and Fourier expansion. *The cover is the quotient of the $c = 2$ cover $\widetilde{\mathrm{GL}}_2^{(2)}$ by the ordinary split scalar subgroup. The inducing character of Θ_v, on the center of the covering torus before this quotient, is*

$$\omega_v\Big(\zeta\, s_v(\mathrm{diag}(x^3, y^3))s_v(zI)\Big) = \iota(\zeta)\left|x^3/y^3\right|_v^{1/6}. \tag{23.1}$$

Here s_v is the ordinary torus section. The normalized inducing norm exponents are $1/6, -1/6$. The local exceptional representation is the irreducible quotient for these exponents and the irreducible subrepresentation for their Weyl conjugates. Every Θ_v has a one-dimensional Whittaker space. The global functional

$$\lambda(\varphi) = \int_{F\backslash\mathbb{A}_F} \theta_\varphi(n(x))\overline{\psi(x)}\,\mathrm{d}x$$

on the automorphic realization $\varphi \mapsto \theta_\varphi$ is nonzero. The representation is spherical at all but finitely many finite places. There is a preliminary pure tensor φ° on which λ is nonzero. Its finitely many nondistinguished finite places lie in the final fixed S, along with the archimedean places, the places over 6, and the places where ψ_v has nonzero conductor.

For a pure tensor $\varphi = \varphi_S \otimes \bigotimes_{v\notin S} \varphi_v$, with the distinguished spherical vector at almost every $v \notin S$, the global Whittaker function factors as

$$\mathcal{W}_\varphi(g) := \lambda(\pi(g)\varphi) = \mathcal{W}_{S,\varphi_S}(g_S) \prod_{v\notin S} W_{v,\varphi_v}(g_v). \tag{23.2}$$

Each exterior spherical functional is normalized at one, so only finitely many factors in this product differ from one for adelic g. The functional at S factors at its complex places; at its finite places it may be retained as a finite sum of products of local Whittaker functionals. For a smooth automorphic vector,

$$\theta_\varphi(g) = \theta_{\varphi,N}(g) + \sum_{\nu\in F^\times} \mathcal{W}_\varphi(a(\nu)g), \qquad \theta_{\varphi,N}(g) = \int_{F\backslash\mathbb{A}_F} \theta_\varphi(n(x)g)\,\mathrm{d}x. \tag{23.3}$$

This expansion converges absolutely and locally uniformly on compact sets of g.

(ii) Exact lifts and exterior local values. *On the quotient the chosen lifts satisfy*

$$\begin{gathered} n(x)a(y) = a(y)n(x/y), \qquad wa(y)w^{-1} = a(y^{-1}), \\ wn(H) = n(H^{-1})a(-H^{-2})\bar{n}(H^{-1}). \end{gathered} \tag{23.4}$$

In the last identity $H \neq 0$. At an exterior prime, when H^{-1} is integral the last factor is compact, and a compact diagonal may absorb the minus sign.

For every exterior prime $\mathfrak{p}$, choose a uniformizer ϖ, put $q_\mathfrak{p} = \mathrm{N}\mathfrak{p}$, and use the compact splitting on $\mathrm{PGL}_2(\mathcal{O}_\mathfrak{p})$. The orientation of ι, equivalently that of the Hilbert-symbol convention, can be fixed so that the scalar by which the torus cocycle acts in a genuine representation is

$$\mathcal{C}_\mathfrak{p}(\varpi, u) = \chi_\mathfrak{p}(u)^{-2} \qquad (u \in \mathcal{O}_\mathfrak{p}^\times). \tag{23.5}$$

The compact splitting agrees with the chosen sections on unit diagonals, Weyl elements, and integral upper and lower root elements. Define

$$g_{2,\mathfrak{p}} = q_\mathfrak{p}^{-1/2} \sum_{x \bmod \mathfrak{p}} \chi_\mathfrak{p}(x)^2 \psi_\mathfrak{p}(x/\varpi).$$

Then $|g_{2,\mathfrak{p}}| = 1$. If $W_\mathfrak{p}$ is the spherical Whittaker function normalized by $W_\mathfrak{p}(1) = 1$, then, for $u \in \mathcal{O}_\mathfrak{p}^\times$,

$$W_\mathfrak{p}(a(\varpi^e u)) = \chi_\mathfrak{p}(u)^{2e} \begin{cases} 0, & e < 0, \\ q_\mathfrak{p}^{-k}, & e = 3k, \quad k \geq 0, \\ q_\mathfrak{p}^{-k-1/2} g_{2,\mathfrak{p}}, & e = 3k+1, \quad k \geq 0, \\ 0, & e = 3k+2, \quad k \geq 0. \end{cases} \tag{23.6}$$

In particular a shift by three nonnegative valuations multiplies the value by $q_\mathfrak{p}^{-1}$.

(iii) Complex local behavior. *At a complex place, Θ_v is the ordinary spherical $\mathrm{PGL}_2(\mathbb{C})$ principal series with normalized characters $(|\cdot|_v^{1/6}, |\cdot|_v^{-1/6})$, where $|y|_v$ is the square of the ordinary modulus. Its spherical Whittaker function is a nonzero constant times*

$$|y|_v^{1/2} K_{1/3}(c_v|y|_v^{1/2}), \qquad c_v > 0. \tag{23.7}$$

The constant term of the automorphic theta function belongs to the Weyl-conjugate torus induction. In particular its dependence on a complex diagonal $a(y)$ is radial, also when the vector on its right has been acted on by a fixed compact element.

In (23.2), the condition $W_{\mathfrak{p}}(1) = 1$ fixes every exterior spherical scalar; the remaining normalization belongs to the S-functional. Section 24 derives this product from the nonzero global functional and local uniqueness, including the correction to the original global formula. The noncompact norm integrations below require the separate absolute bounds in Lemma 23.4.

Corollary 23.2 (The exterior theta coefficient)**.** *At every exterior prime use an allowed prime generator as the uniformizer in Proposition 23.1. Let $A \in R \setminus \{0\}$ be an allowed generator of an integral exterior ideal $\mathfrak{A}$. Then*

$$\prod_{\mathfrak{p}\notin S} W_{\mathfrak{p}}(a(A)) = \begin{cases} \dfrac{\gamma_2(\mathfrak{c})}{q_{\mathfrak{c}}^{1/2} q_{\mathfrak{n}}}, & \mathfrak{A} = \mathfrak{c}\mathfrak{n}^3, \quad \mathfrak{c} \text{ squarefree}, \\ 0, & \text{otherwise}. \end{cases} \tag{23.8}$$

In the first case $\mathfrak{c}, \mathfrak{n}$ are uniquely determined; they need not be coprime.

Proof. The local support in (23.6) permits precisely the valuations $3k$ and $3k+1$. They give the unique decomposition $\mathfrak{A} = \mathfrak{c}\mathfrak{n}^3$, with $\mathfrak{c}$ squarefree, and magnitudes $q_{\mathfrak{p}}^{-k}$ and $q_{\mathfrak{p}}^{-k-1/2}$, respectively. Use allowed product generators $A = cn^3$; any other allowed A differs by an element of U_S^6, which has no effect on the local values. At a prime $\mathfrak{p} \mid \mathfrak{c}$, the local unit factor is $\chi_{\mathfrak{p}}(c/p)^2$. The unit from n^3 has exponent divisible by six in the symbol, including when $\mathfrak{p} \mid \mathfrak{n}$, and disappears. The normalized prime sum $g_{2,\mathfrak{p}}$ equals $\gamma_2(\mathfrak{p})$: the local additive sign from Equation (21.8) has no effect because $\chi_{\mathfrak{p}}(-1) = 1$. The product of the unit factors and these prime sums is $\gamma_2(\mathfrak{c})$ by Equation (21.10). Multiplying the magnitudes gives (23.8). □

Thus theta supplies the exterior factor in B_m. The ideal character will enter through an S-average, and finite upper-unipotent translations will insert the moving sextic twist. The chosen S-Whittaker function first supplies a radial profile; an exact radial average will replace it by P_G. We prepare those S-components next.

The arithmetic power classes also describe the torus factor at S. Put $\mathcal{C}_S = \prod_{v\in S} \mathcal{C}_v$. At an exterior prime, Equation (23.5) and the tame Hilbert-symbol formula give $\mathcal{C}_{\mathfrak{p}}(x, y) = (x, y)_{6,\mathfrak{p}}^2$ for all local x, y. For rational a, b, the product formulas for the chosen torus lifts and for the Hilbert symbol therefore imply

$$\mathcal{C}_S(a, b) = \prod_{\mathfrak{p}\notin S} \mathcal{C}_{\mathfrak{p}}(a, b)^{-1} = H_S(a, b)^2.$$

Both sides are continuous on $F_S^\times \times F_S^\times$, so weak approximation gives

$$\mathcal{C}_S(x, y) = H_S(x, y)^2 \qquad (x, y \in F_S^\times). \tag{23.9}$$

In particular this factor depends only on the classes in the fixed finite group $\mathcal{V} = F_S^\times / F_S^{\times 6}$. The local sections here are the same ones used by the rational splitting.

23.2 Bounds and a prescribed finite Whittaker function

For $v \mid \infty$, define the derivative in the logarithm of the norm by

$$\mathcal{D}_v f(y) = \frac{\mathrm{d}}{\mathrm{d}t} f(e^{t/2} y)\Big|_{t=0}.$$

The factor $1/2$ makes $|e^{t/2} y|_v = e^t |y|_v$.

We give both the Bessel calculation and bounds for the fixed rotations of the vectors that we will use. Choose complex coordinates in which $\psi_v(z) = e^{ic_v \operatorname{Re} z}$, with $c_v > 0$; a harmless rotation and a change of sign give this form. Put $\varrho = |y|_v^{1/2}$. In the dominant normalized principal series set $\nu_0 = 1/2 + 1/6 = 2/3$. Its spherical section, restricted to $wn(x)a(y)$, is a constant times

$$\varrho^{2\nu_0} (\varrho^2 + |x|_{\mathrm{ord}}^2)^{-2\nu_0},$$

where $|x|_{\mathrm{ord}}$ is the ordinary complex modulus. The Jacquet integral is absolutely convergent because $2\nu_0 > 1$. Using the Laplace representation of the negative power and the two-dimensional Gaussian Fourier integral gives

$$\begin{aligned}\varrho^{2\nu_0}\int_{\mathbb{R}^2}\frac{e^{-ic_v x_1}}{(\varrho^2+x_1^2+x_2^2)^{2\nu_0}}\,\mathrm{d}x_1\,\mathrm{d}x_2 &= \frac{\pi\varrho^{2\nu_0}}{\Gamma(2\nu_0)}\int_0^\infty t^{2\nu_0-2}e^{-t\varrho^2-c_v^2/(4t)}\,\mathrm{d}t\\ &= c_v'\varrho K_{2\nu_0-1}(c_v\varrho) = c_v'\varrho K_{1/3}(c_v\varrho), \qquad c_v'\neq 0.\end{aligned}$$

The Laplace and Gaussian integrals may be interchanged absolutely: after integrating the Gaussian without its phase the integrand at zero is $O(t^{2\nu_0-2})$, with exponent greater than -1. This proves (23.7).

For an upper root translate, Whittaker equivariance gives

$$W_{\pi(n(z))\varphi}(a(y)) = \psi_v(yz)W_\varphi(a(y)). \tag{23.10}$$

Suitable complex linear combinations of the two real Lie derivatives in z therefore multiply the spherical function by y and by $\bar{y}$, respectively. Write $\omega_m(y) = (y/|y|_v^{1/2})^m$ for $m \in \{1,-1\}$. After multiplying each derivative vector by a nonzero constant, its Whittaker function is

$$W_{v,m}(a(y)) = |y|_v\,\omega_m(y)K_{1/3}(c_v|y|_v^{1/2}). \tag{23.11}$$

Here are uniform bounds for every vector in a fixed finite list consisting of these derivative vectors and their fixed compact rotations. A rotation of a first Lie derivative of the spherical vector is another first Lie derivative, since the spherical vector is compact invariant and $\pi(k)\,\mathrm{d}\pi(X) = \mathrm{d}\pi(\mathrm{Ad}(k)X)\pi(k)$. For a fixed real Lie algebra element X, differentiate the spherical section using

$$f^\circ(g) = \left(\frac{|\det g|_{\mathrm{ord}}}{|g_{21}|_{\mathrm{ord}}^2+|g_{22}|_{\mathrm{ord}}^2}\right)^{2\nu_0}.$$

On $wn(x)a(y)$, its first Lie derivative and any number of $\mathcal{D}_v$-derivatives are the preceding spherical integrand times a finite sum of bounded homogeneous rational functions of (y,x). More precisely, for every norm derivative order b and every x_1-derivative order h, the resulting derivative is bounded by

$$C_{b,h}\varrho^{2\nu_0}(\varrho^2+|x|_{\mathrm{ord}}^2)^{-2\nu_0-h/2}.$$

This follows by differentiating the displayed quotient: each derivative adds a polynomial numerator of the same degree as its denominator, and an x_1-derivative lowers the homogeneous degree by one. The integral of the bound with $h=0$ is $O(\varrho^{2-2\nu_0}) = O(\varrho^{2/3})$. For $\varrho \geq 1$, integrating by parts h times against $e^{-ic_v x_1}$ instead gives $O_{b,h}(\varrho^{2/3-h})$. These integrable derivative bounds justify the integrations by parts by cutoff and limit. Consequently, for every integer $j \geq 0$ and every $A > 0$, the fixed list satisfies

$$|\mathcal{D}_v^j W(a(y))| \leq C_{A,j}\min(|y|_v^{1/3}, |y|_v^{-A}). \tag{23.12}$$

At a finite place the corresponding elementary bounds hold for every fixed smooth vector. To be explicit, let $W_\varphi(y) = \lambda_v(\pi_v(a(y))\varphi)$. There is an additive open subgroup $J \subset F_v$ for which $n(J)$ fixes φ. The identity

$$W_\varphi(y) = \psi_v(yx)W_\varphi(y) \qquad (x \in J)$$

forces $W_\varphi(y) = 0$ when $|y|_v$ is sufficiently large. For the other end, put $Q_v = \#(\mathcal{O}_v/\mathfrak{p}_v)$ and $z_v = a(\varpi_v^3)$. This element is central in the covering torus. By the two-cell Jacquet calculation for the principal series, the two constituents of the unnormalized ordinary Jacquet module of the principal series have z_v-eigenvalues of absolute values

$$Q_v^{-1} \quad \text{and} \quad Q_v^{-2}.$$

These are the exponents $1/2-1/6$ and $1/2+1/6$ on a triple valuation step. Exactness of the ordinary Jacquet functor gives the same possible eigenvalues on the exceptional subquotient. Hence

a polynomial all of whose roots have those absolute values annihilates the action of z_v on this Jacquet module.

For completeness, this gives a recurrence on the Whittaker values themselves. If P is such an annihilating polynomial, then

$$P(\pi_v(z_v))\varphi = \sum_{i=1}^{h} \big(\pi_v(n(x_i)) - 1\big)\varphi_i$$

for finitely many x_i, φ_i. Apply $\lambda_v \pi_v(a(y))$. For all sufficiently small $|y|_v$ the right side is zero because every $\psi_v(yx_i)$ is one. Cube cocycles are trivial, so the left side is the recurrence $P(E)W_\varphi(y)$, where $EW_\varphi(y) = W_\varphi(y\varpi_v^3)$. On each valuation progression its solutions are sums of a polynomial in the valuation times the powers of the roots of P. The unit variable ranges over only finitely many classes after restricting to a sufficiently small unit subgroup which fixes the vector and consists of cubes. It follows that, for every fixed $\delta < 1/3$,

$$W_\varphi(y) = 0 \quad (|y|_v > C_\varphi), \qquad |W_\varphi(y)| \le C_{\varphi,\delta}|y|_v^{\delta} \quad (|y|_v \le 1). \tag{23.13}$$

All constants in this paragraph may depend on the place, the vector, and the functional. For a fixed finite list one may choose a common positive δ. The local twisted Jacquet quotients of Θ_v are finite dimensional by the quotient argument in Section 24.2. Thus the same conclusions hold term by term for the finite sum of products at S in Proposition 23.1.

Lemma 23.3 (Finite torus cutoff)**.** *Let S_f be a finite set of nonarchimedean places and let $\lambda_f \neq 0$ be a Whittaker functional with character $\prod_{v\in S_f}\psi_v$ on a smooth representation of the product of the local covers. For a vector φ, put*

$$W_\varphi(y) = \lambda_f\Big(\pi\big((a(y_v))_{v\in S_f}\big)\varphi\Big), \qquad y \in \prod_{v\in S_f} F_v^{\times}.$$

Given any open subgroup H_f of the product of the unit groups, there are a compact open product subgroup $U \subset H_f$ and a smooth vector φ_U for which

$$W_{\varphi_U}(y) = \mathbf{1}_U(y) \qquad \left(y \in \prod_{v\in S_f} F_v^{\times}\right).$$

Proof. Choose a vector with $\lambda_f(\varphi) \neq 0$. The map $y \mapsto W_\varphi(y)$ is locally constant near one. Choose a product of sufficiently small groups $1 + \mathfrak{p}_v^{m_v}$, contained in H_f, on which it is the constant $\lambda_f(\varphi)$; call that product U. As a subset of the additive product $\prod F_v$, U is compact and open. By additive Fourier inversion choose a Schwartz–Bruhat function ϕ with

$$\int_{\prod F_v} \phi(t) \prod_v \psi_v(y_v t_v)\, \mathrm{d}t = \mathbf{1}_U(y).$$

The vector $\varphi' = \int \phi(t)\pi(n(t))\varphi \, \mathrm{d}t$ is a well-defined smooth vector: its integrand is locally constant on the compact support of ϕ, so the integral is a finite linear combination of vectors. The split root relation gives

$$W_{\varphi'}(y) = W_\varphi(y) \int \phi(t) \prod_v \psi_v(y_v t_v)\, \mathrm{d}t = W_\varphi(y)\mathbf{1}_U(y).$$

Divide φ' by $\lambda_f(\varphi)$. □

Here is the choice at S that will be used later. Write $S_f = S \setminus S_\infty$. Let $\mathscr{L}$ be any fixed finite family of smooth unitary characters $\mathcal{L}_S$ of $F_S^{\times}$, each trivial on U_S, radially trivial at infinity, and with angular component ω_{m_v}, $m_v \in \{1, -1\}$, at each complex place. This permits the arithmetic character L_S and its finitely many finite-order twists used below. Factor the nonzero global

functional at its complex places, leaving a nonzero functional at the finite places of S. Apply Lemma 23.3 with an open subgroup contained in the sixth powers and in the intersection of the kernels of the finite components of all $\mathcal{L}_S \in \mathscr{L}$. We obtain one product $U = \prod_{v \in S_f} U_v$ and one finite vector. For each $\mathcal{L}_S$, choose at each complex place the derivative of its angular type. The functions in (23.11) are nonzero at one, and their normalization gives an S-Whittaker function

$$W_{S,\mathcal{L}_S}(a(y)) = \mathbf{1}_U(y_f) \prod_{v|\infty} |y_v|_v \, \omega_{m_v}(y_v) K_{1/3}(c_v |y_v|_v^{1/2}). \tag{23.14}$$

In particular $\mathcal{L}_S(y)^{-1} W_{S,\mathcal{L}_S}(a(y))$ is the product of the same radial Bessel factors and $\mathbf{1}_U(y_f)$ for every member of the family. The radial profile and its Gaussian averaging kernel below may therefore be chosen in common for this finite family.

The complex constant term is radial by Proposition 23.1, whereas $\mathcal{L}_S^{-1}$ has a nontrivial angular component. We will use this distinction in Section 25, after the actual weighted theta sums have been shown to descend to the quotient on which they are integrated.

23.3 Integration on norm sheets

Recall that $r = \#S_\infty = [F : \mathbb{Q}]/2$, and define

$$X_u = \{y \in F_S^\times : |y|_S = u\} \qquad (u > 0).$$

For $d \in F_S^\times$, put $q_d = |d|_S$. Multiplication by d sends X_l to $X_{q_d l}$, and inversion sends X_l to $X_{1/l}$. The forward and reflected theta terms will therefore lead to sheet integrals at norms $q_d l$ and q_d/l, respectively. We now bound such one-variable profiles, including absolute envelopes that will justify the ideal sums and the radial average. The two signed orbit substitutions will be proved with their actual integrands in Section 25.

There is at least one complex place. Use the norm section ℓ_u from Section 21; it satisfies $|\ell_u|_S = u$ and $\ell_u \ell_v = \ell_{uv}$. Choose multiplicative Haar measures and Haar measure on X_1 so that

$$\int_{F_S^\times} f(y) \, \mathrm{d}^\times y = \int_0^\infty \int_{X_u} f(y) \, \mathrm{d}_u^\times y \, \frac{\mathrm{d}u}{u}, \tag{23.15}$$

where the measure on X_u is the translate by ℓ_u of that on X_1. At a complex place we take angle measure of total mass one, so its multiplicative measure is a fixed constant times $\mathrm{d}t/t$ times angle measure, with $t = |y_v|_v$. At a finite place every valuation shell has the same multiplicative measure.

Lemma 23.4 (Norm-sheet profiles)**.** *Let $\mathcal{I}$ be a finite index set. For every $\kappa \in \mathcal{I}$, let f_κ be a finite sum of products*

$$f_\kappa(y) = \sum_{\ell=1}^{L_\kappa} \prod_{v \in S} f_{\kappa,\ell,v}(y_v), \qquad y \in F_S^\times.$$

Assume that for one $\delta > 0$ all these local factors satisfy the following conditions.

(a) *At a finite v, $f_{\kappa,\ell,v}$ is locally constant, vanishes for $|y_v|_v > C_v$, and is $O(|y_v|_v^\delta)$ for $|y_v|_v \le 1$.*

(b) *At a complex v, the factor is smooth on $F_v^\times$ and, for every $j \ge 0$ and $A > 0$,*

$$|\mathcal{D}_v^j f_{\kappa,\ell,v}(y_v)| \le C_{A,j} \min(|y_v|_v^\delta, |y_v|_v^{-A}).$$

All constants may depend on the finite family. Define

$$H_\kappa(u) = \int_{X_u} f_\kappa(y) \, \mathrm{d}_u^\times y, \qquad D = u \frac{\mathrm{d}}{\mathrm{d}u}.$$

Fix one complex place v_0, and define the nonnegative absolute derivative profile

$$H_{\kappa,j}^{\mathrm{abs}}(u) = \sum_{\ell=1}^{L_\kappa} \int_{X_u} |\mathcal{D}_{v_0}^j f_{\kappa,\ell,v_0}(y_{v_0})| \prod_{v \ne v_0} |f_{\kappa,\ell,v}(y_v)| \, \mathrm{d}_u^\times y.$$

Then H_κ is smooth, and for every $j \geq 0$ and $\sigma > -\delta$,

$$|D^j H_\kappa(u)| \leq H^{\mathrm{abs}}_{\kappa,j}(u) \leq C_{\sigma,j} u^{-\sigma} \qquad (u > 0). \tag{23.16}$$

In particular, for any $0 < \alpha < \delta$ and every $A > 0$,

$$|D^j H_\kappa(u)| + H^{\mathrm{abs}}_{\kappa,j}(u) \leq C_{A,j} \min(u^\alpha, u^{-A}). \tag{23.17}$$

The conclusions are unchanged if each f_κ is multiplied by a fixed smooth unitary character of $F_S^\times$ and by a bounded function of its class in the finite group $\mathcal{V}$.

Let $\mathcal{K} : (0,\infty) \to \mathbb{C}$ have every absolute moment

$$M_\beta(\mathcal{K}) = \int_0^\infty |\mathcal{K}(T)| T^\beta \, \frac{\mathrm{d}T}{T} < \infty \qquad (\beta \in \mathbb{R}).$$

Both convolutions

$$\int_0^\infty \mathcal{K}(T) H_\kappa(uT) \, \frac{\mathrm{d}T}{T}, \qquad \Phi_\kappa(u) := \int_0^\infty \mathcal{K}(T) H_\kappa(u/T) \, \frac{\mathrm{d}T}{T}$$

and their norm derivatives obey (23.17)*, with constants depending also on the indicated moments. For every j, the two absolute majorants $\int |\mathcal{K}(T)| H^{\mathrm{abs}}_{\kappa,j}(uT^{\pm 1}) \, \mathrm{d}T/T$ satisfy the same two-ended size bound. For any $x_i > 0$, complex a_i, indices $\kappa_i \in \mathcal{I}$, $\tau \in \{1,-1\}$, and $\sigma > -\delta$, one has the absolute bound*

$$\begin{aligned} \int_0^\infty |\mathcal{K}(T)| \sum_i |a_i| H^{\mathrm{abs}}_{\kappa_i,0}(u x_i T^\tau) \, \frac{\mathrm{d}T}{T} \\ \leq C_{\sigma,0} u^{-\sigma} M_{-\tau\sigma}(\mathcal{K}) \sum_i |a_i| x_i^{-\sigma}. \end{aligned} \tag{23.18}$$

In particular the sum and integral may be interchanged if the final series is finite.

Proof. For a finite place, let $A_v(Q_v^{-k})$ be the supremum of the absolute local factor on its k-th valuation shell. Condition (a) implies, for every $\sigma > -\delta$,

$$\sum_{k \in \mathbb{Z}} A_v(Q_v^{-k}) Q_v^{-\sigma k} < \infty :$$

there are only finitely many negative k, and the positive tail is geometric with ratio $Q_v^{-(\delta+\sigma)}$. At a complex place condition (b) likewise gives

$$\int_{F_v^\times} |f_v(y)| \, |y|_v^\sigma \, \mathrm{d}^\times y < \infty, \qquad \sup_{y \in F_v^\times} |y|_v^\sigma |\mathcal{D}_v^j f_v(y)| < \infty.$$

These statements remain valid for all factors in the fixed finite family.

Use $t_{v_0} = |y_{v_0}|_{v_0}$ as the dependent norm in (23.15). On X_u,

$$t_{v_0} = u \prod_{v \neq v_0} |y_v|_v^{-1},$$

and the sheet measure is the product of the angle measure at v_0 and the multiplicative measures at the other places, up to its fixed normalization. Multiplying the absolute integrand by $u^\sigma = \prod_v |y_v|_v^\sigma$, take the weighted supremum of the v_0-factor and the weighted integrals of all the others. This gives (23.16), including its absolute version. On this parametrization D differentiates only the v_0-factor and is exactly $\mathcal{D}_{v_0}$. The same weighted majorants for each derivative justify differentiation under the integral on compact u-intervals. Taking $\sigma = -\alpha$ and then $\sigma = A$ proves (23.17).

At a complex place the sixth-power quotient is trivial, and at a finite place it is finite. Thus a bounded function of $\mathcal{V}$ is a finite linear combination of products of local class indicators. A

unitary character factors over the finite product S; at a complex place its norm derivatives only introduce fixed constants. Multiplying by these functions preserves the hypotheses after splitting into a finite family.

For the convolutions, apply the bound with $\sigma = -\alpha$ or $\sigma = A$ under the T-integral. For example the two bounds for the reflected convolution have additional factors $M_{-\alpha}(\mathcal{K})$ and $M_A(\mathcal{K})$. The corresponding absolute derivative profiles dominate differentiation on compact u-intervals. Finally apply (23.16) to each summand in (23.18); Tonelli and the definition of the moment give the displayed bound. □

The lemma also gives the smooth separation on norm dyads that we will use. If ρ is a fixed smooth compactly supported function on $\mathbb{R}$, integration by parts twice in its ordinary Fourier transform gives

$$\left\|\rho(t)\widehat{\Phi_\kappa(ue^t)}\right\|_{L^1(\mathbb{R})} \le C_{A,\rho}\min(u^\alpha, u^{-A}). \tag{23.19}$$

Indeed the L^1 norm is bounded by a constant times the suprema of the function and its derivatives of orders one and two; these derivatives are controlled by (23.17). Fourier inversion with $t = \log x - \log y$, for x, y in fixed ratio intervals, therefore separates the two norm variables with bounded L^1 cost and retains any chosen large-u power saving.

23.4 Exact radial smoothing

For the chosen S-function (23.14), the angular characters cancel against $\mathcal{L}_S^{-1}$. Define its radial profile by

$$V(u) = \int_{X_u} \mathbf{1}_U(y_f)\prod_{v\mid\infty} |y_v|_v K_{1/3}(c_v|y_v|_v^{1/2})\ \mathrm{d}_u^\times y. \tag{23.20}$$

It is a positive, nonzero profile covered by Lemma 23.4. In this subsection the Mellin convention is $\widehat{f}(s) = \int_0^\infty f(u)u^s\ \mathrm{d}u/u$. The unreflected theta average will contain $V(q_{\mathfrak{Q}}l)$. At $l = T/Z$, the next lemma averages this profile to exactly $P_G(q_{\mathfrak{Q}}/Z)$. After the Weyl operation the norm variable is inverted, and the same kernel instead produces the controlled profiles $\int \mathcal{K}(T)H_\kappa(u/T)\ \mathrm{d}T/T$.

Lemma 23.5 (Exact Gaussian smoothing)**.** *For V in* (23.20)*, there are constants $C_0 > 0$ and $A_0 > 0$ such that*

$$\widehat{V}(s) = C_0 A_0^s\big[\Gamma(s+5/6)\Gamma(s+7/6)\big]^r \qquad (\mathrm{Re}\, s > -5/6). \tag{23.21}$$

There is a smooth function $\mathcal{K} : (0,\infty) \to \mathbb{C}$ such that, for every $N > 0$ and $j \ge 0$,

$$\left|(T\partial_T)^j\mathcal{K}(T)\right| \le C_{N,j}\min(T^N, T^{-N}), \tag{23.22}$$

and, for every $u > 0$,

$$\int_0^\infty \mathcal{K}(T)V(uT)\,\frac{\mathrm{d}T}{T} = P_G(u) = \frac{1}{2\sqrt{\pi}}\exp\left(-\frac{(\log u)^2}{4}\right). \tag{23.23}$$

Its Mellin transform satisfies the exact sign convention

$$\widehat{\mathcal{K}}(-s) = \frac{e^{s^2}}{\widehat{V}(s)}. \tag{23.24}$$

The quotient on the right is interpreted by its entire continuation. For every fixed finite family of rotated S-profiles H_κ covered by Lemma 23.4, the functions $\Phi_\kappa(u) = \int \mathcal{K}(T)H_\kappa(u/T)\ \mathrm{d}T/T$ obey the signed and absolute bounds of that lemma.

Proof. The disintegration (23.15) and Tonelli give the Mellin transform of V as the product of its local Mellin integrals. Each finite factor is the positive volume of U_v, because $|y_v|_v = 1$ there. At a complex place put $t = |y_v|_v$. The classical Bessel integral, or the same Laplace integral used above, gives

$$\int_0^\infty t^{s+1} K_{1/3}(c_v\sqrt{t})\,\frac{\mathrm{d}t}{t} = 2^{2s+1}c_v^{-2s-2}\Gamma(s+5/6)\Gamma(s+7/6), \qquad \operatorname{Re} s > -5/6. \tag{23.25}$$

The angle integrals only change the positive measure constant. Multiplication over the r complex places proves (23.21).

Set

$$\mathcal{R}_V(s) = \frac{e^{s^2}}{\widehat{V}(s)} = C_0^{-1}A_0^{-s}e^{s^2}\left[\Gamma(s+5/6)\Gamma(s+7/6)\right]^{-r}.$$

The reciprocal gamma function is entire, so $\mathcal{R}_V$ is entire. On each fixed closed vertical strip, Stirling's formula implies

$$|\mathcal{R}_V(\sigma+it)| \le C(1+|t|)^C e^{-t^2+\pi r|t|}.$$

The same bound with a different polynomial holds after multiplication by any power of s. Hence

$$\mathcal{K}(T) = \frac{1}{2\pi i}\int_{(\sigma)} \mathcal{R}_V(s)T^s\,\mathrm{d}s \tag{23.26}$$

converges absolutely and is independent of the real line σ: the horizontal sides of a rectangle tend to zero by the Gaussian bound. Differentiation in $\log T$ inserts a power of s. Moving the line to N for $0 < T \le 1$, and to $-N$ for $T \ge 1$, proves (23.22). In particular every $M_\beta(\mathcal{K})$ in Lemma 23.4 is finite.

Fourier inversion in $\log T$ on a vertical line in (23.26) gives $\widehat{\mathcal{K}}(-s) = \mathcal{R}_V(s)$ on that line. Both sides are entire, so (23.24) holds everywhere. For example on any real line $\sigma > 0$, absolute Mellin Fubini is valid because

$$\int_0^\infty |\mathcal{K}(T)|T^{-\sigma}\frac{\mathrm{d}T}{T} < \infty, \qquad \int_0^\infty V(u)u^\sigma\frac{\mathrm{d}u}{u} < \infty.$$

It follows that the Mellin transform of the left side of (23.23) is $\widehat{V}(s)\widehat{\mathcal{K}}(-s) = e^{s^2}$. The function on the right of (23.23) has the same transform, by the elementary Gaussian integral. Mellin inversion proves the identity. The final assertion follows from the convolution part of Lemma 23.4. □

24 The cubic theta normalization

This section supplies the cubic theta specialization and normalization behind Proposition 23.1. We first specialize their cover, inducing datum, and global representation, and derive the normalized Whittaker factorization. We then verify the exact lift identities and compute the spherical Jacquet integral in the chosen sections of the cover. The local calculation fixes the Gauss phase and the unit factor together, and includes the modular factor in the triple-valuation step. The main argument uses the resulting global Fourier expansion with normalized local factors and the exact local rules for the Weyl calculation.

24.1 The selected cover and global representation

Denote the cubic Hilbert symbol at a place v by $(\cdot,\cdot)_{3,v}$. In the convention of Kazhdan and Patterson, the restriction of their determinant-modified cocycle to the diagonal torus is

$$\sigma_{c,v}(\operatorname{diag}(a,b),\operatorname{diag}(a',b')) = (a,b')_{3,v}(ab,a'b')_{3,v}^{c}, \qquad c \in \mathbb{Z}/3\mathbb{Z}. \tag{24.1}$$

The root groups have their canonical splittings. These conventions are those of the classical cubic theta representation theorem.

For the n-fold cover of GL_d, where d is the rank parameter, the center criterion of the classical cubic theta representation theorem requires $z^{d-1+2dc} \in F_v^{\times n}$ for a scalar zI. For $d = 2, n = 3$ the exponent is $1+4c$, which is zero modulo three exactly when $c = 2$. Moreover, (24.1) on two scalars is

$$\sigma_{2,v}(zI, z'I) = (z, z')_{3,v}^{1+4\cdot 2} = 1.$$

Thus their full preimage is central and the ordinary scalar section splits. Quotienting its image gives the asserted local covers of PGL_2. The adelic cover and its rational splitting are obtained by the reciprocity construction in the classical cubic theta representation theorem. The ordinary scalar section agrees with that splitting on rational scalars, so the quotient retains a rational splitting.

We check the datum in (23.1). A Hilbert symbol with a cube argument is one; the cocycle on scalar pairs is also one. Consequently the cube and scalar sections commute and multiply without any remaining cocycle. If an element has two presentations in (23.1), their two cube coordinates differ by a common scalar cube, so the displayed absolute-value ratio agrees. The formula is therefore a genuine character, trivial on the split scalar subgroup. On a coroot cube it gives

$$\omega_v(s_v(\mathrm{diag}(t^3, t^{-3}))) = |t|_v.$$

This is exactly the exceptional condition of the classical cubic theta representation theorem. The exceptional condition alone would only specify the square of the value on $a(\varpi^3)$; the datum in the main text chooses the pure absolute-value value.

Use the idele norm in (23.1) to define the global character. A rational diagonal lying in the adelic cube torus times scalars has a rational ratio which is a cube at every completion. If that ratio were not a cube in F, its nontrivial Kummer extension would have a nonsplit unramified prime by Chebotarev. Thus the ratio is a rational cube. The product formula now shows that our character is trivial on the intersection with the rational lift. This is the compatibility hypothesis in the classical cubic theta representation theorem. Extend the character trivially on rational lifts. The standard extension step for the exceptional representation extends it to the product of local maximal abelian subgroups, agreeing with the rational character on their intersection. The character on the scalar center is unitary.

At finite places, the local exceptional-representation theorem gives the irreducible quotient at the exceptional inducing parameter and the irreducible subrepresentation at its Weyl conjugate. Its unramified spherical-vector assertion applies to the tame normalized datum. The tame Whittaker uniqueness criterion is

$$d = n - 1, \quad 2(c+1) = 0 \pmod{n}, \qquad \text{or} \qquad d = n.$$

It applies to $(d, n, c) = (2, 3, 2)$. The global residue construction realizes the restricted tensor product automorphically. The global nonvanishing and local uniqueness assertions for this construction give a nonzero global Whittaker functional and a one-dimensional local Whittaker space at every place. At complex places the local classification gives the same conclusion. The inducing spaces, their exceptional images, and the automorphic residue inherit the scalar action of ω. That action is trivial, so these representations descend to the scalar quotients.

On the quotient torus the section of $a(y)$ has cocycle $(x, y)_{3,v}^2$. Since -1 is a cube in F, this is an alternating bimultiplicative law. At a tame place it is trivial on pairs of units, and its value on a uniformizer and a unit is the cubic residue symbol or its inverse, with a common orientation at all places. Choose the embedding ι with the orientation giving (23.5). The canonical compact splitting agrees with the sections of unit diagonals, Weyl elements, and integral upper root elements. Conjugating by the Weyl element gives the same agreement for the lower root group. The Whittaker coefficient relations, with the modular factor in normalized induction, reduce the spherical values to their central step. The direct local calculation below proves (23.6) and the lift identities (23.4) in these conventions.

At a complex place the Hilbert symbol and the cover are trivial. The complex local classification identifies the exceptional representation with the ordinary principal series when $d \le n$. Our cube

datum fixes the characters on all of $\mathbb{C}^\times$, because every complex number has a cube root. Triviality on scalars removes the determinant character; equivalently any possible remaining finite-order continuous character of the connected group $\mathbb{C}^\times$ is trivial. This gives the characters in part (iii) of Proposition 23.1. Their Bessel function is computed from the Jacquet integral above. Finally, the constant-term formula for the Eisenstein series has two Bruhat terms. At the exceptional residue the surviving term belongs to the Weyl-conjugate induction. Its complex inducing character, including the modular factor, is $|y|_v^{1/3}$; in particular it has no angular part.

24.2 The normalized global Whittaker expansion

Let λ be the nonzero global functional just obtained. The restricted tensor product is spanned by pure tensors, so there is a pure tensor φ° with $\lambda(\varphi^\circ) \neq 0$. Its nondistinguished finite places form a finite set. The final S contains this set; the arithmetic data are constructed only after that final choice. For $v \notin S$, varying only the v-component of φ° gives a nonzero local Whittaker functional. Normalize it at φ_v°. If other finitely many components have already been varied, the resulting partial functional at v is still a Whittaker functional, and local uniqueness makes it a scalar multiple of the normalized one. Evaluating at φ_v° identifies that scalar. Induction on the number of varied components proves (23.2). The same argument factors the complex components.

At a finite product of remaining places the functional descends to the tensor product of the local twisted Jacquet quotients. The twisted Jacquet quotient of the principal series is finite dimensional by the classical cubic theta representation theorem. Since Θ_v is its quotient by the local exceptional-representation theorem, its twisted Jacquet quotient is a quotient of that finite-dimensional space. A functional on the finite tensor product is therefore a finite sum of products. This is the finite-S assertion in Proposition 23.1.

The rational splitting identifies every nonconstant Fourier coefficient with this Whittaker function. Conjugate $n(x)$ by the rational lift of $a(\nu)$ and change x to νx; the adelic Jacobian is one. This gives (23.3). On a fixed compact set of g, smoothness at infinity and invariance under a fixed finite open subgroup reduce the expansion along $F\backslash\mathbb{A}_F$ to the smooth Fourier expansion on a compact real torus. Integration by parts in sufficiently many archimedean directions gives absolute local uniform convergence there.

24.3 Sections and the spherical induced vector

Fix an exterior finite place $\mathfrak{p}$. In this section write $\mathcal{O} = \mathcal{O}_\mathfrak{p}$, let ϖ be the chosen uniformizer, put $Q = q_\mathfrak{p}$, and abbreviate $\psi = \psi_\mathfrak{p}$, $\chi = \chi_\mathfrak{p}$, and $\mathcal{C} = \mathcal{C}_\mathfrak{p}$. Thus ψ has conductor $\mathcal{O}$, additive measure gives $\mathcal{O}$ volume one, and χ is the sextic residue character on $\mathcal{O}^\times$. In all finite sums it is extended by zero to the zero residue, including when its exponent is zero. The embedding of the cubic symbol has been chosen so that the effective torus cocycle is

$$\mathcal{C}(\varpi, u) = \chi(u)^{-2}, \quad \mathcal{C}(u, \varpi) = \chi(u)^2, \quad \mathcal{C}(u, u') = 1, \quad \mathcal{C}(\varpi, \varpi) = 1 \qquad (u, u' \in \mathcal{O}^\times). \tag{24.2}$$

The last two identities follow from tameness and alternation. In particular pure uniformizer powers multiply without a cocycle. All symbols with -1 are one because $-1 \in F^{\times 6}$.

We first record explicitly why the ordinary rank-one decompositions are valid for these lifts. In the local rank-one construction the determinant-modified cocycle is

$$\sigma_c(g_1, g_2) = \left(\frac{x(g_1 g_2)}{x(g_1)}, \frac{x(g_1 g_2)}{x(g_2)\det g_1}\right)_3 (\det g_1, \det g_2)_3^c, \quad x(g) = \begin{cases} g_{21}, & g_{21} \neq 0, \\ g_{22}, & g_{21} = 0. \end{cases}$$

We take $c = 2$ and then quotient the split scalar subgroup. Substitution in this formula gives

$$n(x)a(y) = a(y)n(x/y), \qquad wa(y) = a(y^{-1})w$$

on the quotient. Before quotienting, the second equality has the extra scalar yI. All its cocycles are symbols between powers of y and -1, hence are one. For $H \neq 0$, the matrix identity

$$wn(H) = n(H^{-1})\,\mathrm{diag}(-H^{-1}, H)\bar{n}(H^{-1})$$

also lifts exactly. The product of the first two factors has cocycle one, and multiplication by the last factor has cocycle $(H^{-1}, -H)_3 = 1$. The middle diagonal is the product of the split scalar HI and $a(-H^{-2})$, again without a cocycle. Thus

$$wn(H) = n(H^{-1})a(-H^{-2})\bar{n}(H^{-1}) \tag{24.3}$$

on the quotient. If $H^{-1} \in \mathcal{O}$, the last factor is in the compact splitting. The latter agrees with the sections on unit diagonals, w, and integral upper root elements by the classical cubic theta representation theorem; conjugation by w gives agreement on the lower root group. The diagonal $a(-1)$ can therefore be absorbed in a compact factor in any spherical calculation.

Let σ^- be the unramified representation of the covering torus with central character the Weyl conjugate of (23.1). Its norm exponent on $a(y)$ is $-1/6$. Modulo scalars, the subgroup of torus elements of valuation divisible by three is maximal abelian: the tame group $F_{\mathfrak{p}}^{\times}/F_{\mathfrak{p}}^{\times 3}$ has one valuation and one unit coordinate over $\mathbb{Z}/3\mathbb{Z}$, and the commutator pairing pairs those coordinates nondegenerately. The torus construction in the classical cubic theta representation theorem realizes σ^- by induction from a character of this subgroup. Its index is three, so the unramified Heisenberg representation has dimension three. The canonical unramified extension is trivial on unit lifts; the induced basis vector supported on that subgroup is therefore unit fixed. Call it e_0, and set

$$e_j = Q^{-j/6}\sigma^-(a(\varpi^j))e_0, \qquad j \in \mathbb{Z}. \tag{24.4}$$

The central element $a(\varpi^3)$ acts by $Q^{1/2}$, so $e_{j+3} = e_j$. These three vectors form a basis. From (24.2) and unit invariance one obtains

$$\sigma^-(a(\varpi^j u))e_0 = Q^{j/6}\chi(u)^{2j}e_j \qquad (j \in \mathbb{Z},\ u \in \mathcal{O}^{\times}). \tag{24.5}$$

Use the vector-valued realization of normalized induction

$$I^- = \mathrm{Ind}_{\widetilde{B}}^{\widetilde{\mathrm{PGL}_2}}(\sigma^-), \qquad f(n(x)a(y)g) = |y|_{\mathfrak{p}}^{1/2}\sigma^-(a(y))f(g).$$

Its spherical vector f° is determined by $f^\circ(k) = e_0$ on the compact subgroup. The total norm exponent in this transformation law is $1/2 - 1/6 = 1/3$. The compact-invariant line of this normalized unramified induction is one dimensional by the unramified local theory. The local exceptional representation is the irreducible subrepresentation of I^- and has a nonzero compact-invariant vector. That vector is a scalar multiple of f°, so the subrepresentation contains f°.

24.4 The shells of the Jacquet integral

For a smooth $f \in I^-$, define the vector Jacquet functional by

$$\mathscr{J}(f) = \lim_{L\to\infty} \int_{\varpi^{-L}\mathcal{O}} f(wn(x))\overline{\psi(x)}\,\mathrm{d}x.$$

This limit stabilizes. Indeed, when $|x|$ is sufficiently large, the right invariance of f makes $\bar{n}(x^{-1})$ fix f. The lifted Bruhat formula then expresses $f(wn(x))$ as

$$|x|_{\mathfrak{p}}^{-1}\sigma^-(a(-x^{-2}))f(1).$$

Choose m so that $1+\varpi^m\mathcal{O}$ consists of cubes and its unit lifts fix the vector $f(1)$ in the smooth torus representation. On a shell $x = \varpi^{-l}v$, the displayed expression is unchanged when v is replaced by vu, $u \in 1 + \varpi^m\mathcal{O}$: the norm is unchanged, all cocycles with u are trivial, and $a(u^{-2})$ fixes $f(1)$.

Each such unit coset is an additive coset of $\varpi^m\mathcal{O}$. For $l > m$, the character $\psi(\varpi^{-l}v)$ has zero integral on it. All sufficiently large shells therefore vanish. This argument applies to every smooth f, including each of its right translates.

If $t \in F_{\mathfrak{p}}$, choose a ball large enough to contain t and to realize the stable values for both f and $\pi(n(t))f$. Translation $x \mapsto x + t$ preserves that ball and gives $\mathscr{J}(\pi(n(t))f) = \psi(t)\mathscr{J}(f)$. Thus each scalar coordinate of $\mathscr{J}$ is a Whittaker functional. This is a direct stable version, at the present inducing parameter, of the Jacquet functionals introduced in the classical cubic theta representation theorem.

For $e \geq 0$ put $J_e = \mathscr{J}(\pi(a(\varpi^e))f^\circ)$; explicitly,

$$J_e = \int_{F_{\mathfrak{p}}} f^\circ(wn(x)a(\varpi^e))\overline{\psi(x)}\,\mathrm{d}x. \tag{24.6}$$

The integral retains the stable interpretation just established. We now evaluate it shell by shell.

Put $x = \varpi^e H$ and use the first two lift identities above. The measure contributes Q^{-e}, the modular factor of $a(\varpi^{-e})$ contributes $Q^{e/2}$, and its torus norm factor contributes $Q^{-e/6}$. Thus every shell has the common factor

$$Q^{-e+e/2-e/6} = Q^{-2e/3}. \tag{24.7}$$

If $H \in \mathcal{O}$, then $wn(H)$ belongs to the compact subgroup, and $\psi(\varpi^e H) = 1$. This region contributes e_{-e} inside the common factor.

Suppose instead that $H = \varpi^{-l}v$, with $l \geq 1$ and $v \in \mathcal{O}^\times$. In (24.3) the last factor is compact, while the first acts trivially in the inducing transformation law. The modular factor of $a(-H^{-2})$ is Q^{-l}, and its torus norm factor is $Q^{l/3}$. Furthermore,

$$a(\varpi^{2l}v^{-2}) = \mathcal{C}(\varpi^{2l}, v^{-2})^{-1}a(\varpi^{2l})a(v^{-2}), \qquad \mathcal{C}(\varpi^{2l}, v^{-2})^{-1} = \chi(v)^{-8l}.$$

The minus sign in $-H^{-2}$ has no effect. Multiplication by $a(\varpi^{-e})$ changes the basis index by $-e$, without a cocycle between the pure uniformizer powers. Finally $\mathrm{d}H = Q^l\,\mathrm{d}v$. It follows that this shell contributes, inside (24.7),

$$Q^{l/3}e_{-e+2l}\int_{\mathcal{O}^\times} \chi(v)^{-8l}\overline{\psi(\varpi^{e-l}v)}\,\mathrm{d}v. \tag{24.8}$$

This accounts separately for the measure, modular, norm, and Hilbert-symbol factors.

Define the normalized prime Gauss sums

$$g_j = Q^{-1/2}\sum_{x \bmod \mathfrak{p}} \chi(x)^j\psi(x/\varpi), \qquad j \in \mathbb{Z}.$$

The powers in these sums retain the unit mask. Since $\chi(-1) = 1$, replacing $\psi(x/\varpi)$ by its complex conjugate has no effect on g_j. For $j \not\equiv 0 \pmod 6$, the usual finite-field Gauss identity gives

$$|g_j| = 1, \qquad g_j g_{-j} = \chi(-1)^j = 1.$$

In particular $g_2 g_4 = 1$. These identities follow, for example, by expanding the product of the two sums, substituting $x = ty$, and using the additive sum in y; the result is $Q\chi(-1)^j$ before division by Q.

Write $I_{e,l}$ for the integral in (24.8). The character there is χ^{-2l}, since $-8l \equiv -2l \pmod 6$. Its exact values are

$$I_{e,l} = \begin{cases} 1 - Q^{-1}, & 1 \leq l \leq e, \quad 3 \mid l, \\ 0, & 1 \leq l \leq e, \quad 3 \nmid l, \\ Q^{-1/2}g_{-2l}, & l = e+1, \quad 3 \nmid l, \\ -Q^{-1}, & l = e+1, \quad 3 \mid l, \\ 0, & l > e+1. \end{cases} \tag{24.9}$$

Indeed, for $l \le e$ the additive character is trivial, so unit orthogonality gives the first two cases. For $l = e+1$, integration over the residue classes gives Q^{-1} times the corresponding prime Gauss sum; the trivial character with its unit mask gives $\sum_{x \neq 0} \psi(x/\varpi) = -1$. For $l > e+1$, split each unit class modulo ϖ^{l-e-1} into its Q lifts modulo ϖ^{l-e}. The multiplicative character is unchanged on these lifts, whereas the additive factors sum to $\sum_{t \bmod \mathfrak{p}} \psi(t/\varpi) = 0$. This also proves the asserted stabilization of (24.6).

We can now sum the shells. For $k \ge 0$, the integral unit ball and the shells $l = 3, 6, \dots, 3k$ have the same basis index and combined coefficient

$$1 + (1 - Q^{-1}) \sum_{j=1}^{k} Q^{j} = Q^{k}.$$

For $e = 3k$, the terminal shell $l = 3k+1$ has coefficient $Q^{k-1/6} g_4$ and basis vector e_2. For $e = 3k+1$, the terminal shell $l = 3k+2$ has coefficient $Q^{k+1/6} g_2$ and basis vector e_0. For $e = 3k+2$, the terminal shell $l = 3k+3$ has coefficient $-Q^{k}$ and the same basis vector e_1 as the preceding combined contribution. Multiplying by (24.7) therefore gives

$$\begin{aligned} J_{3k} &= Q^{-k}(e_0 + Q^{-1/6} g_4 e_2), \\ J_{3k+1} &= Q^{-k}(Q^{-1/2} g_2 e_0 + Q^{-2/3} e_2) = Q^{-k-1/2} g_2 (e_0 + Q^{-1/6} g_4 e_2), \\ J_{3k+2} &= 0. \end{aligned} \tag{24.10}$$

The second equality uses $g_2 g_4 = 1$.

In particular $J_0 = e_0 + Q^{-1/6} g_4 e_2 \neq 0$. Choose a scalar coordinate nonzero on J_0. Its Jacquet functional is nonzero on the exceptional subrepresentation, since that subrepresentation contains f°. The local Whittaker space is one dimensional by the classical cubic theta representation theorem. Hence every spherical Whittaker function normalized at one has the ratios in (24.10):

$$W(a(\varpi^{3k})) = Q^{-k}, \qquad W(a(\varpi^{3k+1})) = Q^{-k-1/2} g_2, \qquad W(a(\varpi^{3k+2})) = 0. \tag{24.11}$$

The middle term is $Q^{-k-1} \sum_{x} \chi(x)^2 \psi(x/\varpi)$, as claimed. This computation agrees with the central-step statement of the classical cubic theta representation theorem: on $a(\varpi^3)$ the Weyl-conjugate inducing character is $Q^{1/2}$ and the modular half-character is $Q^{-3/2}$, so their product is Q^{-1}.

24.5 Negative valuations and the unit normalization

For $e < 0$, choose $x \in \mathcal{O}$ for which $\psi(\varpi^{e} x) \neq 1$. The spherical vector is fixed by $n(x)$, while the split root relation and Whittaker equivariance give

$$W(a(\varpi^{e})) = W(a(\varpi^{e}) n(x)) = \psi(\varpi^{e} x) W(a(\varpi^{e})).$$

Thus the value is zero. For a unit u, the compact section of $a(u)$ fixes the spherical vector. The torus law gives

$$a(\varpi^{e} u) = \mathcal{C}(\varpi^{e}, u)^{-1} a(\varpi^{e}) a(u),$$

where $\mathcal{C}$ includes the scalar action on a genuine vector. Consequently

$$W(a(\varpi^{e} u)) = \chi(u)^{2e} W(a(\varpi^{e})). \tag{24.12}$$

Together with (24.11), this proves (23.6).

There is a useful check that fixes the orientation without referring to a convention for the name of a cubic Gauss sum. At a tame place the residue character is unchanged when ϖ is replaced by ϖu. Substitution $x = uv$ in the residue sum yields

$$Q^{-1} \sum_{x \bmod \mathfrak{p}} \chi(x)^2 \psi(x/(\varpi u)) = \chi(u)^2 Q^{-1} \sum_{v \bmod \mathfrak{p}} \chi(v)^2 \psi(v/\varpi).$$

This is exactly the $e = 1$ transformation in (24.12). Thus the Gauss phase and the unit factor are a single compatible normalization. Changing the sign of an additive Gauss argument is harmless because $\chi(-1) = 1$.

25 Reflection and the quadratic large sieve

We prove the second-moment estimate in Proposition 22.1. Following the reflection method of the coefficient reflection method, we realize B_m as a theta average and apply the rational Weyl element. At a valuation-one row prime the reflected coefficient contributes a first power of the sextic symbol; a squarefree column prime contributes power -4 in the same pairing. Their product has exponent $1-4\equiv 3 \pmod 6$, so a quadratic large sieve applies. The reflected norm gives a column length of order D^2P/Z, and the rapid decay of the norm profile controls the remaining terms.

25.1 An automorphic expression for B_m

All implied constants in this section may depend on the fixed field, S, the target character η, the compact set $\mathscr{K}$, and the chosen finite family of local theta data.

We first replace elements by ideal rows. Choose a balanced allowed generator m_0 of each exterior ideal $\mathfrak{m}$. For every nonzero $m \in R$ generating $\mathfrak{m}$, write $m = \varepsilon_m m_0$ with $\varepsilon_m \in U_S$. When $m \in \mathscr{B}_Z(\mathscr{K})$, the balanced shapes of m_0, the fixed shape set $\mathscr{K}$, and $q_{\mathfrak{m}} \asymp_{\mathscr{K}} Z^{M_0}$ put ε_m in a fixed compact subset of the norm-one group. The S-unit theorem makes U_S discrete there, so one ideal is generated by only boundedly many elements of the row set. Put

$$\Gamma = U_S^6, \qquad U_6 = U_S/\Gamma, \qquad \upsilon_m = \varepsilon_m\Gamma \in U_6.$$

The group U_6 is finite. The value of B_m depends on the unit ε_m only through υ_m, which is fixed when the row m is fixed.

For a nonzero row let $j_{\mathfrak{p}} = v_{\mathfrak{p}}(\mathfrak{m}) \bmod 6$ in $\{0,\dots,5\}$. For an allowed generator A of an integral exterior ideal $\mathfrak{A}$, reciprocity gives

$$\chi_{\mathfrak{A}}(m) = \chi_{\mathfrak{A}}(\varepsilon_m)\prod_{\mathfrak{p}\mid\mathfrak{m}} \chi_{\mathfrak{p}}(A)^{j_{\mathfrak{p}}}. \tag{25.1}$$

For coprime ideals this is the multiplicative reciprocity law from the arithmetic section. At a shared prime both sides vanish by zero extension, including when $j_{\mathfrak{p}} = 0$; the zeroth power there is the unit indicator.

Define $\lambda_{\varepsilon_m}(y) = H_S(\varepsilon_m, y)^{-1}$ on $F_S^\times$. By Equation (21.5), its value at an allowed generator A is $\chi_{\mathfrak{A}}(\varepsilon_m)$, and isotropy of E makes it trivial on U_S. It depends only on υ_m. With the character L_S from Equation (21.14), put $L_{\upsilon_m} = L_S\lambda_{\varepsilon_m}$. Then

$$L_{\upsilon_m}(A) = \vartheta(\mathfrak{A})\chi_{\mathfrak{A}}(\varepsilon_m) \qquad \text{for every allowed generator } A. \tag{25.2}$$

These characters form a fixed finite family. They are smooth, unitary, and trivial on U_S; their radial components at infinity are trivial, and their angular component at each complex place has power 1 or -1.

We insert the moving twists by finite upper-unipotent translations. At an exterior prime $\mathfrak{p}$, write $k_{\mathfrak{p}} = \mathcal{O}_F/\mathfrak{p}$ and use its allowed generator p as a local uniformizer. For $1 \le j \le 5$ put

$$\tau_{\mathfrak{p},j} = \sum_{v\in k_{\mathfrak{p}}^\times} \chi_{\mathfrak{p}}(v)^{-j}\psi_{\mathfrak{p}}(v/p).$$

The character is nontrivial, so $|\tau_{\mathfrak{p},j}| = q_{\mathfrak{p}}^{1/2}$. Define

$$\alpha_{\mathfrak{p},j}(v) = \begin{cases} \tau_{\mathfrak{p},j}^{-1}\chi_{\mathfrak{p}}(v)^{-j}, & 1\le j\le 5,\ v\neq 0,\\ 0, & 1\le j\le 5,\ v=0,\\ 1-q_{\mathfrak{p}}^{-1}, & j=0,\ v=0,\\ -q_{\mathfrak{p}}^{-1}, & j=0,\ v\neq 0. \end{cases} \tag{25.3}$$

For every $t \in \mathcal{O}_{\mathfrak{p}}$,

$$\sum_{v \in k_{\mathfrak{p}}} \alpha_{\mathfrak{p},j}(v)\psi_{\mathfrak{p}}(tv/p) = \chi_{\mathfrak{p}}(t)^j. \tag{25.4}$$

For $j \neq 0$, substitution by t^{-1} proves this for a unit t, and both sides vanish for a nonunit. For $j = 0$, the left side is $1 - q_{\mathfrak{p}}^{-1} \sum_v \psi_{\mathfrak{p}}(tv/p)$, the unit indicator. Thus the formula includes the masked zeroth power.

For each $\upsilon \in U_6$, choose the theta vector in Proposition 23.1 whose finite S-Whittaker function is the indicator of the common small product subgroup contained in sixth powers and in $\ker L_\upsilon$, and whose complex angular type is that of L_υ. Denote its automorphic function by θ_υ. The cutoff and the normalizations are those of Equation (23.14); these choices form a fixed finite family. For the remainder of this subsection fix a nonzero m, put $\upsilon = \upsilon_m$, and write $L = L_\upsilon$.

For $y \in F_S^\times$ and digits $\boldsymbol{v} = (v_{\mathfrak{p}})_{\mathfrak{p}\mid\mathfrak{m}}$, let $g(y, \boldsymbol{v})$ have component $a(y)$ at S, component $n(v_{\mathfrak{p}}/p)$ at $\mathfrak{p} \mid \mathfrak{m}$, and identity at the other exterior places. Use the lifts in Equation (23.4), choose representatives of the residue digits, and set

$$\boldsymbol{\alpha}_m(\boldsymbol{v}) = \prod_{\mathfrak{p}\mid\mathfrak{m}} \alpha_{\mathfrak{p},j_{\mathfrak{p}}}(v_{\mathfrak{p}}), \qquad w_+ = 1, \quad w_- = w.$$

The combined digit sums that we shall integrate are

$$\begin{aligned} \mathscr{D}_m^{\pm}(y) &= \sum_{\boldsymbol{v}} \boldsymbol{\alpha}_m(\boldsymbol{v})\theta_\upsilon(w_\pm g(y, \boldsymbol{v})), \\ \mathscr{D}_{m,N}^{\pm}(y) &= \sum_{\boldsymbol{v}} \boldsymbol{\alpha}_m(\boldsymbol{v})\theta_{\upsilon,N}(w_\pm g(y, \boldsymbol{v})). \end{aligned} \tag{25.5}$$

Rational automorphy gives $\mathscr{D}_m^+ = \mathscr{D}_m^-$. We next show that all four sums descend under Γ.

For $\gamma \in \Gamma$, choose representatives $v'_{\mathfrak{p}} \equiv \gamma v_{\mathfrak{p}} \pmod{\mathfrak{p}}$, and put

$$u_{\mathfrak{p}} = (\gamma v_{\mathfrak{p}} - v'_{\mathfrak{p}})/p \in \mathcal{O}_{\mathfrak{p}}.$$

Let $k_S = 1$, $k_{\mathfrak{p}} = n(u_{\mathfrak{p}})a(\gamma)$ for $\mathfrak{p} \mid \mathfrak{m}$, and $k_{\mathfrak{q}} = a(\gamma)$ at the other exterior places. Because γ is an exterior unit, these are compact lifts fixing the original exterior spherical vector. At S the torus factors involving γ are trivial, since it is a sixth power. At a row prime the exact root relation gives

$$n(v'_{\mathfrak{p}}/p)n(u_{\mathfrak{p}})a(\gamma) = a(\gamma)n(v_{\mathfrak{p}}/p).$$

The rational diagonal and Weyl lifts therefore satisfy

$$\begin{aligned} g(\gamma y, \boldsymbol{v}') &= a(\gamma)g(y, \boldsymbol{v})k^{-1}, \\ wg(\gamma y, \boldsymbol{v}') &= a(\gamma^{-1})wg(y, \boldsymbol{v})k^{-1}. \end{aligned} \tag{25.6}$$

The second identity has the same compact factor at the far right.

The upper constant term has the required invariance at these right arguments as well. For $q \in F^\times$, any adelic h, and any such exterior compact k, the root relation and the product formula give

$$\begin{aligned} \theta_{\upsilon,N}(a(q)hk^{-1}) &= \int_{F\backslash\mathbb{A}_F} \theta_\upsilon(a(q)n(x/q)hk^{-1})\,\mathrm{d}x \\ &= \theta_{\upsilon,N}(h). \end{aligned} \tag{25.7}$$

Indeed $n(x)a(q) = a(q)n(x/q)$; rational automorphy removes $a(q)$, and right invariance removes k^{-1} at the far right. The substitution $x \mapsto x/q$ preserves Haar measure on $F\backslash\mathbb{A}_F$. This argument applies in particular when $h_{\mathfrak{p}} = wn(v/p)$. Finally $\chi_{\mathfrak{p}}(\gamma) = 1$, so $\alpha_{\mathfrak{p},j}(\gamma v) = \alpha_{\mathfrak{p},j}(v)$, including the zero digit. Reindexing by $\boldsymbol{v}' = \gamma\boldsymbol{v}$ in Equations (25.6) and (25.7) proves the asserted descent of the combined sums before and after w.

Recall $X_l = \{y \in F_S^\times : |y|_S = l\}$ and the sheet measures from Equation (23.15). The quotient X_l/Γ is compact; choose a measurable fundamental set $\Omega_1 \subset X_1$ and put $\Omega_l = \ell_l \Omega_1$, using the norm section from that disintegration. Every integral over X_l/Γ means the integral over Ω_l with the restricted sheet measure, equivalently the quotient measure induced using counting measure on Γ. The factors $\mathbf{1}_B([y])$ and $L(y)^{-1}$ also descend, since $[\gamma] = 1$ in $\mathcal{V}$ and $L(\gamma) = 1$. The resulting integrands are integrable on the quotient: the digit sums are finite and the factors are smooth or locally constant.

Both combined constant terms have zero weighted integral. To see this, rotate y by a complex number z of modulus one at one complex place. In the plus sum this acts by the left diagonal $a(z)$; in the minus sum it acts by $a(z^{-1})$. The constant term belongs to the torus induction in Proposition 23.1. Its complex inducing law is radial for every right argument, so both sums are unchanged, including at the exterior arguments $wn(v/p)$. The class condition is unchanged because $\mathbb{C}^\times$ is sixth-divisible, while $L(y)^{-1}$ transforms by a nontrivial angular character. Haar invariance therefore gives

$$\int_{X_l/\Gamma} \mathbf{1}_B([y]) L(y)^{-1} \mathscr{D}^{\pm}_{m,N}(y) \, \mathrm{d}_l^\times y = 0. \tag{25.8}$$

We may now define the quotient average

$$\begin{aligned} \mathcal{T}_m(l) &= \int_{X_l/\Gamma} \mathbf{1}_B([y]) L(y)^{-1} \mathscr{D}^{+}_{m}(y) \, \mathrm{d}_l^\times y \\ &= \int_{X_l/\Gamma} \mathbf{1}_B([y]) L(y)^{-1} \mathscr{D}^{-}_{m}(y) \, \mathrm{d}_l^\times y. \end{aligned} \tag{25.9}$$

Lemma 25.1 (Theta representation of the smoothed sum)**.** *Let V and $\mathcal{K}$ be the common unrotated profile and Gaussian averaging weight from Lemma 23.5. For $l, Z > 0$,*

$$\mathcal{T}_m(l) = \sum_{\substack{\mathfrak{c} \text{ squarefree} \\ \mathfrak{n}}} \frac{\gamma_2(\mathfrak{c})}{q_{\mathfrak{c}}^{1/2} q_{\mathfrak{n}}} L(A) \prod_{\mathfrak{p} \mid \mathfrak{m}} \chi_{\mathfrak{p}}(A)^{j_{\mathfrak{p}}} V(q_{\mathfrak{A}} l), \tag{25.10}$$

$$B_m(Z) = \int_0^\infty \mathcal{K}(T) \mathcal{T}_m(T/Z) \, \frac{\mathrm{d}T}{T}. \tag{25.11}$$

Here $\mathfrak{A} = \mathfrak{cn}^3$ and $A = cn^3$ is an allowed product generator. The ideal sums are over integral exterior ideals. Both formulas converge absolutely after the sheet integrations, and the second converges absolutely after the T-integration.

Proof. Use the Fourier expansion and normalized factorization in Equations (23.2) and (23.3) in the plus average. The constant term vanishes by Equation (25.8). At a nonconstant rational frequency ν, a row-prime factor is

$$W_{\mathfrak{p}}(a(\nu) n(v/p)) = \psi_{\mathfrak{p}}(\nu v/p) W_{\mathfrak{p}}(a(\nu)).$$

Spherical support forces $\nu \in R$; summing the digits with Equation (25.4) supplies $\prod_{\mathfrak{p}\mid\mathfrak{m}} \chi_{\mathfrak{p}}(\nu)^{j_{\mathfrak{p}}}$.

Write $f_v(v)$ for the chosen unrotated S-Whittaker function. The S-factor of the frequency is $\mathcal{C}_S(\nu, y) f_v(\nu y)$. Its cutoff has $[\nu y] = 1$ in $\mathcal{V}$. Together with $[y] \in B$, this gives $[\nu] \in B$. By alternation and Equation (23.9),

$$\mathcal{C}_S(\nu, y) = H_S(\nu, \nu^{-1})^2 H_S(\nu, \nu y)^2 = 1$$

on that support. If A is an allowed generator of the exterior ideal of ν, then $\nu/A \in U_S$. The direct sum $E \oplus B$ and the injection $U_6 \longrightarrow E$ now imply $\nu = \gamma A$ for $\gamma \in \Gamma$. The grouped exterior factor is unchanged by this multiplication, by the local unit rule and $\chi_{\mathfrak{p}}(\gamma) = 1$.

We justify the orbit substitution with an absolute integral first. The sets $\gamma\Omega_l$ partition X_l up to null sets. Tonelli, multiplicative Haar invariance, and Equation (23.16) give for $\sigma > 1/2$

$$\begin{aligned} \sum_{\gamma \in \Gamma} \int_{\Omega_l} |f_v(\gamma A y)| \, \mathrm{d}_l^\times y &= \int_{X_{q_{\mathfrak{A}} l}} |f_v(v)| \, \mathrm{d}_{q_{\mathfrak{A}} l}^\times v \\ &\ll_\sigma (q_{\mathfrak{A}} l)^{-\sigma}. \end{aligned} \tag{25.12}$$

The signed substitution $v = \gamma A y$ is therefore legitimate. Since $[y] = [A]^{-1}[v]$ and $L(y)^{-1} = L(A)L(v)^{-1}$, it gives

$$\begin{aligned}
&\sum_{\gamma\in\Gamma}\int_{\Omega_l} \mathbf{1}_B([y])L(y)^{-1}f_v(\gamma A y)\ \mathrm{d}_l^{\times}y \\
&\quad = L(A)\int_{X_{q_{\mathfrak{A}}l}} \mathbf{1}_B([A]^{-1}[v])L(v)^{-1}f_v(v)\ \mathrm{d}_{q_{\mathfrak{A}}l}^{\times}v \\
&\quad = L(A)V(q_{\mathfrak{A}}l).
\end{aligned} \tag{25.13}$$

For the last equality, $[v] = 1$ on the cutoff, $[A] \in B$, and the angular characters cancel as in Equation (23.20). Thus the integral depends on A only through its norm. Multiplicative Haar measure introduces no norm Jacobian.

By Corollary 23.2, the exterior spherical coefficient vanishes unless $\mathfrak{A} = \mathfrak{c}\mathfrak{n}^3$ with $\mathfrak{c}$ square-free, and then equals $\gamma_2(\mathfrak{c})/(q_{\mathfrak{c}}^{1/2}q_{\mathfrak{n}})$. This includes primes shared by $\mathfrak{c}$ and $\mathfrak{n}$. Together with Equation (25.13) it proves Equation (25.10).

Here are the absolute bounds for the exchanges. Summing Equation (25.12) with the exterior magnitudes is at most $l^{-\sigma}$ times the convergent expression in Equation (22.3). Before evaluating the finite digit sums, their total absolute mass is at most $q_{\mathfrak{p}}^{1/2}$ for $j \neq 0$ and is less than 2 for $j = 0$. For a fixed row their finite product multiplies the same spherical majorant, which justifies that earlier exchange. Finally Lemma 23.5 gives every absolute moment of $\mathcal{K}$. The averaged absolute sum is bounded by a constant times

$$Z^{\sigma}M_{-\sigma}(\mathcal{K})$$

times the finite sum in Equation (22.3). We may therefore apply Equation (23.23) term by term. Equations (25.1) and (25.2) and the definition of B_m give Equation (25.11). □

25.2 The local Weyl calculation

The second expression in Equation (25.9) is now the starting point. At S, $wa(y) = a(y^{-1})w$ replaces the vector by a fixed rotation and the argument by y^{-1}. Its constant term has already been removed in Equation (25.8). At an exterior place with zero or no digit, w is compact. We compute the effect of each nonzero digit.

Lemma 25.2 (Local effect of reflection)**.** *Let $\mathfrak{p}$ be an exterior prime and put $W_e = W_{\mathfrak{p}}(a(p^e))$. For $j \in \{0,\dots,5\}$, a unit t, and an integer e, define*

$$\mathcal{R}_{\mathfrak{p},j,e}(t) = \sum_{v\in k_{\mathfrak{p}}^{\times}} \alpha_{\mathfrak{p},j}(v)W_{\mathfrak{p}}(a(p^{e-2}t)wn(v/p)).$$

It vanishes for $e < 0$. For $e \geq 0$,

$$\mathcal{R}_{\mathfrak{p},j,e}(t) = W_e\chi_{\mathfrak{p}}(t)^{4+2e}\sum_{v\in k_{\mathfrak{p}}^{\times}} \alpha_{\mathfrak{p},j}(v)\chi_{\mathfrak{p}}(v)^4\psi_{\mathfrak{p}}(p^{e-1}t/v). \tag{25.14}$$

For $j = 1$, it vanishes for $e > 0$, while

$$\begin{aligned}
\mathcal{R}_{\mathfrak{p},1,0}(t) &= \omega_{\mathfrak{p}}\chi_{\mathfrak{p}}(t), \qquad |\omega_{\mathfrak{p}}| = 1, \\
\omega_{\mathfrak{p}} &= \frac{\sum_{x\neq 0}\chi_{\mathfrak{p}}(x)^3\psi_{\mathfrak{p}}(x/p)}{\tau_{\mathfrak{p},1}}.
\end{aligned} \tag{25.15}$$

For all j, its magnitude at $e = 0$ is at most 1, and for $e > 0$ is at most $q_{\mathfrak{p}}^{1/2}|W_e|$. More precisely, for $e \geq 0$ put $s_{\mathfrak{p},j,e} = \mathcal{R}_{\mathfrak{p},j,e}(1)$ and choose the exponents modulo 6 by

$$b_{j,0} \equiv 8 - j, \qquad b_{j,e} \equiv 4 + 2e\ (e > 0), \qquad \mathcal{R}_{\mathfrak{p},j,e}(t) = s_{\mathfrak{p},j,e}\chi_{\mathfrak{p}}(t)^{b_{j,e}}. \tag{25.16}$$

For $e > 0$, $s_{\mathfrak{p},j,e} = 0$ unless $j = 4$, and it also vanishes for $e \equiv 2 \pmod 3$. The zero-digit term is available only for $j = 0$; at a frequency ν it is $(1 - q_{\mathfrak{p}}^{-1})W_{\mathfrak{p}}(a(\nu))$, with ordinary spherical support.

Proof. For a unit digit v, the exact lift identity in Equation (23.4), with $H = v/p$, is

$$wn(H) = n(H^{-1})a(H^{-2})k_v, \qquad k_v = a(-1)\bar{n}(p/v)$$

with k_v in the spherical compact group. Only the diagonal minus sign has been absorbed in k_v; its symbols are trivial because F contains μ_{12}. Commuting the upper root past $a(p^{e-2}t)$ gives the positive additive argument $\psi_{\mathfrak{p}}(p^{e-1}t/v)$. The effective torus law gives, for integers e_1, e_2 and units u, v,

$$\mathcal{C}_{\mathfrak{p}}(p^{e_1}u, p^{e_2}v) = \chi_{\mathfrak{p}}(u)^{2e_2}\chi_{\mathfrak{p}}(v)^{-2e_1}.$$

The unit–unit and uniformizer–uniformizer factors are trivial. Thus torus multiplication contributes

$$\mathcal{C}_{\mathfrak{p}}(p^{e-2}t, p^2v^{-2}) = \chi_{\mathfrak{p}}(t)^4\chi_{\mathfrak{p}}(v)^{4(e-2)}.$$

The remaining spherical argument is $p^e tv^{-2}$. It is zero for $e < 0$; otherwise Equation (23.6) contributes $\chi_{\mathfrak{p}}(t)^{2e}\chi_{\mathfrak{p}}(v)^{-4e}W_e$. Their product proves Equation (25.14).

For $e > 0$ the additive character is trivial. Unit-character orthogonality makes the digit sum zero unless $j = 4$, including the case $j = 0$. For every j its absolute value is at most $q_{\mathfrak{p}}^{1/2}$: use $|\alpha_{\mathfrak{p},j}(v)| = q_{\mathfrak{p}}^{-1/2}$ for $j \neq 0$ and $q_{\mathfrak{p}}^{-1}$ for $j = 0$. Its unit dependence is the power $4 + 2e$, and $W_e = 0$ for $e \equiv 2 \pmod 3$.

For $e = 0$, substitute $x = t/v$. If $j \neq 0$, this leaves the power $\chi_{\mathfrak{p}}(t)^{8-j}$, divided by $\tau_{\mathfrak{p},j}$, and the Gauss sum of $\chi_{\mathfrak{p}}^{j-4}$ against $\psi_{\mathfrak{p}}(x/p)$. That sum has magnitude $q_{\mathfrak{p}}^{1/2}$, except for $j = 4$, when it has magnitude 1. For $j = 1$ its character is quadratic and $8 - j \equiv 1 \pmod 6$, giving Equation (25.15). For $j = 0$, the same substitution leaves $\chi_{\mathfrak{p}}(t)^8$ times a nontrivial Gauss sum divided by $-q_{\mathfrak{p}}$, again of magnitude at most one. These observations prove Equation (25.16) and the remaining bounds. Finally, compactness of w gives the zero-digit assertion by spherical right invariance. □

For a row $\mathfrak{m} = \mathfrak{h}\mathfrak{d}$, call a prime active when its digit is nonzero. Every prime of $\mathfrak{d}$ is active, because there $j_{\mathfrak{p}} = 1$. For $\mathfrak{p} \mid \mathfrak{h}$ put $j_{\mathfrak{p}} = v_{\mathfrak{p}}(\mathfrak{h}) \bmod 6$. The exact set of possible active products in $\mathfrak{h}$ is

$$\mathcal{F}(\mathfrak{h}) = \{\mathfrak{f} \text{ squarefree} : \mathfrak{f} \mid \mathrm{rad}(\mathfrak{h}),\ \mathfrak{p} \mid \mathfrak{f} \text{ if } \ \mathfrak{p} \mid \mathfrak{h},\ j_{\mathfrak{p}} \neq 0\}. \tag{25.17}$$

Only a prime with $j_{\mathfrak{p}} = 0$ can be inactive. For $\mathfrak{f} \in \mathcal{F}(\mathfrak{h})$, its exact inactive scalar is

$$I_{\mathfrak{h},\mathfrak{f}} = \prod_{\substack{\mathfrak{p}\mid\mathfrak{h}\\ \mathfrak{p}\nmid\mathfrak{f}}}(1 - q_{\mathfrak{p}}^{-1}),$$

and powerfulness gives

$$\mathfrak{f} \mid \mathrm{rad}(\mathfrak{h}), \qquad q_{\mathfrak{f}}^2 \leq q_{\mathfrak{h}}. \tag{25.18}$$

Let $\mathcal{A}(\mathfrak{f})$ be the integral exterior ideals $\mathfrak{a}'_{\mathfrak{f}}$ supported on $\mathfrak{f}$, with every exponent congruent to 0 or 1 modulo 3; set $\mathcal{A}(1) = \{1\}$. This is a convenient common support: the actual coefficient $s_{\mathfrak{p},j_{\mathfrak{p}},e}$ remains zero in the locally vanishing cases of Lemma 25.2.

The local support shows that every frequency with a nonzero grouped exterior coefficient occurs among

$$\nu = \varepsilon\frac{a'_{\mathfrak{f}}c'(n')^3}{(df)^2}, \qquad \mathfrak{a}' = \mathfrak{a}'_{\mathfrak{f}}\mathfrak{c}'(\mathfrak{n}')^3, \qquad \varepsilon \in U_S, \tag{25.19}$$

where $\mathfrak{a}'_{\mathfrak{f}} \in \mathcal{A}(\mathfrak{f})$, $\mathfrak{c}'$ is squarefree, and $(\mathfrak{c}'\mathfrak{n}', \mathfrak{d}\mathfrak{f}) = 1$. The displayed elements $d, f, a'_{\mathfrak{f}}, c', n'$ are allowed generators. The ideals $\mathfrak{c}'$ and $\mathfrak{n}'$ may share primes, and primes of $\mathfrak{h}$ omitted from $\mathfrak{f}$ may occur in either. The valuation-one formula forces numerator exponent zero at each prime of $\mathfrak{d}$. Away from $\mathfrak{d}\mathfrak{f}$, the valuations $3k$ and $3k + 1$ determine $\mathfrak{c}'$ and $\mathfrak{n}'$ uniquely. The remaining ambiguity is the displayed S-unit, since allowed product generators are unique modulo Γ.

For later absolute estimates define

$$\beta_{\mathfrak{p}}(e)=\begin{cases}1, & e=0,\\ q_{\mathfrak{p}}^{-k}, & e=1+3k,\ k\ge 0,\\ q_{\mathfrak{p}}^{1/2-k}, & e=3k,\ k\ge 1,\\ 0, & \text{otherwise,}\end{cases}\qquad \beta_{\mathfrak{f}}(\mathfrak{a}'_{\mathfrak{f}})=\prod_{\mathfrak{p}\mid\mathfrak{f}}\beta_{\mathfrak{p}}(v_{\mathfrak{p}}(\mathfrak{a}'_{\mathfrak{f}})). \tag{25.20}$$

The local lemma and the spherical magnitudes give $|s_{\mathfrak{p},j,e}|\le\beta_{\mathfrak{p}}(e)$. The other exterior magnitudes together are at most $q_{\mathfrak{c}'}^{-1/2}q_{\mathfrak{n}'}^{-1}$.

25.3 The full reflected expansion

We now assemble these local factors. The S-integrals form a fixed family of one-variable profiles. Let $f_{w,\upsilon}(v)$ be the S-Whittaker function of the chosen vector after the fixed Weyl rotation. Choose once a presentation

$$f_{w,\upsilon}(v)=\sum_{\ell=1}^{J_\upsilon}\prod_{a\in S}f_{\upsilon,\ell,a}(v_a).$$

The finite S-functional and the fixed cutoff vector provide such presentations by Proposition 23.1 and Equation (23.14); their lengths are bounded over the finite row-class family. All fixed scalar coefficients are included in the displayed local factors. Write H_S also for its induced pairing on $\mathcal{V}$. For $\kappa\in\mathcal{V}$, define

$$\begin{aligned}H_{\upsilon,\kappa,\ell}(x)&=\int_{X_x}\mathbf{1}_B(\kappa[v]^{-1})H_S(\kappa,[v])^2L_\upsilon(v)\\&\qquad\qquad\times\prod_{a\in S}f_{\upsilon,\ell,a}(v_a)\ \mathrm{d}_x^\times v,\\ \Phi_{\upsilon,\kappa,\ell}(x)&=\int_0^\infty\mathcal{K}(T)H_{\upsilon,\kappa,\ell}(x/T)\,\frac{\mathrm{d}T}{T},\\ \Phi_{\upsilon,\kappa}(x)&=\sum_{\ell=1}^{J_\upsilon}\Phi_{\upsilon,\kappa,\ell}(x).\end{aligned}\tag{25.21}$$

The class multiplier is bounded and depends only on $\mathcal{V}$, by Equation (23.9). The local bounds for the fixed rotated vectors and Lemmas 23.4 and 23.5 therefore give one $\alpha>0$ such that, for every $N>0$ and integer $j\ge 0$,

$$|(x\partial_x)^j\Phi_{\upsilon,\kappa}(x)|\ll_{N,j}\min(x^\alpha,x^{-N})\qquad(x>0). \tag{25.22}$$

The constants are uniform in $\upsilon\in U_6$ and $\kappa\in\mathcal{V}$. This is a fixed finite family: the functional and tensor terms have been combined in $\Phi_{\upsilon,\kappa}$, while the bounded class function stays inside its integral. In particular those fixed coefficients will not enter the local majorant $\beta_{\mathfrak{f}}$.

Choose representatives ε_u for the separate frequency-unit classes $u\in U_6$. For

$$\mathfrak{t}=(\mathfrak{h},\mathfrak{f},\mathfrak{a}'_{\mathfrak{f}},\mathfrak{n}',u),\qquad \mathfrak{f}\in\mathcal{F}(\mathfrak{h}),\quad \mathfrak{a}'_{\mathfrak{f}}\in\mathcal{A}(\mathfrak{f}),\quad(\mathfrak{n}',\mathfrak{f})=1,$$

and class slices $s_d,s_c\in B$, put

$$\kappa_6(\mathfrak{t};s_d,s_c)=[\varepsilon_u][a'_{\mathfrak{f}}]s_c[n']^3s_d^{-2}[f]^{-2}\in\mathcal{V}. \tag{25.23}$$

This is a selected class, rather than a further summation index. For $\mathfrak{p}\mid\mathfrak{f}$, abbreviate

$$e_{\mathfrak{p}}=v_{\mathfrak{p}}(\mathfrak{a}'_{\mathfrak{f}}),\qquad s_{\mathfrak{p}}=s_{\mathfrak{p},j_{\mathfrak{p}},e_{\mathfrak{p}}},\qquad b_{\mathfrak{p}}=b_{j_{\mathfrak{p}},e_{\mathfrak{p}}}.$$

With $\upsilon,\mathfrak{t},s_d,s_c$ fixed, the next three factors organize the dependence on the two varying ideals: ω depends on neither, ρ depends on the row $\mathfrak{d}$ alone, and λ depends on the column $\mathfrak{c}'$ alone. All

three are independent of Z. The quadratic symbol and norm profile will contain the remaining joint dependence on $\mathfrak{d}$ and $\mathfrak{c}'$. For $L = L_\upsilon$ and $\varepsilon = \varepsilon_u$, define

$$\begin{aligned}\omega_{\upsilon,\mathfrak{t}} &= I_{\mathfrak{h},\mathfrak{f}} L(a'_\mathfrak{f})^{-1} L(n')^{-3} L(f)^2 \\ &\quad \times \prod_{\mathfrak{p}\mid\mathfrak{f}} s_\mathfrak{p}\chi_\mathfrak{p}\left(\varepsilon \frac{a'_\mathfrak{f}}{p^{e_\mathfrak{p}}} (n')^3 (f/p)^{-2}\right)^{b_\mathfrak{p}}.\end{aligned} \tag{25.24}$$

For a squarefree $\mathfrak{d}$ with $(\mathfrak{d}, \mathfrak{h}\mathfrak{n}') = 1$ and $[d] = s_d$, put

$$\begin{aligned}\rho_{\upsilon,\mathfrak{t},s_d}(\mathfrak{d}) &= L(d)^2 \\ &\quad \times \prod_{\mathfrak{p}\mid\mathfrak{d}} \omega_\mathfrak{p}\chi_\mathfrak{p}\Big(\varepsilon a'_\mathfrak{f} (n')^3 (d/p)^{-2} f^{-2}\Big) \\ &\quad \times \prod_{\mathfrak{p}\mid\mathfrak{f}} \chi_\mathfrak{p}(d)^{-2b_\mathfrak{p}},\end{aligned} \tag{25.25}$$

and put it equal to zero otherwise. For a squarefree $\mathfrak{c}'$ with $(\mathfrak{c}', \mathfrak{f}) = 1$ and $[c'] = s_c$, put

$$\begin{aligned}\lambda_{\upsilon,\mathfrak{t},s_c}(\mathfrak{c}') &= L(c')^{-1} \prod_{\mathfrak{p}\mid\mathfrak{f}} \chi_\mathfrak{p}(c')^{b_\mathfrak{p}} \\ &\quad \times \prod_{\mathfrak{q}\mid\mathfrak{c}'} g_{2,\mathfrak{q}}\chi_\mathfrak{q}\Big(\varepsilon a'_\mathfrak{f} (c'/q) f^{-2}\Big)^2,\end{aligned} \tag{25.26}$$

and put it equal to zero otherwise. All character arguments in these formulas are units at their primes on the stated supports. Thus negative character powers are well-defined. These definitions give one coherent row function and one coherent column function for each frozen tuple and class slice.

Lemma 25.3 (The reflected expansion)**.** *For every nonzero $m \in R$ and $Z > 0$, let $(m)_S = \mathfrak{h}\mathfrak{d}$ and $\upsilon = \upsilon_m$. Then*

$$\begin{aligned}B_m(Z) = &\sum_{\mathfrak{f}\in\mathcal{F}(\mathfrak{h})} \sum_{\substack{\mathfrak{a}'_\mathfrak{f}\in\mathcal{A}(\mathfrak{f}) \\ \mathfrak{n}' \\ (\mathfrak{n}',\mathfrak{f})=1}} \sum_{\substack{u\in U_6 \\ s_d,s_c\in B}} \frac{\omega_{\upsilon,\mathfrak{t}}}{q_{\mathfrak{n}'}} \\ &\times \rho_{\upsilon,\mathfrak{t},s_d}(\mathfrak{d}) \Bigg[\sum_{\mathfrak{c}'} \sum_{\text{squarefree}} \frac{\lambda_{\upsilon,\mathfrak{t},s_c}(\mathfrak{c}')}{q_{\mathfrak{c}'}^{1/2}} \chi_\mathfrak{d}(c')^3 \\ &\times \Phi_{\upsilon,\kappa_6(\mathfrak{t};s_d,s_c)}\left(\frac{Z q_{\mathfrak{a}'_\mathfrak{f}} q_{\mathfrak{c}'} q_{\mathfrak{n}'}^3}{q_\mathfrak{d}^2 q_\mathfrak{f}^2}\right)\Bigg].\end{aligned} \tag{25.27}$$

All ideal sums are over integral exterior ideals, and the expansion is absolutely convergent. The coefficients satisfy

$$|\omega_{\upsilon,\mathfrak{t}}| \le \beta_\mathfrak{f}(\mathfrak{a}'_\mathfrak{f}), \qquad |\rho_{\upsilon,\mathfrak{t},s_d}| \le 1, \qquad |\lambda_{\upsilon,\mathfrak{t},s_c}| \le 1.$$

The row-unit class υ is fixed; u is the independently summed frequency-unit class.

Proof. We use the minus average in Equation (25.9) and its nonconstant Fourier terms. We first supply the absolute bound needed before unfolding any signed unit orbit. Define

$$F^{\mathrm{abs}}_{w,\upsilon}(v) = \sum_{\ell=1}^{J_\upsilon} \prod_{a\in S} |f_{\upsilon,\ell,a}(v_a)|, \qquad H^{\mathrm{abs}}_{w,\upsilon}(x) = \int_{X_x} F^{\mathrm{abs}}_{w,\upsilon}(v)\ \mathrm{d}^\times_x v.$$

This is an absolute profile of Lemma 23.4; in particular $H^{\mathrm{abs}}_{w,\upsilon}(x) \ll_\sigma x^{-\sigma}$ for $\sigma > 1/2$, uniformly over the fixed row-class family. For any $\nu_0 \in F^\times$, the sets $\gamma\Omega_l^{-1}$ partition $X_{1/l}$. Tonelli and

multiplicative Haar invariance give

$$\sum_{\gamma\in\Gamma}\int_{\Omega_l} F^{\mathrm{abs}}_{w,v}(\gamma\nu_0/y)\ \mathrm{d}_l^{\times}y = H^{\mathrm{abs}}_{w,v}(|\nu_0|_S/l) \ll_\sigma (|\nu_0|_S/l)^{-\sigma}. \tag{25.28}$$

For a fixed row, this also justifies the exchanges before evaluating the digits. At one nonzero digit, the absolute Whittaker factor in the proof of Lemma 25.2 is $|W_e|$ at numerator valuation $e \geq 0$. Summing the absolute digit weights costs at most $q_{\mathfrak{p}}^{1/2}$ for $j \neq 0$ and is less than 2 for $j = 0$. For $\sigma > 1/2$, the weighted spherical series is

$$\sum_{e\geq 0} |W_e| q_{\mathfrak{p}}^{-\sigma e} = \frac{1 + q_{\mathfrak{p}}^{-1/2-\sigma}}{1 - q_{\mathfrak{p}}^{-1-3\sigma}} < \infty.$$

The denominator shift $e - 2$ multiplies this by $q_{\mathfrak{p}}^{2\sigma}$. There are only finitely many row primes, so their resulting product is finite for the fixed row. Here the numerator valuations at the primes of $\mathfrak{d}$ are still summed. The product of ordinary spherical series at the other primes is bounded by $\zeta_F^S(\sigma + 1/2)\zeta_F^S(3\sigma + 1)$. Partitioning the units into the finitely many U_6-classes and applying Equation (25.28) bounds the absolute sheet sum. The T-average then uses $M_\sigma(\mathcal{K}) < \infty$. Thus these preliminary exchanges are valid for each fixed row. The uniform estimates below use the coefficients after the finite digit sums have been evaluated.

Partition the zero and nonzero digits by $\mathfrak{f} \in \mathcal{F}(\mathfrak{h})$, apply Lemma 25.2, and parameterize the frequencies by Equation (25.19). For the fixed representative $\varepsilon = \varepsilon_u$, write

$$\nu = \gamma\nu_0, \qquad \nu_0 = \varepsilon\frac{a'_{\mathfrak{f}}c'(n')^3}{(df)^2}, \qquad \gamma \in \Gamma.$$

The valuations and then the frequency-unit class give a unique such parameterization on the stated support. At $\mathfrak{p} \mid \mathfrak{d}$, the local unit of $\nu_0 = p^{-2}t$ is

$$t = \varepsilon a'_{\mathfrak{f}}c'(n')^3(d/p)^{-2}f^{-2}.$$

At $\mathfrak{p} \mid \mathfrak{f}$, the local unit of $\nu_0 = p^{e_{\mathfrak{p}}-2}t$ is

$$t = \varepsilon(a'_{\mathfrak{f}}/p^{e_{\mathfrak{p}}})c'(n')^3d^{-2}(f/p)^{-2}.$$

At $\mathfrak{q} \mid \mathfrak{c}'$, put $k = v_{\mathfrak{q}}(\mathfrak{n}')$. The spherical valuation is $1+3k$, and its unit exponent is 2 modulo 6. The cube unit from $(n'/q^k)^3$ disappears from the symbol. A prime of $\mathfrak{n}'$ outside $\mathfrak{c}'$ has unit exponent divisible by 6. Consequently the exact grouped exterior factor on $(\mathfrak{d}, \mathfrak{h}\mathfrak{n}') = (\mathfrak{c}', \mathfrak{f}) = (\mathfrak{d}, \mathfrak{c}') = 1$ is

$$\begin{aligned}\mathcal{E}_{\mathfrak{t}}(\mathfrak{d}, \mathfrak{c}') = \frac{I_{\mathfrak{h},\mathfrak{f}}}{q_{\mathfrak{c}'}^{1/2}q_{\mathfrak{n}'}} &\times \prod_{\mathfrak{p}\mid\mathfrak{d}} \omega_{\mathfrak{p}}\chi_{\mathfrak{p}}\Big(\varepsilon a'_{\mathfrak{f}}c'(n')^3(d/p)^{-2}f^{-2}\Big) \\ &\times \prod_{\mathfrak{p}\mid\mathfrak{f}} s_{\mathfrak{p}}\chi_{\mathfrak{p}}\Big(\varepsilon(a'_{\mathfrak{f}}/p^{e_{\mathfrak{p}}})c'(n')^3d^{-2}(f/p)^{-2}\Big)^{b_{\mathfrak{p}}} \\ &\times \prod_{\mathfrak{q}\mid\mathfrak{c}'} g_{2,\mathfrak{q}}\chi_{\mathfrak{q}}\Big(\varepsilon a'_{\mathfrak{f}}(c'/q)(df)^{-2}\Big)^2.\end{aligned} \tag{25.29}$$

The factor $I_{\mathfrak{h},\mathfrak{f}}$ is present whether or not an inactive prime occurs in $\mathfrak{c}'$ or $\mathfrak{n}'$. Thus the formula includes their overlap at such a prime as well as every other $\mathfrak{c}'$–$\mathfrak{n}'$ overlap. The local zeros $s_{\mathfrak{p}} = 0$ keep the broader active numerator support exact.

We identify the only factor involving both row and column variables. At a pair $\mathfrak{p} \mid \mathfrak{d}$, $\mathfrak{q} \mid \mathfrak{c}'$, the row factor supplies $\chi_{\mathfrak{p}}(q)$, and the denominator d^{-2} in the spherical column factor supplies $\chi_{\mathfrak{q}}(p)^{-4}$. Allowed-generator reciprocity gives

$$\chi_{\mathfrak{p}}(q)\chi_{\mathfrak{q}}(p)^{-4} = \chi_{\mathfrak{p}}(q)^{1-4} = \chi_{\mathfrak{p}}(q)^{3}. \tag{25.30}$$

Multiplying over the pairs gives $\chi_{\mathfrak{d}}(c')^3$. There is no further pairing from a prime occurring only in $\mathfrak{n}'$, since its unit exponent is divisible by 6. The separated S-character is

$$L(\nu_0)^{-1} = L(a'_{\mathfrak{f}})^{-1}L(c')^{-1}L(n')^{-3}L(d)^2L(f)^2,$$

because $L(\varepsilon) = 1$. Comparing Equation (25.29) with Equations (25.24) to (25.26) therefore gives, on the indicated class slices,

$$\mathcal{E}_{\mathfrak{t}}(\mathfrak{d}, \mathfrak{c}')L(\nu_0)^{-1} = \frac{\omega_{v,\mathfrak{t}}\rho_{v,\mathfrak{t},s_d}(\mathfrak{d})\lambda_{v,\mathfrak{t},s_c}(\mathfrak{c}')}{q_{\mathfrak{c}'}^{1/2}q_{\mathfrak{n}'}}\chi_{\mathfrak{d}}(c')^3. \tag{25.31}$$

The local bounds prove the three coefficient bounds in the lemma. The zero extension of the quadratic symbol makes the right side zero when $\mathfrak{d}$ and $\mathfrak{c}'$ meet, as the valuation-one support requires. A prime shared by $\mathfrak{d}$ and $\mathfrak{n}'$ must instead be removed by the separate row support of ρ. Together with $(\mathfrak{n}', \mathfrak{f}) = 1$ and the column support of λ, these conditions reproduce exactly $(\mathfrak{c}'\mathfrak{n}', \mathfrak{d}\mathfrak{f}) = 1$.

It remains to evaluate the S-integral. The grouped digit sums $\mathcal{R}_{\mathfrak{p},j,e}(t)$ are unchanged under $t \mapsto \gamma t$ for $\gamma \in \Gamma$, by Equation (25.16). The same is true of each ordinary spherical factor and of the zero-digit factor, since $\chi_{\mathfrak{p}}(\gamma) = 1$. This includes active $j = 0$ terms and the zero digit. Thus the grouped exterior coefficient may be taken outside the signed Γ-orbit. That orbit is absolutely dominated by Equation (25.28).

The reflected S-factor is $\mathcal{C}_S(\gamma\nu_0, y^{-1})f_{w,v}(\gamma\nu_0/y)$. In the legitimate substitution $v = \gamma\nu_0/y$, $[y] = [\nu_0][v]^{-1}$ and $L(y)^{-1} = L(\nu_0)^{-1}L(v)$. Alternation and Equation (23.9) give

$$\mathcal{C}_S(\gamma\nu_0, y^{-1}) = H_S(\gamma\nu_0, v/(\gamma\nu_0))^2 = H_S(\nu_0, v)^2.$$

We obtain the exact identity

$$\begin{aligned}\sum_{\gamma\in\Gamma}\int_{\Omega_l} \mathbf{1}_B([y])L(y)^{-1}\mathcal{C}_S(\gamma\nu_0, y^{-1})f_{w,v}(\gamma\nu_0/y)\ \mathrm{d}_l^{\times}y \\ = L(\nu_0)^{-1}\int_{X_{|\nu_0|_S/l}} \mathbf{1}_B([\nu_0][v]^{-1})H_S(\nu_0, v)^2L(v)f_{w,v}(v)\ \mathrm{d}^{\times}_{|\nu_0|_S/l}v.\end{aligned} \tag{25.32}$$

The sheet norm is

$$\frac{|\nu_0|_S}{l} = \frac{q_{\mathfrak{a}'_{\mathfrak{f}}}q_{\mathfrak{c}'}q_{\mathfrak{n}'}^3}{q_{\mathfrak{d}}^2q_{\mathfrak{f}}^2l};$$

there is no norm Jacobian. Inside the integral ν_0 occurs only through its class in $\mathcal{V}$. On the supports of the two class indicators this class is exactly $\kappa_6(\mathfrak{t}; s_d, s_c)$. Thus no continuous shape of the row or column generator remains inside the profile.

Insert the two class partitions, use Equation (25.31), and split the fixed S-function into its ℓ-terms. Putting $l = T/Z$ in Equation (25.11) produces precisely the combined profile in Equation (25.21), and hence Equation (25.27). For a fixed row class and fixed ideal data $(\mathfrak{h}, \mathfrak{f}, \mathfrak{a}'_{\mathfrak{f}}, \mathfrak{n}')$, there are only $|U_6||B|^2$ choices of the frequency unit and two slices. After those choices, the class κ_6 is determined and there is one combined profile. The active sets and the ideal sums are the explicit outer sums in the expansion.

We finish by verifying the uniform absolute bound after grouping. For $\sigma \geq 0$,

$$E_{\mathfrak{p}}(\sigma) := \sum_{e\geq 0}\beta_{\mathfrak{p}}(e)q_{\mathfrak{p}}^{-\sigma e} = 1 + \frac{q_{\mathfrak{p}}^{-\sigma} + q_{\mathfrak{p}}^{-1/2-3\sigma}}{1 - q_{\mathfrak{p}}^{-1-3\sigma}} \leq 3 + \sqrt{2}. \tag{25.33}$$

In particular, for every $\delta_1 > 0$, $\prod_{\mathfrak{p}|\mathfrak{f}} E_{\mathfrak{p}}(\sigma) \ll_{F,\delta_1} q_{\mathfrak{f}}^{\delta_1}$, uniformly for $\sigma \geq 0$. Absorb the finitely many primes of norm below $(3+\sqrt{2})^{1/\delta_1}$ into the constant and bound each remaining factor by $q_{\mathfrak{p}}^{\delta_1}$.

Put $Q = q_{\mathfrak{a}'_{\mathfrak{f}}} q_{\mathfrak{c}'} q_{\mathfrak{n}'}^3$. Dropping squarefreeness and the unfrozen coprimality restrictions in the nonnegative majorant gives

$$\begin{aligned} \mathcal{D}_{\mathfrak{f}}(\sigma) &:= \sum_{\mathfrak{a}'_{\mathfrak{f}},\mathfrak{c}',\mathfrak{n}'} \beta_{\mathfrak{f}}(\mathfrak{a}'_{\mathfrak{f}}) q_{\mathfrak{c}'}^{-1/2} q_{\mathfrak{n}'}^{-1} Q^{-\sigma} \\ &\leq \zeta_F^S(\sigma+1/2)\zeta_F^S(3\sigma+1) \prod_{\mathfrak{p}|\mathfrak{f}} E_{\mathfrak{p}}(\sigma) < \infty \qquad (\sigma > 1/2). \end{aligned} \tag{25.34}$$

The frozen ideal still ranges over $\mathcal{A}(\mathfrak{f})$. For $t_{\mathfrak{d}} = q_{\mathfrak{d}}^2 q_{\mathfrak{f}}^2 / Z$, the absolute profile above dominates every class multiplier and yields

$$\begin{aligned} \int_0^\infty |\mathcal{K}(T)| \sum_{\mathfrak{a}'_{\mathfrak{f}},\mathfrak{c}',\mathfrak{n}'} \beta_{\mathfrak{f}}(\mathfrak{a}'_{\mathfrak{f}}) q_{\mathfrak{c}'}^{-1/2} q_{\mathfrak{n}'}^{-1} H_{w,v}^{\mathrm{abs}}\Big(\frac{Q}{t_{\mathfrak{d}} T}\Big) \frac{\mathrm{d}T}{T} \\ \ll_\sigma t_{\mathfrak{d}}^\sigma M_\sigma(\mathcal{K}) \mathcal{D}_{\mathfrak{f}}(\sigma) < \infty. \end{aligned} \tag{25.35}$$

This justifies all exchanges in the grouped expansion. Its constants are uniform in the ideal data, apart from the displayed product of local majorants. Since $\mathcal{F}(\mathfrak{h})$ is finite for each fixed row, the full expansion is absolutely convergent. □

The case $m = 0$ remains the separate identity $B_0(Z) = P_G(1/Z)$ from the coefficient-family section.

The expansion also displays the length used in the second moment. On dyads $P \leq q_{\mathfrak{h}} < 2P$ and $D \leq q_{\mathfrak{d}} < 2D$, put

$$\begin{aligned} Q &= q_{\mathfrak{a}'_{\mathfrak{f}}} q_{\mathfrak{c}'} q_{\mathfrak{n}'}^3, \qquad & t_{\mathfrak{d}} &= \frac{q_{\mathfrak{d}}^2 q_{\mathfrak{f}}^2}{Z}, \\ \mathcal{L} &= \frac{8D^2 P}{Z}, \qquad & R_0 &= Z^\delta \mathcal{L}, \end{aligned} \tag{25.36}$$

where $\delta > 0$ will be chosen small. By Equation (25.18), $t_{\mathfrak{d}} \leq \mathcal{L}$. We shall retain $Q \leq R_0$, which in particular gives $q_{\mathfrak{c}'} \leq R_0$. If $R_0 < 1$, this range is empty because Q is the norm of an integral ideal. The tail estimate below justifies this truncation for every dyad, including that case.

25.4 The quadratic large sieve and the moment estimate

We use the quadratic large sieve established over $\mathbb{Q}$ by Heath-Brown and extended to Hecke families by Goldmakher–Louvel the quadratic large-sieve theorem. We state the latter result in the form needed for Equation (25.27) and verify its family hypotheses for the present generator convention.

Lemma 25.4 (Quadratic large sieve for the exterior symbols)**.** *For $C, D \geq 1$ and complex coefficients $z_{\mathfrak{c}}$ supported on squarefree exterior ideals with $q_{\mathfrak{c}} \leq C$, one has, for every $\epsilon > 0$,*

$$\sum_{\substack{\mathfrak{d} \text{ squarefree} \\ q_{\mathfrak{d}} \leq D}} \Bigg| \sum_{\substack{\mathfrak{c} \text{ squarefree} \\ q_{\mathfrak{c}} \leq C}} z_{\mathfrak{c}} \chi_{\mathfrak{d}}(c)^3 \Bigg|^2 \ll_{F,S,\epsilon} (C+D)(CD)^\epsilon \sum_{\mathfrak{c}} |z_{\mathfrak{c}}|^2. \tag{25.37}$$

Arbitrary restrictions on the rows and separate restrictions or unitary factors on the columns are allowed by deleting rows and changing the coefficients.

Proof. We verify the quadratic Hecke family axioms before applying the quadratic Hecke large-sieve theorem. Choose an ordinary integral ideal $\mathfrak{q}_S$ supported on the finite places of S, with a positive exponent m_v at each such place large enough that $1 + \mathfrak{p}_v^{m_v} \mathcal{O}_v \subset F_v^{\times 2}$. Every local quadratic character at v is trivial on this subgroup, so its conductor exponent is at most m_v. Exterior ideals are exactly the ordinary ideals prime to $\mathfrak{q}_S$, with the same norms.

For a squarefree exterior ideal $\mathfrak{c}$, let $\Xi_{\mathfrak{c}}$ be the primitive Hecke character of the possibly trivial quadratic Kummer extension $F(\sqrt{c})/F$. It is independent of the allowed generator, since two such generators differ by a sixth power of an S-unit. At an exterior prime its local conductor exponent is one if the prime divides $\mathfrak{c}$ and zero otherwise: at the odd residue characteristics under consideration an odd valuation gives a tamely ramified quadratic extension, whereas a unit gives an unramified or trivial extension. Thus

$$\operatorname{cond}(\Xi_{\mathfrak{c}}) = \mathfrak{c}\mathfrak{s}_{\mathfrak{c}}, \qquad \mathfrak{s}_{\mathfrak{c}} \mid \mathfrak{q}_S.$$

Its infinite type is trivial because all infinite places of F are complex. On an exterior ideal $\mathfrak{d}$,

$$\Xi_{\mathfrak{c}}(\mathfrak{d}) = \chi_{\mathfrak{d}}(c)^3.$$

This includes the zero at a shared prime, which is in the conductor of $\Xi_{\mathfrak{c}}$. These are the primitive global characters, not a character with additional artificially deleted Euler factors.

Let B_2 be the image of B in $F_S^{\times}/F_S^{\times 2}$ and set $\mathrm{cl}_2(\mathfrak{c}) = [c]_{S,2} \in B_2$. Allowed generators multiply modulo U_S^6, so this extends to a homomorphism on the group of all fractional exterior ideals. It has finite image. It is also ray-class data for $\mathfrak{q}_S$: for a principal exterior ideal (x) its value is the B_2-projection of $[x]_{S,2}$ in the direct sum induced by $E \oplus B$; if $x \equiv 1 \pmod{\mathfrak{q}_S}$, all its local classes at S are squares, so this projection is trivial.

For coprime squarefree exterior ideals, the exact reciprocity of allowed generators gives

$$\Xi_{\mathfrak{c}}(\mathfrak{d}) = \chi_{\mathfrak{d}}(c)^3 = \chi_{\mathfrak{c}}(d)^3 = \Xi_{\mathfrak{d}}(\mathfrak{c}).$$

Thus the finite-class reciprocity factor in that definition may be taken identically one. If $\mathfrak{c}_1, \mathfrak{c}_2$ are coprime squarefree exterior ideals and have the same cl_2, their quotient c_1/c_2 is a square at every place of S. The primitive character inducing $\Xi_{\mathfrak{c}_1}\overline{\Xi_{\mathfrak{c}_2}}$ is consequently trivial there. At every prime of $\mathfrak{c}_1\mathfrak{c}_2$ it is tamely ramified with conductor exponent one, and at every other exterior prime it is unramified. Its *global* conductor is therefore exactly $\mathfrak{c}_1\mathfrak{c}_2$. The product here is understood as the primitive inducing character, with the convention specified in the quadratic Hecke large-sieve theorem. This verifies the conductor-product axiom, including the part at S.

These conductor and reciprocity checks make this a quadratic Hecke family. The quadratic Hecke large-sieve theorem therefore gives (25.37). Apply it to the full family and then delete unwanted rows or set unwanted coefficients to zero; a single class slice need not itself be a Hecke family. □

Proof of Proposition 22.1. The initial element-to-ideal reduction lets us fix one row-unit class $\upsilon \in U_6$ and sum over ideal rows at a bounded cost. Fix dyads $P \le q_{\mathfrak{h}} < 2P$ and $D \le q_{\mathfrak{d}} < 2D$ which meet the row set. They satisfy

$$PD \ll_{\mathscr{K}} Z^{M_0}, \qquad P, D \ge 1. \tag{25.38}$$

There are $O_F(P^{1/2})$ powerful ideals on the first dyad. In fact every powerful ideal has a unique expression $\mathfrak{r}^2\mathfrak{s}^3$ with $\mathfrak{s}$ squarefree (choose its exponent as the parity of each prime exponent). The fixed-field bound for the number of ideals of norm at most t is $O_F(t)$, hence

$$\#\{\mathfrak{h} \text{ powerful} : q_{\mathfrak{h}} < 2P\} \ll_F P^{1/2} \sum_{\mathfrak{s}} q_{\mathfrak{s}}^{-3/2} \ll_F P^{1/2}. \tag{25.39}$$

Fix $\mathfrak{h}$, and use $Q, t_{\mathfrak{d}}, \mathcal{L}, R_0$ from Equation (25.36). For each $\mathfrak{f} \in \mathcal{F}(\mathfrak{h})$ and $\mathbf{s} = (u, s_d, s_c) \in U_6 \times B^2$, split its partial sum in Equation (25.27) according to $Q \le R_0$ or $Q > R_0$. Denote these sums, which still include $\mathfrak{a}'_{\mathfrak{f}}, \mathfrak{n}', \mathfrak{c}'$, by $\mathrm{Ret}_{\mathfrak{f},\mathbf{s},\mathfrak{d}}$ and $\mathrm{Tail}_{\mathfrak{f},\mathbf{s},\mathfrak{d}}$. Sum them over the admitted $\mathfrak{f}$ and the finite $\mathbf{s}$ to define $\mathrm{Ret}_{\mathfrak{h},\mathfrak{d}}$ and $\mathrm{Tail}_{\mathfrak{h},\mathfrak{d}}$. Thus, on a row with class υ,

$$B_m(Z) = \mathrm{Ret}_{\mathfrak{h},\mathfrak{d}} + \mathrm{Tail}_{\mathfrak{h},\mathfrak{d}}\,.$$

Set both parts to zero outside the squarefree rows $\mathfrak{d}$ coprime to $\mathfrak{h}$. The threshold R_0 is common to the dyad and independent of the individual row. When $R_0 < 1$ the retained part is empty, so no large sieve at a length below one is required.

We first bound the discarded terms, uniformly in this case as well. Put $\mathcal{J} = \max(Z^\delta, \mathcal{L}^{-1})$, and fix $N > \sigma > 1/2$. For $Q > R_0$ we have both $Q/t_\mathfrak{d} > Z^\delta$ and $Q/t_\mathfrak{d} \geq \mathcal{L}^{-1}$, because $Q \geq 1$. The profile estimate therefore gives

$$\mathbf{1}_{Q>R_0} |\Phi_{\upsilon,\kappa}(Q/t_\mathfrak{d})| \ll_N \mathcal{J}^{-(N-\sigma)} (t_\mathfrak{d}/Q)^\sigma.$$

Summing absolutely with Equation (25.34), for one active product and one fixed $\mathbf{s}$, yields

$$|\operatorname{Tail}_{\mathfrak{f},\mathbf{s},\mathfrak{d}}| \ll_{N,\sigma,F} \mathcal{L}^\sigma \mathcal{J}^{-(N-\sigma)} \zeta_F^S(\sigma + 1/2) \zeta_F^S(3\sigma + 1) \prod_{\mathfrak{p}\mid\mathfrak{f}} E_\mathfrak{p}(\sigma). \tag{25.40}$$

This one estimate includes all column, cube, and frozen-exponent tails. If $R_0 < 1$, then $\mathcal{J} = \mathcal{L}^{-1}$ and the corresponding entire partial sum is $O(\mathcal{L}^N \prod E_\mathfrak{p}(\sigma))$.

For every $\delta_1 > 0$, Equation (25.33) also gives

$$\sum_{\mathfrak{f}\mid\operatorname{rad}(\mathfrak{h})} \prod_{\mathfrak{p}\mid\mathfrak{f}} E_\mathfrak{p}(\sigma) = \prod_{\mathfrak{p}\mid\mathfrak{h}} (1 + E_\mathfrak{p}(\sigma)) \ll_{F,\delta_1} q_\mathfrak{h}^{\delta_1} \qquad (\sigma \geq 0). \tag{25.41}$$

The same finite-small-prime argument proves this inequality; including all divisors only adds active choices that may actually be forbidden. Sum Equation (25.40) over them and over the fixed finite class list, square, and use $O_F(D)$ for the number of row ideals on the second dyad and Equation (25.39). The total squared tail is

$$\ll_{N,\sigma,F,\delta_1} P^{1/2+2\delta_1} D\, \mathcal{L}^{2\sigma} \mathcal{J}^{-2(N-\sigma)}. \tag{25.42}$$

Here Equation (25.38) implies $P \ll Z^{M_0}$ and $\mathcal{L} \ll Z^{2M_0-1}$, while $\mathcal{J} \geq Z^\delta$. Given any $A_0 > 0$, choosing N with

$$2\delta(N - \sigma) > A_0 + 2M_0\delta_1 + 2\sigma(2M_0 - 1)$$

makes Equation (25.42) $O(Z^{-A_0} P^{1/2} D)$. This proves the promised absolute domination of the discarded range, including all dyads for which $R_0 < 1$.

It remains to bound the retained terms when $R_0 \geq 1$. Fix $\mathfrak{f}, \mathbf{s}, \mathfrak{a}'_\mathfrak{f}, \mathfrak{n}'$. Then $\kappa = \kappa_6(\mathfrak{t}; s_d, s_c)$ is fixed, while $\rho_{\upsilon,\mathfrak{t},s_d}$ and $\lambda_{\upsilon,\mathfrak{t},s_c}$ are coherent functions of all rows and columns in the two slices. Split $\mathfrak{c}'$ into dyads $C \leq q_{\mathfrak{c}'} < 2C$, $C \geq 1$. Put

$$v_0 = \frac{Z q_{\mathfrak{a}'_\mathfrak{f}} q_{\mathfrak{n}'}^3 C}{D^2 q_\mathfrak{f}^2}, \qquad x = \log(q_{\mathfrak{c}'}/C) - 2\log(q_\mathfrak{d}/D).$$

Then $x \in [-2\log 2, \log 2]$, and the profile in Equation (25.27) is $\Phi_{\upsilon,\kappa}(v_0 e^x)$. Choose a fixed $\varphi \in C_c^\infty(\mathbb{R})$ equal to one on that interval and set $g_{v_0}(x) = \varphi(x)\Phi_{\upsilon,\kappa}(v_0 e^x)$. Applying Equation (23.19) with this cutoff gives

$$\int_\mathbb{R} |\widehat{g}_{v_0}(\tau)|\, \mathrm{d}\tau \ll_N \min(v_0^\alpha, v_0^{-N}) \ll_N 1. \tag{25.43}$$

Fourier inversion now separates the norm factors as $(q_{\mathfrak{c}'}/C)^{i\tau}(q_\mathfrak{d}/D)^{-2i\tau}$ with the bounded integral cost in Equation (25.43). The retained indicator $\mathbf{1}_{q_{\mathfrak{a}'_\mathfrak{f}} q_{\mathfrak{c}'} q_{\mathfrak{n}'}^3 \leq R_0}$ is a column coefficient after the two frozen ideals are fixed.

Use Minkowski's inequality for this Fourier integral and apply Lemma 25.4 to each τ. Separate row factors have modulus at most one, and the column square mass is

$$\sum_{C \leq q_{\mathfrak{c}'} < 2C} q_{\mathfrak{c}'}^{-1} \ll_F 1$$

by ideal counting. For any $\delta_2 > 0$, the row ℓ^2 norm of this one column dyad, before the factor $\beta_{\mathfrak{f}}(\mathfrak{a}'_{\mathfrak{f}})/q_{\mathfrak{n}'}$, is therefore at most

$$\ll_{N,\delta_2} (D+C)^{1/2}(DC)^{\delta_2/2}\min(v_0^{\alpha}, v_0^{-N}). \tag{25.44}$$

The restrictions $(\mathfrak{d}, \mathfrak{h}\mathfrak{n}') = 1$ merely delete rows. The zero extension of the symbol already treats shared row and column primes, so no joint restriction was removed in applying the sieve.

For the retained range, $C \le R_0$ and $q_{\mathfrak{n}'} \le R_0^{1/3}$. There are $O(\log(2R_0))$ such column dyads, and partial summation of the fixed-field ideal count gives

$$\sum_{q_{\mathfrak{n}'} \le R_0^{1/3}} q_{\mathfrak{n}'}^{-1} \ll_F \log(2R_0).$$

Drop the final minimum in Equation (25.44) and use Minkowski successively for the column dyads, $\mathfrak{n}'$, and $\mathfrak{a}'_{\mathfrak{f}}$. Equation (25.33) at $\sigma = 0$ shows that their retained row norm, for fixed $\mathfrak{h}, \mathfrak{f}, \mathbf{s}$, is

$$\ll_{\delta_2} \left(\prod_{\mathfrak{p}\mid\mathfrak{f}} E_{\mathfrak{p}}(0)\right) \log^2(2R_0)(D+R_0)^{1/2}(DR_0)^{\delta_2/2}.$$

Sum these norms over the admitted $\mathfrak{f}$ and the finite $\mathbf{s}$, using Equation (25.41) for the former. Square the resulting estimate for each fixed $\mathfrak{h}$, and only then sum the squares over $\mathfrak{h}$. Since each row has a unique powerful part, no triangle inequality across different powerful parts is necessary. Equation (25.39) yields

$$\sum_{\substack{\mathfrak{h}\ \text{powerful} \\ P \le q_{\mathfrak{h}} < 2P}} \sum_{D \le q_{\mathfrak{d}} < 2D} |\operatorname{Ret}_{\mathfrak{h},\mathfrak{d}}|^2 \ll_{\delta_1,\delta_2} P^{1/2+2\delta_1}\log^4(2R_0)(D+R_0)(DR_0)^{\delta_2}. \tag{25.45}$$

All the finite unit and S-class choices have only multiplied this by a fixed constant.

We make the power losses explicit. On the range Equation (25.38),

$$D + R_0 \ll Z^{\delta}\left(D + \frac{D^2P}{Z}\right), \qquad DR_0 \ll Z^{3M_0-1+\delta}, \qquad P \ll Z^{M_0}.$$

For any $\delta_3 > 0$, $\log^4(2R_0) \ll_{\delta_3} Z^{\delta_3}$ because $1 \le R_0 \ll Z^{2M_0-1+\delta}$. Choose the positive auxiliaries so that

$$\delta + 2M_0\delta_1 + (3M_0 - 1 + \delta)\delta_2 + \delta_3 < \epsilon.$$

Then Equation (25.45) is bounded by the right side of Equation (22.5). The retained mass is zero for $R_0 < 1$, and Equation (25.42) is smaller than that right side in all cases after N is chosen as above. Adding the two parts and restoring the bounded element multiplicities proves Equation (22.5).

Finally, for every nonempty dyad, $P \ge 1$ and Equation (25.38) give

$$P^{1/2}D \le PD \ll Z^{M_0} \le Z^{2M_0-1}, \qquad P^{1/2}\frac{D^2P}{Z} = \frac{(PD)^2}{ZP^{1/2}} \ll Z^{2M_0-1},$$

where the first comparison uses $M_0 > 1$. There are only $O((\log Z)^2)$ possible dyads. Applying the dyadic estimate with a smaller preliminary loss absorbs this factor and proves Equation (22.6). Multiplying $\mathscr{K}$ by any element of a fixed compact norm-one set only enlarges the fixed compact shape set used in the initial multiplicity bound, which proves the asserted uniformity. □

26 The Poisson probe and its Euler product

We insert the theta sum $B_m(Z)$ of Equation (22.2) into an additive average. The reflected second moment bounds this average from above. Poisson summation will give a second expression whose principal frequency contains the reciprocal of the target L-function. The use of an auxiliary additive polynomial while retaining the unrestricted cube factors follows the method of the coefficient reflection method; the calculation here is over the fixed field F and uses its sextic characters.

Keep the field F, the set S, the target η, and the allowed generators fixed as in Section 21. In particular $\vartheta = \overline{\Lambda}\,\overline{G}\,\eta$, and every character power is zero on nonunits, including the zeroth power. The scale notation a, b, M_0, h_0, X, Y is fixed in Equation (22.4). All implied constants in this and the next section may depend on the fixed F, S, η and the fixed weights. Exponents displayed as numerical constants do not have that dependence. An arbitrary positive power loss may always be chosen smaller before an estimate is applied.

26.1 Weights and the estimate from reflection

Choose a nonzero nonnegative function $W_1 \in C_c^\infty((0,\infty))$, and set

$$\widehat{W}_1(w) = \int_0^\infty W_1(t) t^w \, \frac{dt}{t}.$$

Choose a tensor Bruhat–Schwartz function W_0 on F_S, compactly supported in $F_S^\times$, such that

$$\widehat{W}_0(0) = 0, \qquad M(1/6) \neq 0, \qquad M(z) = \int_{F_S^\times} \widehat{W}_0(y) |y|_S^z \, d^\times y. \tag{26.1}$$

The condition $\widehat{W}_0(0) = 0$ will remove the zero additive frequency after Poisson summation. The condition $M(1/6) \neq 0$ will ensure that the weight does not kill the residue at $z = 1/6$ in that expansion. These are separate from the support condition on W_0, which removes the original row $m = 0$ from the probe. Here the Fourier sign is the one fixed in Section 21. We explain the choice and the analytic properties that will be needed. At each place the local zeta functional equation gives, for $0 < s < 1$,

$$\int_{F_v^\times} \widehat{W}_v(y) |y|_v^s \, d^\times y = c_v(s) \int_{F_v^\times} W_v(y) |y|_v^{1-s} \, d^\times y, \qquad c_v(1/6) \neq 0;$$

the nonzero constant incorporates the chosen local measures and Fourier sign. At one complex place take a smooth annular function whose additive integral is zero but whose pairing with $|y|_v^{5/6} d^\times y$ is nonzero. Such a function is obtained by subtracting suitable multiples of two smooth bumps supported in different radial subannuli: the additive integral is a constant multiple of the pairing with $|y|_v d^\times y$, and the two radial weights are not proportional. At all other places take nonnegative nonzero annular functions. The local identity then proves both requirements in Equation (26.1).

For every $\sigma > 0$, each local integral of $|\widehat{W}_v(y)||y|_v^\sigma$ is finite. At a finite place this follows by summing the geometric series of shells near zero and using compact support at infinity; at a complex place it follows from boundedness at zero and Schwartz decay at infinity. The same assertions hold after any number of derivatives in a complex logarithmic norm. Thus M is holomorphic for $\operatorname{Re} z > 0$, and integration by parts in that norm gives, for every compact interval $J \subset (0,\infty)$ and every $A > 0$,

$$|M(\sigma + i\tau)| \ll_{J,A} (1 + |\tau|)^{-A} \quad (\sigma \in J). \tag{26.2}$$

The analogous estimate holds for $\widehat{W}_1$ on every fixed real strip.

For an exterior ideal $\mathfrak{s}$, with an allowed generator s, put

$$g_{\mathfrak{s}}(k) = \sum_{d \bmod \mathfrak{s}} \chi_{\mathfrak{s}}(d) \mathrm{e}(kd/s).$$

The value is independent of the allowed generator: a change by ϵ^6, $\epsilon \in U_S$, is absorbed by the same change of the residue variable. Use product multiplicative Haar measures for which the norm decomposition of $F_S^\times$ has the form

$$d^\times y = \frac{dT}{T}\,dt, \qquad y = t\ell_T, \quad t \in F_S^1.$$

The measure on F_S^1/U_S^6 below is the quotient measure induced by dt. This choice also fixes the normalization in M. Define

$$I(X,Y,Z) = \int_{F_S^1/U_S^6} \frac{1}{Y} \sum_{\mathfrak{s}} W_1(q_{\mathfrak{s}}/Y) \frac{1}{\sqrt{q_{\mathfrak{s}} X}} \sum_{m \in R} q_{\mathfrak{s}}^{-1/2} g_{\mathfrak{s}}(-m) W_0\left(\frac{m}{t_0 \ell_{q_{\mathfrak{s}} X}}\right) B_m(Z)\,dt_0. \tag{26.3}$$

All undecorated ideal sums in this section are over nonzero integral exterior ideals. The summand for $m = 0$ vanishes because W_0 is supported in $F_S^\times$. For $\epsilon \in U_S^6$, $\chi_{\mathfrak{A}}(\epsilon) = \chi_{\mathfrak{s}}(\epsilon) = 1$; changing m to ϵm shows that the integrand descends to the quotient. That quotient is compact by Lemma 21.2, and we use representatives in a fixed bounded subset of F_S^1. For each such representative the m-sum has finite support. The Gaussian weight in B_m makes its ideal sum absolutely convergent.

We first record the additive estimate in the form also used later for conductor characters. If $\mathfrak{s} = R$, its one residue class is understood to give the single fraction 0.

Lemma 26.1 (Separated additive fractions)**.** *Let K be a fixed compact subset of F_S, and let $N, V \geq 2$. Choose balanced generators s for the exterior ideals with $q_{\mathfrak{s}} \leq V$. Let $\mathcal{P}$ be any subset of the distinct points*

$$\left\{ \frac{d}{s} + R : q_{\mathfrak{s}} \leq V,\ d \in (R/\mathfrak{s})^\times \right\} \subset F_S/R.$$

For any complex b_α,

$$\sum_{m \in R \cap \ell_N K} \left| \sum_{\alpha \in \mathcal{P}} b_\alpha \mathrm{e}(m\alpha) \right|^2 \ll_{F,S,K} (N + V^2) \sum_{\alpha \in \mathcal{P}} |b_\alpha|^2. \tag{26.4}$$

The dual inequality, with the sums over m and α interchanged, has the same constant.

Proof. Reduced fractions with distinct denominator ideals are distinct modulo R. Indeed, at a prime dividing its denominator the valuation of a reduced fraction is the negative of the denominator valuation; at any other exterior prime it is nonnegative. If the denominator valuations differ, their difference therefore has negative valuation there. If all agree, congruence modulo R says the residue classes agree. For distinct fractions and any $r \in R$, the numerator in

$$\frac{d}{s} - \frac{d'}{s'} + r = \frac{ds' - d's + rss'}{ss'}$$

is a nonzero element of R. The norm identity of Lemma 21.1 gives

$$\left| \frac{d}{s} - \frac{d'}{s'} + r \right|_S \geq (q_{\mathfrak{s}} q_{\mathfrak{s}'})^{-1} \geq V^{-2}. \tag{26.5}$$

For $r \in R \setminus \{0\}$ it also gives $|r|_S \geq 1$.

Put $d_F = [F : \mathbb{Q}]$ and $\delta = c \min(N^{-1/d_F}, V^{-2/d_F})$, with $c > 0$ fixed sufficiently small. There is a nonnegative Bruhat–Schwartz function Φ which is at least one on $\ell_N K$, has $\widehat{\Phi}(0) \ll \delta^{-d_F}$, and whose Fourier transform is supported in a product of fixed finite-place additive subgroups and complex disks of ordinary radius $C\delta$. Here is a direct construction. At each complex place take the inverse transform of a smooth bump supported in a disk, evaluated at δy_v. Its value stays bounded below when $\delta |y_v|$ is sufficiently small. At finite places use the inverse transform of the indicator of a sufficiently small compact open subgroup, whose annihilator contains the finite projection of K. The product, multiplied by a fixed constant and squared in modulus, majorizes the box. Fourier

support follows from convolution, and a change of scale gives integral $O(\delta^{-d_F})$. Shrinking c makes this support contain no translate by R of a difference of distinct points in $\mathcal{P}$, by Equation (26.5): the product of the complex squared absolute values on that support is $O(\delta^{d_F})$, and its finite-place product is bounded by a fixed constant. It contains no nonzero point of R for the same reason.

Expand the square after majorizing it by Φ, and apply Poisson summation on R. For $\alpha, \beta \in \mathcal{P}$,

$$\sum_{m \in R} \Phi(m) \mathrm{e}(m(\alpha - \beta)) = \sum_{r \in R} \widehat{\Phi}(r - \alpha + \beta) = \mathbf{1}_{\alpha=\beta} \widehat{\Phi}(0).$$

Since $\delta^{-d_F} \ll N + V^2$, this proves Equation (26.4). Equality of the operator norms of a finite matrix and its adjoint gives the dual assertion. □

Proposition 26.2. *For the weights above, for every $\epsilon > 0$ and $Z \geq 2$,*

$$I(Z^a, Z^b, Z) \ll_{F,S,\eta,W_0,W_1,\epsilon} Z^{(2M_0-1-b)/2+\epsilon} = Z^{933/2000+\epsilon}. \tag{26.6}$$

Proof. Write $v = q_{\mathfrak{s}}/Y$ and $y = m/(t_0 \ell_{XY})$. For v in the support of W_1, the support of $W_0(y/\ell_v)$ puts y in one fixed compact annular set. On this set the smooth function

$$v^{-1/2} W_1(v) W_0(y/\ell_v)$$

has a Fourier series in $\log v$ whose coefficients $c_j(y)$ satisfy $\sum_j \|c_j\|_\infty < \infty$, uniformly in y. To see this, take a compact interval containing the logarithmic support, extend by zero to a slightly larger period, and integrate twice by parts. All the required derivatives are bounded uniformly on the compact y-set. Thus we can separate this weight from the additive sums with a bounded total coefficient cost.

For each Fourier term, expansion of $g_{\mathfrak{s}}$ gives a coefficient of modulus $Y^{-1} q_{\mathfrak{s}}^{-1/2}$ at each reduced fraction d/s, with $q_{\mathfrak{s}} \asymp Y$. Their total squared mass is at most

$$Y^{-2} \sum_{q_{\mathfrak{s}} \asymp Y} q_{\mathfrak{s}}^{-1} |(R/\mathfrak{s})^\times| \ll_F Y^{-1},$$

using the fixed-field ideal count. Take $N = XY$ and $V \ll Y$ in Lemma 26.1. The bounded set of representatives t_0 only enlarges the fixed compact row box. By the triangle inequality for the Fourier series, the squared ℓ^2 norm in m of the whole additive factor in Equation (26.3), including $(q_{\mathfrak{s}} X)^{-1/2}$, is

$$\ll (XY)^{-1} \frac{XY + Y^2}{Y} \ll Y^{-1}.$$

The last inequality uses $b < a$. On the same enlarged compact annular row box, Proposition 22.1 gives

$$\sum_m |B_m(Z)|^2 \ll_\epsilon Z^{2M_0-1+\epsilon}.$$

Cauchy–Schwarz in m, followed by integration over the compact quotient, proves Equation (26.6), after decreasing the preliminary loss. □

26.2 Poisson summation and three Mellin variables

For an exterior ideal $\mathfrak{A}$ with allowed generator A, write

$$g_{\chi_{\mathfrak{A}}}(A, k) = \sum_{v \bmod \mathfrak{A}} \chi_{\mathfrak{A}}(v) \mathrm{e}(kv/A)$$

and, for $h \in R$, define

$$F(s, A, h) = \sum_{\substack{d \bmod \mathfrak{s} \\ h \equiv Ad \pmod{\mathfrak{s}}}} \chi_{\mathfrak{s}}(d)\, g_{\chi_{\mathfrak{A}}}\left(A, \frac{h - Ad}{s}\right). \tag{26.7}$$

The argument of the last Gauss sum is in R by the congruence. The definitions for the unit ideal mean a single residue and the value 1 for its character.

The finite Fourier transform of the row coefficient modulo $\mathfrak{s}\mathfrak{A} = sAR$ is exactly

$$\sum_{m \bmod \mathfrak{s}\mathfrak{A}} q_{\mathfrak{s}}^{-1/2} g_{\mathfrak{s}}(-m)\chi_{\mathfrak{A}}(m)\mathrm{e}(hm/(sA)) = q_{\mathfrak{s}}^{1/2} F(s, A, h). \tag{26.8}$$

In fact expand $g_{\mathfrak{s}}$, write $m = v + Aj$ with $v \bmod \mathfrak{A}$ and $j \bmod \mathfrak{s}$, and sum over j. The sum of $\mathrm{e}((h - Ad)j/s)$ is $q_{\mathfrak{s}}$ precisely when $h \equiv Ad \pmod{\mathfrak{s}}$, and is zero otherwise. The remaining sum over v is the one in Equation (26.7). This argument does not require $\mathfrak{s}$ and $\mathfrak{A}$ to be coprime.

The dilation $t_0\ell_{q_{\mathfrak{s}}X}$ has additive modulus $q_{\mathfrak{s}}X$, and the lattice sAR has index $q_{\mathfrak{s}}q_{\mathfrak{A}}$. Apply Poisson summation to the m-sum and use Equation (26.8). Its dilation divided by the index is $X/q_{\mathfrak{A}}$; after the two square root factors in the probe the multiplier is

$$(q_{\mathfrak{s}}X)^{-1/2}\frac{X}{q_{\mathfrak{A}}}q_{\mathfrak{s}}^{1/2} = \frac{X^{1/2}}{q_{\mathfrak{A}}}.$$

The Fourier argument at h is

$$t_0\ell_{q_{\mathfrak{s}}X}\frac{h}{sA}, \qquad \left|\ell_{q_{\mathfrak{s}}X}\frac{h}{sA}\right|_S = \frac{Xq_h}{q_{\mathfrak{A}}}.$$

These norm factors will determine all three Mellin exponents.

For $\epsilon \in U_S^6$, changing d to ϵd and the Gauss residue variable v to $\epsilon^{-1}v$ in Equation (26.7) shows $F(s, A, \epsilon h) = F(s, A, h)$. Sum the nonzero h's in such orbits and unfold the quotient F_S^1/U_S^6. If

$$V_0(T) = \int_{F_S^1} \widehat{W}_0(t\ell_T)\, dt,$$

the orbit of h contributes $F(s, A, h)V_0(Xq_h/q_{\mathfrak{A}})$. Multiplication by the norm-one part of $\ell_{q_{\mathfrak{s}}X}h/(sA)$ preserves the sheet measure, so no shape factor remains. This unfolding is absolutely justified: for each $\sigma > 0$, the local integrability used above gives an absolute sheet majorant $O_\sigma(T^{-\sigma})$, by using one complex weighted supremum and the other weighted local integrals. Taking $\sigma > 1$ sums the $q_u^{-\sigma}q_{\mathfrak{r}}^{-6\sigma}$ majorant, while the Gaussian handles the ideal $\mathfrak{A}$; this permits Tonelli's theorem in the orbit sum. The zero frequency contributes nothing because $\widehat{W}_0(0) = 0$.

By the measure disintegration, the Mellin transform of V_0 is M. The same weighted derivative majorants in the complex norm variables justify differentiation of the sheet integral in $\log T$, so $V_0(T)$ is smooth for $T > 0$. Its absolute weighted integrability and Equation (26.2) then give pointwise Mellin inversion by Fourier inversion in $\log T$ on each line $\mathrm{Re}\, z > 0$. By Equation (21.7), write the orbits uniquely as $h = ur^6$, where $u \in \mathcal{U}$ and $\mathfrak{r}$ is an exterior ideal with allowed generator r. The principal row $u = 1$ consists of the surviving nonzero sixth-power frequencies $h = r^6$. Hence the precise Poisson expression is

$$\begin{aligned} I(X, Y, Z) = \frac{X^{1/2}}{Y} \sum_{\substack{\mathfrak{c}\ \text{squarefree} \\ \mathfrak{n},\mathfrak{s}}} \frac{\gamma_2(\mathfrak{c})\vartheta(\mathfrak{A})}{q_{\mathfrak{c}}^{1/2} q_{\mathfrak{n}} q_{\mathfrak{A}}} P_G(q_{\mathfrak{A}}/Z) W_1(q_{\mathfrak{s}}/Y) \\ \times \sum_{u\in\mathcal{U}} \sum_{\mathfrak{r}} F(s, A, ur^6) V_0\left(\frac{Xq_u q_{\mathfrak{r}}^6}{q_{\mathfrak{A}}}\right), \qquad \mathfrak{A} = \mathfrak{c}\mathfrak{n}^3. \end{aligned} \tag{26.9}$$

Here P_G is the Gaussian of Equation (22.1).

Use Mellin inversion for P_G, W_1, V_0, with transforms $e^{t^2}, \widehat{W}_1(w), M(z)$, respectively. The factors involving $\mathfrak{A}$ are $q_{\mathfrak{A}}^{-1}q_{\mathfrak{A}}^{-t}q_{\mathfrak{A}}^{z}$. Set $t = \xi + z - 1$. We obtain

$$\begin{aligned} I(X, Y, Z) = \frac{1}{(2\pi i)^3} \int_{(4)}\int_{(4)}\int_{(2)} X^{1/2-z} Z^{\xi+z-1} Y^{w-1} e^{(\xi+z-1)^2} M(z)\widehat{W}_1(w) \\ \times \sum_{u\in\mathcal{U}} q_u^{-z}\mathcal{F}_u(\xi, w, z)\, dz\, dw\, d\xi, \end{aligned} \tag{26.10}$$

where the innermost line is $\operatorname{Re} z = 2$, the middle line is $\operatorname{Re} w = 4$, and the outer line is $\operatorname{Re} \xi = 4$, and

$$\mathcal{F}_u(\xi, w, z) = \sum_{\substack{\mathfrak{c} \text{ squarefree} \\ \mathfrak{n},\mathfrak{s},\mathfrak{r}}} \gamma_2(\mathfrak{c})\vartheta(\mathfrak{A})F(s, A, ur^6)q_{\mathfrak{c}}^{-1/2-\xi} q_{\mathfrak{n}}^{-1-3\xi} q_{\mathfrak{s}}^{-w} q_{\mathfrak{r}}^{-6z}, \qquad \mathfrak{A} = \mathfrak{c}\mathfrak{n}^3. \tag{26.11}$$

All interchanges used for Equation (26.10) are in absolute convergence. For example, $|F(s, A, h)| \leq q_{\mathfrak{s}} q_{\mathfrak{A}}$, $|\gamma_2(\mathfrak{c})| = 1$, and the ideal count, including the finitely many S-unit classes for u, give absolute convergence on these lines. The factors $M(z)$ and $\widehat{W}_1(w)$ have arbitrary vertical polynomial decay, while the Gaussian has Gaussian decay in $\operatorname{Im}(\xi + z)$. Changing height variables to $(\operatorname{Im}(\xi + z), \operatorname{Im} z, \operatorname{Im} w)$ proves absolute integrability as well.

26.3 The local correction

We extend the notation $L^S(s, \eta)$ from Section 22 to a Hecke character on exterior primes: the factors at S are omitted, and any additional zero prime values are retained in the Euler product. Thus, for example,

$$L^S(w, \chi_\bullet(u)) = \prod_{\mathfrak{p} \notin S} (1 - \chi_{\mathfrak{p}}(u) q_{\mathfrak{p}}^{-w})^{-1} \quad (\operatorname{Re} w > 1).$$

As before, $\zeta_F^S(s) = \prod_{\mathfrak{p} \notin S} (1 - q_{\mathfrak{p}}^{-s})^{-1}$.

Proposition 26.3. *For every $u \in \mathcal{U}$, the series in Equation* (26.11) *has the Euler factorization*

$$\mathcal{F}_u(\xi, w, z) = \frac{\zeta_F^S(6z) L^S(w, \chi_\bullet(u))}{L^S(\xi, \eta\overline{\chi_\bullet(u)})} \mathcal{H}_u(\xi, w, z) \tag{26.12}$$

initially in absolute convergence. The correction $\mathcal{H}_u$ is holomorphic in open neighborhoods of both real-part regions

$$\begin{aligned} &\operatorname{Re}\xi \geq .998, \quad \operatorname{Re} w \geq .003, \quad \operatorname{Re} z \geq .4, \\ &\operatorname{Re}\xi \geq .998, \quad \operatorname{Re} w \geq .95, \quad \operatorname{Re} z \geq .16. \end{aligned} \tag{26.13}$$

For every $\epsilon > 0$ it satisfies

$$|\mathcal{H}_u(\xi, w, z)| \ll_{F,S,\eta,\epsilon} q_u^{\epsilon}$$

there, uniformly in all heights. If S contains every prime of norm at most a sufficiently large fixed constant, then $\mathcal{H}_1$ is bounded and nowhere zero on both neighborhoods.

The two regions serve different contour movements in the next section. The first permits the small and intermediate nonprincipal rows to place w at .003 and z at .4, while ξ can reach .998 once the reciprocal denominator has been controlled. The second contains the principal numerator poles $w = 1$, $z = 1/6$ and the nearby lines $w = .95$, $z = .16$ used to isolate their double residue. The nowhere-zero conclusion for $u = 1$ will ensure that its correction does not cancel a singularity of $1/L^S(\xi, \eta)$.

Proof. We first factor the finite expression, keeping the units that Chinese remaindering introduces. Use products of allowed prime generators for s, A, r. They differ from any other allowed choices by sixth powers of S-units, which have no effect on the expressions. At an exterior prime $\mathfrak{p}$, with generator p and norm $Q = q_{\mathfrak{p}}$, write

$$\begin{gathered} t = v_{\mathfrak{p}}(\mathfrak{A}) = e_0 + 3l, \quad e_0 = v_{\mathfrak{p}}(\mathfrak{c}) \in \{0, 1\}, \quad l = v_{\mathfrak{p}}(\mathfrak{n}), \quad k = v_{\mathfrak{p}}(\mathfrak{s}), \\ j = v_{\mathfrak{p}}(u), \quad m' = v_{\mathfrak{p}}(\mathfrak{r}), \quad J = j + 6m'. \end{gathered}$$

Write $s = p^k s_p$, $A = p^t A_p$, and $ur^6 = p^J h_p$; the subscripts mean that the full p-power has been omitted. These three subscripted quantities are units at $\mathfrak{p}$. Put $\psi_{\mathfrak{p}}^\circ(x) = \psi_{\mathfrak{p}}(-x)$. For $x \in F$, the global additive product gives $\mathrm{e}(x) = \prod_{\mathfrak{p} \notin S} \psi_{\mathfrak{p}}^\circ(x)$, with only finitely many nontrivial factors.

Expand the inner Gauss sum in Equation (26.7) and apply the Chinese remainder decomposition to each residue ring. Before changing variables, the local factor of $F(s, A, ur^6)$ is

$$\sum_{\substack{d \bmod p^k \\ p^J h_p \equiv p^t A_p d \pmod{p^k}}} \chi_{\mathfrak{p}}(d)^k \sum_{v \bmod p^t} \chi_{\mathfrak{p}}(v)^t \psi_{\mathfrak{p}}^{\circ}\left(\frac{(p^J h_p - p^t A_p d)v}{p^{k+t} s_p A_p}\right).$$

Here an absent modulus p^0 has one residue and character value 1. For a positive modulus the character retains its unit mask, even when its exponent is divisible by six. Now make the changes

$$d = (h_p/A_p)d', \qquad v = (s_p A_p/h_p)v'.$$

They are bijections of the relevant residue rings. The congruence becomes $p^J \equiv p^t d' \pmod{p^k}$, and the local additive argument becomes $(p^J - p^t d')v'/p^{k+t}$. The unit multiplier removed by these changes is exactly

$$\chi_{\mathfrak{p}}(h_p)^{k-t} \chi_{\mathfrak{p}}(s_p)^t \chi_{\mathfrak{p}}(A_p)^{t-k}. \tag{26.14}$$

There are no hidden coprimality assumptions here: these changes only use the units left after omitting the full p-powers. Moreover $\chi_{\mathfrak{p}}(h_p) = \chi_{\mathfrak{p}}(u/p^j)$, because the remaining factors of r^6 are sixth powers.

For completeness define the stripped local sum by

$$C_{t,k}(J) = \sum_{\substack{d \bmod p^k \\ p^J \equiv p^t d \pmod{p^k}}} \chi_{\mathfrak{p}}(d)^k \sum_{v \bmod p^t} \chi_{\mathfrak{p}}(v)^t \psi_{\mathfrak{p}}^{\circ}\left(\frac{(p^J - p^t d)v}{p^{k+t}}\right). \tag{26.15}$$

When $k = 0$ the outer modulus is one, its character is one, and its single residue may be taken to be zero. The analogous convention applies to $t = 0$ in the inner sum. When $k > 0$ or $t > 0$, the corresponding character is supported on units even if its exponent is divisible by six. These conventions distinguish the absent modulus from the zeroth power character on a nontrivial modulus.

We check explicitly that the remaining factors in different primes separate. Expand Equation (26.14) using the product generators, and consider an unordered pair of distinct primes $\mathfrak{p}, \mathfrak{p}'$. In their common symbol $\chi_{\mathfrak{p}}(p') = \chi_{\mathfrak{p}'}(p)$, the contributions from $\chi_{\mathfrak{p}}(s_p)^t$ and $\chi_{\mathfrak{p}}(A_p)^{t-k}$, together with the analogous factors at $\mathfrak{p}'$, have exponent

$$tk' + t'k + (t-k)t' + (t'-k')t = 2tt'.$$

In particular all s-A interactions cancel. By Equation (21.10), the Chinese remainder factors in $\gamma_2(\mathfrak{c})$ add $4e_0e_0'$. Their total is

$$2(e_0 + 3l)(e_0' + 3l') + 4e_0e_0' \equiv 0 \pmod 6.$$

Thus all cross-prime terms cancel, including when both moduli have powers of the same primes; those shared powers were retained inside Equation (26.15). The h_p-factors remain precisely as characters of u/p^j. If $\theta_{\mathfrak{p}} = \chi_{\mathfrak{p}}(u/p^j) \in \mu_6$, the exact local factor is consequently

$$\begin{aligned}\mathcal{F}_{u,\mathfrak{p}}(\xi, w, z) = \sum_{\substack{e_0=0,1 \\ l,k,m' \geq 0}} &\gamma_2(\mathfrak{p})^{e_0} \vartheta(\mathfrak{p})^{e_0+3l} \theta_{\mathfrak{p}}^{k-e_0-3l} C_{e_0+3l,k}(j+6m') \\ &\times Q^{-e_0(1/2+\xi)-l(1+3\xi)-kw-6m'z}.\end{aligned} \tag{26.16}$$

It follows initially by absolute convergence that $\mathcal{F}_u = \prod_{\mathfrak{p} \notin S} \mathcal{F}_{u,\mathfrak{p}}$.

Here are explicit evaluations of every $C_{t,k}(J)$. Put

$$T_v(y) = \sum_{x \bmod p} \chi_{\mathfrak{p}}(x)^v \psi_{\mathfrak{p}}^{\circ}(yx/p), \qquad \tau_v = T_v(1).$$

The unit mask is retained for $v \equiv 0 \pmod 6$. Thus

$$T_v(y) = \begin{cases} \overline{\chi_{\mathfrak{p}}(y)}^{v}\tau_v, & p \nmid y, \\ Q-1, & p \mid y,\ v \equiv 0 \pmod 6, \\ 0, & p \mid y,\ v \not\equiv 0 \pmod 6, \end{cases} \qquad |\tau_v| = \sqrt{Q}\ (v \not\equiv 0 \pmod 6), \quad \tau_0 = -1.$$

For $t = 0$ the value is $C_{0,0}(J) = 1$, and $C_{0,k}(J) = \mathbf{1}_{J=0}$ for $k > 0$. For $t > 0$ it is

$$\begin{aligned} C_{t,0}(J) &= \mathbf{1}_{J \ge t-1} Q^{t-1} T_t(p^{J-t+1}), \\ C_{t,1}(J) &= \mathbf{1}_{J \ge t} \tau_1 Q^{t-1} T_{t-1}(p^{J-t}), \\ C_{t,k}(J) &= 0 \qquad (k \ge 2). \end{aligned} \tag{26.17}$$

For the first formula sum the Q^{t-1} lifts of a residue $v \bmod p$ in the inner sum; they cancel unless $J \ge t-1$. For $k = 1$ the congruence first requires $J \ge 1$. Sum $d \bmod p$ before v. Because v is a unit, the sum is $\tau_1\overline{\chi_{\mathfrak{p}}(v)}$; $\chi_{\mathfrak{p}}(-1) = 1$ removes the sign. The remaining sum over lifts of v is zero unless $J \ge t$, and then gives the second formula. For $k \ge 2$, varying d by $p^{k-1}\lambda$, $\lambda \bmod p$, preserves the congruence and its unit character. For every unit v the phase varies by $\psi_{\mathfrak{p}}^{\circ}(-\lambda v/p)$, whose sum in λ is zero. This proves the last formula. In particular, for $t > 0$,

$$|C_{t,0}(J)| \le Q^t, \qquad |C_{t,1}(J)| \le Q^{t+1/2}, \tag{26.18}$$

with the support restrictions visible in Equation (26.17).

Suppose first that $\mathfrak{p} \nmid (u)R$, so $j = 0$. Set

$$V = Q^{-6z}, \quad W = \chi_{\mathfrak{p}}(u)Q^{-w}, \quad D_1 = \eta(\mathfrak{p})\overline{\chi_{\mathfrak{p}}(u)}Q^{-\xi}, \quad E_1 = \vartheta(\mathfrak{p})^6 Q^{4-6\xi}V.$$

The $t = 0$ values give $(1-V)^{-1} + W/(1-W)$: a positive k requires $J = 0$. The first squarefree theta term has $e_0 = 1$, $l = k = m' = 0$, hence $t = 1$, $J = 0$, and $C_{1,0}(0) = \tau_1$. Its contribution is

$$\begin{aligned} &\gamma_2(\mathfrak{p})\vartheta(\mathfrak{p})\overline{\chi_{\mathfrak{p}}(u)}Q^{-1/2-\xi}\tau_1 \\ &\quad = \gamma_1(\mathfrak{p})\gamma_2(\mathfrak{p})\vartheta(\mathfrak{p})\overline{\chi_{\mathfrak{p}}(u)}Q^{-\xi} = -D_1. \end{aligned}$$

Indeed $\tau_1 = \sqrt{Q}\,\gamma_1(\mathfrak{p})$ in the chosen additive sign, and Equation (21.9) gives $\gamma_1\gamma_2\vartheta = -\eta$. This negative term is the first nonconstant term of the reciprocal local L-factor $1 - D_1$; it is the reason that $L^S(\xi, \eta\overline{\chi_\bullet(u)})$ occurs in the denominator of Equation (26.12). We extract $(1-V)^{-1}(1-W)^{-1}(1-D_1)$, whose first two factors are those of $\zeta_F^S(6z)$ and $L^S(w, \chi_\bullet(u))$. The mixed terms and the higher theta terms will be included in the correction.

For $t > 0$ and $J = 6m'$, the formulas show that only $t \equiv 0, 1 \pmod 6$ occur. For $t = 6d$, $d \ge 1$, the $k = 0$ sum is

$$E_1^d \frac{1 - Q^{-1}}{1 - V},$$

and the $k = 1$ term is $E_1^d W$. For the latter, the only possible valuation is $J = t$, and $\tau_1\tau_5 = Q$. For $t = 6d+1$, $d \ge 0$, the $k = 0$ term is $-D_1E_1^d$, extending the first squarefree term just computed. Finally its $k = 1$ sum is

$$-(Q-1)D_1E_1^d\,\frac{WV}{1-V}.$$

Here $J > t$ and $T_0(p^{J-t}) = Q - 1$; the first possible extra m' gives the displayed V. Summing in d proves the closed formula

$$\begin{aligned} \mathcal{F}_{u,\mathfrak{p}} = \frac{1}{1-V} + \frac{W}{1-W} + \frac{E_1}{1-E_1}\left(\frac{1-Q^{-1}}{1-V} + W\right) \\ - \frac{D_1}{1-E_1}\left(1 + \frac{(Q-1)WV}{1-V}\right). \end{aligned} \tag{26.19}$$

This derivation also explains the vanishing of the allowed theta exponents $t \equiv 3, 4 \pmod 6$: the possible boundary values of J in Equation (26.17) have the wrong residue modulo six, and the remaining T-sum is a nontrivial character sum at zero.

Define at every prime

$$\mathcal{H}_{u,\mathfrak{p}} = (1 - V)\frac{1 - W}{1 - D_1}\mathcal{F}_{u,\mathfrak{p}},$$

where $W = D_1 = 0$ at $\mathfrak{p} \mid (u)R$, in accordance with the zero extension. We verify normal convergence on explicit open neighborhoods: in each line of Equation (26.13) decrease all three lower endpoints by 10^{-5}, and use the resulting strict inequalities. On these neighborhoods all of V, W, D_1, E_1 have modulus bounded away from one once Q exceeds a fixed constant. Subtracting 1 after multiplying Equation (26.19) by $(1 - V)(1 - W)/(1 - D_1)$ gives

$$|\mathcal{H}_{u,\mathfrak{p}} - 1| \ll |VW| + |D_1V| + |D_1W| + Q|D_1WV| + |E_1| \ll Q^{-1-\delta}, \qquad \delta = \frac{1}{4000}, \tag{26.20}$$

uniformly for $\mathfrak{p} \nmid (u)R$. For clarity, before the 10^{-5} enlargement the weakest of the first three terms is $|D_1W| \le Q^{-1.001}$, in the first region. The exponents in the last two terms are at most

$$1 - \operatorname{Re}\xi - \operatorname{Re} w - 6\operatorname{Re} z \le -1.908, \qquad 4 - 6\operatorname{Re}\xi - 6\operatorname{Re} z \le -2.948$$

on the union of the two regions. The enlargement leaves respectively the margins 0.00098, 0.90792, and 1.94788 beyond exponent -1, which suffice for the stated δ. More explicitly, the specialization at $E_1 = 0$ satisfies

$$\mathcal{H}_{u,\mathfrak{p}}|_{E_1=0} - 1 = \frac{D_1(W + V - WV) - WV - (Q - 1)D_1(1 - W)WV}{1 - D_1}.$$

This accounts for the $Q|D_1WV|$ term as well as the first three terms in Equation (26.20). The difference from this specialization is $O(|E_1|)$ on the stated neighborhoods: the remaining denominators are bounded, and $Q|D_1WV| < 1$ once Q exceeds the fixed cutoff.

At $\mathfrak{p} \mid (u)R$, $1 \le j \le 5$. The $t = 0$ contribution is $(1 - V)^{-1}$. The term $t = 1, k = 0$ vanishes, since $J \ge 1$ and $T_1(p^J) = 0$. By Equation (26.18), the sum for $t = 1, k = 1$ has modulus at most

$$\frac{Q^{1-\operatorname{Re}\xi-\operatorname{Re} w}}{1 - |V|}.$$

Its exponent is at most -0.001 on the first region and at most -0.948 on the second. For $l \ge 1$, before the factor $V^{m'}$, the bound for the summand with $e_0 \in \{0, 1\}$ is at most a constant times Q to the power

$$(2 - 3\operatorname{Re}\xi)l + (1/2 - \operatorname{Re}\xi)e_0 + \max(0, 1/2 - \operatorname{Re} w). \tag{26.21}$$

This is at most -0.497 for $l = 1, e_0 = 0$ on the first region and decreases by at least 0.994 for each further l; the second region has a larger saving. The 10^{-5} enlargement still leaves a saving greater than 0.49 in this sum, and leaves 0.00098 in the $t = 1, k = 1$ term. The geometric sums in l, m', followed by multiplication by $1 - V$, therefore give

$$\mathcal{H}_{u,\mathfrak{p}} = 1 + O(Q^{-\delta}) \quad (\mathfrak{p} \mid (u)R), \qquad \delta = \frac{1}{4000}, \tag{26.22}$$

on the same neighborhoods. These estimates prove absolute local convergence there as well.

The product of the factors in Equation (26.20) converges normally, since $\sum_{\mathfrak{p}} q_{\mathfrak{p}}^{-1-\delta} < \infty$. The remaining primes are precisely the divisors of $(u)R$. For every $\epsilon > 0$,

$$\prod_{\mathfrak{p}|(u)R} (1 + Cq_{\mathfrak{p}}^{-\delta}) \ll_{F,S,\epsilon} q_u^{\epsilon} :$$

for sufficiently large Q each local factor is at most Q^{ϵ}, and the finitely many smaller primes contribute a fixed constant. Consequently $\mathcal{H}_u = \prod_{\mathfrak{p} \notin S} \mathcal{H}_{u,\mathfrak{p}}$ is holomorphic and obeys the asserted

bound. Multiplication of its defining local factors gives Equation (26.12) in the initial region and then as a meromorphic identity on either neighborhood.

For $u = 1$ every prime satisfies Equation (26.20). The local estimate determines a fixed norm cutoff beyond which its uniform bound is smaller than $1/2$. Choose S to contain all primes up to that cutoff from the outset, as permitted in Section 21. Every remaining $\mathcal{H}_{1,\mathfrak{p}}$ is nonzero, and its logarithm defined near 1 has an absolutely and uniformly convergent sum. Its exponential is $\mathcal{H}_1$, which is therefore bounded and nowhere zero on both neighborhoods. This proves the proposition. □

For the remainder of the proof, write

$$\mathcal{H}_\eta(s) := \mathcal{H}_1(s, 1, 1/6).$$

This is the principal specialization announced in Section 22. Since $(w, z) = (1, 1/6)$ lies in the second region of Equation (26.13), $\mathcal{H}_\eta$ is holomorphic, bounded, and nowhere zero on an open neighborhood of the closed half-plane $\operatorname{Re} s \geq .998$, uniformly in height.

27 Zero detection and Mellin continuation

We now prove Theorem 2.2 for the fixed but arbitrary F and η. The factorization in Proposition 26.3 singles out the frequency $u = 1$: it is the only row whose numerator L-function has a pole. We shall bound the other rows by a conductor large sieve, isolate the two residues of that principal row, and compare them with Proposition 26.2. The resulting bound for one Mellin integral will exclude zeros of the target function.

27.1 A conductor large sieve and a zero detector

Let ω_u denote the primitive Hecke character inducing $\eta\overline{\chi_\bullet(u)}$. By Lemma 21.3, on a dyad $U \leq q_u < 2U$,

$$\mathrm{N}\mathfrak{f}_{\omega_u} \ll_{F,S,\eta} U, \qquad \#\{u \in \mathcal{U} : U \leq q_u < 2U\} \ll_{F,S} U. \tag{27.1}$$

The second bound follows from the ideal count and the finite number of S-unit classes modulo sixth powers. The characters ω_u in this family are distinct: equality cancels the fixed twist and forces u/v to be a sixth power in F; the selected exterior valuations then agree, and the quotient is in U_S^6. Their local components at S run through a fixed finite set, since the Kummer part depends on $F_v^\times/F_v^{\times 6}$. The principal numerator occurs only for $u = 1$. The primitive denominator ω_u may be principal for a different row, so it will be treated separately in zero detection.

We give the conductor estimate required for that detection. In the next lemma and in the detector, ideals without the adjective exterior are integral ideals of $\mathcal{O}_F$, and $q_\mathfrak{n} = \mathrm{N}\mathfrak{n}$ also for those ideals.

Lemma 27.1 (Primitive conductor large sieve)**.** *Fix $C_0 > 0$ and a finite set $\mathscr{L}_S$ of possible local components at S. Let $\mathscr{X}(U)$ be any family of distinct primitive finite-order Hecke characters of F whose conductor norms are at most C_0U and whose components at S belong to that set. For $N, U \geq 2$, any complex coefficients $b_\mathfrak{n}$, and every $\epsilon > 0$,*

$$\sum_{\omega \in \mathscr{X}(U)} \left| \sum_{q_\mathfrak{n} \leq N} b_\mathfrak{n} \omega(\mathfrak{n}) \right|^2 \ll_{F,S,C_0,\mathscr{L}_S,\epsilon} (N + U^2)(NU)^\epsilon \sum_{q_\mathfrak{n} \leq N} |b_\mathfrak{n}|^2. \tag{27.2}$$

As usual a primitive character is extended by zero on ideals meeting its conductor.

Proof. First restrict the coefficient ideals to one exterior norm dyad and the characters to one choice of their components at S. If $\mathfrak{q}$ is the exterior part of the conductor, principality of R and the global character relation write the value at $\mathfrak{n} = nR$, for its selected balanced generator n, as

$$\omega(\mathfrak{n}) = \lambda_S(n)\lambda_\mathfrak{q}(n).$$

Here λ_S is a common fixed unitary function of the S-components, and $\lambda_{\mathfrak{q}}$ is a primitive residue character modulo $\mathfrak{q}$. The orientation of the local characters is absorbed in these two symbols. The formula includes zero on nonunits. It follows directly by evaluating the idele character at the rational element n: the unramified exterior components give $\omega(\mathfrak{n})$, leaving the ramified exterior components and the components in S. In particular, for this fixed S-choice the residue character determines ω.

Let q be a balanced generator of $\mathfrak{q}$. By Lemma 21.1, $(n,d) \mapsto \mathrm{e}(nd/q)$ is a perfect additive pairing on $R/\mathfrak{q}$. Put

$$\tau(\overline{\lambda_{\mathfrak{q}}}) = \sum_{d \bmod \mathfrak{q}} \overline{\lambda_{\mathfrak{q}}(d)}\, \mathrm{e}(d/q).$$

For a primitive residue character, this Gauss sum has modulus $q_{\mathfrak{q}}^{1/2}$, and

$$\lambda_{\mathfrak{q}}(n) = \frac{1}{\tau(\overline{\lambda_{\mathfrak{q}}})} \sum_{d \bmod \mathfrak{q}} \overline{\lambda_{\mathfrak{q}}(d)}\, \mathrm{e}(nd/q).$$

For a unit n this follows by a change of residue variable. For a nonunit the Gauss sum is zero: locally at a prime dividing n the additive character factors through a proper quotient of the conductor, and multiplication by a unit in the kernel of that quotient on which the primitive multiplicative character is nontrivial forces the sum to vanish. Finite additive Parseval now gives

$$\sum_{n \bmod \mathfrak{q}} \left| \sum_{d \bmod \mathfrak{q}} \overline{\lambda_{\mathfrak{q}}(d)} \mathrm{e}(nd/q) \right|^2 = q_{\mathfrak{q}} |(R/\mathfrak{q})^{\times}|.$$

The left side is $|(R/\mathfrak{q})^{\times}|\,|\tau(\overline{\lambda_{\mathfrak{q}}})|^2$, which proves the asserted modulus. Put

$$T_{d/q} = \sum_{\mathfrak{n}} b_{\mathfrak{n}} \lambda_S(n) \mathrm{e}(nd/q).$$

Sum the square of the preceding Gauss expansion over the characters of conductor $\mathfrak{q}$. Extending that sum to all residue characters and using character orthogonality bounds it by

$$\frac{|(R/\mathfrak{q})^{\times}|}{q_{\mathfrak{q}}} \sum_{d \in (R/\mathfrak{q})^{\times}} |T_{d/q}|^2 \le \sum_{d \in (R/\mathfrak{q})^{\times}} |T_{d/q}|^2.$$

The same assertion holds for the unit conductor with its single residue. Sum now over $q_{\mathfrak{q}} \le C_0 U$. The fractions are distinct reduced fractions. The balanced generators of the coefficient ideals on a norm dyad lie in $\ell_{N'}K$ for one fixed compact K, with $N' \le N$, by Lemma 21.2. The dual assertion of Lemma 26.1, with $V = \max(2, C_0 U)$, bounds this sum by $(N + U^2) \sum |b_{\mathfrak{n}}|^2$.

To pass to all exterior lengths at most N, split into dyads and use Cauchy–Schwarz, at a logarithmic cost. For general $\mathcal{O}_F$ ideals write $\mathfrak{n} = \mathfrak{n}_S \mathfrak{n}^S$, with the first factor supported in S_f. There are $O_S((1+\log N)^{|S_f|})$ possibilities for $\mathfrak{n}_S$. Within a fixed S-choice the value $\omega(\mathfrak{n}_S)$ is fixed (or zero). Cauchy–Schwarz over these possibilities and the preceding exterior estimate introduce only powers of $\log N$; the square masses over the disjoint pieces sum to the original square mass. The finite partition by S-components costs a constant. Absorbing all logarithms into $(NU)^{\epsilon}$ proves Equation (27.2). □

We use standard Hecke continuation and the functional equation in a deliberately conservative convexity bound. For a nonprincipal primitive finite-order ω with $\mathrm{N}\mathfrak{f}_\omega \ll U$, there is an $A_F > 0$, depending on the fixed field, such that on any of the fixed strips below, in particular for $.003 \le \sigma \le 4$,

$$|L_F(\sigma + it, \omega)| \ll_{F,S,\eta,\epsilon} U^{1/2+\epsilon}(2+|t|)^{A_F}. \tag{27.3}$$

This follows from the functional equation, the absolutely convergent Euler product on a right line, and the Phragmén–Lindelöf principle; the gamma factors contribute the field-dependent height

power. The same estimates give polynomial height bounds on any fixed strip for $(s-1)\zeta_F(s)$. Omitting factors at a fixed set or at primes dividing u changes a positive-real-part estimate by q_u^{ϵ}: for a fixed $\sigma_* > 0$, each of

$$\prod_{\mathfrak{p}\mid(u)R} (1+q_{\mathfrak{p}}^{-\sigma_*}), \qquad \prod_{\mathfrak{p}\mid(u)R} (1-q_{\mathfrak{p}}^{-\sigma_*})^{-1}$$

is $O_{\epsilon,\sigma_*}(q_u^{\epsilon})$, by bounding a sufficiently large prime's factor by $q_{\mathfrak{p}}^{\epsilon}$ and absorbing the finitely many other primes into the constant.

Proposition 27.2. *For the fixed F, S, η there are $\rho > 0$ and U_0 with the following property. For every $U \geq U_0$, there is a set $\mathcal{E}(U) \subset \{u \in \mathcal{U} : U \leq q_u < 2U\}$ of cardinality*

$$|\mathcal{E}(U)| \ll_{F,S,\eta} U^{1/2}$$

which contains every row for which ω_u is principal or $L_F(s,\omega_u)$ has a zero with $\operatorname{Re} s \geq .996$ and $|\operatorname{Im} s| \leq 4U^{\rho}$. For $u \notin \mathcal{E}(U)$ and every $\epsilon > 0$,

$$\left| L^S(\xi, \eta\overline{\chi_\bullet(u)})^{-1} \right| \ll_{F,S,\eta,\epsilon} U^{\epsilon} \qquad (\operatorname{Re}\xi \geq .998,\ |\operatorname{Im}\xi| \leq 2U^{\rho}). \tag{27.4}$$

The exponents $.996, .998, 1/2$ are absolute. The height exponent ρ, the constants, and U_0 may depend on the fixed data.

Proof. Set $\sigma_0 = .996$. For a primitive nonprincipal character ω in the family of Equation (27.1), define the truncated inverse and its product coefficients by

$$D_U(s,\omega) = \sum_{q_{\mathfrak{d}} \leq U^2} \mu_F(\mathfrak{d})\omega(\mathfrak{d}) q_{\mathfrak{d}}^{-s}, \qquad \alpha_U(\mathfrak{n}) = \sum_{\substack{\mathfrak{d}\mid\mathfrak{n} \\ q_{\mathfrak{d}} \leq U^2}} \mu_F(\mathfrak{d}).$$

Here μ_F is the Möbius function of integral $\mathcal{O}_F$ ideals. In absolute convergence,

$$L_F(s,\omega) D_U(s,\omega) = \sum_{\mathfrak{n}} \alpha_U(\mathfrak{n})\omega(\mathfrak{n}) q_{\mathfrak{n}}^{-s}.$$

Complete multiplicativity with the primitive zero extension justifies this identity also on ideals meeting the conductor. The coefficients are independent of ω and satisfy

$$\alpha_U(1) = 1, \qquad \alpha_U(\mathfrak{n}) = 0 \quad (1 < q_{\mathfrak{n}} \leq U^2), \qquad |\alpha_U(\mathfrak{n})| \leq \tau_F(\mathfrak{n}) \ll_{F,\epsilon} q_{\mathfrak{n}}^{\epsilon}. \tag{27.5}$$

The coefficients with $1 < q_{\mathfrak{n}} \leq U^2$ have vanished. We shall show that at a zero in the stated rectangle the full exponentially smoothed sum is $o(1)$. After its exponential tail is discarded, the unit coefficient must therefore be cancelled, up to $o(1)$, by terms with $U^2 < q_{\mathfrak{n}} \leq U^{11}$. A dyadic prefix of those terms is consequently large. The conductor sieve will bound the number of characters for which such a prefix can be large at any allowed height.

Suppose $L_F(\beta + i\gamma, \omega) = 0$ with $\beta \geq \sigma_0$, $|\gamma| \leq 4U^{\rho}$. Absolute convergence of the Euler product implies $\beta \leq 1$. Mellin inversion of the exponential gives, with $c > 1$,

$$\sum_{\mathfrak{n}} \alpha_U(\mathfrak{n})\omega(\mathfrak{n}) q_{\mathfrak{n}}^{-\beta-i\gamma} e^{-q_{\mathfrak{n}}/U^{10}} = \frac{1}{2\pi i} \int_{(c)} L_F(\beta+i\gamma+v,\omega) D_U(\beta+i\gamma+v,\omega)\Gamma(v) U^{10v}\, dv. \tag{27.6}$$

Move the v-line to $\operatorname{Re} v = 1/2 - \beta$, which lies between $-1/2$ and $-.496$. The only crossed gamma pole is $v = 0$, and its residue vanishes because $L_F(\beta + i\gamma, \omega) = 0$. The character is nonprincipal, so its L-function has no pole in the shift. The gamma decay and the polynomial height bounds justify the horizontal limits.

On the new line the fixed-field ideal count gives

$$|D_U(1/2 + it, \omega)| \leq \sum_{q_{\mathfrak{d}} \leq U^2} q_{\mathfrak{d}}^{-1/2} \ll_F U.$$

Use Equation (27.3) and integrate the gamma factor. The right side of Equation (27.6) is at most

$$\ll U^{10(1/2-\beta)+1/2+1+\epsilon}(1+|\gamma|)^{A_F} \ll U^{-3.46+A_F\rho+\epsilon}. \tag{27.7}$$

Choose once and for all

$$0 < \rho < \min\left\{\frac{1}{10}, \frac{1}{A_F+1}\right\}$$

and then take the preliminary loss small. The last bound is $o(1)$.

The first term on the left of Equation (27.6) is $e^{-U^{-10}} = 1 + o(1)$, and the intervening terms through U^2 vanish by Equation (27.5). The portion with $q_{\mathfrak{n}} > U^{11}$ is exponentially small, uniformly in $\beta \geq \sigma_0$: the divisor bound and ideal count majorize it by a fixed power of U times $e^{-U/2}$. It follows that the sum over $U^2 < q_{\mathfrak{n}} \leq U^{11}$ has modulus bounded below by a positive constant. Decompose this interval into $O(\log U)$ dyads. On any such dyad the remaining real weight is

$$t^{\sigma_0-\beta}e^{-t/U^{10}},$$

which is nonnegative, decreasing, and at most one. Partial summation therefore bounds its weighted sum by twice the maximum modulus of the unweighted prefix sums. Consequently for at least one dyadic $U^2/2 < N \leq U^{11}$ and one $y \in [N, 2N]$,

$$\left|\sum_{N<q_{\mathfrak{n}}\leq y} \alpha_U(\mathfrak{n})\omega(\mathfrak{n})q_{\mathfrak{n}}^{-\sigma_0-i\gamma}\right| \gg (\log U)^{-1}. \tag{27.8}$$

We spell out the two maximal estimates needed to use the conductor sieve on this assertion. Order the ideals on a dyad by norm, with any fixed order for ties. A prefix is the disjoint union of at most $O(\log N)$ binary index intervals. Cauchy–Schwarz and summation over those intervals bound the squared maximum of all prefixes by the sum of their squared interval sums, with $O((\log N)^2)$ total cost after the large sieve is applied; every coefficient occurs in one interval at each level. For a fixed interval its Dirichlet polynomial $P_\omega(t)$ satisfies on a unit interval the elementary bound

$$\sup_{t\in[j,j+1]} |P_\omega(t)|^2 \ll \int_{j-1}^{j+2} (|P_\omega(t)|^2 + |P'_\omega(t)|^2)\, dt.$$

The derivative only multiplies its coefficients by $\log q_{\mathfrak{n}} \ll \log N$. Apply Lemma 27.1 at each t, and sum these integrals over $|t| \leq 4U^\rho$. This costs U^ρ and powers of logarithms. The squared coefficient mass on a dyad is

$$\sum_{N<q_{\mathfrak{n}}\leq 2N} |\alpha_U(\mathfrak{n})|^2 q_{\mathfrak{n}}^{-2\sigma_0} \ll_{F,\epsilon} N^{1-2\sigma_0+\epsilon}.$$

Since $N > U^2/2$ and $N \leq U^{11}$, the nonlogarithmic cost in Equation (27.2) is bounded by

$$(N+U^2)N^{1-2\sigma_0+\epsilon} \ll N^{2-2\sigma_0+\epsilon} \ll U^{.088+O(\epsilon)}. \tag{27.9}$$

Here $2 - 2\sigma_0 = .008$, and $11(.008) = .088$; the factor $(NU)^\epsilon$ in the large sieve is included in $U^{O(\epsilon)}$. Chebyshev's inequality with Equation (27.8), followed by summation over the $O(\log U)$ dyads, gives

$$\#\{\omega \text{ with a detected zero}\} \ll U^{.088+\rho+O(\epsilon)}.$$

The choice $\rho < .1$ leaves more than 0.31 before exponent $1/2$. Take the losses small and include the at most one principal character. Distinctness of the ω_u's now gives the asserted bound for $\mathcal{E}(U)$.

It remains to obtain the reciprocal bound from the zero-free rectangles just proved. Let $T = U^\rho$ and $u \notin \mathcal{E}(U)$. For $|t| \leq 2T$, consider disks centered at $2 + it$ of radii

$$R_2 = 1.0035, \qquad r_2 = 1.003, \qquad r_0 = .999, \qquad r_1 = 1.002.$$

For large U the outer disk lies within $\operatorname{Re} s > .996$, $|\operatorname{Im} s| < 4T$; its left edge is .9965. The nonprincipal $L_F(s, \omega_u)$ is entire and nonzero on the disk. It has a branch of $\log L_F$ there agreeing at the center with the Euler logarithm. The convexity bound gives $\operatorname{Re} \log L_F = \log |L_F| \le C_F \log U$ on the outer disk, and the value at the center is $O_F(1)$. Borel–Carathéodory consequently bounds $|\log L_F|$ by $O_F(\log U)$ on radius r_2. On radius r_0 the real part is at least 1.001, so the absolutely convergent Euler logarithm is $O_F(1)$. The three-circles theorem applied to the analytic function $\log L_F$ now gives, on radius r_1,

$$|\log L_F(s, \omega_u)| \ll_F (\log U)^\theta, \qquad \theta = \frac{\log(1.002/.999)}{\log(1.003/.999)} < 1.$$

These disks reach the line $\operatorname{Re} s = .998$; absolute convergence covers the larger real parts outside them. Hence for $\operatorname{Re} s \ge .998$ and $|\operatorname{Im} s| \le 2T$,

$$|L_F(s, \omega_u)^{-1}| \le \exp(O_F((\log U)^\theta)) \ll_{F,\epsilon} U^\epsilon.$$

The estimate for omitted factors preceding the proposition gives the same bound for the incomplete function, proving Equation (27.4). □

27.2 The principal component and the other frequency rows

Use the scales in Equation (22.4), and let $\sigma_\xi, \sigma_w, \sigma_z$ denote the real parts of ξ, w, z. Write $E_*(\sigma_\xi, \sigma_w, \sigma_z)$ for the real power of Z outside the Dirichlet factors in Equation (26.10). In the principal row, taking the residues at $w = 1$ and $z = 1/6$ leaves the exponent $C_*(\xi)$ in the second line:

$$\begin{aligned} E_*(\sigma_\xi, \sigma_w, \sigma_z) &= \frac{a}{2} + \sigma_\xi - 1 + h_0\sigma_z + b(\sigma_w - 1), \\ C_*(\xi) &= \frac{a}{2} + \xi - 1 + \frac{h_0}{6} = \xi - \frac{8}{15}, \qquad C_*(1) = \frac{7}{15}. \end{aligned} \tag{27.10}$$

Thus $C_*(1) = 7/15$ is the reference power for comparing the principal remainders and the other frequency rows.

We specify a decay observation that justifies the contour limits and, later, the truncated shifts. On any fixed real strips used below, for every $A > 0$,

$$\left| e^{(\xi+z-1)^2} M(z) \widehat{W}_1(w) \right| \ll_A e^{-(\operatorname{Im}\xi + \operatorname{Im} z)^2} (1 + |\operatorname{Im} z|)^{-A} (1 + |\operatorname{Im} w|)^{-A}. \tag{27.11}$$

The constants can depend on those strips. Away from the poles extracted below, all numerator Hecke and zeta functions on them have polynomial height growth, and $\mathcal{H}_u$ is uniformly bounded by q_u^ϵ. After increasing A, their height powers are absorbed in Equation (27.11). In particular for every $A_1 > 0$,

$$\int_{\mathbb{R}} e^{-(T+v)^2} (1 + |v|)^{-A} \, dv \ll_{A_1} (1 + |T|)^{-A_1}$$

with A sufficiently large. Thus integration in z, w gives arbitrary polynomial decay in the remaining ξ-height as well. On lines $\operatorname{Re}\xi > 1$ the reciprocal denominator is globally bounded by its absolutely convergent Euler product. These facts justify all infinite contour shifts on such lines by their horizontal limits.

Let $I_1(Z)$ denote the contribution of $u = 1$ to the initially absolutely convergent formula Equation (26.10), on the scales $(X, Y) = (Z^a, Z^b)$. Its numerator is $\zeta_F^S(w)$. Put $\delta_1 = 10^{-7}$, $\sigma_1 = 1 + \delta_1$, and $\zeta_1 = 1/6 + \delta_1$. Move the initial lines to

$$\operatorname{Re}\xi = \sigma_1, \qquad \operatorname{Re} w = \sigma_1, \qquad \operatorname{Re} z = \zeta_1.$$

These moves cross no poles: both zeta arguments remain to the right of one, and the reciprocal denominator is in absolute convergence. The second region of Equation (26.13) contains all the

moves. Now move w to .95, crossing its simple pole at 1, and call the remaining triple integral $R_w(Z)$. In the $w = 1$ residue move z to .16, crossing the pole of $\zeta_F^S(6z)$ at $1/6$, and call the remaining residue integral $R_z(Z)$. Their contours are

$$\begin{aligned} R_w: &\quad (\operatorname{Re}\xi, \operatorname{Re} w, \operatorname{Re} z) = (\sigma_1, .95, \zeta_1), \\ R_z: &\quad (\operatorname{Re}\xi, w, \operatorname{Re} z) = (\sigma_1, 1, .16). \end{aligned}$$

The other term is the double residue. With its evaluation $c_0 f_\eta(Z)$ below, the principal component is exactly

$$I_1(Z) = c_0 f_\eta(Z) + R_w(Z) + R_z(Z).$$

The powers of the two remainders relative to $C_*(1)$ are

$$\begin{aligned} E_*(\sigma_1, .95, \zeta_1) - C_*(1) &= (1 + h_0)\delta_1 - .05b = -.00664989, \\ E_*(\sigma_1, 1, .16) - C_*(1) &= \delta_1 + h_0(.16 - 1/6) = -.000666566666\ldots. \end{aligned}$$

The height bounds just given and Proposition 26.3 therefore give

$$\begin{aligned} R_w(Z) &= O_{F,S,\eta,W_0,W_1}(Z^{C_*(1)-.0005}), \\ R_z(Z) &= O_{F,S,\eta,W_0,W_1}(Z^{C_*(1)-.0005}). \end{aligned} \tag{27.12}$$

To evaluate the double residue, write $R_S = \operatorname{Res}_{s=1} \zeta_F^S(s) > 0$. It has the form $c_0 f_\eta(Z)$, where

$$\begin{aligned} c_0 &= \frac{R_S^2}{6} M(1/6)\widehat{W_1}(1) \neq 0, \\ f_\eta(Z) &= \frac{1}{2\pi i}\int_{(2)} Z^{C_*(\xi)} e^{(\xi-5/6)^2} \frac{\mathcal{H}_\eta(\xi)}{L^S(\xi,\eta)}\, d\xi. \end{aligned} \tag{27.13}$$

Indeed the residue of $\zeta_F^S(6z)$ is $R_S/6$, and $\widehat{W_1}(1) = \int_0^\infty W_1(t)\, dt > 0$. The other factor is nonzero by Equation (26.1). The norm measures were fixed before the probe, so this is the exact scalar in its normalization. The line of the double residue was initially $\operatorname{Re}\xi = \sigma_1$; we moved it right to 2 to write Equation (27.13). This uses only absolute convergence of $1/L^S(\xi,\eta)$ for $\operatorname{Re}\xi > 1$.

It remains to bound the rows with $u \neq 1$. Split their sum on the initial lines into dyads $U \le q_u < 2U$, $U = 1, 2, 4, \ldots$, before shifting any of them. Each dyad contains finitely many rows; the full sum on the initial lines is absolutely convergent. Denote by $\mathcal{J}_{\mathcal{C}}(Z)$ the triple integral of a subset $\mathcal{C}$ of such a dyad, and set

$$\mathcal{J}_U^{\mathrm{np}}(Z) = \mathcal{J}_{\{u\neq 1 : U \le q_u < 2U\}}(Z).$$

The numerator for each of these rows is entire. From Equation (27.3) and the omitted-factor bound,

$$L^S(w, \chi_\bullet(u)) \ll_\epsilon U^{1/2+\epsilon}(2 + |\operatorname{Im} w|)^{A_F} \qquad (.003 \le \operatorname{Re} w \le 4). \tag{27.14}$$

On $\operatorname{Re}\xi = \sigma_1$, independently of u and the height,

$$|L^S(\xi, \eta\overline{\chi_\bullet(u)})^{-1}| \le \prod_{\mathfrak{p}\notin S}(1 + q_{\mathfrak{p}}^{-\sigma_1}) \le \zeta_F(\sigma_1).$$

For each row move the initial lines to

$$\operatorname{Re}\xi = \sigma_1, \qquad \operatorname{Re} w = .003, \qquad \operatorname{Re} z = .4.$$

One may first move ξ, then z, then w. The z-line remains above $1/6$; the numerator is entire; and the denominator stays in absolute convergence. The first region of Equation (26.13) contains the whole move. If $|\mathcal{C}| \ll U^\alpha$, the bounds above give

$$|\mathcal{J}_{\mathcal{C}}(Z)| \ll_\epsilon Z^{E_*(\sigma_1,.003,.4)} U^{\alpha+1/2-.4+\epsilon}. \tag{27.15}$$

The factor $U^{-.4}$ is the q_u^{-z} in Equation (26.10). The height integral is bounded by Equation (27.11).

To see how the size of the frequency determines the contours, rewrite Equation (27.15) as

$$|\mathcal{J}_{\mathcal{C}}(Z)| \ll_\epsilon Z^{a/2+\sigma_1-1+b(.003-1)} U^{\alpha+1/2+\epsilon} \left(\frac{Z^{h_0}}{U}\right)^{.4}.$$

For a general z-line, the corresponding factor is $(Z^{h_0}/U)^{\operatorname{Re} z}$. Its dependence on $\operatorname{Re} z$ changes direction at $U = Z^{h_0} = Z^{.1}$: below that scale a rightward z-shift enlarges this factor, while above it such a shift reduces the factor. We retain the current low-w contour through $U \le Z^{h_0+.00001}$, and use a rightward z-shift beyond that enlarged threshold.

The common Z-exponent on the current contour, relative to the principal scale, is

$$E_*(\sigma_1, .003, .4) - C_*(1) = 10^{-7} + \frac{7}{300} - .132601 = -.109267566666\ldots.$$

For $U \le Z^{.09}$, use Equation (27.15) with $\alpha = 1$. The $U^{1.1}$ factor adds at most $.09(1.1) = .099$ to this relative exponent, so these rows are already smaller than the principal scale.

We use Proposition 27.2 only in the remaining interval

$$Z^{.09} < U \le Z^{h_0+.00001} = Z^{.10001}, \tag{27.16}$$

where its threshold U_0 is met for all sufficiently large Z. For the rows in $\mathcal{E}(U)$, use the same contour estimate with $\alpha = 1/2$. The $U^{.6}$ factor then adds at most $.10001(.6) = .060006$ to the same relative exponent.

For each remaining row truncate its ξ-line at $|\operatorname{Im}\xi| \le U^\rho$, and move that finite portion to $\operatorname{Re}\xi = .998$. The rectangle and its two horizontal sides satisfy Equation (27.4); hence the new central integral satisfies

$$\ll_\epsilon Z^{E_*(.998,.003,.4)} U^{1+1/2-.4+\epsilon} \tag{27.17}$$

after summing all the good rows of the dyad. The estimate does not use the reciprocal outside the specified rectangle. Since E_* is linear in $\operatorname{Re}\xi$ with coefficient one, this move gains $Z^{.998-\sigma_1} = Z^{-.0020001}$. Including the $U^{1.1}$ cost up to the upper endpoint of Equation (27.16), the relative exponent is at most

$$-.002 + \frac{7}{300} - .132601 + .10001(1.1) = -\frac{377}{300000} = -.001256666666\ldots. \tag{27.18}$$

To bound the discarded and horizontal portions, integrate in z, w first in Equation (27.11). For every prescribed $A_1 > 0$ the discarded tails on $\operatorname{Re}\xi = \sigma_1$ and the horizontal portions at $\operatorname{Im}\xi = \pm U^\rho$ are, in total over a good dyad, at most

$$\ll_{A_1,\epsilon} Z^{E_*(\sigma_1,.003,.4)} U^{1+1/2-.4+\epsilon-\rho A_1}. \tag{27.19}$$

On the horizontal portions the real exponent is bounded by its value at σ_1; the reciprocal is bounded by Equation (27.4). On the discarded original tails it is bounded by absolute convergence. The convolution estimate following Equation (27.11), used with pointwise exponent $A_1 + 2$ before integrating the discarded ξ-tails, supplies $U^{-\rho A_1}$ in both cases. Since $U > Z^{.09}$, choosing $A_1 > 1/\rho$, after ρ is fixed, gives more than enough additional saving for these terms.

All these comparisons leave a fixed margin after the arbitrary small power losses. The sums over the $O(\log Z)$ dyads in these ranges cost only another arbitrarily small power.

Finally consider $U > Z^{.10001}$. Now $Z^{h_0}/U \le Z^{-.00001}$, so a fixed rightward z-shift suppresses the whole dyad. Keep $\operatorname{Re}\xi = \operatorname{Re} w = 4$ and move z to the fixed right line $\operatorname{Re} z = R_0 = 350000$. The move stays in the Euler region, crosses no pole, and uses the holomorphy of M on $\operatorname{Re} z > 0$. Both Hecke factors are bounded by absolute convergence on these lines, so Proposition 26.3 and Equation (27.11) give

$$|\mathcal{J}_{\mathcal{C}}(Z)| \ll_\epsilon Z^{3.849} U^{1+\epsilon} \left(\frac{Z^{.1}}{U}\right)^{R_0} = Z^{3.849+.1R_0} U^{1-R_0+\epsilon}.$$

The dyads sum geometrically. At their lower endpoint the power of Z is

$$3.849 + .1R_0 + .10001(1 - R_0 + \epsilon) = .44901 + .10001\epsilon. \tag{27.20}$$

This is strictly less than $C_*(1) - .01$ for small ϵ. The very large z-line is fixed; its weight constants may be large, but are independent of Z and U.

Choose the preliminary losses small enough that their total cost in the finite frequency ranges, including the dyadic count, is less than 10^{-4} in the Z-exponent. This is possible since $U \le Z^{.10001}$ there. The smallest central saving in Equation (27.18) is larger than 0.0012; Equation (27.19) and Equation (27.20) have larger savings. We have therefore proved

$$\sum_{j \ge 0} \mathcal{J}^{\mathrm{np}}_{2^j}(Z) \ll Z^{C_*(1)-.0004}. \tag{27.21}$$

Each dyadic integral here is obtained from its initial absolutely convergent integral, and the preceding bounds justify their sum.

27.3 Completion of the proof of Theorem 2.2

The numerical exponents established above, before the indicated small losses, are collected in Table 1. In particular the exponent from reflection is

$$\frac{2M_0 - 1 - b}{2} = \frac{933}{2000} = C_*(1) - \frac{1}{6000}.$$

Contribution	exponent relative to $C_*(1) = 7/15$
Probe bound from reflection	$-1/6000 = -.000166666\ldots$
Principal remainder R_w	$-.00664989$
Principal remainder R_z	$-.000666566\ldots$
Nonprincipal rows $U \le Z^{.09}$	$-.010267566\ldots$
Exceptional rows $U \le Z^{.10001}$	$-.049261566\ldots$
Good central rows $U \le Z^{.10001}$	$-.001256666\ldots$
Rows $U > Z^{.10001}$, $R_0 = 350000$	$-.017656666\ldots$

Table 1: The fixed exponent margins in the probe comparison. The row bounds precede arbitrary small power losses.

Combining the principal decomposition $I_1 = c_0 f_\eta + R_w + R_z$, Equations (27.12) and (27.21), and Equation (26.10) gives, for sufficiently large Z,

$$I(Z^a, Z^b, Z) = c_0 f_\eta(Z) + O(Z^{C_*(1)-.0004}).$$

Use Proposition 26.2 with, for example, a loss 10^{-5}. Since $1/6000 - 10^{-5} > 1/20000$ and $c_0 \neq 0$, the preceding identity implies the safe bound

$$f_\eta(Z) \ll_{F,S,\eta,W_0,W_1} Z^{7/15-1/20000} \qquad (Z \text{ sufficiently large}). \tag{27.22}$$

Set

$$\mathscr{A}(\xi) = e^{(\xi-5/6)^2} \frac{\mathcal{H}_\eta(\xi)}{L^S(\xi, \eta)}.$$

For $\operatorname{Re}\xi > 1$ this is holomorphic. In the definition of f_η in Equation (27.13), its ξ-line may be moved right from 2 to any fixed real part $\sigma_R > 2$. The reciprocal there is in absolute convergence, the correction is holomorphic and bounded, and the Gaussian gives rapid vertical decay. It follows that for every $A_0 > 0$,

$$f_\eta(Z) = O_{A_0}(Z^{A_0}) \qquad (0 < Z \le 1), \tag{27.23}$$

by taking $\sigma_R > A_0 + 8/15$. Thus this endpoint estimate has used no zero exclusion to the left of one.

Consider

$$\mathcal{T}(s) = \int_0^\infty f_\eta(Z) Z^{-C_*(s)} \, \frac{dZ}{Z}. \tag{27.24}$$

By Equations (27.22) and (27.23), it converges locally uniformly, and hence defines a holomorphic function, for

$$\operatorname{Re} s > 1 - \frac{1}{20000}.$$

Indeed at infinity the real exponent is $1 - 1/20000 - \operatorname{Re} s < 0$, and at zero Equation (27.23) allows an arbitrarily large exponent. The same majorants with powers of $\log Z$ justify differentiation under the integral.

We identify this transform on its original line without a formal contour interchange. Put $t = \log Z$. From Equation (27.13),

$$e^{-C_*(2)t} f_\eta(e^t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\tau t} \mathscr{A}(2 + i\tau) \, d\tau.$$

The function $\mathscr{A}(2 + i\tau)$ is continuous and rapidly decreasing by the Gaussian and the bounded Euler factors. The endpoint bounds on f_η make the left side integrable in t. Fourier inversion therefore gives

$$\mathcal{T}(2 + i\tau) = \mathscr{A}(2 + i\tau) \qquad (\tau \in \mathbb{R}). \tag{27.25}$$

For $\operatorname{Re} s > 1 - 1/20000$, the second neighborhood in Proposition 26.3 makes the numerator

$$e^{(s-5/6)^2} \mathcal{H}_\eta(s)$$

holomorphic and nowhere zero. The Hecke continuation makes $\mathscr{A}(s)$ meromorphic there. The identity theorem, first on $\operatorname{Re} s > 1$ using Equation (27.25) and then for meromorphic functions on the connected larger half-plane, identifies $\mathscr{A}$ with the holomorphic function $\mathcal{T}$. A zero of $L^S(s, \eta)$ in that half-plane would be a pole of $\mathscr{A}$, since its numerator is nonzero, which is impossible. A pole of the principal L-function simply makes its reciprocal vanish.

The finite factors removed from $L_F(s, \eta)$ are either one or $1 - \eta(\mathfrak{p}) q_\mathfrak{p}^{-s}$ with $|\eta(\mathfrak{p})| = 1$; their zeros have real part zero. Thus the same zero exclusion holds for the full $L_F(s, \eta)$. The obtained half-plane contains $\operatorname{Re} s > 1 - 10^{-6}$. All onsets and constants in the proof were allowed to depend on F, S, η, but the numerical saving $1/20000$ is absolute. A field-dependent onset in Equation (27.22) affects only a finite portion of the integral and does not change its half-plane of convergence. Since the initial field and finite-order character were arbitrary in the scope of Theorem 2.2, this proves that theorem.

28 Conclusion

The square condition (1.1) is necessary and sufficient for a nonempty finite set of integers at least 2 to be primitive roots simultaneously at a positive lower proportion of primes. Sufficiency follows by constructing quadratic-nonresidue classes, counting candidates with a large prime factor of $p-1$, and removing the possible order obstructions. Necessity follows from the Legendre symbol of an odd product.

The lower bound holds in every sufficiently large dyadic interval, so the result gives infinitely many suitable primes for each fixed admissible set. It also gives the prime-base conclusion of Corollary 8.2. No assertion about the existence of an exact limiting density or its value is required.